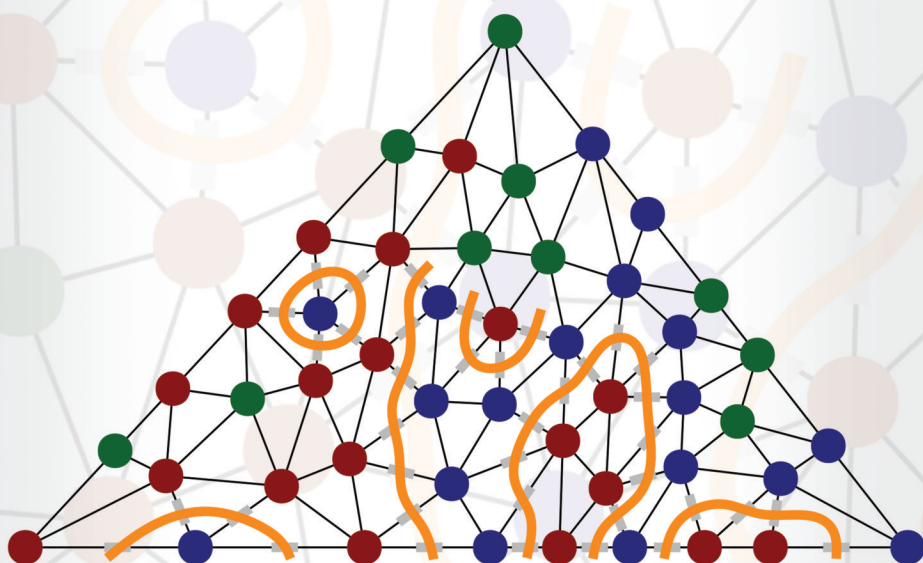


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Volume 80

Game Theory

A Playful Introduction

Matt DeVos
Deborah A. Kent



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Dedicated to my family, MD.
For Mom, in memory with love, DK.

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Preface

The story of this book began in 2002 when Matt, then a postdoc at Princeton University, was given the opportunity to teach an undergraduate class in game theory. Thanks largely to the 2001 release of a Hollywood movie on the life of the famous Princeton mathematician and (classical) game theorist John Nash, this course attracted a large and highly diverse audience. Princeton's mathematics department featured not only Nash, but also John Conway, the father of modern combinatorial game theory. So it seemed only natural to blend the two sides of game theory, combinatorial and classical, into one (rather ambitious) class. The varied backgrounds of the students and the lack of a suitable textbook made for an extremely challenging teaching assignment (that sometimes went awry). However, the simple fun of playing games, the rich mathematical beauty of game theory, and its significant real-world connections still made for an amazing class.

Deborah adopted a variant of this material a few years later and further developed it for a general undergraduate audience. Over the ensuing years, Deborah and Matt have both taught numerous incarnations of this course at various universities. Through exchange and collaboration, the material has undergone a thorough evolution, and this textbook represents the culmination of our process. We hope it will provide an introductory course in mathematical game theory that you will find inviting, entertaining, mathematically interesting, and meaningful.

Combinatorial game theory is the study of games like Chess and Checkers in which two opponents alternate moves, each trying to win the game. This part of game theory focuses on deterministic games with full information and is thus highly amenable to recursive analysis. Combinatorial game theory traces its roots to Charles Bouton's theory of the game Nim and a classification theorem attributed independently to Roland Sprague and Patrick Grundy. The 1982 publication of the classic *Winning Ways for Your Mathematical Plays* by Elwyn Berlekamp, John Conway, and Richard Guy laid a modern foundation for the subject—now a thriving branch of combinatorics. In contrast, classical game theory is an aspect of applied mathematics frequently taught in departments of economics. Classical game theory is the study of strategic decision-making in situations with two or more players, each of whom may affect the outcome. John von Neumann and Oskar Morgenstern are commonly credited with the foundation of classical game theory in their groundbreaking work *Theory of Games and Economic Behavior* published in 1944. This treatise established a broad mathematical framework for reasoning about rational decision-making in a wide variety of contexts and it launched a new branch of academic study. Although there have been many significant developments in this theory, John Nash merits mention for his mathematical contributions, most notably the Nash Equilibrium Theorem.

Traditionally, the classical and combinatorial sides of game theory are separated in the classroom. A strong theme of strategic thinking nonetheless connects them and we have found the combination to result in a rich and engaging class. The great fun we have had teaching this broad mathematical tour through game theory undergirded our decision to write this book. From the very beginning of this project, our goal has been to give an honest introduction to the mathematics of game theory (both combinatorial and classical) that is accessible to an early undergraduate student. Over the years, we have developed an approach to teaching combinatorial game theory that avoids some of the set-theoretic complexities found in advanced treatments yet still holds true to the subject. As a result, we achieve the two cornerstones of the Sprague-Grundy Theorem and the Simplicity Principle in an efficient and student-friendly

manner. The classical game theory portion of the book contains numerous carefully sculpted and easy-to-follow proofs to establish the theoretical core of the subject (including the Minimax Theorem, Nash arbitration, Shapley Value, and Arrow's Paradox). Most significantly, Chapter 9 is entirely devoted to an extremely gentle proof of Nash's Equilibrium Theorem. For the sake of concreteness, the chapter focuses on 2×2 matrices, but each argument generalizes and Appendix C contains full details. Sperner's Lemma appears in this chapter as the first step of our proof and we offer an intuitive exposition of this lemma by treating it as a game of solitaire. More broadly, Sperner's Lemma provides a touchstone through other chapters. In addition to using it to prove Nash's Equilibrium Theorem, we also call on it to show that the combinatorial game Hex cannot end in a draw. Later still, Sperner's Lemma allows us to construct an envy-free division of cake.

Beyond including both combinatorial and classical theory, we have sought to provide a broad overview of (both sides of) the subject. Within the world of combinatorial game theory, we begin at a very high level of generality with game trees and Zermelo's Theorem—concepts that apply to Chess, Checkers, and many other 2-player games. We also introduce some widely applicable ideas such as symmetry and strategy stealing before specializing in normal-play games to develop the heart of the theory. On the classical side, in addition to the essential mathematical concepts, we tour a variety of exciting supplementary topics including the Folk Theorem, cake cutting, and stable marriages. Furthermore, we have devoted considerable effort to connecting the theory with applications. Chapter 7 focuses on the modeling capability of a game-theoretic framework in the context of sports, biology, business, politics, and more!

One of our primary goals in this book is to enhance the mathematical development of our student readers. Indeed, we aim to take advantage of the naturally stimulating subject of game theory to teach mathematics. We have found that blending combinatorial and classical game theory has great pedagogical advantages. Beginning with combinatorial games means that student pairs are playing and recursively analyzing games right from the start. These games are not only fun to play, but they provide a perfect environment for working with game trees, proving theorems by induction, and starting to think strategically. This part

of the book features numerous rich examples of proofs by induction and also a number of interesting proofs by contradiction. Turning to classical game theory, we encounter basic probability, linear algebra, and convexity in our study of zero-sum matrix games. Our later chapters on general games continue to emphasize probability and geometric methods but also introduce questions of modeling as well as plentiful applications. The proof of Nash's Equilibrium Theorem involves a nice blend of combinatorial and continuous mathematics in addition to a taste of topology. Whenever a significant new mathematical concept is required, we pause to introduce it; accordingly, this book contains elementary introductions to proofs by induction, proofs by contradiction, probability, and convexity.

We have constructed this textbook for a one-semester undergraduate course aimed at students who have already taken courses in differential calculus and linear algebra. However, we have found this material adaptable to a variety of situations and a range of audiences. In particular, most of the book does not directly call upon either calculus or linear algebra and is thus suitable for students who lack these prerequisites but have a similar level of sophistication. Indeed, calculus is used very rarely, and for a capable student without linear algebra, only the proofs of the Minimax and Equilibrium Theorems would be out of reach after a quick introduction to matrix multiplication. The complete book is likely more material than can be comfortably covered in a standard undergraduate semester 3-credit course. To allow the instructor considerable flexibility in content choices, we have limited dependencies between the chapters (see the diagram on page xv). These limited dependencies also allow for portions of this book to be used in other contexts. For instance, the first four chapters on combinatorial games provide an appealing theme for an introductory proofs course, Chapters 5 and 6 on zero-sum matrix games together with Appendix B on linear programming make a nice addition to a linear algebra course, and all three sections in Chapter 12 can be taught independently.

Further to assist the instructor, each chapter ends with a generous supply of exercises. We have sought to include problems at a variety of levels from basic skills all the way up to challenging proofs, with espe-

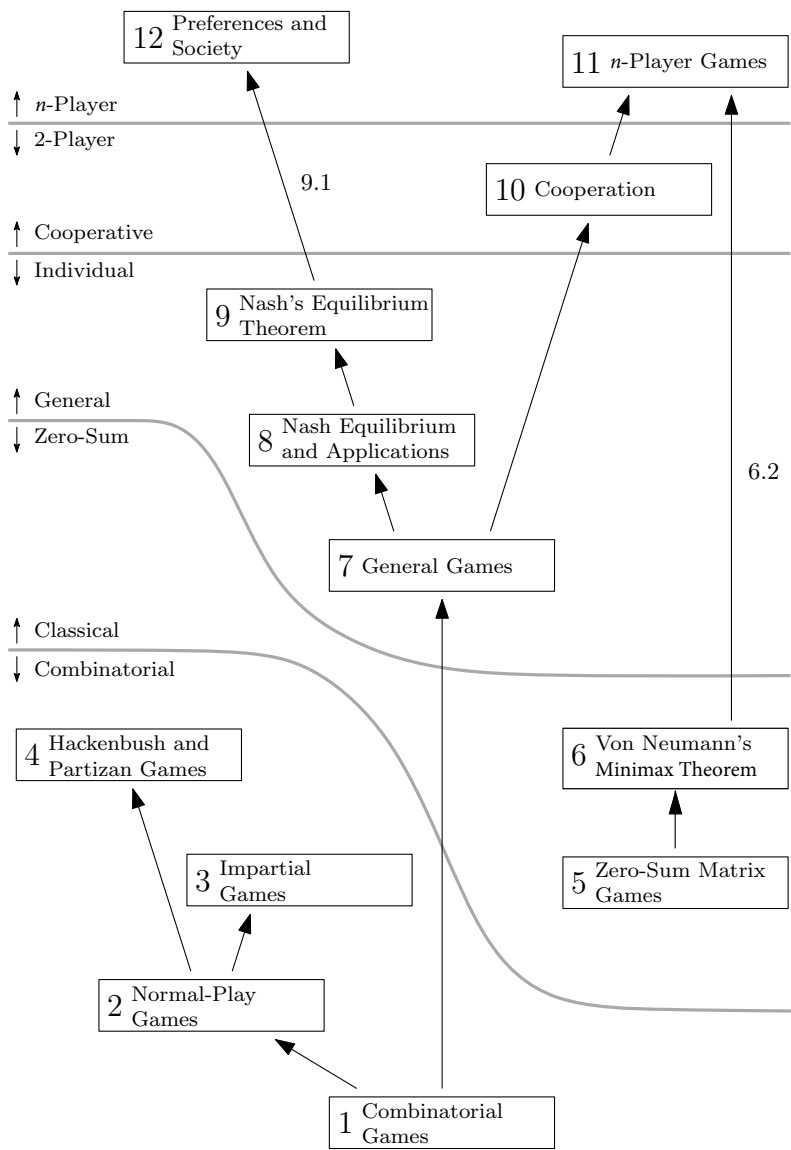


Figure 0.1. Implication Diagram

cially difficult exercises marked with the symbol *. References to exercises in the same chapter are by exercise number, while those to exercises in another chapter also include the chapter number. In addition, game boards and further supplementary material can be found online at

www.ams.org/bookpages/stml-80.

This book owes its existence to the many amazing teachers from whom we have been fortunate to learn. Matt's genesis as a combinatorialist is thanks to his incomparable PhD supervisor, Paul Seymour. He also benefited from an inspiring introduction to combinatorial games from John Conway and a detailed initiation to the mathematics of classical game theory under the guidance of Hale Trotter. Deborah deeply appreciates her inimitable dissertation advisor, Karen Parshall, who introduced her to the joys and labors of academic mathematics. She also thanks Tom Archibald for his generous support of this and her other postdoctoral projects. We are so grateful to many of our friends and colleagues who have influenced the development of this book either directly or indirectly: Derek Smith, Drago Bokal, Francis Su, Claude Tardif, and Dave Muraki top this list, but there are countless others. We owe a debt of gratitude to the universities that made it possible for us to teach versions of this class and to the many students who helped to shape this material with their questions, comments, and corrections. We would also like to thank Ina Mette, Arlene O'Sean, Courtney Rose, and the rest of the editorial staff at the AMS whose careful work on our manuscript dramatically improved the final product. Finally, we thank our friends and especially our families for their amazing support throughout the extensive process of creating this book. Although it has taken far more effort and energy than we could ever have foreseen, writing this book has been a labor of love for us. We hope you will enjoy it, too!

Chapter 1

Combinatorial Games

The best way to get a feel for combinatorial games is to play some! Try playing these games in which two players alternate making moves.

Game 1.1 (Pick-Up-Bricks). This game is played with a pile of bricks. Each move consists of removing 1 or 2 bricks from the pile. The game ends when the pile is empty, and the last player to take a brick wins.

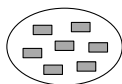


Figure 1.1. A 7-brick position in Pick-Up-Bricks

Game 1.2 (Chop). Start with an $m \times n$ array viewed as a plank that is secured only at the lower left corner. On each turn, a player must either make a vertical or horizontal chop of the plank, and then any piece no longer connected to the lower left corner falls off into the water. The last player to make a move wins.

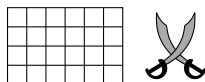


Figure 1.2. A 4×6 position in Chop

Game 1.3 (Chomp). Start with an $m \times n$ array viewed as a chocolate bar, but with the lower left corner square poisoned. On each turn, a player chooses a square and eats this square and all other squares that lie above and to the right of this one (i.e. the northeast corner). The last player to eat a nonpoison square wins.

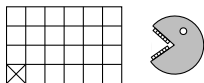


Figure 1.3. A 4×6 position in Chomp

Game 1.4 (Hex). This is a game played on the board pictured in Figure 1.4.¹ Similar to the widely familiar game Tic-Tac-Toe, on each turn a player marks an empty hexagon. One player uses $*$ as his mark, while the other uses $\circ$. If the player using $*$ can form a chain of hexagons with this mark connecting the left and right sides of the board, that player wins. The player using $\circ$ will win by forming a chain of hexagons with this mark connecting the top and bottom.

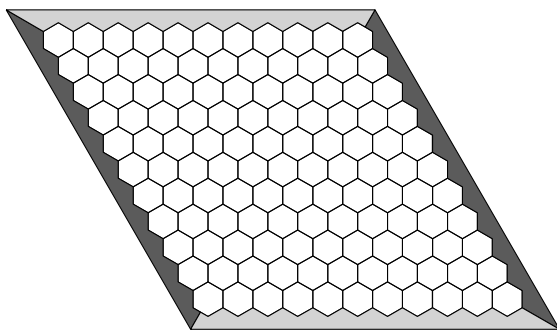


Figure 1.4. The board for Hex

Now that we have a handful of combinatorial games in mind, let's introduce some formal definitions that will allow us to treat these objects mathematically.

¹A large version of this game board (as well as the pictured positions in Chop and Chomp) can be found at the end of this book and also online at www.ams.org/bookpages/stml-80.

Definition 1.5. A *combinatorial game* is a 2-player game played between Louise (for **Left**) and Richard (for **Right**).² The game consists of the following:

- (1) A set of possible *positions*. These are the states of the game (e.g. a 2×3 board in Chop).
- (2) A *move rule* indicating for each position what positions Louise can move to and what positions Richard can move to.
- (3) A *win rule* indicating a set of *terminal positions* where the game ends. Each terminal position has an associated *outcome*, either Louise wins and Richard loses (denoted $+-$), Louise loses and Richard wins ($-+$), or it is a draw (00).

Observe that the definition of a combinatorial game does not indicate which player moves first, nor does it explicitly state which position is the starting position. To *play* one of these games, we choose a starting position and designate a player to move first. From then on, the players alternate making moves until a terminal position is reached and the game ends. In three of the games we have seen so far, Pick-Up-Bricks, Chop, and Chomp, the loser is the first player to have no available move. This is a common win rule, and we call a combinatorial game with this win rule a *normal-play* game.

Although numerous common games like Checkers and Chess are combinatorial games, there also exist many games that are not combinatorial games. Notably, combinatorial games have no element of randomness—so die rolls or spinners cannot be used to determine actions. Combinatorial games also require each player to have full information about the position of the game. Later in Chapter 7, we will broaden our horizons and introduce some of these variations. For now, though, we restrict ourselves to combinatorial games.

1.1. Game Trees

In this section, we will introduce a powerful tool called a game tree, which will be helpful for understanding the play of a game. We will

²These names are chosen in honor of Richard Guy, one of the founders of modern combinatorial game theory, and his wife, Louise.

begin by seeing how to model the play of any combinatorial game in this manner.

Modeling Play. There is a natural way to depict all possible sequences of moves in the play of a game using a tree, where each branch node models a choice point for one of the players and every terminal node indicates an outcome. This construct will prove extremely useful for us as we start to think strategically. We first introduce the simple game of Tic and consider its game tree.

Game 1.6 (Tic). This is a combinatorial game similar to Tic-Tac-Toe but played on a 1×3 array. To move, Louise marks an empty square with a $\circ$ and Richard marks an empty square with a $\times$. If Richard or Louise gets two adjacent squares marked with his or her symbol, then he or she wins. If all squares get marked without this happening, the games ends in a draw.

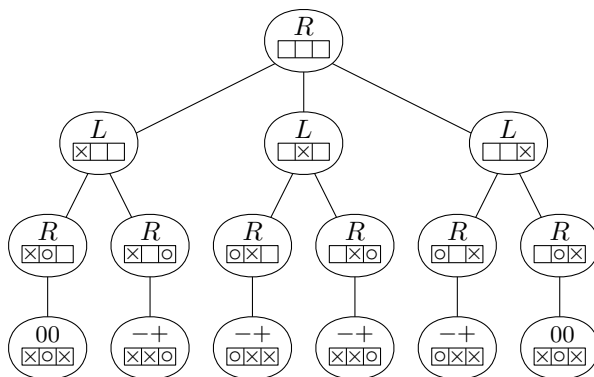


Figure 1.5. A game tree for Tic

Our game tree in Figure 1.5 depicts every possible sequence of moves starting from a blank board with Richard moving first. Observe that each node of the game tree contains the current position of the game, so we can see the positions updating as moves are made and we work downward in the tree. Each branch node also contains either an L or R to indicate that it is either Louise or Richard's turn to play. The terminal

nodes indicate that the game has ended with an outcome, either a win for Richard $-+$, a win for Louise $+-$, or a draw 00 . The topmost node containing the starting position is called the *root* node.

More generally, we can model any combinatorial game with this process. We will call these figures *game trees* and they will prove quite helpful for our analysis.

Procedure 1.7 (Build a Game Tree). To make a game tree starting at position α with Louise moving first, we begin by making a root node containing an L (since Louise is first) and an α (since this is the starting position). If Louise can move to positions $\alpha_1, \dots, \alpha_k$, then we join k new nodes to the root node; each one of the new nodes contains one of these α_i positions. If any of these positions is terminal, then we put the appropriate outcome (either $+-$, $-+$, or 00) in this node. For the other nonterminal positions, it will be Richard's turn to play, so all of these nodes will contain an R . Continue this process until it is complete.

W-L-D Game Trees. We introduced the game tree in Figure 1.5 as a way to model the play of Tic. However, once we have this tree in hand, we could actually play it instead of the game. Rather than starting with an empty 1×3 array and having Louise and Richard alternately mark boxes with their symbols, we could start at the root node of this game tree and descend it by having Louise choose at the nodes marked L and having Richard choose at the nodes marked R . As you can see, these two different ways of playing are essentially equivalent.

Once we are operating in this game tree model, the position information that is contained in each node is superfluous. Indeed, all that our players need to play this game tree is the information concerning which player has a choice to make at each branch node and what the outcome is at each terminal node. Ignoring the position information for the game Tic gives us the tree depicted in Figure 1.6.

This type of game tree without the position information is extremely useful, so we shall give it a name. We define a **W-L-D game tree** (for **Win-Lose-Draw**) to be a tree with a distinguished root node (as the starting position) in which each terminal node contains an outcome ($+-$, $-+$, or 00) and each branch node contains either an L or an R , indicating which

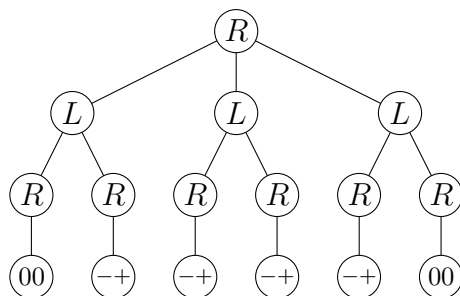


Figure 1.6. A W-L-D game tree for Tic

player moves by making a choice at that node. The nice property of W-L-D game trees is that they give us a unified way to think about the play of combinatorial games. So, instead of having to consider players taking tokens and marking squares and eating chocolate, we can always view play in terms of descending a W-L-D game tree.

It is possible to have a W-L-D game tree consisting of just a single terminal node (so neither player has any decisions to make and the outcome is already decided). This wouldn't be much fun to play in practice, but it will be convenient for our theory. There are three such trees with no moves (see Figure 1.7) and we will call them *trivial* game trees.



Figure 1.7. The trivial W-L-D game trees

There is another extreme to consider... it is possible that our game tree could go on forever and never end! Although the definition does not force combinatorial games to end, we will restrict our attention only to games that must end after a finite number of moves. Accordingly, we will always assume that W-L-D game trees are finite.

Strategy. When playing a game, we like to win! To do so, we will want a plan. The idea of strategy formalizes this notion of a plan. The term strategy is familiar to game players everywhere and can be used to indicate a general principle of play—in chess, for example, one may wish to

“control the center”—but we will adopt a more refined and specific usage of this term. We define a *strategy* for a player in a W-L-D game tree to be a set of decisions indicating which move to make at each node where that player has a choice. In the game tree in Figure 1.8, we have depicted a strategy for Richard by boldfacing the edges indicating his choices.

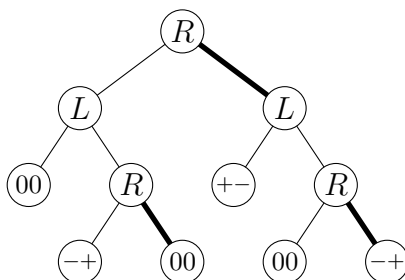


Figure 1.8. A strategy for Richard

In the play of our game, a player may *follow* a strategy by using it to make every decision. Note that the strategy indicated for Richard in Figure 1.8 is not a terribly good one since it gives Louise the opportunity to move to a terminal node with outcome $+-$ where Richard will lose. Richard would do better to follow a strategy that makes the opposite choice at the root node since then it will be impossible for him to lose.

We will generally be interested in finding good strategies for our players. The best we could hope for is a strategy that guarantees a win for a player who follows it. We will call any such strategy a *winning* strategy. Next best would be a strategy that guarantees a player doesn't lose. A strategy with the property that following it guarantees either a win or a draw is called a *drawing* strategy. Note that a player following a drawing strategy could end up winning; the only guarantee is that this player will not lose.

Let us note that there may be some extraneous information included in a strategy as we have defined it. For example, in the strategy for Richard depicted in Figure 1.8, Richard will choose the right branch as his first move. As a result, he will never encounter the node in the lower left part of the tree labeled R . Since he will never encounter this node, it

may seem unnecessary for him to decide what to do there. Nevertheless, our definition of strategy includes the decision Richard would make at every node labeled R . The simplicity of this definition makes strategies easier to work with.

Working Backwards. Let's now assume that our players are highly rational with perfect foresight and consider how they might play in a large W-L-D game tree. From the nodes at the top of the tree, it is not clear what choices either player would prefer since there are so many decisions still ahead. In contrast, for the nodes close to the bottom it is much easier to see how best to play. Consider the choice for Richard in the game tree depicted in Figure 1.9. It is clear that from here Richard will choose the right branch for a win.

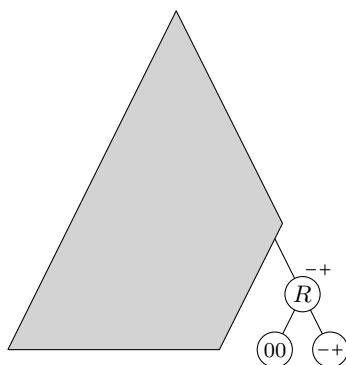


Figure 1.9. An easy decision for Richard

Since we now know that reaching Richard's decision node in Figure 1.9 will result in a win for Richard, we have written a " $-+$ " next to this node. In fact, we can now think of this as a new terminal node and then apply a similar process elsewhere in the game tree. Next we formalize this approach.

Procedure 1.8 (Working Backwards). Suppose one of our players has a decision to make at node N . Suppose also that we have already determined what the outcome will be under rational play for all possible nodes from node N . Then choose a best possible outcome for this player, indicate this choice by darkening this edge, and then mark the node N

with the resulting outcome. Continue this process until the root node has been marked with an outcome.

Figure 1.10 shows the result of carrying this procedure to completion on a larger game tree. Note in this figure that the root node has been labelled $-+$. This means that under rational play, Richard will win this game.

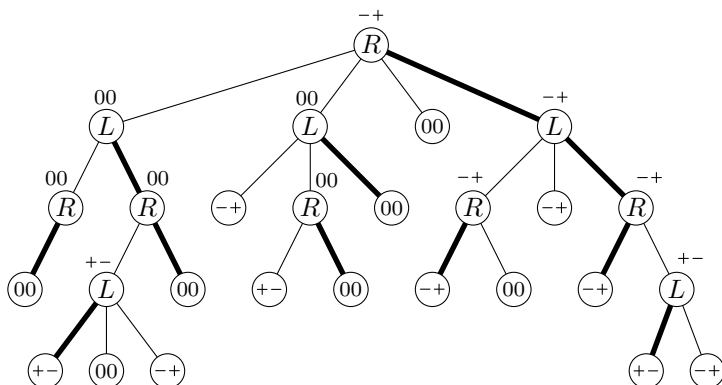


Figure 1.10. Rational decision-making

In fact, as you may readily verify, the strategy for Richard indicated in Figure 1.10 is a winning strategy. In the following section, we will show that this procedure works more generally for any W-L-D game tree. More precisely, when we apply our working backwards procedure to any W-L-D game tree, we either get a $+-$ on the root node and a winning strategy for Louise or a $-+$ on the root node and a winning strategy for Richard or a 00 on the root node and a drawing strategy for each player.

1.2. Zermelo's Theorem

In this section, we will prove a famous theorem due to Ernst Zermelo. This theorem tells us that in every combinatorial game, either Louise has a winning strategy or Richard has a winning strategy or both players have a strategy to guarantee them a draw. In the process of proving this theorem, we will develop the tools to prove that the working backwards procedure described above really does work as claimed on game trees of

all sizes. The proof of Zermelo's Theorem is based on the mathematical principle of induction, so we begin with a brief discussion of this important concept.

Mathematical Induction. Induction is an extremely powerful tool for proving theorems. The simplest proofs by induction are used to prove properties of the nonnegative integers. For instance, suppose that $P(n)$ is a certain property of the number n that we wish to prove holds true for every $n \geq 0$. Since there are infinitely many nonnegative integers, it would be impossible to make a new proof for each individual one! Mathematical induction instead provides a general method to prove that $P(n)$ holds true for every $n \geq 0$.

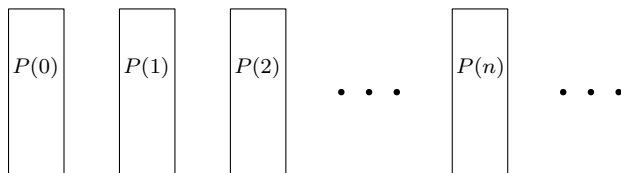


Figure 1.11. A proof by induction

The inductive approach is to view $P(0)$, $P(1)$, $P(2)$, ... as dominoes. We will think of knocking a domino over as showing that property P holds true for integer n . The proof involves two stages. We first show that the first domino falls over—that is, we prove that $P(0)$ is true. This part is called the *base case*. The second part is to prove that for every $n \geq 1$, if all the dominos before the n^{th} domino fall, then the n^{th} domino also falls. That is, we must prove that if $P(k)$ is true for all $k < n$ (this is the *inductive hypothesis*), then $P(n)$ is also true.³ This part is called the *inductive step*. Of course, if this happens, every domino will be knocked over.

Base Case: $P(0)$ is true.

Inductive Step: If $P(k)$ is true for all $k < n$, then $P(n)$ is true.

³Technically speaking, we are introducing “strong induction” here since our inductive assumption is that $P(k)$ holds for all $k < n$. In “weak induction” this is replaced by the weaker assumption that $P(n - 1)$ holds. These two principles are logically equivalent, but this text frequently utilizes the strong form, so that is what we will adopt throughout.

Let's consider a straightforward example that exhibits a nice property of positive integers.

Example 1.9. For every $n \geq 0$, the sum of the first n odd integers is n^2 . For the proof, we proceed by induction on n . To verify the base case, observe that $0^2 = 0$ is the sum of the first 0 odd integers. For the inductive step, let $n \geq 1$ be an arbitrary integer, and assume that our formula holds true for all nonnegative integers less than n . In particular, the formula holds for $n - 1$, which means

$$1 + 3 + 5 + \cdots + (2(n - 1) - 1) = (n - 1)^2.$$

Starting with this equation and adding $2n - 1$ to both sides gives us

$$1 + 3 + 5 + \cdots + (2n - 3) + (2n - 1) = (n - 1)^2 + (2n - 1) = n^2$$

and this completes the proof.

Notice in this example that property $P(n)$ is never formally defined in the proof. Nevertheless, the idea is there: $P(n)$ is the property that the sum of the first n odd numbers is n^2 . This proof handles the base case by verifying $P(0)$, or showing that the sum of the first 0 integers is 0^2 (as usual, the base case was the easy part). For the inductive step, we assumed $P(k)$ to be true for all $k < n$ (our inductive hypothesis) and then used this to prove that $P(n)$ holds true for every $n \geq 1$. The inductive hypothesis gave us the advantage of starting with the equation for $P(n - 1)$, from which we then deduced $P(n)$.

Mathematical induction enjoys extremely wide application, well beyond proving nice properties of positive integers. Indeed, we will later see more involved instances of induction at work in our investigations of combinatorial games. The general context in which induction applies involves some property P , which we want to show holds true for infinitely many things. To proceed by induction, we will want to organize these things into different *sizes* (i.e. some things have size 0, some have size 1, etc.). Now, instead of trying to prove all at once that all of these things satisfy property P , we can proceed by induction on the size. The base case will be to prove that P is true for all things of size 0. Then, for the inductive step, we will assume (the inductive hypothesis) that P is true for all things of size less than n and use this to show that P then holds true for an arbitrary thing of size n .

In some cases where induction applies, the smallest relevant size might not be 0 and could instead be 1 or 2 or something else. Most generally, the base case involves proving the result for the smallest size that makes sense in context. The inductive step handles all things of larger size.

Proof of Zermelo’s Theorem. With the concept of induction in hand, we are now ready to give a proof of Zermelo’s famous theorem. Our proof will rely upon an inductive argument that applies to W-L-D game trees, so we will need to decide upon a “size” for these trees. We will use a quantity called depth as the size of a game tree. The *depth* of a game tree is the maximum number of possible moves from the start to the end of the game (e.g. the tree in Figure 1.10 has depth 4). Since every game tree is finite, every game tree has some depth and that depth will always be a nonnegative integer. This sets us up to use induction on depth to prove common properties of all game trees.

To prove a property P of trees by induction on depth, we first prove the base case, that P is true for all trees of depth 0. For the inductive step, we need to prove that P holds for an arbitrary tree of depth $n > 0$ under the inductive assumption that P holds true for all trees with depth less than n . Next we’ll see this idea in action in the proof of Zermelo’s Theorem. This theorem introduces a new definition, the *type* of a W-L-D game tree, and it establishes that every possible W-L-D game tree has one of three types.

Theorem 1.10 (Zermelo). *Every W-L-D game tree is one of*

Type	Description
$+-$	<i>Louise has a winning strategy.</i>
$-+$	<i>Richard has a winning strategy.</i>
00	<i>Both players have drawing strategies.</i>

Proof. We proceed by induction on the depth of the game tree. As a base case, observe that if the tree has depth 0 (i.e. it is trivial), then the game is already decided and it is either $+-$ in which case Louise has a winning strategy, $-+$ in which case Richard has a winning strategy, or 00 in which case both players have drawing strategies.

For the inductive step, let T be a W-L-D game tree of depth $n > 0$ and assume the theorem holds for every tree with smaller depth. Suppose that Richard has the first move in T (the case where Louise has the first move follows from a similar argument) and he can move to one of the nodes $N_1, N_2, \dots, N_\ell$.

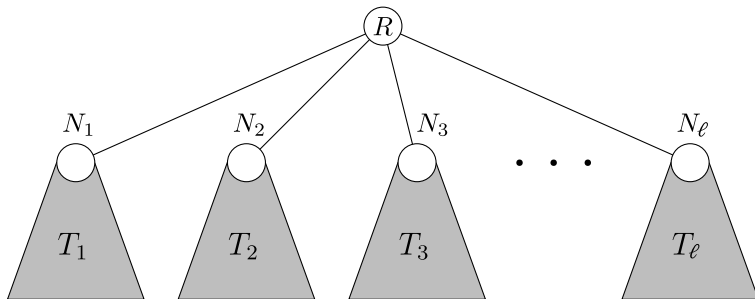


Figure 1.12. The first move in a game tree

We can consider each node N_i as the root of a new W-L-D game tree T_i , consisting of N_i and all the nodes below it. Since each T_i has depth $< n$, our inductive hypothesis tells us that all of these games must satisfy the theorem (i.e. must be type $+-$, $-+$, or 00). So, for every $1 \leq i \leq \ell$ we may choose strategies $\mathcal{L}_i$ for Louise and $\mathcal{R}_i$ for Richard in the game tree T_i with the property that either one of these strategies is winning or both are drawing. We form a strategy $\mathcal{L}$ for Louise in the original game tree by combining $\mathcal{L}_1, \dots, \mathcal{L}_\ell$. To form a strategy $\mathcal{R}$ for Richard, we will combine $\mathcal{R}_1, \dots, \mathcal{R}_\ell$ but we will also need to make a decision at the root node. Next we split into cases.

Case 1. *At least one of $T_1, \dots, T_\ell$ is type $-+$.*

Let T_i be type $-+$. Then Richard's strategy $\mathcal{R}_i$ is winning in T_i and we may form a winning strategy $\mathcal{R}$ in the original game by having Richard play to node N_i .

Case 2. *All of $T_1, \dots, T_\ell$ are type $+-$.*

In this case, every $\mathcal{L}_i$ strategy is winning, so $\mathcal{L}$ is a winning strategy for Louise.

Case 3. *None of $T_1, \dots, T_\ell$ is type $-+$, but at least one is type 00 .*

Let T_i be type 00. Then Richard's strategy $\mathcal{R}_i$ is drawing and we may form a drawing strategy $\mathcal{R}$ in the original game by having Richard play to N_i . Since none of $T_1, \dots, T_\ell$ is type $-+$, each of Louise's strategies $\mathcal{L}_1, \dots, \mathcal{L}_\ell$ is drawing or winning, and it follows that $\mathcal{L}$ is a drawing strategy for Louise. $\square$

So, Zermelo's Theorem gives us a classification of game trees into types $+-$, $-+$, and 00. Furthermore, the proof of this result implies that our "Working Backwards" technique from the previous section will always have one of the following results.

Corollary 1.11. *For every W-L-D game tree applying the Working Backwards procedure results in one of the following:*

<i>root label</i>	<i>result</i>
$+-$	<i>A winning strategy for Louise</i>
$-+$	<i>A winning strategy for Richard</i>
00	<i>Drawing strategies for both players</i>

For the purposes of analyzing small games, constructing a game tree is a convenient way to determine whether one of the players has a winning strategy or if they both have drawing strategies. For large games like Chess, it is theoretically possible to construct a game tree⁴. However, the number of positions in Chess has been estimated at approximately 10^{120} while the number of atoms in the universe is around 10^{80} ... so our universe isn't big enough for such analysis! The fact that such a game tree does exist for Chess nevertheless means that Zermelo's Theorem applies to it. So, in Chess, either one of the two players has a winning strategy or both have drawing strategies ... we just don't know which of these it is.

1.3. Strategy

Game trees can be a useful tool to determine who will win when playing from a particular position in a game such as Chop. However, Chop boards come in infinitely many sizes, so there are infinitely many different game trees, and this approach will be limited at best! In this section

⁴Although it is possible to repeat positions in a game of chess, there are certain lesser-known rules that make it a finite game.

we will introduce a pair of useful ideas—namely symmetry and strategy stealing—that can help us to determine which player has a winning strategy without having to analyze the game tree. In particular, we will learn exactly which player has a winning strategy in every Chop position. Although these techniques do not apply to all games, they are powerful when they work.

Before we introduce these new ideas, let us pause to discuss how we represent or define a strategy. Suppose that we are interested in playing a certain position in a combinatorial game (with someone going first). To describe a strategy for a player, it is possible to construct the associated game tree and then depict the strategy there. However, this is a bit tedious, so it will be helpful for us to adopt a more relaxed treatment. Accordingly, we will generally describe a strategy for a player (in words) by giving a rule that tells this player what to do at each possible position. Given a strategy described in this manner, we could form the game tree and depict it there, but there is generally no need to do this.

Symmetry. The key to finding winning strategies in certain games is symmetry of the positions. Indeed, this simple idea is the key to understanding both Chop and Pick-Up-Bricks. By definition, every position in the game of Chop is a rectangle of the form $m \times n$. Some of these rectangles have the additional symmetry of being square (so $m = n$) and these positions are the key to understanding this game.

Proposition 1.12. *Consider an $m \times n$ position in Chop.*

- (1) *If $n = m$, the second player has a winning strategy.*
- (2) *If $n \neq m$, the first player has a winning strategy.*

Proof. First we prove that the second player has a winning strategy whenever the initial position is square. This winning strategy for the second player is easy to describe: On each turn, move the position to a square one. Assuming the second player does this, every time the first player has a move to make (including the first) the position will be a square and any move is to a nonsquare position. Note that the second player can always move a nonsquare position to a square one. Assuming this is done, the second player will eventually move to a 1×1 position and win the game.

A similar idea reveals a first-player winning strategy when the initial position is not square. On the first turn, the first player may move the board to a square position. From there, that player may adopt the above strategy (always moving to a square position). This will guarantee the first player a win. $\square$

In Pick-Up-Bricks, the positions where the number of bricks is a multiple of 3 are the symmetric positions, and they play the same role as the square positions in Chop.

Proposition 1.13. *Consider a Pick-Up-Bricks position of n bricks.*

- (1) *If 3 divides n , the second player has a winning strategy.*
- (2) *Otherwise, the first player has a winning strategy.*

Proof. For the first part, the following strategy is winning for the second player: On each turn, do the opposite of the first player's move. So, if the first player picks up one brick, then the second player picks up two, and if the first player picks up two, then the second player picks up one. This ensures that after both players have played, the total number of bricks is three fewer and the new position is again a multiple of 3. Following this, the second player will eventually take the last brick and win.

The first player can win when the starting position is not a multiple of 3. To start, the first player may remove either one or two bricks to bring the position to a multiple of 3. Now the first player may adopt the second player strategy described above and win from here. $\square$

Strategy Stealing. Strategy stealing is another tool to approach the general question of which player has a winning strategy. Much how symmetry sometimes works well to understand strategy for games, strategy stealing also is very effective when it applies. However, unlike the symmetry arguments that explicitly construct winning strategies, strategy-stealing arguments prove the existence of a winning strategy without giving any indication of what this strategy is! This is because strategy-stealing arguments employ proofs by contradiction.

To create a proof by contradiction, we begin by assuming the opposite of what we are trying to prove. We then argue deductively that a necessary consequence of that assumption is something impossible.

It follows that our assumption generating this contradiction must have been false, and therefore, the opposite is true. The strategy-stealing arguments that we will introduce here rely on somewhat subtle proofs by contradiction, which will benefit from careful contemplation. Before taking them on, we begin with a warm-up proof by contradiction in the following example.

Example 1.14. Let n be an integer and assume that n^2 is even. Then n must also be even. We will prove this (admittedly easy) fact by contradiction. So, our first step will be to assume that n is odd (i.e. assume the negation of what we are trying to prove). Since we are now assuming that n is odd, we may express it as $n = 2k + 1$ for another integer k . This gives us

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

We conclude from the above equation that n^2 is odd, but this contradicts the hypothesis that n^2 is even. So, our initial assumption that n is odd has lead us to a contradiction, and thus we can conclude that n must be even, as desired.

We next use proof by contradiction to determine which player has a winning strategy in Chomp and in Hex. These proofs are called strategy-stealing arguments because we will assume (for a contradiction) that one of the players has a winning strategy, and then the other player will try to steal this strategy to win. Note again that both of these arguments will only prove the existence of a winning strategy. The proofs give no information about the specific moves that could be used to win the game.

Proposition 1.15. *For every rectangular position in Chomp except 1×1 , the first player has a winning strategy.*

Proof. Chomp cannot end in a draw, so it follows from Zermelo's Theorem that one of the two players has a winning strategy. To prove that it is the first player who has a winning strategy, we will employ a proof by contradiction. For our proof, we will assume the first player does *not* have a winning strategy. It then follows from Zermelo's Theorem that the second player does have a winning strategy, which we will call S . Now, consider what would happen if the first player removed just the

upper rightmost square and then the second player chose her move according to $\mathcal{S}$. The resulting position must be some version of one of the shapes in Figure 1.13.



Figure 1.13. Simple positions in Chomp

This position must be one from which the second player to move has a winning strategy (since the second player followed the winning strategy $\mathcal{S}$ to get here). However, this very same position could also be reached in one step! The first player could have moved the board to this position on his or her first move. Then the first player would be second to play from this position, and this means that the first player has a winning strategy. This contradicts our assumption that the first player did *not* have a winning strategy. We conclude that our assumption is false, which means this game does indeed have a winning strategy for the first player. $\square$

In Chapter 9, Exercise (11), we will show that the game of Hex cannot end in a draw. Our next theorem uses this property together with a strategy-stealing argument to show that the first player has a winning strategy.

Proposition 1.16. *The first player has a winning strategy in Hex (starting from an empty board).*

Proof. Assuming Hex cannot end in a draw (this fact is proved in Exercise (11) of Chapter 9), Zermelo's Theorem tells us that one of the two players must have a winning strategy. Let's assume (for the sake of contradiction) that the first player does not have a winning strategy. In this case we may choose a winning strategy $\mathcal{S}$ for the second player.

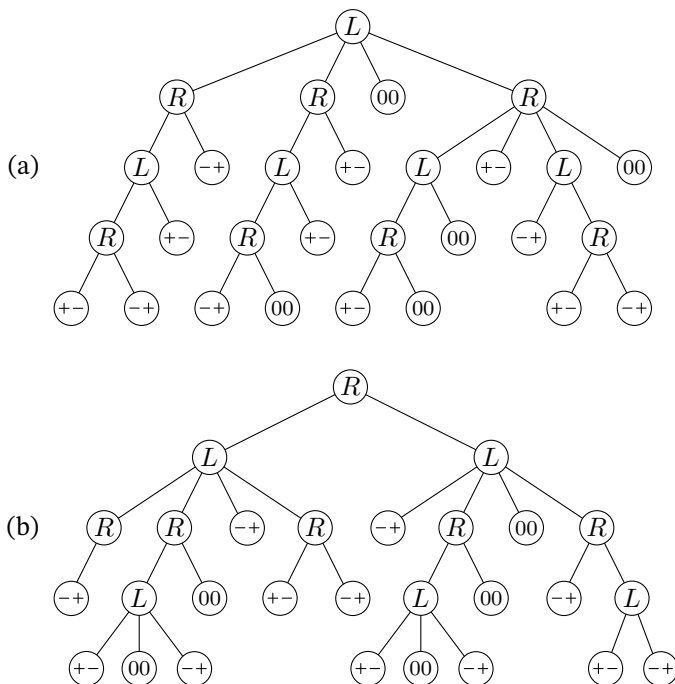
Now we will take control of the first player and we'll take advantage of the strategy $\mathcal{S}$ to win (this will give us a contradiction, thus proving $\mathcal{S}$ cannot exist). On our first move, we choose an arbitrary hexagon h and mark it with our symbol $*$. However, we will pretend that the hexagon h

has no mark and that we are the second player. This allows us to adopt the strategy $\mathcal{S}$ to make our moves. By following this winning strategy, we are guaranteed to win in the pretend version of our game. Since the true game just has one extra hexagon marked with $*$, this also gives us a win in the true game.

There is one slight complication we ignored in the above argument, but it isn't hard to fix. Namely, it might be the case that at some point the strategy $\mathcal{S}$ we are following in the pretend game instructs us to mark the hexagon h with $*$. Since h is already occupied by a $*$, this is not a possible move for us in the true game. In this case, we just choose another unoccupied hexagon h' , mark it with $*$ and now pretend that h' is empty. This permits us to keep following the strategy $\mathcal{S}$ in our pretend game. We may later end up with other pretend-empty hexagons h'' , h''' , and so on. But in any case, it follows from the assumption that $\mathcal{S}$ is a winning strategy that we win our pretend game. Since any win in the pretend game guarantees us a win in the true game, we have constructed a winning strategy for the first player. This contradiction shows that the second player does not have a winning strategy, so the first player does. $\square$

Exercises

- (1) For each game below, construct a game tree with Louise moving first from the indicated starting position.
 - (a) Tic starting from a blank board.
 - (b) Pick-Up-Bricks starting with 4 bricks.
 - (c) Chop starting from a 2×3 board.
- (2) In each W-L-D game tree, use Procedure 1.8 to find a winning strategy for one of the players or a drawing strategy for both.



- (3) For each position below, construct a game tree with Louise moving first. Then use Procedure 1.8 to find a winning strategy for one of the players.

(a) A 5-brick position in Pick-Up-Bricks.

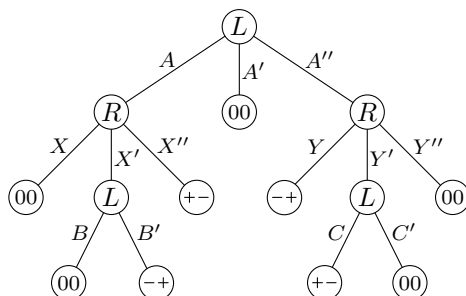
(b) The Chomp position .

- (4) Consider the labeled W-L-D game tree below. To describe a strategy for Louise, we can indicate the label corresponding to each of her choices. For instance, $AB'C$ and $A''BC'$ are strategies for Louise. Similarly, XY' and $X''Y$ are strategies for Richard.

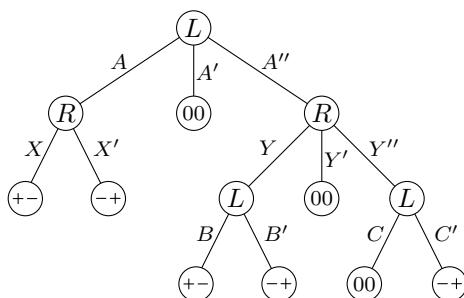
(a) Find all of Louise's strategies.

(b) Find all of Richard's strategies.

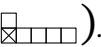
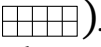
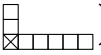
(c) Find a pair of strategies, one for Louise and one for Richard, that will produce outcome $+-$. Repeat for the outcomes $-+$ and 00 .



- (5) As in the previous problem, we may describe strategies in the game tree below using labels so $AB'C$ is a strategy for Louise and XY'' is a strategy for Richard.
- Find all of Louise's strategies.
 - Find all of Richard's strategies.
 - Which of Louise's strategies are drawing?
 - Which of Richard's strategies are drawing?



- (6) Determine if each of the following games is a combinatorial game. If a given game is not a combinatorial game, explain why not.
- Connect-Four.
 - Battleship.
 - Backgammon.
 - Go.
- (7) For every pair m, n of positive integers, determine the depth of the game tree for
- an m -brick position in Pick-Up-Bricks,

- (b) an $m \times n$ position in Chop, and
 (c) an $m \times n$ position in Chomp.
- (8) Prove the following formulas by induction on n :
 (a) $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.
 (b) $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$.
- (9) For every positive integer n let α_n be the two-row Chomp position where the top row has a single square and the bottom row has n squares (e.g. α_5 is ). Let β_n be the position in Chop given by a $2 \times n$ array (e.g. β_5 is ). Prove that the game trees for the positions α_n and β_n have the same number of nodes for every $n \geq 1$.
- (10) For any pair m, n of positive integers, define $\alpha_{m,n}$ to be the “L” shaped position in Chomp consisting of a column of m squares with the poison square at the bottom and a row of n squares with the poison square on the left (e.g. $\alpha_{3,6}$ is ). For all possible values of m and n find a winning strategy for either the first or second player.
- (11) Let T be a W-L-D game tree with the property that every time a player has a choice to make there are exactly two options. If t is the number of terminal nodes and b is the number of branch nodes (i.e. non-terminal nodes), prove that $t = b + 1$.
- (12) Consider a game tree T where Louise has n_2 nodes where she has a choice between 2 options, n_3 nodes where she has a choice between 3 options, ..., n_k nodes where she has a choice between k options (and no nodes where she chooses among more than k options). What is the formula for the total number of strategies that Louise has in T ?
- (13) Let T be a W-L-D game tree and assume that every time Louise has a decision to make in T she has exactly two options, and also assume that none of Richard’s choices can bring the game to a terminal position. Let ℓ be the number of nodes marked L and let n be the total number of nodes. Prove that $n = 3\ell$ if Louise makes a choice at the root node, and otherwise $n = 3\ell + 1$.
- (14) If N and N' are two nodes of a game tree and a player with a choice at N can move to N' , then N and N' are connected by an edge (in our drawings, this edge is realized by a line segment). Prove every game tree has exactly one more node than edge.

- (15) Let a and b be positive integers and let T be a W-L-D game tree with the property that whenever Louise has a choice to make she has exactly a options and whenever Richard has a choice to make he has exactly b options. Suppose that n is the total number of nodes, ℓ is the number of nodes marked L , and r is the number of nodes marked R . Prove that $a\ell + br + 1 = n$.
- (16) For every integer $n \geq 1$ let c_n be the number of nodes in the game tree for a $1 \times n$ position in Chop. Find (and prove) a formula for c_n for every $n \geq 1$.
- (17) The Fibonacci Sequence is an infinite sequence that starts off 0, 1, 1, 2, 3, 5, 8, 13, 21, It is defined by the following rules: $f_0 = 0$, $f_1 = 1$, and $f_{n+2} = f_{n+1} + f_n$ for every $n \geq 0$.
- (a) Let φ, ψ be the solutions to the equation $x^2 = x + 1$ where $\varphi > 0$ (here φ is the Golden Ratio). Prove by induction that $f_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$ for every $n \geq 0$.
- (b) For every real number x , let $[x]$ denote the closest integer to x . Prove that $f_n = \left[\frac{\varphi^n}{\sqrt{5}} \right]$ holds for every $n \geq 0$.
- (c) Let p_n be the number of nodes in a game tree for a Pick-Up-Bricks position with n bricks. Find (and prove) a formula for p_n .
- (18) Consider a Chomp position that consists of just two rows, with the bottom row of length n (including the poison square) and the other row of length m and assume $m \leq n$. Determine which player has a winning strategy for all possible values of m and n and prove your answer. (Hint: Consider the special case $n = m + 1$.)
- (19) The game Kayles is played with a $1 \times n$ array, each square of which is either empty or is marked by a P indicating that it contains a bowling pin. On each turn we think of a player as bowling a ball toward this line-up of bowling pins. Each player has perfect aim and may choose to knock down any one pin or any two consecutive pins (i.e. erase the P from either one box or from two adjacent boxes). The first player to clear the board wins. For every positive integer n find a winning strategy for either the first or second player when the starting position is a $1 \times n$ array where every square contains a P .

- (20) Consider playing d -dimensional Tic-Tac-Toe on a board with dimensions $\underbrace{n \times n \times \cdots \times n}_d$. To win, a player needs to get all n boxes in some line marked with his or her symbol ($\times$ or $\circ$). Prove that for every $n, d \geq 1$ the second player does not have a winning strategy.
- (21*) The game SOS is played on a $1 \times n$ array. On his turn a player may choose any empty square and mark either an “S” or an “O” in it. If a player manages to get the letters “SOS” appearing in three consecutive squares and he is the first one to achieve this, then he wins. If the game ends without this arrangement, it is a draw. Prove that from each of the following starting positions, the first player has a winning strategy.
- (a) A 1×4 array where the leftmost square is marked “S” and the other three squares are empty.
 - (b) A blank 1×7 array.
 - (c) A blank $1 \times n$ array where $n \geq 7$ is odd.

Chapter 2

Normal-Play Games

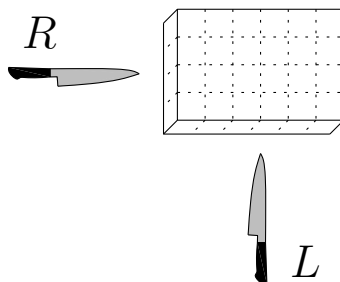


Figure 2.1. A game of Cut-Cake

Combinatorial games encompass a wide assortment of games, including those we played in the previous chapter and many others such as Chess, Go, and Checkers. Many of these games have quite complicated rules and, apart from Zermelo's Theorem, there is little general theory to apply. Normal-play games, however, have some additional structure that facilitates further analysis. Recall that a combinatorial game is normal-play if the win rule dictates that the winner is the last player to move. This win rule gives rise to both a type of addition on normal-play games and a rich notion of equivalence. We have already encountered several normal-play games, namely Pick-Up-Bricks, Chop, and Chomp. Let's play another one now.

Game 2.1 (Cut-Cake). In this normal-play game played between Louise and Richard, each position consists of a collection of uncut pieces of cake. Each uncut piece of cake is rectangular and has dotted lines running horizontally and vertically indicating where it can be cut. On her turn, Louise chooses any piece of cake and makes a vertical cut, while on Richard's turn he chooses any piece and makes a horizontal cut. The last player to make a move wins.

From a position in Cut-Cake, the available moves for Louise and Richard are generally quite different. This is in sharp contrast to the games Pick-Up-Bricks, Chop, and Chomp, where the available moves from every position are always the same for either player. We define a normal-play game to be *impartial* if, from every position, the set of moves available to either player is the same. Otherwise, we say that the game is *partizan*. In the following chapter, we will study impartial games in depth and Chapter 4 will focus on partizan games.

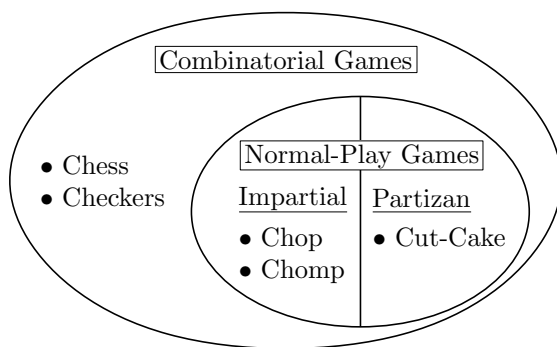


Figure 2.2. A Venn diagram of games

The Venn diagram in Figure 2.2 shows our present division of combinatorial games. Note that Chess and Checkers are not normal-play games since both of these games can end in a draw.

2.1. Positions and Their Types

Now we begin our mathematical investigation of normal-play games. We will develop notation to represent positions, introduce a classification of positions into types, and then learn how to determine the type of a given position.

Positions. Normal-play games are generally simple to describe. They consist of a set of positions together with a rule that dictates, for each position, to which positions Louise can move and to which positions Richard can move. That's it! The diagram below depicts an example position in Cut-Cake, together with the available moves for Louise and for Richard.

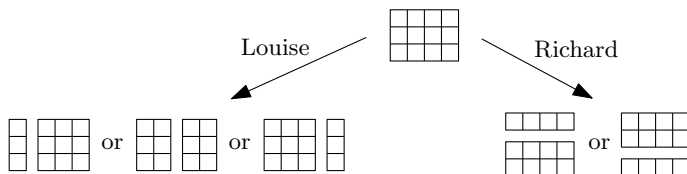


Figure 2.3. Moves for Louise and Richard in a Cut-Cake position

The key property of any position is to which other positions each player can move, so our notation will emphasize this. We will represent a position by a pair of set braces containing all moves Louise can make and then a bar followed by all possible moves Richard can make. So, we would represent the above position in Cut-Cake by Figure 2.4.

$$\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} = \left\{ \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \mid \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \right\}$$

Figure 2.4. Position notation for Cut-Cake

In Figure 2.4, the two positions to which Richard can move are essentially the same; each consists of one 1×4 piece and one 2×4 piece. Likewise, two of Louise's possible positions consist of one 3×1 piece and one 3×3 piece. Going forward, we will usually simplify such expressions by listing just one of the two.

As a convention, we will generally reserve the equals symbol for use in position notation $\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$. If we wish to express that γ and γ' are positions from the same game that are exactly the same, we will say that γ is *identical to* γ' .

Types of Positions. Let us consider playing a normal-play game starting from a given position. If Louise is first to play, then Zermelo's Theorem says either Louise or Richard will have a winning strategy (since normal-play games cannot end in a draw). Similarly, if Richard plays first, then one of our two players is guaranteed to have a winning strategy. Thus, Zermelo's Theorem gives us the following classification of positions in normal-play games into four types.¹

Corollary 2.2. *Every position in a normal-play game is one of the following types:*

Type	Description
L	Louise has a winning strategy whoever goes first.
R	Richard has a winning strategy whoever goes first.
N	The N ext player to play has a winning strategy.
P	The second (or P revious) player has a winning strategy.

We use familiar games (Pick-Up-Bricks and Cut-Cake) to illustrate one position of each type in Figure 2.5.

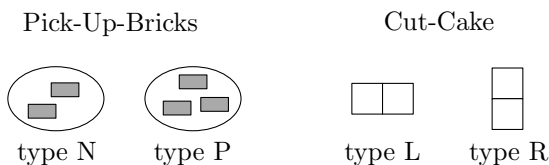


Figure 2.5. A position of each type

Determining Type. The type of a position is its most important characteristic, so we focus on determining it. As before, we rely on principles of recursive analysis, although now the game trees are generally suppressed. Consider a position α in a normal-play game. How might we

¹We have chosen to use the term “type” instead of the standard term “outcome class”. This is to keep this concept separate from that of outcome, which is used extensively in the classical game theory part of this book.

determine its type? Suppose that Louise will move first from α and that she has a move to a position β of type L or type P. By the definition of type, she has a winning strategy playing second from β , and this gives her a winning strategy playing first from α . Conversely, if every possible move for Louise results in a position of type R or type N, then Richard will have a winning strategy whenever she plays first. A similar analysis for Richard gives the following result.

Proposition 2.3. *If $\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$, the type of γ is given by the following chart:*

	<i>some β_j is type R or P</i>	<i>all of $\beta_1, \dots, \beta_n$ are types L or N</i>
<i>some α_i is type L or P</i>	N	L
<i>all of $\alpha_1, \dots, \alpha_m$ are types R or N</i>	R	P

This proposition gives us a straightforward recursive procedure to determine the type of a position. We will demonstrate this procedure by finding the types of some small positions in Cut-Cake. We shall begin with the simplest positions and build up to determine the type of a 2×3 position.

Notice first that whenever a Cut-Cake position has all pieces of size 1×1 , neither player has any available moves, so such a position will be type P. Another easily understood position is one in which just one player has available moves. If Richard can move but Louise cannot, the position is type R. If Louise can move but Richard cannot, the position is type L. So, in particular, we can determine the types of the positions in Figure 2.6.



Figure 2.6. Cut-Cake positions where only one player can move

Let's determine the types of a couple more simple positions. In each of the positions in Figure 2.7, whoever moves first puts the game in a

position from which only the opponent can move. So, Richard can only bring the position to one of type L, whereas Louise can only bring the position to one of type R. It's easy to see that any such position will be type P (you may also deduce this by applying Proposition 2.3). Since 1×1 pieces are irrelevant in Cut-Cake positions, the position notation ignores them for simplicity.

$$\begin{aligned} \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} &= \left\{ \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \mid \begin{array}{|c|} \hline \square \\ \hline \end{array} \right\} \Rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \\[10pt] \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} &= \left\{ \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \mid \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right\} \Rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \end{aligned}$$

Figure 2.7. Types of some Cut-Cake positions

In Figure 2.8, Proposition 2.3 is used to complete our example. We have determined that a 2×3 position in Cut-Cake has type P. More generally, anytime we want to determine the type of a position in a normal-play game, we can use Proposition 2.3 to work backwards.

$$\begin{aligned} \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} &= \left\{ \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \mid \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right\} \Rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \\[10pt] \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} &= \left\{ \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \mid \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}, \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right\} \Rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \\[10pt] \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} &= \left\{ \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \mid \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right\} \Rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \end{aligned}$$

Figure 2.8. Types of more Cut-Cake positions

2.2. Sums of Positions

This section continues to develop our algebraic treatment of games. We will define a sum operation on normal-play games and begin to explore the properties of positions under this operation. This is a significant departure from the usual way we think about games—we will now treat positions like numbers.

Sums. In fact, we have already seen sums in the game of Cut-Cake. Even if we begin with a single piece of cake, it will break into many pieces over the course of the game. If we call each piece a component, then we can view a Cut-Cake position as a sum of these components. Each player, in turn, must select one component in which to move.

Although we will continue to focus exclusively on normal-play games, it is worth noting that a similar phenomena is encountered in some well-known (nonnormal-play) combinatorial games. For instance, in the late stages of a game of Go, the board will be divided into many small regions, and on each turn a player chooses one of these “components” in which to move.

We next introduce a general notion of sum that allows addition of positions even from different games.

Definition 2.4. If α and β are positions in normal-play games, then we define $\alpha + \beta$ to be a new position consisting of the *components* α and β . To move in $\alpha + \beta$, a player chooses one of the components in which to move. So, for instance if a player may move from α to α' , then the player may move from $\alpha + \beta$ to $\alpha' + \beta$. Similarly, if a player can move from β to β' , then he or she may move from $\alpha + \beta$ to $\alpha + \beta'$.

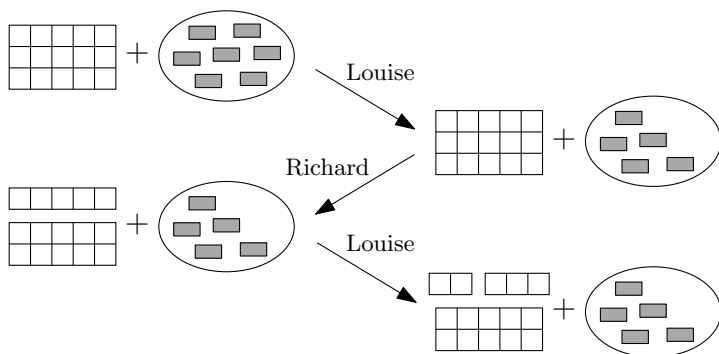


Figure 2.9. Three moves in a sum of Cut-Cake and Pick-Up-Bricks

Example 2.5. Figure 2.9 exhibits a sequence of three moves in a position that is the sum of two positions, one from Cut-Cake and the other from

Pick-Up-Bricks. As you can see, Louise chooses to play from the Pick-Up-Bricks component for her first move. The next two moves are both made in the Cut-Cake component.

Determinate Sums. Now that we can add positions, we want to understand strategy in sums of positions. More precisely, we will be interested in understanding the behavior of our types (L, R, P, N) under sums. First we will consider some instances where we can determine the type of a sum based on the types of its components.

Let's imagine that we have a particular position α that is type R, so as first player, Richard has a winning strategy $\mathcal{S}$. What would you advise Richard to do if he and Louise were to play the position shown in Figure 2.10 with Richard going first?

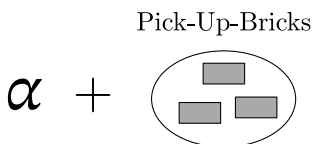


Figure 2.10. What should Richard do?

In fact, the component consisting of three bricks in Pick-Up-Bricks doesn't change things significantly. Richard can just ignore this component and follow his strategy $\mathcal{S}$, pretending just to play against Louise in the position α . At some point, Louise may choose to pick up either 1 or 2 bricks for a move in the Pick-Up-Bricks component. At that point, Richard should respond to Louise by making the opposite play in the Pick-Up-Bricks component. This sequence would cost Louise and Richard one move each, but would return the game to the α component. From there, Richard can continue to follow the strategy $\mathcal{S}$, which guarantees him a win. The next proposition replaces this Pick-Up-Bricks position with an arbitrary position of type P to generalize the result.

Proposition 2.6. *If β is type P, then α and $\alpha + \beta$ are the same type.*

Proof. To prove this, we will need to show that whenever a certain player (Louise or Richard) has a winning strategy moving either first or second in the game α , then the same holds true in the game $\alpha + \beta$. We will show

below that whenever Louise has a winning strategy as second player in α , then she also has a winning strategy as the second player in $\alpha + \beta$. The other cases are similar.

Let $\mathcal{S}$ be a winning strategy for Louise playing second in α and let $\mathcal{T}$ be a second player winning strategy in β . Here is Louise's strategy for playing $\alpha + \beta$ as second player: If Richard moves in component α , then Louise follows strategy $\mathcal{S}$ to respond in this component; if Richard moves in the component β , then Louise follows strategy $\mathcal{T}$ to respond in β . Since the strategies $\mathcal{S}$ and $\mathcal{T}$ guarantee Louise that she will make the last move, it follows that Louise will make the last move while following this strategy in $\alpha + \beta$, thus guaranteeing her a win. $\square$

A similar argument (Exercise (12)) establishes the following.

Proposition 2.7. *If α and β are both type L, then $\alpha + \beta$ is type L. Similarly, if α and β are both type R, then $\alpha + \beta$ is type R.*

Our current state of knowledge concerning the behavior of types under summation is given by the table in Figure 2.11.

+	L	R	N	P
L	L	?	?	L
R	?	R	?	R
N	?	?	?	N
P	L	R	N	P

Figure 2.11. Types of sums

It turns out that none of the question marks in Figure 2.11 can be truthfully replaced by one of our types. For instance, if we know that α is type R and β is type N, it is not in general possible to determine the type of $\alpha + \beta$. In the next part of this section we will introduce a new game to demonstrate this.

Indeterminate Sums. To assist us in exploring sums, we now introduce a game called Domineering.

Game 2.8 (Domineering). This is a normal-play game played using some squares from a rectangular array. On Louise's turn, she may place a 2×1 domino over two unoccupied squares. On Richard's turn, he may place a 1×2 domino over two unoccupied squares. Since this is a normal-play game, the last player to move wins.

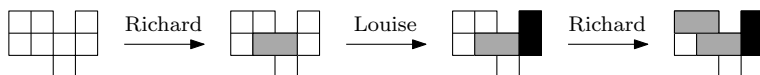


Figure 2.12. A sequence of moves in Domineering

It is most appealing to play Domineering with actual dominoes, but for the purposes of calculation, it is easier to imagine that the players move by deleting squares (so Louise deletes two vertically adjacent squares while Richard deletes two horizontally adjacent squares). We can treat disconnected parts of the grid separately and may ignore any 1×1 component. This allows us to view the game from Figure 2.12 as seen in Figure 2.13.

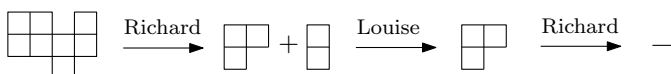


Figure 2.13. Domineering notation

Consider the simple Domineering positions in Figure 2.14. The type of each of these positions is straightforward to verify: The position on the left is type N since both players have an available move, but after one player makes a first move, the other player has no move. The other positions are type R since Richard can move, but Louise cannot.

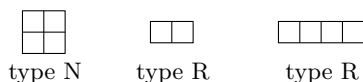


Figure 2.14. Domineering positions

Now we will consider sums of positions from Figure 2.14. Were Richard to play first in the Figure 2.15 position, he could win by playing in the left component (after which Louise would have no move available). On the other hand, if Louise were to play first, then she would play in the left component and then win the game. It follows that this position is type N.

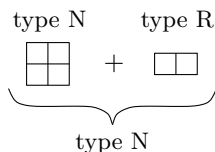


Figure 2.15. An $N + R$ position that is type N

Nevertheless, not every $N + R$ position is type N. Both a 1×2 array in Domineering and a 1×4 array are type R, but the 1×4 position gives Richard a greater advantage than the 1×2 position. In fact, this difference is enough so that the sum from Figure 2.16 behaves differently from the sum in Figure 2.15.

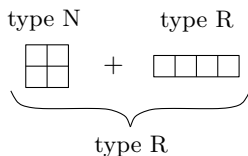


Figure 2.16. An $N + R$ position that is type R

In the above position, Richard can win as first player by moving first in the 2×2 component; after that, Louise has no available move so Richard wins. On the other hand, if Louise moves first, she must move in the 2×2 component; after this, Richard can move in the 1×4 component so as to leave himself another move. Assuming he does so, he will win. We conclude that this position is type R.

Now we have seen an $N + R$ position of type N and an $N + R$ position of type R. This example illustrates that it is not always possible to determine the type of a sum from the types of its components. Exercise

(14) demonstrates that all of the other question marks Figure 2.11 are likewise ambiguous. In the next section we will introduce a finer notion than that of type that will allow us to understand sums better.

2.3. Equivalence

Consider the two positions in Figure 2.17, one from Domineering and one from Pick-Up-Bricks.

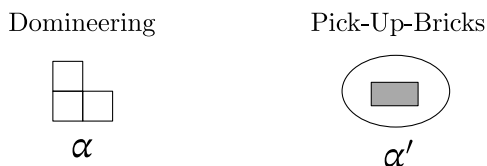


Figure 2.17. Two different positions that play the same

As you can see, the player who moves first in either α or α' will leave a position with no available moves for either player. So, although α and α' are from two completely different games, the positions are very much the same. Moreover, if we were to take any other position β , then the positions $\alpha + \beta$ and $\alpha' + \beta$ would play similarly, too.

We would like our theory to express this intuition that the positions α and α' are essentially the same. With a meaningful notion of equivalence that allows us to assert that two positions (possibly in different games) are strategically the same, we can move beyond investigating one game at a time. We will be able to identify similar strategic features in different games and prove broad theorems that apply to many games at once.

To do this, we will need the following definition of equivalence for two positions. Although this definition is straightforward to state, the reader should be cautious of its subtleties.

Definition 2.9. We say that two positions α and α' in (possibly different) normal-play games are *equivalent* if for every position β in any normal-play game, the two positions $\alpha + \beta$ and $\alpha' + \beta$ have the same type. We write $\alpha \equiv \alpha'$ to indicate this.

In other words, two positions are equivalent if they behave the same under summation. Our positions α and α' from Figure 2.17 are indeed equivalent. To prove this, we need to show that for every position β , the positions $\alpha + \beta$ and $\alpha' + \beta$ have the same type. This is easy to check since one may immediately translate a winning strategy from one of these positions into one for the other.

Be aware that this instance of equivalence is a very simple one. In fact, there are many other positions that are also equivalent to α and α' , many of which have lots of available moves for either player. The power of equivalence is that it allows us to treat all of these positions as the same.

Equivalence Relations. There are some fundamental properties of equivalence that follow directly from our definitions. The next proposition features three important properties known as reflexivity, symmetry, and transitivity. We leave it to the reader to verify these.

Proposition 2.10. *If α, β, γ are positions in normal-play games, then*

- (1) $\alpha \equiv \alpha$ (reflexivity),
- (2) $\alpha \equiv \beta$ implies $\beta \equiv \alpha$ (symmetry),
- (3) $\alpha \equiv \beta$ and $\beta \equiv \gamma$ implies $\alpha \equiv \gamma$ (transitivity).

The term *equivalence relation* is used to describe any relation that is reflexive, symmetric, and transitive. A few relatively well-known equivalence relations include similarity for triangles, congruence modulo n for the integers, and row equivalence for matrices. As the term “equivalence relation” suggests, these properties behave much like equals does for numbers. Indeed, $=$ is an equivalence relation—one of fundamental importance.

In addition to the equivalence relation on positions defined above, we have already implicitly introduced two other equivalence relations for positions. Specifically, note that—although we have not given them symbols—the notions “is identical to” and “is the same type as” are also reflexive, symmetric, and transitive and thus also give equivalence relations for positions.

We now have three equivalence relations for positions. How do they compare with one another? If two positions α and β are identical, then

it is straightforward to check that they must be equivalent and they must also have the same type. So the relation “is identical to” implies both of the other equivalence relations. Next we will investigate the relationship between equivalence and type.

Equivalence vs. Type. It is a direct consequence of the definitions that anytime two positions are equivalent, they must have the same type. This is formalized in the following proposition.

Proposition 2.11. *If $\alpha \equiv \alpha'$, then α and α' have the same type.*

Proof. Let β be a position in a normal-play game with no moves left for either player (for instance β could be a position in Pick-Up-Bricks with 0 bricks). We claim that the following holds:

$$\alpha \begin{array}{c} \xleftarrow{\text{same}} \\ \text{type} \end{array} \alpha + \beta \begin{array}{c} \xleftarrow{\text{same}} \\ \text{type} \end{array} \alpha' + \beta \begin{array}{c} \xleftarrow{\text{same}} \\ \text{type} \end{array} \alpha'$$

Since β has no moves, it is immediate that α and $\alpha + \beta$ have the same type. Similarly, $\alpha' + \beta$ and α' have the same type. To complete the argument we need to show that $\alpha + \beta$ and $\alpha' + \beta$ have the same type. This follows from the assumption $\alpha \equiv \alpha'$. $\square$

Could there be two positions that have the same type but are not equivalent? To answer this question, consider the Domineering sums from the previous section shown in Figure 2.18.

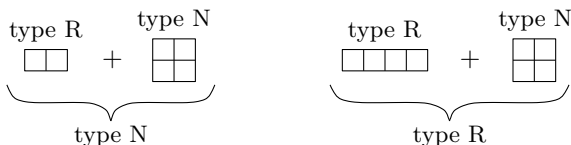


Figure 2.18. Two sums in Domineering with different type

In Figure 2.18, we see that the sum of a 1×2 array and 2×2 array in Domineering has a different type than the sum of a 1×4 array and a 2×2 array. Therefore, a 1×2 array and a 1×4 array are not equivalent, even though they are both type R.

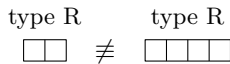


Figure 2.19. Inequivalent Domineering positions of the same type

In short, equivalent positions must have the same type, but not all positions of the same type are equivalent. We have just exhibited two inequivalent positions of type R, and Exercises (6) and (9) demonstrate the existence of inequivalent positions of types L and N. In contrast to this, we shall prove at the end of this section that any two positions of type P are equivalent. In preparation for that, we will need to establish some basic properties of $+$ and $\equiv$.

The Algebra of Sums and Equivalence. In this part of the section we will prove a handful of properties which demonstrate that positions (with $+$ and $\equiv$) behave very much like numbers (with $+$ and $=$). The first such result follows immediately from our definitions, so we leave the reader to verify it.

Proposition 2.12. *If α, β, γ are positions in normal-play games, then*

- (1) $\alpha + \beta \equiv \beta + \alpha$ (commutativity),
- (2) $(\alpha + \beta) + \gamma \equiv \alpha + (\beta + \gamma)$ (associativity).

When working with numbers, we know that anytime we have two expressions that give equal numbers, we can substitute one for the other in an equation. Next we establish a similar result for positions.

Lemma 2.13. *All α and β terms in the following expressions are assumed to be positions in normal-play games.*

- (1) *If $\alpha \equiv \alpha'$, then $\alpha + \beta \equiv \alpha' + \beta$.*
- (2) *If $\alpha_i \equiv \alpha'_i$ for $1 \leq i \leq n$, then $\alpha_1 + \cdots + \alpha_n \equiv \alpha'_1 + \cdots + \alpha'_n$.*
- (3) *If $\alpha_i \equiv \alpha'_i$ for $1 \leq i \leq m$ and $\beta_j \equiv \beta'_j$ for $1 \leq j \leq n$, then $\{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\} \equiv \{\alpha'_1, \dots, \alpha'_m \mid \beta'_1, \dots, \beta'_n\}$.*

Proof. To prove part (1), let γ be an arbitrary position in a normal-play game. We must show that the positions $(\alpha + \beta) + \gamma$ and $(\alpha' + \beta) + \gamma$ have the same type. By associativity, the first of these positions is equivalent

to $\alpha + (\beta + \gamma)$ and the second is equivalent to $\alpha' + (\beta + \gamma)$. It now follows from the assumption $\alpha \equiv \alpha'$ that these two positions have the same type.

For part (2), we repeatedly apply part (1) as follows:

$$\begin{aligned}\alpha_1 + \cdots + \alpha_n &\equiv \alpha_1 + \cdots + \alpha_{n-2} + \alpha_{n-1} + \alpha'_n \\ &\equiv \alpha_1 + \cdots + \alpha_{n-2} + \alpha'_{n-1} + \alpha'_n \\ &\quad \vdots \\ &\equiv \alpha'_1 + \cdots + \alpha'_n.\end{aligned}$$

Part (3) is proved in Exercise (22). □

As the lemmas in this part of the section demonstrate, positions really do behave like numbers!

Type P. Positions of type P play a very special role in the world of normal-play games. In fact, these positions behave under sums just like the number 0 does under addition. As we prove next, whenever a position of type P is added to a position α , the resulting position is still equivalent to α (just as $a + 0 = a$).

Lemma 2.14. *If β is type P, then $\alpha + \beta \equiv \alpha$.*

Proof. Let γ be an arbitrary position in a normal-play game. By Proposition 2.6, we find that $\alpha + \beta + \gamma$ and $\alpha + \gamma$ have the same type. It follows that $\alpha + \beta \equiv \alpha$, as desired. □

Below, we prove that any two positions of type P are equivalent. Recall that this differs from the behavior of positions of type N, L, and R.

Proposition 2.15. *If α and α' are type P, then $\alpha \equiv \alpha'$.*

Proof. For every position γ , the previous lemma gives us

$$\alpha + \gamma \equiv \gamma \equiv \alpha' + \gamma.$$

Since $\alpha + \gamma$ and $\alpha' + \gamma$ are equivalent, they have the same type. This holds for every position γ , so we have $\alpha \equiv \alpha'$ as desired. □

The final result in this chapter gives another special property of type P positions analogous to the number zero. The next lemma parallels for positions the following simple property for numbers: If $a + b = 0$ and $a' + b = 0$, then $a = a'$.

Lemma 2.16. *If $\alpha + \beta$ and $\alpha' + \beta$ are both type P, then $\alpha \equiv \alpha'$.*

Proof. Lemma 2.14 implies $\alpha \equiv \alpha + (\alpha' + \beta) \equiv \alpha' + (\alpha + \beta) \equiv \alpha'$. $\square$

At this point we have established the basic principles of type, sums, and equivalence for normal-play games. We have begun to think of positions in a manner similar to numbers, and we have learned that type P positions (which are all equivalent to one another) behave like the number 0 does. In the next two chapters we will call upon these ideas to delve more deeply into the structure of impartial and partizan games.

Exercises

- (1) Use the $\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$ notation to represent the following positions:
 - (a) a 2×3 position in Chop,
 - (b) a 4×5 position in Cut-Cake,
 - (c) a blank 2×3 board in Domineering.
- (2) Use the $\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$ notation to represent the following positions:
 - (a) a 3×3 piece in Cut-Cake plus 4 bricks in Pick-Up-Bricks,
 - (b) a blank 2×3 board in Domineering plus 5 bricks in Pick-Up-Bricks.
- (3) Use Proposition 2.3 to find the types of the positions below:
 - (a) a 2×4 position in Cut-Cake,
 - (b) a blank 2×3 board in Domineering.
- (4) Prove that each of the following positions is type N by finding a winning strategy for each player when he or she moves first. (Don't construct the entire game tree. Instead, indicate the first move and explain how to proceed from there).
 - (a) The sum of a blank 2×2 board in Domineering and a Pick-Up-Bricks position with 2 bricks.
 - (b) The sum of a 3×3 piece in Cut-Cake and a 5-brick position in Pick-Up-Bricks.

- (5) Using the definition of $\equiv$ and Lemma 2.16, indicate for each case below whether $\alpha \equiv \alpha'$ or $\alpha \not\equiv \alpha'$ or if the provided information is inconclusive.
- $\alpha + \beta$ and $\alpha' + \beta$ are both type N.
 - $\alpha + \beta$ is type N and $\alpha' + \beta$ is type L.
 - $\alpha + \beta$ and $\alpha' + \beta$ are both type P.
 - $\alpha + \beta$ is type L and $\alpha' + \beta$ is type R.
 - $\alpha + \beta$ and $\alpha' + \beta$ are both type R.
- (6) Consider the Domineering positions α and β .
- 
 α

 
 β
- Show that both α and β are type N by finding the first move of a winning strategy for the first player and giving a brief explanation of how to play from there.
 - Find the types of $\alpha + \alpha$ and $\beta + \alpha$ and provide a short proof in each case.
 - Show that $\alpha \not\equiv \beta$.
- (7) Find a Domineering position of type L with at most four squares. Repeat for types R, N, and P.
- (8) Use Lemma 2.16 to prove that the Domineering positions $\square\square$ and $\square\square + \square\square$ are equivalent.
- (9) Prove that the Cut-Cake positions $1 \times s$ and $1 \times t$ are not equivalent whenever $s \neq t$. (Hint: Find another Cut-Cake position β so that adding β to the $1 \times s$ position and adding β to the $1 \times t$ position give positions of different types.)
- (10) This exercise shows a way in which positions and type P behave differently from real numbers and 0. If $a, b \in \mathbb{R}$ and $a + a + b + b = 0$, then $a + b = 0$. For the Domineering positions α and β , prove that $\alpha + \alpha + \beta + \beta$ has type P. Then determine the type of $\alpha + \beta$.
- (11) For every position α and positive integer n , construct a position α' equivalent to α so that α' has at least n available moves for whichever player plays first.
- (12) Prove Proposition 2.7 by showing that the sum of two positions of type L is another position of type L. (A similar argument shows that the sum of two type R positions is type R.)

- (13) Show that a player with a winning strategy playing first in α and a winning strategy when playing second in β also has a winning strategy playing first in $\alpha + \beta$. What can you deduce about the sum of a position of type L and a position of type N? How about the sum of a position of type R and one of type N?
- (14) Find Domineering positions α and β of the indicated types that sum to a position $\alpha + \beta$ of the indicated type:

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
α	L	L	R	R	L	L	L	L	N	N	N	N
β	N	N	N	N	R	R	R	R	N	N	N	N
$\alpha + \beta$	L	N	R	N	L	R	N	P	L	R	N	P

- (15) For each question mark in Figure 2.11, determine all of the possible types of positions that can be obtained by a sum of the given types. So, for instance, in the entry corresponding to (N, R) enter all possible types of positions that can be obtained by adding a position of type N to one of type R. Justify that your answer is complete using the results of prior exercises.
- (16) Let α, β be positions in impartial games.
- Prove that $\alpha + \alpha$ is type P.
 - Prove that $\alpha \equiv \beta$ if and only if $\alpha + \beta$ is type P.
- (17) Let P_n denote an n -brick position in Pick-Up-Bricks. Use the previous exercise to prove that for every positive integer n , exactly one of the following holds: $P_n \equiv P_0$ or $P_n \equiv P_1$ or $P_n \equiv P_2$.
- (18) Prove that the sum of a 1×2 piece in Cut-Cake with any Chop position is always type L.
- (19) Let α, β be positions in Domineering, and define $-\alpha$ to be the position obtained from α by reflecting each component about the line $y = x$ (thus interchanging vertical and horizontal directions).
- Prove that $\alpha + (-\alpha)$ is type P.
 - Prove that $\alpha \equiv \beta$ if and only if $\beta + (-\alpha)$ is type P.
- (20) A *partition* of a set X is a collection $\mathcal{P}$ of nonempty subsets of X with the property that every element in X is contained in exactly one member of $\mathcal{P}$. For example $\mathcal{P} = \{\{1, 2, 7\}, \{3, 8\}, \{4\}, \{5, 6\}\}$ is a partition of $X = \{1, \dots, 8\}$. If $\sim$ is an equivalence relation on X , the *equivalence class* of $x \in X$ is $[x] = \{x' \in X \mid x \sim x'\}$.

- (a) If two equivalence classes have nonempty intersection, prove they are equal. Deduce that $\mathcal{P}_\sim = \{[x] \mid x \in X\}$ is a partition of X .
 - (b) Prove that for every partition $\mathcal{P}$ of X , there is an equivalence relation $\sim$ on X so that $\mathcal{P} = \mathcal{P}_\sim$.
 - (c) We have three equivalence relations for positions: identical to, equivalence, and same type. Describe the equivalence classes for these three equivalence relations.
- (21) Define a relation on positions by the rule that $\alpha \geq \beta$ if there exists δ so that $\alpha + \delta$ has type L or P and $\beta + \delta$ has type P.
- (a) Prove that $\alpha \geq \beta$ and $\beta \geq \alpha$ implies that $\alpha \equiv \beta$. Hint: Choose δ, δ' so that $\alpha + \delta$ and $\beta + \delta'$ are type P and $\alpha + \delta'$ and $\beta + \delta$ have type L or P and consider $\alpha + \beta + \delta + \delta'$.
 - (b) Prove that $\alpha \geq \beta$ and $\beta \geq \gamma$ implies that $\alpha \geq \gamma$.
- (22) Prove part (3) of Lemma 2.13.
- (23*) Euclid's Game is an impartial game in which each position consists of a set of two distinct nonnegative integers $\{a, b\}$. If one of the numbers is zero, there are no available moves. If both are positive and, say, $a > b$, then a player may move to any position of the form $\{a - tb, b\}$, where t is a positive integer and $tb \leq a$.
- (a) Show that whenever a and b are relatively prime, the numbers $a - tb$ and b are also relatively prime.
 - (b) Prove that whenever the game starts with two positive relatively prime numbers, the ending position is always $\{0, 1\}$.
 - (c) Let $\varphi = \frac{1+\sqrt{5}}{2}$ be the Golden Ratio. Define a position $\{a, b\}$ with $a > b$ to be *controlled* if either $b = 0$ or $a < \varphi b$ and *non-controlled* otherwise. Show that from every controlled position of nonzero numbers, there is only one available move, and it results in a noncontrolled position. Show that from every non-controlled position there exists a move to a controlled one.
 - (d) Determine the type of every position consisting of two relatively prime positive integers.

Chapter 3

Impartial Games

This chapter focuses on impartial games (recall that these are normal-play games in which the available moves for Louise and Richard are always the same). Although these games appear simple in form, they exhibit fascinating structure. Our central result, the Sprague-Grundy Theorem, will provide a thorough understanding of equivalence for impartial games. The proof of this well-known theorem was foundational to the subject of combinatorial game theory.

The particular impartial game Nim will be of central importance in our investigation. Let's get started by playing it!

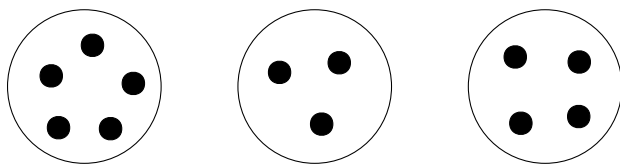


Figure 3.1. Play Nim!

Game 3.1 (Nim). This is an impartial game in which a position consists of ℓ piles of stones of sizes $a_1, a_2, \dots, a_\ell$. To make a move, a player removes one or more (up to all) stones from a chosen pile. The last player to take a stone wins.

In an impartial game, the identity of the player does not affect the available moves. So if, for instance, Richard has a winning strategy as the first player, then Louise will also have a winning strategy as the first player. This means impartial games have no positions of type L or type R, so Corollary 2.2 gives us the following.

Corollary 3.2. *Every position in an impartial game is one of the following:*

Type N The next player to play has a winning strategy.

Type P The second (or previous) player has a winning strategy.

In the previous chapter, we established Proposition 2.3 to determine the type of a position γ from the types of the possible positions to which Louise and Richard can move from γ . Incorporating the fact that all impartial game positions have type N or P yields the following essential corollary.

Corollary 3.3. *A position in an impartial game is*

Type N if there exists a move to a position of type P,

Type P if there is no move to a position of type P.

Can you determine the type of a Nim position with two equal piles? What about a Nim position with two unequal piles?

3.1. Nim

In this section, we will follow the work of Charles Bouton to develop the theory of Nim. In particular, we will determine the type of every position in Nim and learn to implement winning strategies. Then we will prove a theorem which tells us exactly when two Nim positions are equivalent. Much of this analysis is based upon properties of powers of two, so that is where we begin.

Binary Expansion. Here we will establish the basic properties of binary expansion necessary for our analysis of Nim. We begin with a simple but crucial fact.

Proposition 3.4. *For every nonnegative integer ℓ ,*

$$2^0 + 2^1 + 2^2 + \cdots + 2^\ell = 2^{\ell+1} - 1.$$

Proof. This follows from the equation

$$\begin{aligned}
 2^0 + 2^1 + \cdots + 2^\ell &= 2(2^0 + 2^1 + \cdots + 2^\ell) - (2^0 + 2^1 + \cdots + 2^\ell) \\
 &= 2^1 + 2^2 + \cdots + 2^{\ell+1} - 2^0 - 2^1 - \cdots - 2^\ell \\
 &= 2^{\ell+1} - 2^0. \quad \square
 \end{aligned}$$

For a nonnegative number n , the *binary expansion* of n is a representation of n as a sum of distinct powers of two. To construct the binary expansion of n , repeatedly take out the largest possible power of two.

Example 3.5. Here is the binary expansion of 45:

$$\begin{aligned}
 45 &= 32 + 13 \\
 &= 32 + 8 + 5 \\
 &= 32 + 8 + 4 + 1.
 \end{aligned}$$

Next, we look at a formal proof that binary expansions always exist and are unique.

Proposition 3.6. *Every nonnegative integer n has a unique binary expansion.*

Proof. We proceed by induction on n . The base case $n = 0$ holds because 0 is (uniquely expressed as) a sum of no powers of 2. For the inductive step, assume $n > 0$ and that every integer smaller than n has a unique binary expansion. Choose k so that $2^k \leq n < 2^{k+1}$. No power of 2 greater than 2^k can appear in a binary expansion of n since it would already be too big. The biggest number we can represent without 2^k is $1 + 2 + \cdots + 2^{k-1} = 2^k - 1 < n$. Therefore, every binary expansion of n must consist of 2^k plus a representation of $n' = n - 2^k$. By induction n' has a unique binary expansion and we thus conclude that n has a unique binary expansion. $\square$

With this basic understanding of binary expansion, we will turn our attention to positions in Nim.

Types of Positions. Let's introduce some notation to work with Nim positions. For any nonnegative integer a let $*a$ denote a Nim position consisting of a single pile of a stones. Then a position with piles of sizes

$a_1, a_2, \dots, a_\ell$ may be denoted in our sum notation as $*a_1 + *a_2 + \dots + *a_\ell$. Figure 3.2 depicts the position $*11 + *13 + *7 + *12$.

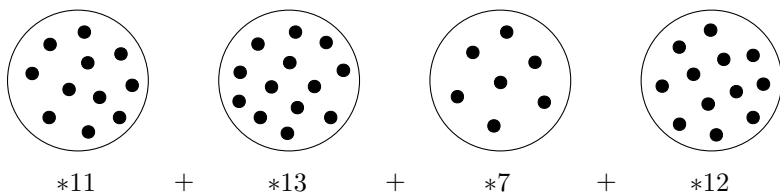


Figure 3.2. A position in Nim

Now we will break each pile into subpiles based on the binary expansion of the size of the pile. For example, since $13 = 8 + 4 + 1$, we will divide the pile of size 13 into subpiles with sizes 8, 4, and 1.

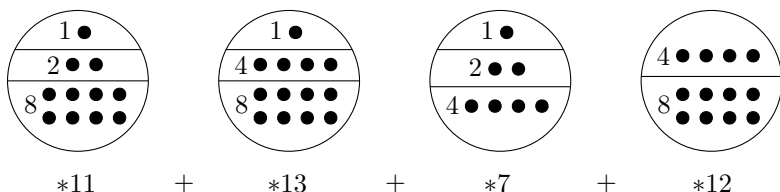


Figure 3.3. Division into subpiles

Now we are ready to introduce a crucial concept in Nim. Define a position $*a_1 + *a_2 + \dots + *a_k$ to be *balanced* if, for every power of 2, the total number of subpiles of that size is even. For example, the position depicted in Figure 3.3 is unbalanced since there are three subpiles of size 8. In contrast, the position $*3 + *6 + *5$ is balanced since $*(1 + 2) + *(2 + 4) + *(1 + 4)$ has two subpiles of sizes 1, 2, and 4 and zero subpiles of every other size. The ability to move from an unbalanced position to a balanced position figures prominently in a winning Nim strategy.

Procedure 3.7 (Balancing an Unbalanced Position). Let $*a_1 + \dots + *a_k$ be an unbalanced game of Nim. Suppose 2^m is the largest power of 2 for which there are an odd number of subpiles of that size. Choose a pile, say $*a_i$, which has a subpile of size 2^m . Now (temporarily—this isn't the move!) pick up all subpiles of this pile which have size less than or equal

to 2^m . At the moment, there are an even number of subpiles of size 2^j for every $j \geq m$. Now, for every 2^j with $j < m$, if the number of subpiles of size 2^j is odd, return 2^j stones. Since at least 2^m stones were initially picked up and at most $2^0 + 2^1 + \dots + 2^{m-1} < 2^m$ stones were put back, this is indeed a legal move. It is straightforward to verify that the resulting position is balanced.

Let's apply the balancing procedure to the position depicted in Figures 3.2 and 3.3. Here, 8 is the largest power of 2 appearing an odd number of times, so we can move on any pile with a subpile of size 8. Let's choose the first one. Now we pick up the subpile of size 8 and all smaller subpiles (in this case, the entire pile). At this point, there remain an even number of subpiles of sizes 8 and 1, but an odd number of subpiles of sizes 2 and 4. To return to a balanced position, we put back subpiles of sizes 2 and 4. This moves the game to the balanced position shown in Figure 3.4.

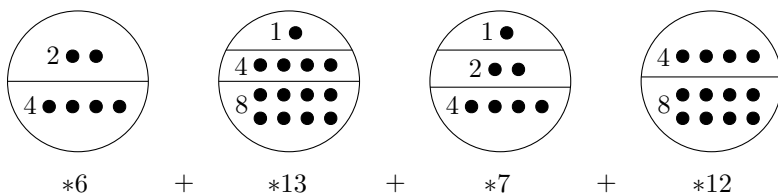


Figure 3.4. A balanced position

Proposition 3.8. *Every balanced Nim position is type P and every unbalanced Nim position is type N. In both cases, the balancing procedure provides a winning strategy.*

Proof. We start by showing that every balanced position is type P. To do so, consider an arbitrary balanced position. Let's adopt the role of the second player and show that the balancing procedure gives us a winning strategy. Any move the first player makes will leave an unbalanced position since that player alters just one pile, which, consequently, will divide into different subpiles. On our turn as the second player, we may apply the balancing procedure and return the position to a balanced one. This pattern continues. After each move that the first player makes the

position will be unbalanced and we, as the second player, can again move to make the position balanced. The game-ending position with no stones is balanced. It follows that we will be the player to move to that position, so we have a winning strategy and every balanced position is type P.

Starting from an unbalanced position, the first player may apply the balancing procedure to move the position to one of type P. Then the above strategy gives the first player a winning strategy, so every unbalanced position is type N. $\square$

This proposition gives us a simple classification of Nim positions by type. In answer to a question from the start of this chapter, we find that a Nim position with two piles $*a + *b$ will be type P if $a = b$ and type N otherwise. Now that we understand types of Nim positions, our next goal will be to understand equivalence of Nim positions.

Equivalence. Moving forward, it will be helpful to think about positions in Nim like numbers and, to this end, we will call a position of the form $*a$ a *number*. We determined above which Nim positions are type P and which are type N. More precisely, we showed that balanced positions (those with an even number of subpiles of each size) are type P and all other positions are type N. This means $*0$ is type P and, since any two type P positions are equivalent, this tells us that every balanced position is equivalent to $*0$. In this section, our consideration of equivalence for general Nim positions will lead to a proof of the surprising fact that every Nim position is equivalent to a single number!

Let's consider a Nim position given by $*a_1 + *a_2 + \cdots + *a_\ell$ from the perspective of the second player. Imagine that before the start of play the second player could add one entirely new pile, say $*b$, to the position. The second player would want the new position $*a_1 + \cdots + *a_\ell + *b$ to be balanced so that he or she would have a winning strategy. So the pile $*b$ should have a subpile of size 2^j if and only if there are an odd number of subpiles of that size in $*a_1, \dots, *a_\ell$. This gives rise to a new type of summation as follows.

Definition 3.9. The *Nim-sum* of the nonnegative integers $a_1, \dots, a_\ell$, denoted $a_1 \oplus \cdots \oplus a_\ell$, is the nonnegative integer b with the property that

2^j appears in the binary expansion of b if and only if this term appears an odd number of times in the expansions of $a_1, \dots, a_\ell$.

To compute b , find the binary expansion of each a_i and then cross off equal terms in pairs. When there are no more pairs of equal terms, the sum of the remaining terms is b . Here is an example:

$$\begin{aligned} 13 \oplus 19 \oplus 10 &= (8 + 4 + 1) \oplus (16 + 2 + 1) \oplus (8 + 2) \\ &= (\cancel{8} + 4 + \cancel{1}) \oplus (16 + \cancel{2} + \cancel{1}) \oplus (\cancel{8} + \cancel{2}) \\ &= 4 + 16 \\ &= 20. \end{aligned}$$

We are now ready to prove the main result from this section.

Theorem 3.10. *If $a_1, \dots, a_\ell$ are nonnegative integers and $b = a_1 \oplus a_2 \cdots \oplus a_\ell$, then*

$$*a_1 + *a_2 + \cdots + *a_\ell \equiv *b.$$

Proof. Consider the game $*a_1 + *a_2 + \cdots + *a_\ell + *b$. It follows from the above definition of Nim-sum that this game is balanced. Therefore, it is type P and equivalent to $*0$. This gives us $*a_1 + *a_2 + \cdots + *a_\ell + *b \equiv *0$. Together with the fact that $*b + *b \equiv *0$, this gives us

$$\begin{aligned} *a_1 + \cdots + *a_\ell &\equiv *a_1 + \cdots + *a_\ell + *0 \\ &\equiv *a_1 + \cdots + *a_\ell + *b + *b \\ &\equiv *0 + *b \\ &\equiv *b. \end{aligned}$$

□

This result may be viewed as a refinement of Proposition 3.8. Every Nim position $*a_1 + \cdots + *a_\ell$ is equivalent to a number $*b$. If $*a_1 + \cdots + *a_\ell$ is balanced, then $b = 0$, and the position is type P. Otherwise, $*a_1 + \cdots + *a_\ell$ is unbalanced, so $b \neq 0$ and the position (which is equivalent to $*b$) will be type N.

We can now view all Nim positions according to their equivalent number as in Figure 3.5.

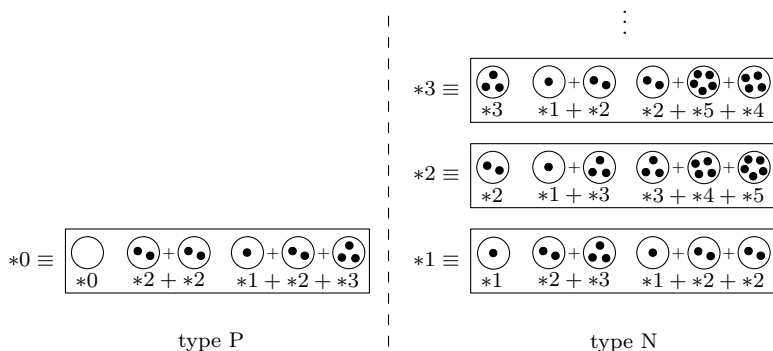


Figure 3.5. Equivalence of Nim positions

3.2. The Sprague-Grundy Theorem

We showed above that every Nim position is equivalent to a number. In fact, every position in every impartial game is equivalent to a number! In this section we give a proof of this surprising result.

First, we introduce some simplified notation for working with positions in impartial games. For general normal-play games, we denote a position α by $\alpha = \{\beta_1, \dots, \beta_\ell \mid \gamma_1, \dots, \gamma_m\}$ if Louise can move to one of $\beta_1, \dots, \beta_\ell$ and Richard can move to one of $\gamma_1, \dots, \gamma_m$. The moves for both players are always the same in impartial games, so this notation is redundant. Accordingly, if α is a position in an impartial game and $\alpha_1, \dots, \alpha_k$ are the moves available to either player, then we write $\alpha = \{\alpha_1, \dots, \alpha_k\}$. For a Nim example, we have $*4 = \{*0, *1, *2, *3\}$. Figure 3.6 depicts a representation of a 2×4 position in Chop.

$$\begin{array}{|c|c|c|c|} \hline & & & \\ \hline & & & \\ \hline \end{array} = \left\{ \begin{array}{|c|c|c|c|} \hline & & & \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline & & \\ \hline \end{array}, \begin{array}{|c|c|} \hline & \\ \hline \end{array}, \begin{array}{|c|} \hline \\ \hline \end{array} \right\}$$

Figure 3.6. Notation for a position in Chop

Definition 3.11. For a set $S = \{a_1, a_2, \dots, a_n\}$ of nonnegative integers, we define the **Minimal EXcluded value**, abbreviated **MEX**, of S to be the smallest nonnegative integer b which is not one of $a_1, \dots, a_n$. So, for instance, the MEX of the set $\{0, 1, 2, 5, 8\}$ is 3.

Theorem 3.12 (The MEX Principle). *Let $\alpha = \{\alpha_1, \dots, \alpha_k\}$ be a position in an impartial game. Suppose that*

$$\alpha_i \equiv *a_i \text{ for every } 1 \leq i \leq k.$$

*Then $\alpha \equiv *b$ where b is the MEX of the set $\{a_1, \dots, a_k\}$.*

Proof. If we can prove that $\alpha + *b$ is type P, then $\alpha + *b \equiv *0$ and then adding $*b$ to both sides gives the desired result $\alpha \equiv *b$. To complete the proof, then, it will suffice to show that $\alpha + *b$ is type P. To do this, we need to show that the second player can always win from this position.

First, suppose that the first player moves the component $*b$ to $*b'$ for some $b' < b$. Since b is the MEX of the set $\{a_1, \dots, a_k\}$, there must exist $1 \leq i \leq k$ so that $a_i = b'$. Now the second player may move to bring the component α to α_i . The position of the entire game is now $*b' + \alpha_i \equiv *b' + *a_i \equiv *0$. Therefore this position is type P, and the second player has a winning strategy from here.

Next, suppose that the first player moves the α component to α_j . After this move the position is $*b + \alpha_j \equiv *b + *a_j$. Since $a_j \neq b$ (this follows from the fact that b is the MEX of $\{a_1, \dots, a_k\}$), this position is equivalent to the nonzero number $*(b \oplus a_j)$. It follows from this that the second player has a winning strategy, and this completes the proof. $\square$

Given the MEX Principle, we need just one additional definition for the main result of this chapter, the Sprague-Grundy Theorem. For a position α in an arbitrary combinatorial game, we define the *depth* of α to be the maximum number of moves before α reaches a terminal position, where any sequence of moves is permitted (i.e. instead of alternating, either Louise or Richard is permitted to make multiple moves in a row). As usual, we will assume that all positions have finite depth. Just as we used the depth of a game tree to prove theorems about these trees by induction, we will use the depth of a position in our inductive proof of the following theorem.

Theorem 3.13 (Sprague-Grundy). *Every position in an impartial game is equivalent to a nimber.*

Proof. Let α be a position in an impartial game. We will prove this theorem by induction on the depth of α . As a base case, if α has depth 0, then

$\alpha \equiv *0$ since our game is normal-play. For the inductive step, suppose that α has depth > 0 and assume that the theorem holds for every position of depth smaller than α . Using our notation, $\alpha = \{\alpha_1, \dots, \alpha_\ell\}$. Each of the positions α_i has depth less than that of α . By induction, then, every α_i is equivalent to $*a_i$ for some a_i . Take b to be the MEX of $\{a_1, \dots, a_\ell\}$ and apply the MEX Principle to conclude

$$\alpha = \{\alpha_1, \dots, \alpha_\ell\} \equiv \{*\alpha_1, \dots, *\alpha_\ell\} \equiv *b.$$

This proves that α is equivalent to a number. □

The Sprague-Grundy Theorem gives us a thorough understanding of equivalence in impartial games. Furthermore, its proof gives an algorithm for finding number equivalents for any position. The key idea is to work recursively using the MEX Principle. To find a number equivalent to the position α , we first need to find number equivalents for the positions to which we can move from α . So, if $\alpha = \{\alpha_1, \dots, \alpha_\ell\}$ and we have numbers $*a_i$ so that $\alpha_i \equiv *a_i$ for $1 \leq i \leq \ell$, then α will be equivalent to $*b$ where b is the MEX of $\{a_1, \dots, a_\ell\}$. The next section gives several examples of this procedure in action.

3.3. Applying the MEX Principle

So far, we have encountered the impartial games Nim, Pick-Up-Bricks, Chop, and Chomp. The Sprague-Grundy Theorem tells us that every position in every one of these games is equivalent to some number, and the MEX Principle gives us a recipe for finding that equivalent number.

Small Positions. Let's first see the MEX Principle in action with a couple of small positions.

Example 3.14 (Chop). Here we work out the number equivalents for some small positions in Chop.

$$\begin{aligned}
\Box &= \{ \} \equiv *0 \\
\Box\Box &= \{ \Box \} \equiv \{ *0 \} \equiv *1 \\
\Box\Box\Box &= \{ \Box, \Box\Box \} \equiv \{ *0, *1 \} \equiv *2 \\
\Box\Box\Box\Box &= \{ \Box, \Box\Box, \Box\Box\Box \} \equiv \{ *0, *1, *2 \} \equiv *3 \\
\Box \\ \Box &= \{ \Box \} \equiv \{ *0 \} \equiv *1 \\
\Box\Box \\ \Box &= \{ \Box\Box, \Box \\ \Box \} \equiv \{ *1 \} \equiv *0 \\
\Box\Box\Box \\ \Box &= \{ \Box\Box, \Box \\ \Box\Box\Box \} \equiv \{ *0, *1, *2 \} \equiv *3 \\
\Box\Box\Box\Box \\ \Box &= \{ \Box\Box\Box, \Box\Box, \Box \\ \Box\Box\Box\Box \} \equiv \{ *0, *1, *3 \} \equiv *2
\end{aligned}$$

Figure 3.7. Equivalence in Chop

Example 3.15 (Chomp). Here we work out the number equivalents for some small positions in Chomp.

$$\begin{aligned}
\boxtimes &= \{ \} \equiv *0 \\
\boxtimes\boxtimes &= \{ \boxtimes \} \equiv \{ *0 \} \equiv *1 \\
\boxtimes \\ \boxtimes &= \{ \boxtimes \} \equiv \{ *0 \} \equiv *1 \\
\boxtimes\boxtimes\boxtimes &= \{ \boxtimes, \boxtimes\boxtimes \} \equiv \{ *0, *1 \} \equiv *2 \\
\boxtimes \\ \boxtimes\boxtimes &= \{ \boxtimes\boxtimes, \boxtimes \\ \boxtimes \} \equiv \{ *1 \} \equiv *0 \\
\boxtimes\boxtimes\boxtimes &= \{ \boxtimes, \boxtimes\boxtimes, \boxtimes\boxtimes\boxtimes \} \equiv \{ *0, *1, *2 \} \equiv *3 \\
\boxtimes\boxtimes \\ \boxtimes &= \{ \boxtimes\boxtimes, \boxtimes \\ \boxtimes\boxtimes \} \equiv \{ *0, *1 \} \equiv *2 \\
\boxtimes\boxtimes\boxtimes &= \{ \boxtimes, \boxtimes\boxtimes\boxtimes, \boxtimes\boxtimes\boxtimes\boxtimes, \boxtimes\boxtimes\boxtimes \} \equiv \{ *1, *2, *3 \} \equiv *0 \\
\boxtimes\boxtimes\boxtimes\boxtimes &= \{ \boxtimes, \boxtimes\boxtimes\boxtimes\boxtimes, \boxtimes\boxtimes\boxtimes\boxtimes\boxtimes, \boxtimes\boxtimes\boxtimes\boxtimes\boxtimes, \boxtimes\boxtimes\boxtimes\boxtimes\boxtimes\boxtimes \} \equiv \{ *0, *1, *2, *3 \} \equiv *4
\end{aligned}$$

Figure 3.8. Equivalence in Chomp

Next, we'll see how our theory allows us to compute number equivalents to sums of positions efficiently.

Example 3.16. Find a number equivalent to the position in Figure 3.9.

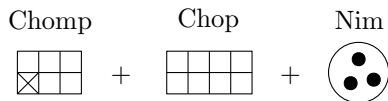


Figure 3.9. A sum of impartial games

We have shown that a 2×3 array in Chomp is equivalent to $*4$ and a 2×4 grid in Chop is equivalent to $*2$. Therefore the position in Figure 3.9 is equivalent to $*4 + *2 + *3 \equiv *(4 \oplus 2 \oplus 3) \equiv *5$.

While knowing a number equivalent is useful, what we really want to know is how to win! If we are moving in an impartial game from a position α , we want to move to a position β of type P, if possible. This is because, by the definition of type P, we will have a winning strategy playing second from β . Accordingly, we define a *winning move* in an impartial game to be any move to a position of type P. Note that winning moves only exist from positions of type N.

In Nim, the balancing procedure always provides a winning move. We can also apply this procedure to play sums of positions involving other impartial games. For instance, if we wish to play the sum $\alpha + \beta + \gamma$ and we have found that $\alpha \equiv *a$ and $\beta \equiv *b$ and $\gamma \equiv *c$, then we just operate as if we were playing the Nim position $*a + *b + *c$.

Example 3.16 (Continued). For the position in Figure 3.9, we have shown that the Chomp component is equivalent to $*4$ and the Chop component is equivalent to $*2$, so this position behaves like $*4 + *2 + *3$. The balancing procedure dictates that in the position $*4 + *2 + *3$, the $*4$ component is moved to $*1$. Therefore, a winning move for this Figure 3.9 position is to move the Chomp component to a 2×1 array which is equivalent to $*1$. At this point the full position will be equivalent to $*1 + *2 + *3 \equiv *0$ and thus of type P.

Pick-Up-Bricks and Chop. Now we will revisit the games of Pick-Up-Bricks and Chop. With the MEX Principle and induction, we can determine the number equivalent to any position in one of these games.

Theorem 3.17. *Let $n = 3\ell + k$ where $0 \leq k \leq 2$. Then a Pick-Up-Bricks position of n bricks is equivalent to $*k$.*

Proof. For every nonnegative integer m , define P_m to be a position in Pick-Up-Bricks with m bricks. Our goal is to prove $P_n = P_{3\ell+k} \equiv *k$ by induction on n . First consider both $n = 0$ and $n = 1$. When $n = 0$, the position has no moves, so it is equivalent to $*0$, or, using our notation, $P_0 \equiv *0$. When $n = 1$, apply the MEX Principle and $P_1 = \{P_0\} \equiv \{*0\} \equiv *1$. This verifies our base cases. For the inductive step, assume $n \geq 2$ and that our formula holds for all smaller values. From a position with $n \geq 2$ bricks, a player may remove either one or two bricks, so

$$P_n = \{P_{n-1}, P_{n-2}\}.$$

As above, assume $n = 3\ell + k$ with $0 \leq k \leq 2$ and consider cases depending on k . In all of the cases, we may utilize our inductive hypothesis to determine both P_{n-1} and P_{n-2} :

$$\begin{aligned} k = 2, \quad P_{3\ell+2} &= \{P_{3\ell+1}, P_{3\ell}\} \equiv \{*1, *0\} \equiv *2, \\ k = 1, \quad P_{3\ell+1} &= \{P_{3\ell}, P_{3(\ell-1)+2}\} \equiv \{*0, *2\} \equiv *1, \\ k = 0, \quad P_{3\ell} &= \{P_{3(\ell-1)+2}, P_{3(\ell-1)+1}\} \equiv \{*2, *1\} \equiv *0. \end{aligned}$$

So we see that in all cases, $P_{3\ell+k} \equiv *k$, as desired. $\square$

Next we turn our attention to the game of Chop. Given an $m \times n$ position in Chop, a move by either player will reduce either the number of rows or the number of columns, but not both. So, from an $m \times n$ position, a player may move either to an $m' \times n$ position with $1 \leq m' < m$ or to an $m \times n'$ position with $1 \leq n' < n$ (i.e. a player can either decrease m or decrease n). In fact, this behaves just like a two-pile game of Nim given by $*(m-1) + *(n-1)$. Pause to convince yourself of this equivalence before reading the formal proof appearing next.

Theorem 3.18. *For every $m, n \geq 1$, an $m \times n$ position in Chop is equivalent to $*(m-1) + *(n-1)$.*

Proof. We let $C_{m,n}$ denote an $m \times n$ position in Chop and will prove by induction on $m + n$ that $C_{m,n} \equiv *(m-1) + *(n-1)$. As a base case, when $m + n = 2$, we have $m = n = 1$ and $C_{1,1}$ has no moves for either player, so $C_{1,1} \equiv *0 \equiv *0 + *0$ and the formula holds. For the inductive step, $m + n > 2$, and we may assume the formula holds true for all smaller values of $m + n$. This gives us the following equation (here

we use position notation = twice, and the one instance of $\equiv$ comes from our inductive assumption).

$$\begin{aligned}
 C_{m,n} &= \{C_{1,n}, \dots, C_{(m-1),n}, \\
 &\quad C_{m,1}, \dots, C_{m,(n-1)}\} \\
 &\equiv \left\{ (*0 + *(n-1)), \dots, (*(m-2) + *(n-1)), \right. \\
 &\quad \left. (*(m-1) + *0), \dots, (*(m-1) + *(n-2)) \right\} \\
 &= *(m-1) + *(n-1). \quad \square
 \end{aligned}$$

Let's combine our results to analyze a more complicated sum.

Example 3.19. Find a number equivalent for the position in Figure 3.10 and a winning move if it exists.

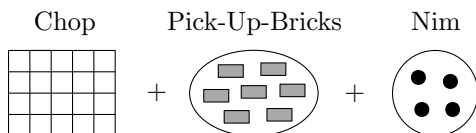


Figure 3.10. A complicated sum

By Theorem 3.18, a 4×5 game of Chop is equivalent to $*3 + *4 \equiv *7$ and, by Theorem 3.17, a Pick-Up-Bricks game with 7 bricks is equivalent to $*1$. Therefore, the position in Figure 3.10 is equivalent to $*7 + *1 + *4 \equiv *(7 \oplus 1 \oplus 4) \equiv *2$. The balancing procedure applied to $*7 + *1 + *4$ instructs us to move the $*7$ component to $*5$. Therefore, a winning move in the given position is to move the Chop position from one equivalent to $*7$ to one equivalent to $*5$. This can be achieved by moving the 4×5 array to a 2×5 array.

Exercises

- (1) Find the binary expansion for the following integers:
 - (a) 23,
 - (b) 47,
 - (c) 163.
- (2) Evaluate:
 - (a) $7 \oplus 4 \oplus 3$,
 - (b) $14 \oplus 24 \oplus 32$,
 - (c) $19 \oplus 13 \oplus 23 \oplus 57$.
- (3) Building upon the analysis in the text, find a number which is equivalent to a 2×4 grid in Chomp.
- (4) Use the balancing procedure to find a winning move in each of the following Nim positions:
 - (a) $*3 + *4 + *5$,
 - (b) $*7 + *9 + *14 + *6$,
 - (c) $*19 + *37 + *28 + *33$.
- (5) For each position, find an equivalent number and a winning move if it exists:
 - (a) a 2×3 array in Chomp plus a 2×4 array in Chop plus $*5$,
 - (b) a 4-brick position in Pick-Up-Bricks plus a 5×3 array in Chop plus $*7$,
 - (c) an 11-brick position in Pick-Up-Bricks plus a 18×24 array in Chop plus $*20$.
- (6) Show that for all integers $a, b \geq 0$, the number $(a + b) - (a \oplus b)$ is both nonnegative and even.
- (7) For the game Kayles, introduced in Exercise (19) in Chapter 1, find a number which is equivalent to a position consisting of k pins in a row for $k = 0, 1, \dots, 8$.
- (8) *Empty and Divide* is an impartial game played with two boxes each of which contains a positive number of chips. To move, a player chooses one of the boxes and removes all of the chips from this box. Then they divide the chips from the other box between the two boxes (so that both still have a positive number). Note that

1	1
---	---

 is therefore a terminal position.

- (a) Find a number which is equivalent to the Empty and Divide position $\boxed{4} \boxed{5}$.
- (b) Find all winning moves from the position below:

Empty and Divide



+

Nim



- (9) *Even-Nim* and *Odd-Nim* are like Nim in that they are played with piles of stones. However, in *Even-Nim* a move consists of removing a positive even number of stones from a pile, while in *Odd-Nim* a move removes an odd number. (Note that a single pile of 1 stone is thus a terminal position in *Even-Nim*.)
- (a) Find a sequence of integers $a_0, a_1, \dots$ so that a pile of size i in *Odd-Nim* is equivalent to $*a_i$.
- (b) Find a sequence of integers $b_0, b_1, \dots$ so that a pile of size i in *Even-Nim* is equivalent to $*b_i$.
- (10) For any finite set S of positive integers, *S-Pick-Up-Bricks*¹ is an impartial game where every position consists of a pile of bricks. To move, a player removes k bricks from the pile for some $k \in S$. So $\{1, 2\}$ -Pick-Up-Bricks is the standard Pick-Up-Bricks game. For the given set S , find a sequence of integers $a_0, a_1, \dots$ so that a position in *S-Pick-Up-Bricks* with i bricks is equivalent to $*a_i$. Prove that your formula holds by induction.
- (a) $S = \{1, 2, \dots, k\}$.
- (b) $S = \{2, 3\}$.
- (c) $S = \{1, 4, 5\}$.
- (11) *Wythoff's King* is an impartial game played with a single king on an otherwise empty chessboard (of any dimension). On each turn a player moves the king one square left, one square down, or one square diagonally down and left.
- (a) For each square on a 6×6 board, find a number equivalent to a king in this position.
- (b) Determine (and prove!) a general formula for the number equivalent to a king in position (i, j) .

¹This is more commonly known as a subtraction game.

- (12) *3D Chop* is an impartial game played with a 3-dimensional array. To move, a player makes a chop to decrease the array in one dimension. So, if the position is an $\ell \times m \times n$ array, then a move will bring it to an $\ell' \times m \times n$ array with $1 \leq \ell' < \ell$ or an $\ell \times m' \times n$ array with $1 \leq m' < m$ or an $\ell \times m \times n'$ array with $1 \leq n' < n$. Find (and prove!) a formula for the nimber equivalent to an $\ell \times m \times n$ array in this game.
- (13) The game *Empty and Divide* was introduced in Exercise 8. Determine (and prove!) what the type P positions are in this game.
- (14) For integers s, t with $t \geq s \geq 0$ and $t \geq 1$ let $\alpha_{s,t}$ denote a position in Chomp consisting of two rows where the top row has length s and the bottom length t (for example $\alpha_{3,7}$ is ). For every such s, t let $a_{s,t} \in \mathbb{Z}$ satisfy $*a_{s,t} \equiv \alpha_{s,t}$. Find and prove a formula in each case below:
- $a_{0,t}$ for $t \geq 1$,
 - $a_{1,t}$ for $t \geq 1$,
 - $a_{2,t}$ for $t \geq 2$.
- (15) The base 2 representation of 13 is $(1101)_2$ since $13 = 8 + 4 + 1 = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0$. More generally we write $n = (a_k \cdots a_1 a_0)_2$ if $n = a_k 2^k + \cdots + a_1 2^1 + a_0 2^0$ where each coefficient a_i is either 0 or 1. If $c = (c_k \cdots c_0)_2$ and $d = (d_k \cdots d_0)_2$, what is the formula for the i^{th} digit of the base 2 representation of $c \oplus d$?
- (16) *Turning Turtles* is an impartial game played on a $1 \times n$ array where each square contains a coin showing either heads or tails. To move, a player turns a coin from heads to tails and then, if she wishes, she may turn over one other coin to the left of it (either from heads to tails or from tails to heads). Prove that a position with heads in positions $n_1, n_2, \dots, n_k$ is equivalent to $*n_1 + *n_2 + \cdots + *n_k$.
- (17) *Infinite Nim* is a game like Nim, except that we allow for one or more infinite piles, each denoted $*\infty$. To move $*\infty$, a player may reduce this pile to any finite size pile he wishes. Extending our notation, we write $*\infty = \{ *0, *1, *2, *3, \dots \}$. For every possible Infinite Nim position $*a_1 + \cdots + *a_k$, determine (and prove!) its type.
- (18) *Heap* is an impartial game in which each position consists of a heap of n tokens for some $n \geq 1$. To move from a position with n tokens,

a player may remove a number of tokens which is at least one and at most $\frac{n}{2}$ from the heap (so the game ends when the heap is down to one token since no moves are possible). For every positive integer i let H_i denote the position with a heap of size i , and let h_i be an integer so that $*h_i \equiv H_i$. Find (and prove) a formula for each of the terms in the sequence $h_0, h_2, h_4, \dots$

- (19) *Wythoff's Queen* is an impartial game played with a single queen on an otherwise empty chessboard (of any size). On each turn a player moves the queen any number of squares either left, down, or diagonally down and to the left.
- For each square on a 6×6 board, find a number equivalent to a queen in this position.
 - Construct a sequence of pairs $(0, 0), (1, 2), (3, 5), (4, 7), \dots$ according to the following rule: Let $(a_0, b_0) = (0, 0)$. For every $n \geq 1$ define a_n to be the smallest positive integer not in the set $\{a_0, b_0, a_1, b_1, \dots, a_{n-1}, b_{n-1}\}$ and then define $b_n = a_n + n$.²
 - Prove that $a_0 < a_1 < a_2 < \dots$.
 - Prove that $b_0 < b_1 < b_2 < \dots$.
 - Prove that every positive integer occurs exactly once in the sequence $a_1, b_1, a_2, b_2, a_3, b_3, \dots$.
 - Prove that a queen in position (i, j) is a type P position if and only if $\{i, j\} = \{a_n, b_n\}$ for some $n \geq 0$.
- (20*) This exercise calls upon the game *S-Pick-Up-Bricks* as defined in Exercise (10). Let S be a finite set of positive integers, and let $a_0, a_1, a_2, \dots$ be integers with the property that a position of n bricks in *S-Pick-Up-Bricks* is equivalent to $*a_n$. Prove that the sequence $a_0, a_1, a_2, \dots$ is eventually periodic. That is, prove that there exist numbers N, p so that $a_i = a_{i+p}$ for every $i > N$.

²Wythoff proved that $(a_n, b_n) = (\lfloor \varphi n \rfloor, \lfloor \varphi^2 n \rfloor)$ for every $n \geq 0$ where $\varphi = \frac{1+\sqrt{5}}{2}$ is the Golden Ratio.

Chapter 4

Hackenbush and Partizan Games

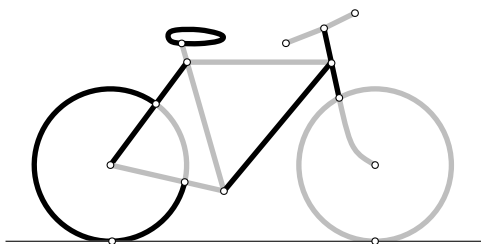


Figure 4.1. A Hackenbush bicycle

The central goal in combinatorial game theory is to understand positions. How can we determine a position's type? What's a good way to play? In the previous chapter we developed a powerful theory for impartial games which provides answers to these questions. We proved the Sprague-Grundy Theorem, which says that every position in an impartial game is equivalent to a nimber. Further, we established the MEX Principle, which gave us a way to compute these numbers.

In this chapter, we turn to partizan games to develop a roughly analogous theory. Where our analysis of impartial games centered on the game Nim, here a game called Hackenbush takes center stage. We will

assign numbers (similar to nimbers) to certain special positions in Hackenbush. We will then prove the Simplicity Principle, which can be used (like the MEX Principle) to show that some (but not all!) positions in other partizan games are equivalent to these special Hackenbush positions.

Since the game Hackenbush will play such a key role in this chapter, let's play it!

Game 4.1 (Hackenbush). This is a normal-play game played between Louise and Richard. The game consists of a graph drawn with black and gray edges, some of which are attached to the ground. A move for Louise is to erase a bLack edge, while a move for Richard is to erase a gRay edge.¹ After an erasure, any part of the graph no longer connected to the ground floats away, out of the game. Since this game is normal-play, the last player to make a move wins.

4.1. Hackenbush

To study Nim, we assigned numbers to certain positions (the nimbers) and then used them to understand more general positions. We adopt a similar approach here by assigning numbers to certain Hackenbush positions and then using these to understand more general positions. To start, define $\bullet 0$ to be the Hackenbush position with no edges (so there are no available moves). The position $\bullet 0$ is type P, so Proposition 2.15 gives us the following.

Proposition 4.2. *A Hackenbush position α satisfies $\alpha \equiv \bullet 0$ if and only if α is type P.*

The sum operation defined in Chapter 2 works on every normal-play game, so we can use it to add any two Hackenbush positions. Since the position $\bullet 0$ has no available moves for either player, we can see that, for every Hackenbush position α , we have $\alpha + \bullet 0 \equiv \alpha$. Indeed, the position $\bullet 0$ in Hackenbush behaves like the position $\ast 0$ in Nim and the number 0 in the integers.

¹This is a grayscale version of the usual definition featuring blue and red edges, with Louise deleting bLue edges and Richard deleting Red edges.

Negation. Something new for us in the world of games is a notion of negation. In Hackenbush, negation arises naturally since we can reverse the roles of the players in any position α by switching the colors of all the edges. This new position is denoted $-\alpha$ and is called the *negative* of α .

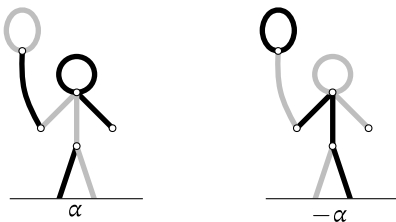


Figure 4.2. A position and its negative

The next proposition shows that negation of Hackenbush positions behaves just like negation of numbers.

Proposition 4.3. *If α and β are Hackenbush positions, then*

- (1) $-(-\alpha) \equiv \alpha$,
- (2) $\alpha + (-\alpha) \equiv \bullet 0$, and
- (3) $\beta + (-\alpha) \equiv \bullet 0$ implies $\alpha \equiv \beta$.

Proof. The first part follows immediately from the definition of negation since $-(-\alpha)$ is identical to α . The previous proposition helps with the second part—it suffices to show that the position $\alpha + (-\alpha)$ is type P. So we will assume the role of the second player and show that we have a winning strategy. Symmetry is the key idea here. We call a Hackenbush position *symmetric* if it has the form $\gamma + (-\gamma)$. If we play second from a symmetric position, then whichever edge our opponent erases from one component, say γ , we will erase the corresponding edge from the other component, $-\gamma$. This pair of moves returns the position to another symmetric one. Repeating this procedure guarantees us the last move and

the win. Finally, the last part of the proposition follows from

$$\beta + (-\alpha) \equiv \bullet 0 \quad \Rightarrow \quad \beta + (-\alpha) + \alpha \equiv \alpha \quad \Rightarrow \quad \beta \equiv \alpha. \quad \square$$

In the world of Hackenbush, we have a zero element, addition, and negation. These concepts all behave in ways familiar from the integers. Further, the last part of the previous proposition gives us a very convenient tool for proving that two Hackenbush positions are equivalent. This will be quite useful moving forward.

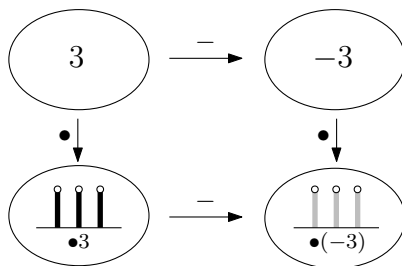
Integer Positions. Our next step will be to give names to some more Hackenbush positions. For every positive integer n , define $\bullet n$ to be the Hackenbush position consisting of n isolated black edges. For a negative integer $-m$, we define $\bullet(-m)$ to be the Hackenbush position consisting of m isolated gray edges.



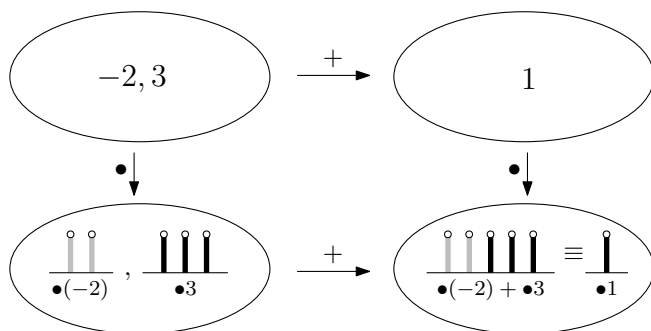
Figure 4.3. Some integral positions

At this point, we have a Hackenbush position $\bullet n$ associated to every integer n , and we call any position of this form an *integral position*. Just as we can negate integers and add integers, we can also negate Hackenbush positions and add Hackenbush positions. We next demonstrate that addition and negation on integral Hackenbush positions correspond naturally to addition and negation of integers.

We may view $\bullet$ as an operation that takes an integer and outputs a special kind of Hackenbush position. In this context, imagine starting with the integer n and applying negation and the $\bullet$ operation. We can do these two operations in two different orders (since we can negate both integers and Hackenbush positions). Specifically, we can go from n to $-n$ to $\bullet(-n)$ or we can go from n to $\bullet n$ to $-(\bullet n)$. Our definitions happily imply that the positions $\bullet(-n)$ and $-(\bullet n)$ are equivalent (and, in fact, $\bullet(-n)$ and $-(\bullet n)$ are identical). In short, we have $-\bullet n \equiv \bullet(-n)$.

Figure 4.4. Negation and $\bullet$.

What about the relationship between the operations of $\bullet$ and addition? Given two integers m and n , we can perform the operations of $\bullet$ and $+$ in two different orders. We can first add integers m and n to $m+n$ and then apply the $\bullet$ operation to the sum to get $\bullet(m+n)$. Alternately, we can first apply the $\bullet$ operation to integers m and n to get $\bullet m$ and $\bullet n$ and then add these to get $\bullet m + \bullet n$. We claim that the two positions resulting from this process will always be equivalent, or, more succinctly, $\bullet m + \bullet n \equiv \bullet(m+n)$.

Figure 4.5. Sum and $\bullet$.

If m and n have the same sign, then these two positions will be identical (and they will consist entirely of edges of the same color) and the result is automatic. Suppose next that m and n have different signs and set $\ell = |m - n|$. In this case, $\bullet(m+n)$ will consist of ℓ isolated edges of the same color, while $\bullet m + \bullet n$ will have edges of both colors, but ℓ

more of one color than the other. Fortunately, Proposition 4.3 and the above analysis of negation imply that $\bullet k + \bullet(-k) \equiv \bullet 0$ for every integer k . Therefore, modifying a position by deleting the same number of isolated gray edges and isolated black edges results in an equivalent position. It follows that $\bullet(m + n) \equiv \bullet m + \bullet n$, as desired. The following theorem summarizes these results.

Theorem 4.4. *For any integers m and n , we have*

- (1) $\bullet(-n) \equiv \bullet(-n)$,
- (2) $(\bullet m) + (\bullet n) \equiv \bullet(m + n)$.

The above theorem provides a complete understanding of the behavior of integral Hackenbush positions under sums and negation. These positions behave just like the integers with which they are named. For instance, the position $\bullet a + \bullet(-b) + \bullet(c) + \bullet(-d)$ will be equivalent to the position $\bullet(a - b + c - d)$. It is straightforward to determine the type of an integer position since in each such position some player has no moves. The position $\bullet n$ will be type L when $n > 0$, type P when $n = 0$, and type R when $n < 0$.

We have just observed that for $n > 0$, the integral position $\bullet n$ will be type L and thus have an advantage for Louise. In fact, it is reasonable to quantify this advantage and say that the position $\bullet n$ gives Louise an advantage of n . We would likewise consider the integral position $\bullet(-n)$ as one that has an advantage of n for Richard. Of course, this interpretation of these integral positions applies more broadly also to positions which are equivalent to integral positions. We would thus consider an arbitrary Hackenbush position α to offer Louise an advantage of n if $\alpha \equiv \bullet n$ and we would consider α to offer Richard an advantage of n if $\alpha \equiv \bullet(-n)$.



Figure 4.6. A position with an advantage of 2 for Louise

What happens to this concept of advantage when we add positions? Suppose that α offers Louise an advantage of m and β offers her an advantage of n . Then we have $\alpha \equiv \bullet m$ and $\beta \equiv \bullet n$, so $\alpha + \beta \equiv \bullet m + \bullet n \equiv \bullet(m+n)$. Thus, $\alpha + \beta$ will give Louise an advantage of $m+n$. So, Louise's advantage in $\alpha + \beta$ is the sum of her advantages in α and β .

Fractional Positions? Could there be a Hackenbush position α that gives Louise an advantage of $\frac{1}{2}$? If so, we would expect $\alpha + \alpha$ to offer Louise an advantage of 1, and then the position $\alpha + \alpha + \bullet(-1)$ would be equivalent to $\bullet 0$ (of type P), with no advantage for either player. Test this for the position α consisting of one gray edge on top of one black edge, as seen in Figure 4.7.

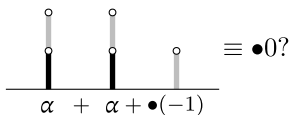


Figure 4.7. Is α worth $\frac{1}{2}$ to Louise?

A bit of analysis reveals that the full position in Figure 4.7 is of type P, so it is indeed equivalent to $\bullet 0$. So, the position α is one that gives Louise an advantage of $\frac{1}{2}$! Using negation, we would consider $-\alpha$ to be one which offers Richard an advantage of $\frac{1}{2}$. Could there be a position β which offers Louise an advantage of $\frac{1}{4}$? If so, we would expect $\beta + \beta + (-\alpha) \equiv \bullet 0$. Test this by considering the position β in Figure 4.8.

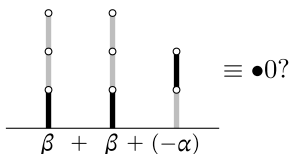


Figure 4.8. Is β worth $\frac{1}{4}$ to Louise?

Again, a bit of analysis reveals that the full position in Figure 4.8 is type P and thus is equivalent to $\bullet 0$. So we consider β to be a position with an advantage for Louise of $\frac{1}{4}$. We now introduce some terminology

to help formalize these new ideas. For every positive integer k , define $\bullet \frac{1}{2^k}$ to be the Hackenbush position shown in Figure 4.9.

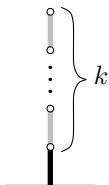
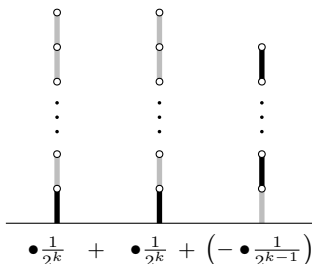


Figure 4.9. The position $\bullet \frac{1}{2^k}$

Now extend the analysis we used on Figures 4.7 and 4.8 to the more general $\bullet \frac{1}{2^k}$ positions defined in Figure 4.9.

Lemma 4.5. *For every positive integer k , we have $\bullet \frac{1}{2^k} + \bullet \frac{1}{2^k} \equiv \bullet \frac{1}{2^{k-1}}$.*

Proof. It suffices to prove $\bullet \frac{1}{2^k} + \bullet \frac{1}{2^k} + (-\bullet \frac{1}{2^{k-1}}) \equiv \bullet 0$ for all $k \geq 1$, and we do this by induction on k . The base case $k = 1$ was handled in our earlier discussion. For the inductive step ($k \geq 2$), we will prove the desired equation by showing that $\bullet \frac{1}{2^k} + \bullet \frac{1}{2^k} + (-\bullet \frac{1}{2^{k-1}})$ is type P.



We first consider the case that Richard plays first and will show that Louise has a winning strategy as the second player. If Richard chooses the gray edge from the $-\bullet \frac{1}{2^{k-1}}$ component, then Louise will win no matter what. Suppose instead that Richard chooses a gray edge from one of the $\bullet \frac{1}{2^k}$ components. Since it cannot hurt his future options and does not affect Louise's, we may assume that he chooses the topmost edge

(convince yourself this makes sense). Now Louise can remove the bottom edge from the other $\bullet \frac{1}{2^k}$ component to bring the full position to $\bullet \frac{1}{2^{k-1}} + (-\bullet \frac{1}{2^{k-1}}) \equiv \bullet 0$. As second player from this position, Louise has a winning strategy.

We next consider the case that Louise plays first and will show that Richard has a winning strategy as the second player. If Louise removes the black edge from one of the $\bullet \frac{1}{2^k}$ components, then Richard may remove the topmost gray edge from the other $\bullet \frac{1}{2^k}$ component to bring the position to $\bullet \frac{1}{2^{k-1}} + (-\bullet \frac{1}{2^{k-1}}) \equiv \bullet 0$. From there, Richard has a winning strategy as second player. If Louise removes a black edge from the $-\bullet \frac{1}{2^{k-1}}$ component, we may assume that she removes the topmost, bringing this to $-\bullet \frac{1}{2^{k-2}}$. Now, let Richard remove the topmost gray edge from one of the $\bullet \frac{1}{2^k}$ components. The position is then $\bullet \frac{1}{2^k} + \bullet \frac{1}{2^{k-1}} + (-\bullet \frac{1}{2^{k-2}})$. By induction, $\bullet \frac{1}{2^{k-1}} + \bullet \frac{1}{2^{k-1}} + (-\bullet \frac{1}{2^{k-2}}) \equiv \bullet 0$, so Richard has a winning strategy as second player in $\bullet \frac{1}{2^{k-1}} + \bullet \frac{1}{2^{k-1}} + (-\bullet \frac{1}{2^{k-2}})$. Since the present position is obtained from this one by adding a gray edge on top, this strategy also guarantees Richard a win in the present position. $\square$

For every positive integer k , we have now established a position $\bullet \frac{1}{2^k}$ that may be interpreted as having an advantage of $\frac{1}{2^k}$ for Louise. Although it is clear that these type L positions offer Louise an advantage, the precise quantification of this advantage is somewhat surprising. We can use these curious new fractional positions to analyze more general Hackenbush positions, but first we need to introduce an important set of numbers.

4.2. Dyadic Numbers and Positions

This section introduces the dyadic numbers and establishes some of their basic properties. With the requisite properties of these numbers in place, we will then define some special Hackenbush positions based on them.

The Dyadic Numbers. Any number that can be expressed as a fraction where the denominator is a power of 2 (and the numerator is an integer) is called a *dyadic number*. So for instance, $\frac{17}{32}$ and $\frac{531}{64}$ are dyadic numbers, but $\frac{2}{7}$ and π are not.

As before, the *binary expansion* of a number q is a representation of q as a sum of distinct powers of 2. We now will also use negative powers of 2 (or, equivalently, fractions of the form $\frac{1}{2^d}$ with $d > 0$). Finding binary expansions of dyadic numbers is straightforward given the procedure for integers, as illustrated by the following example:

$$\begin{aligned}\frac{83}{64} &= \frac{1}{64}(83) \\ &= \frac{1}{64}(64 + 19) \\ &= \frac{1}{64}(64 + 16 + 3) \\ &= \frac{1}{64}(64 + 16 + 2 + 1) \\ &= 1 + \frac{1}{4} + \frac{1}{32} + \frac{1}{64}.\end{aligned}$$

Next is a formal proof that this procedure always works.

Proposition 4.6. *Every dyadic number has a unique (finite) binary expansion.*

Proof. Let $q = \frac{n}{2^k}$ be a dyadic number. Since n is an integer, it has a binary expansion $n = 2^{d_1} + 2^{d_2} + \dots + 2^{d_\ell}$. Therefore, q has binary expansion given by $q = 2^{d_1-k} + 2^{d_2-k} + \dots + 2^{d_\ell-k}$. To see that this is unique, suppose (for a contradiction) that we have two different binary expansions of q . If the smallest power of 2 appearing in either one is 2^{-d} , then multiplying both expressions through by 2^d gives two different binary expansions of the integer $2^d q$, which is impossible. $\square$

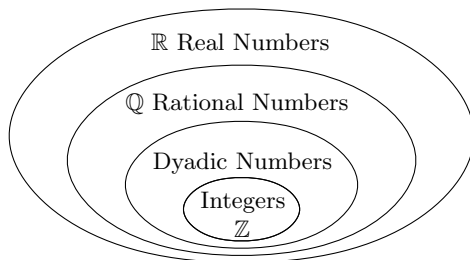


Figure 4.10. Sets of numbers

For every dyadic number $\frac{m}{2^k}$, its negation $-\frac{m}{2^k}$ is another dyadic number. Similarly, for a pair of dyadic numbers, $\frac{m}{2^k}$ and $\frac{n}{2^h}$, the sum is likewise another dyadic number, $\frac{2^h m + 2^k n}{2^{k+h}}$. This property is known as *closure*, so we say that the dyadic numbers are *closed* under negation and addition. Many familiar and fundamental sets of numbers such as $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are closed under these operations. There are other, lesser-known, sets like the dyadic numbers that also exhibit these properties. The Venn diagram in Figure 4.10 shows the relationship between these various closed subsets of numbers.

Birthdays of Dyadic Numbers. Although we generally view the dyadic numbers as static, we are going to take a very different perspective here. We will imagine time moving forward one day at a time, starting with day 0, and we will think of each dyadic number as appearing, or being born, on a certain day. This gives each dyadic number a *birthday*, and these birthdays will be a key concept moving forward. Before formalizing the procedure, let's start by seeing which numbers are born on the first few days. For each day n , Figure 4.11 shows in black the new numbers born on day n , while the numbers born on days before n are gray.

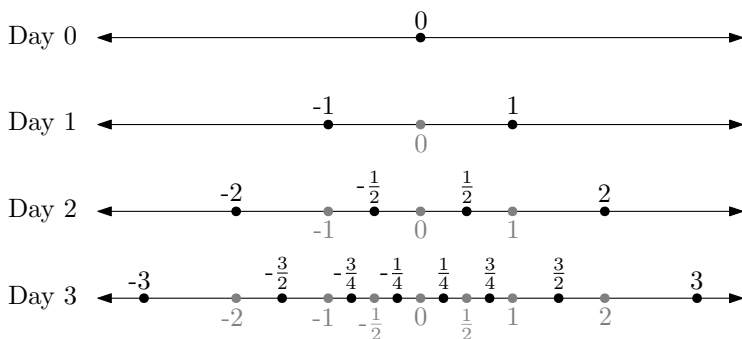


Figure 4.11. Birthdays of some dyadic numbers

Here is the general rule: On day 0 the number 0 is born. If $a_1 < a_2 < \dots < a_\ell$ are the numbers born on days $0, 1, \dots, n$, then on day $n + 1$ the following new numbers are born:

- the largest integer which is less than a_1 ,
- the smallest integer which is greater than a_ℓ ,
- the number $(a_i + a_{i+1})/2$ for every $1 \leq i \leq \ell - 1$.

Exercise (9) proves that the set of all numbers created in the above manner is precisely the set of dyadic numbers. The other necessary property we require is the following.

Proposition 4.7. *Every open interval of real numbers (a, b) , (a, ∞) , $(-\infty, b)$, or even $(-\infty, \infty)$, contains a unique oldest dyadic number.*

Proof. First, observe that every interval of length $> \frac{1}{2^k}$ will contain a dyadic number which can be expressed as a fraction with denominator 2^k . It follows that every open interval contains at least one dyadic number. To complete the proof, we need to show that there cannot be two distinct dyadic numbers q_1, q_2 in the interval I so that both q_1 and q_2 are oldest numbers in I . Suppose (for a contradiction) that this is true. If q_1 and q_2 are both born on day n , then by the construction, there must exist a dyadic number q born on a day before n which is between q_1 and q_2 . However, then q is also in the interval I and is older than q_1 or q_2 , which is a contradiction. $\square$

Note, tangentially, that this recursive process with which we have constructed the dyadic rationals can be continued to day infinity and beyond! This results in an amazing number system called the surreal numbers, discussed in Appendix A.

Dyadic Positions. Now that we have a good handle on dyadic numbers, we will introduce associated Hackenbush positions called dyadic positions. For every dyadic number $q > 0$ with binary expansion $2^{d_1} + 2^{d_2} + \dots + 2^{d_\ell}$ (here $d_1 > d_2 > \dots > d_\ell$ are integers which may be positive

or negative), we define the position²

$$\bullet q = \bullet 2^{d_1} + \bullet 2^{d_2} + \cdots + \bullet 2^{d_\ell}.$$

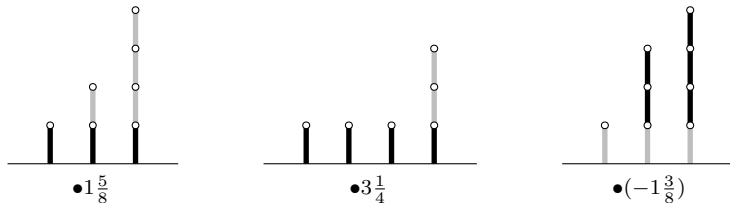


Figure 4.12. Some dyadic positions

To get negative positions, we define $\bullet(-q) = -(\bullet q)$. We will call any position of the form $\bullet q$ a *dyadic position*. In the previous section, we proved that addition and negation for integral Hackenbush positions behave just like addition and negation does for the integers. The next step is to extend this property to the dyadic numbers. In preparation for this we first prove the following lemma.

Lemma 4.8. *Let $a_1, \dots, a_n$ be numbers, each of which is either 0 or has the form $\pm 2^k$ for some integer k . If $a_1 + a_2 + \cdots + a_n = 0$, then $\bullet a_1 + \bullet a_2 + \cdots + \bullet a_n \equiv \bullet 0$.*

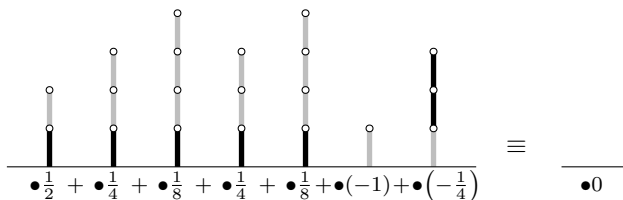


Figure 4.13. A Hackenbush position equivalent to $\bullet 0$

Proof. We proceed by induction on n . If $n = 1$, there is just one number a_1 , so we must have $a_1 = 0$, and thus $\bullet a_1 \equiv \bullet 0$, as desired. For the

²Technically speaking, we are redefining some integer positions here. For instance 7 is an integer, but also a dyadic rational, so we are redefining $\bullet 7 = \bullet 4 + \bullet 2 + \bullet 1$. Fortunately, this definition agrees with our earlier one, so all is well.

inductive step, let $a_1, \dots, a_n$ be a list with $n \geq 2$ satisfying the assumptions, and assume the lemma is true for any list with $< n$ terms. If all of the numbers in our list are integers, then Theorem 4.4 implies that $\bullet a_1 + \dots + \bullet a_n = \bullet(a_1 + \dots + a_n) = \bullet 0$ and we are done. Otherwise, choose k to be the largest number so that one of $a_1, \dots, a_n$ is equal to $\pm \frac{1}{2^k}$. The ordering of the numbers in our list does not matter, so by reordering we may assume $a_n = \pm \frac{1}{2^k}$. If all of the terms in $a_1, \dots, a_{n-1}$ can be expressed as fractions with denominator 2^{k-1} , then it is not possible for $a_1 + \dots + a_n = 0$. So, there must be another term in $a_1, \dots, a_{n-1}$ that is equal to $\pm \frac{1}{2^k}$. By reordering, we may assume $a_{n-1} = \pm \frac{1}{2^k}$. Define $b = a_{n-1} + a_n$ and note that either $b = 0$ or $b = \pm \frac{1}{2^{k-1}}$. Now our lemma applies by induction to the sequence $a_1, \dots, a_{n-2}, b$. This gives the equation

$$\bullet a_1 + \dots + \bullet a_{n-2} + \bullet b \equiv \bullet 0.$$

If a_n and a_{n-1} have opposite signs, then $b = 0$ and $\bullet a_{n-1} + \bullet a_n \equiv \bullet 0 \equiv \bullet b$, which combines with the above equation to give the desired result. On the other hand, if a_n and a_{n-1} have the same sign, then Lemma 4.5 implies that $\bullet a_{n-1} + \bullet a_n \equiv \bullet b$ and again we may combine this with the above equation to get the desired result. $\square$

Since each dyadic position $\bullet q$ is a sum of positions of the form $\bullet(\pm 2^d)$, the previous lemma is all we need to establish the behavior of dyadic positions under sums. For instance, the proof that the equation in Figure 4.14 holds true follows from the equation in Figure 4.13.

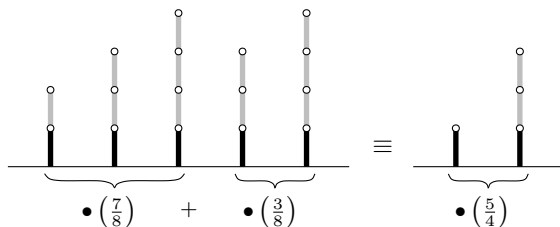


Figure 4.14. Adding two dyadic positions

Theorem 4.9. *If p, q are dyadic numbers, then*

- (1) $-(\bullet p) \equiv \bullet(-p)$,
- (2) $(\bullet p) + (\bullet q) \equiv \bullet(p + q)$.

Proof. The first part of this theorem follows from the definitions of negation and our dyadic positions. For the second part, apply the previous lemma to obtain $\bullet p + \bullet q + (-\bullet(p + q)) \equiv \bullet 0$. Then adding $\bullet(p + q)$ to both sides of this equation yields the desired result. $\square$

It is rather straightforward to determine the type of a dyadic position. For a positive rational number $q > 0$ the position $\bullet q$ is a sum of positions of type L and therefore $\bullet q$ is type L. Conversely, the position $\bullet(-q)$ is a sum of positions of type R, so $\bullet(-q)$ will also be type R. This gives us the following observation.

Observation 4.10. *For a dyadic number q we have*

$$\bullet q \text{ is type } \begin{cases} \text{L} & \text{if } q > 0, \\ \text{P} & \text{if } q = 0, \\ \text{R} & \text{if } q < 0. \end{cases}$$

The above observation and previous theorem give us an appealing extension to our previous interpretation of advantage. Namely, we may now consider a position α with $\alpha \equiv \bullet q$ as one which offers an advantage of q for Louise. Then adding two positions which give Louise an advantage of p and q yields a position which gives Louise an advantage of $p + q$.

4.3. The Simplicity Principle

Chapter 3 introduced the MEX Principle, which gave us a recursive procedure to determine the nimber equivalent of any position in an impartial game. This section introduces an analogous principle for impartial games called the Simplicity Principle. This new principle provides a procedure to determine a dyadic position equivalent to a given one under certain assumptions.

The Proof. While the available moves for both players are the same in impartial games, in Hackenbush they are generally quite different. Accordingly, we will use the position notation introduced in Section 2.1 to denote Hackenbush positions (and more generally positions in partizan games). As in Figure 4.15, the available moves for Louise are to the left of the bar, and those for Richard are to the right.

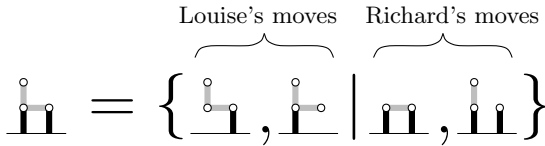


Figure 4.15. Position notation

Our proof of the Simplicity Principle involves some basic properties about positions of the form $\bullet \cdot \frac{1}{2^k}$. Using our position notation with dyadic positions, for every $k > 0$ we have

$$\bullet \cdot \frac{1}{2^k} = \{ \bullet \cdot 0 \mid \bullet \cdot 1, \bullet \cdot \frac{1}{2}, \dots, \bullet \cdot \frac{1}{2^{k-1}} \}.$$

So, if a player moves $\bullet \cdot \frac{1}{2^k}$ to $\bullet \cdot c$, then either Richard moved and c is at least $\frac{1}{2^k}$ larger than $\frac{1}{2^k}$, or Louise moved and c is $\frac{1}{2^k}$ smaller. This principle immediately yields the following.

Lemma 4.11. *Let $c = \frac{n}{2^k}$ with $k \geq 1$ and suppose a player moves the position $\bullet \cdot c$ to the new position $\bullet \cdot c'$.*

- (1) *If Louise moved, then $c' \leq c - \frac{1}{2^k}$.*
- (2) *If Richard moved, then $c' \geq c + \frac{1}{2^k}$.*

We are now ready to introduce the Simplicity Principle that we will use to prove that certain positions in partizan games are equivalent to dyadic positions.

Theorem 4.12 (The Simplicity Principle). *Consider a position in a partizan game given by $\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$ and suppose*

$$\alpha_i \equiv \bullet \cdot a_i \text{ for } 1 \leq i \leq m,$$

$$\beta_j \equiv \bullet \cdot b_j \text{ for } 1 \leq j \leq n.$$

If there do not exist a_i and b_j with $a_i \geq b_j$, then $\gamma \equiv \bullet c$ where c is the oldest number larger than all of $a_1, \dots, a_m$ and smaller than all of $b_1, \dots, b_n$.

Proof. Assume first that both players have available moves and (by possibly reordering) that $a_m \leq \dots \leq a_1$ and $b_1 \leq \dots \leq b_n$. Then c is the oldest number in the interval (a_1, b_1) . To simplify, assume further that $c = \frac{\ell}{2^k}$ where $\ell, k \geq 1$ (the other cases are similar). Since both $c + \frac{1}{2^k}$ and $c - \frac{1}{2^k}$ are older than c , we have

$$c - \frac{1}{2^k} \leq a_1 < c < b_1 \leq c + \frac{1}{2^k}.$$

To prove that $\gamma \equiv \bullet c$, it suffices to show that $\gamma + \bullet(-c) \equiv \bullet 0$ (then adding $\bullet c$ to both sides yields our desired equation). We then need to prove that the position $\gamma + \bullet(-c)$ is type P. That is, for $\gamma + \bullet(-c)$, Richard has a winning strategy when Louise plays first, and Louise has a winning strategy when Richard plays first.

Assume Louise plays first and moves the γ component to α_i . The new position $\alpha_i + \bullet(-c)$ is equivalent to $\bullet a_i + \bullet(-c)$. Since $a_i - c \leq a_1 - c < 0$, Richard has a winning strategy from here. Suppose next that Louise plays first and moves in the $\bullet(-c)$ component. In this case, Lemma 4.11 implies that the new position will be equivalent to $\gamma + \bullet(-c')$ where $(-c') \leq (-c) - \frac{1}{2^k}$. So, now Richard can move γ to β_1 bringing the game to a position equivalent to $\bullet b_1 + \bullet(-c')$. Since $b_1 - c' \leq b_1 - c - \frac{1}{2^k} \leq 0$, this new position is of either type R or P and in either case, Richard has a winning strategy.

Similarly, if Richard plays first and moves in γ , the new position is equivalent to $\bullet b_j + \bullet(-c)$, and since $b_j - c > 0$, Louise has a winning strategy from here. If Richard plays first and moves $\bullet(-c)$ to $\bullet(-c')$, then Louise can move γ to α_1 bringing the full position to one equivalent to $\bullet a_1 + \bullet(-c')$. Since $(-c') \geq (-c) + \frac{1}{2^k}$, we have $a_1 - c' \geq a_1 - c + \frac{1}{2^k} \geq 0$, so playing second from here, Louise has a winning strategy. $\square$

Example. Here we use the Simplicity Principle to find a dyadic position equivalent to a chair-shaped Hackenbush position.

$$\begin{aligned}
\underline{\quad} &= \{ \mid \} \equiv \{\} \equiv \bullet 0 \\
\underline{\bullet} &= \{ \underline{\quad} \mid \} \equiv \{\bullet 0 \mid \} \equiv \bullet 1 \\
\underline{\bullet\bullet} &= \{ \underline{\bullet} \mid \} \equiv \{\bullet 1 \mid \} \equiv \bullet 2 \\
\underline{\circ} &= \{ \underline{\quad} \mid \underline{\bullet} \} \equiv \{\bullet 0 \mid \bullet 1\} \equiv \bullet \frac{1}{2} \\
\underline{\bullet\circ} &= \{ \underline{\circ} \mid \underline{\bullet\bullet} \} \equiv \{\bullet \frac{1}{2} \mid \bullet 2\} \equiv \bullet 1 \\
\underline{\circ\circ} &= \{ \underline{\quad} \mid \underline{\circ} \} \equiv \{\bullet 0 \mid \bullet \frac{1}{2}\} \equiv \bullet \frac{1}{4} \\
\underline{\circ\bullet} &= \{ \underline{\quad} \mid \underline{\circ}, \underline{\bullet} \} \equiv \{\bullet 0 \mid \bullet \frac{1}{2}, \bullet 1\} \equiv \bullet \frac{1}{4} \\
\underline{\bullet\circ\bullet} &= \{ \underline{\bullet}, \underline{\circ} \mid \underline{\bullet\bullet} \} \equiv \{\bullet 1, \bullet \frac{1}{2} \mid \bullet 2\} \equiv \bullet \frac{3}{2} \\
\underline{\circ\bullet\circ} &= \{ \underline{\circ\bullet}, \underline{\circ\circ} \mid \underline{\bullet\bullet}, \underline{\bullet\circ} \} \equiv \{\bullet \frac{1}{4} \mid \bullet 1, \bullet \frac{3}{2}\} \equiv \bullet \frac{1}{2}
\end{aligned}$$

Figure 4.16. Applying the Simplicity Principle in Hackenbush

Applying the Simplicity Principle. In the previous chapter, we established the MEX Principle, which can be used (recursively) to show that a position in an arbitrary impartial game is equivalent to a number. So even though the MEX Principle is based on Nim, it can be more broadly applied. The Simplicity Principle acts similarly. Although dyadic positions are Hackenbush positions, the Simplicity Principle can be used (recursively) to show that many positions in other partizan games are equivalent to dyadic positions. Let's do a small example from Domineering, as in Figure 4.17.

$$\begin{aligned}
- &= \{ | \} \equiv \bullet 0 \\
\Box\Box &= \{ | - \} \equiv \{ | \bullet 0 \} \equiv \bullet(-1) \\
\Box &= \{ - | \} \equiv \{ \bullet 0 | \} \equiv \bullet 1 \\
\Box\Box\Box &= \{ \Box\Box | -, \Box \} \equiv \{ \bullet(-1) | \bullet 0, \bullet 1 \} \equiv \bullet(-\tfrac{1}{2})
\end{aligned}$$

Figure 4.17. Domineering positions equivalent to dyadic positions

The Simplicity Principle is also the key to understanding Cut-Cake. To start, note that a $1 \times n$ piece in Cut-Cake is equivalent to $\bullet(n-1)$ since in this position Louise has $n-1$ moves—which she can make in any order—but Richard has no available moves. Similarly an $n \times 1$ piece in Cut-Cake is equivalent to $\bullet(-n+1)$. Next we apply the Simplicity Principle to analyze a small position in this game, frequently taking advantage of our knowledge of sums.

$$\begin{aligned}
\Box &= \{ \Box + \Box | \Box + \Box \} \equiv \{ \bullet(-2) | \bullet 2 \} \equiv \bullet 0 \\
\Box\Box &= \{ \Box + \Box | \Box\Box + \Box\Box \} \equiv \{ \bullet(-1) | \bullet 4 \} \equiv \bullet 0 \\
\Box &= \{ \Box + \Box | \Box + \Box \} \equiv \{ \bullet(-4) | \bullet 1 \} \equiv \bullet 0 \\
\Box\Box &= \{ \Box + \Box | \Box\Box + \Box\Box \} \equiv \{ \bullet(-2) | \bullet 2 \} \equiv \bullet 0 \\
\Box\Box\Box &= \{ \Box + \Box\Box, \Box\Box + \Box | \Box\Box\Box + \Box\Box\Box \} \equiv \{ \bullet(-1), \bullet 0 | \bullet 6 \} \equiv \bullet 1 \\
\Box\Box\Box &= \{ \Box + \Box\Box, \Box\Box + \Box | \Box\Box\Box + \Box\Box\Box \} \equiv \{ \bullet(-2), \bullet 0 | \bullet 4 \} \equiv \bullet 1
\end{aligned}$$

Figure 4.18. Cut-Cake positions equivalent to dyadic positions

Since we understand how dyadic positions behave under sums, we can also evaluate sums of positions which are equivalent to dyadic positions. For instance if α, β are positions equivalent to $\bullet a, \bullet b$, then $\alpha + \beta \equiv \bullet a + \bullet b$. Let's do an example.

Example 4.13. By earlier analysis, the position in Figure 4.19 is equivalent to $\bullet \frac{3}{4}$, so it is type L and gives Louise an advantage of $\frac{3}{4}$. See Exercise (7) for a discussion of winning moves.

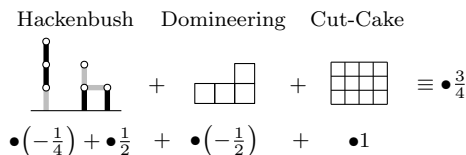


Figure 4.19. Sums of positions

The Big Picture. In the great world of normal-play games, we have now established two important sets of positions: the numbers and the dyadic positions. We can use the MEX Principle to show that any position in an impartial game is equivalent to a number. Likewise, we can use the Simplicity Principle to show that some positions in partizan games are equivalent to dyadic positions. Our theory in both cases also allows us to understand sums. If two positions are equivalent to numbers, say $*a$ and $*b$, then their sum will be equivalent to $*(a \oplus b)$. On the other hand, if two positions are equivalent to dyadic positions $\bullet a$ and $\bullet b$, their sum will be equivalent to $\bullet(a + b)$.

Numbers	Dyadic Positions
MEX Principle to show equivalence	Simplicity Principle to show equivalence
Nim-sum to add: $*a + *b \equiv *(a \oplus b)$	Usual sum to add: $\bullet a + \bullet b \equiv \bullet(a + b)$
Applies to all impartial games	Applies to some partizan games

As we have mentioned, the Simplicity Principle does not apply to every position in a partizan game. For the position

$$\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\} \equiv \{\bullet a_1, \dots, \bullet a_m \mid \bullet b_1, \dots, \bullet b_n\}$$

the Simplicity Principle is only applicable when $\max\{a_1, \dots, a_m\} < \min\{b_1, \dots, b_n\}$. As a result, there is a wide world of partizan-game positions that are not equivalent to dyadic positions.

Earlier in this section, we succeeded in using the Simplicity Principle to show that a certain position in Domineering is equivalent to $\bullet(-\frac{1}{2})$. That calculation worked since the Simplicity Principle applied at each step. However, there are other positions in Domineering that behave differently. For instance, consider a 2×2 array as in Figure 4.20. Any move by Louise brings the game to a 1×2 array which is equivalent to $\bullet 1$, while any move by Richard brings the game to a 2×1 array which is equivalent to $\bullet(-1)$.

$$\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array} = \left\{ \begin{array}{|c|} \hline \\ \hline \end{array} \mid \begin{array}{|c|c|} \hline & \\ \hline \end{array} \right\} \equiv \{\bullet 1 \mid \bullet(-1)\} \equiv ?$$

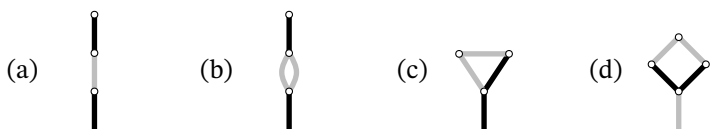
Figure 4.20. A position in Domineering

Our analysis demonstrates that this 2×2 position is equivalent to $\{\bullet 1 \mid \bullet(-1)\}$. Yet, since the maximum of the set of numbers on the left-hand side is greater than the minimum of those on the right, the Simplicity Principle does not apply here. What distinguishes this position from any dyadic position is the fact that each player would prefer to move first—this is a position of type N.

In fact, there is a vast and vibrant theory of partizan games featuring a wide variety of games and positions, some fascinating theoretical results, and numerous tantalizing unsolved problems. Material in Appendix A prepares the interested reader for further study of the subject in the brilliant classic *Winning Ways* [7] and the modern update *Lessons in Play* [1].

Exercises

- (1) Find the binary expansion for each of the following fractions:
 - (a) $15/16$,
 - (b) $61/32$,
 - (c) $317/128$.
- (2) Draw each of the given dyadic positions:
 - (a) $\bullet(5/8)$,
 - (b) $\bullet(23/32)$,
 - (c) $\bullet(-121/64)$.
- (3) Which dyadic numbers are born on days 4 and 5?
- (4) Find an equivalent dyadic position for each given Hackenbush position:



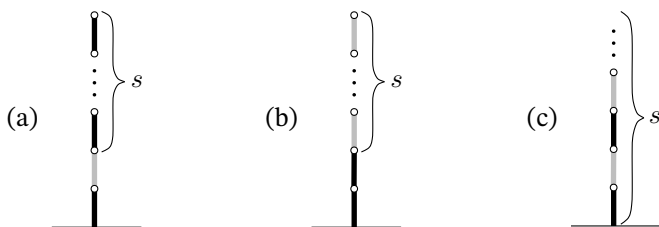
- (5) Find a dyadic position equivalent to the following given positions:
- (a) A 3×6 board in Cut-Cake. (Hint: Use the text.)
 - (b) The Domineering position .
 - (c) The sum of a 3×6 board in Cut-Cake, the Domineering position from part (b), and $\bullet(-5/4)$.
- (6) *Push* is a partizan game played on a $1 \times n$ array. Each cell is either empty or contains an L or an R. On Louise's turn, she may choose any L to push one cell to the left. Similarly, on Richard's turn, he may choose any R to push one cell to the left. Each cell can contain at most one letter, so if a letter gets pushed into an already occupied cell, the letter in that cell gets pushed one cell further to the left, and so on. Any letter which gets pushed from the leftmost square falls out of the array and disappears. Find a dyadic position equivalent to the Push position

R	R	L
---	---	---

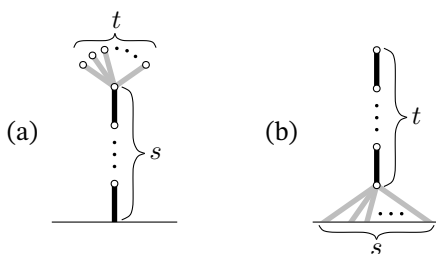
.
- (7) For a position γ in a partizan game, a *winning move* for Louise is any move to a position of type L or type P, while a *winning move* for Richard is any move to a position of type R or type P (in both cases, the player has a winning strategy playing second from the resulting position). Let $a_1, \dots, a_n$ be dyadic numbers, and consider the position $\alpha = \bullet a_1 + \bullet a_2 + \dots + \bullet a_n$.
- (a) If $a_1 + a_2 + \dots + a_n \geq 1$, what are Louise's winning moves from α ?
 - (b) If $0 < a_1 + a_2 + \dots + a_n < 1$, what are Louise's winning moves from α ?
 - (c) More generally, when we have positions which are equivalent to dyadic ones, we can use the same principle to find winning moves. Find all winning moves for Louise in the position from Figure 4.19.
- (8) For every nonnegative integer n , let D_n denote the set of dyadic numbers born on day n . Prove the following properties of D_n for every $n \geq 0$:
- (a) $\max D_n = n$ and $\min D_n = -n$,
 - (b) D_n is closed under negation (if $q \in D_n$ then $-q \in D_n$).
- (9) Fix an integer $n \geq 0$ and consider the real interval $[n, n+1]$. For every nonnegative integer i let C_i be the set of numbers in the interval

$[n, n + 1]$ which have been born on or before day i . Find and prove a formula for C_i for every $i \geq 0$. Note: This exercise together with Exercise (8) establishes that the dyadic numbers are precisely those numbers with finite birthdays.

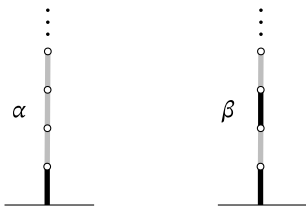
- (10) For every pair s, t of positive integers, let $\delta_{s,t}$ be the “L”-shaped position in Domineering made up from a column of s squares and a row of t squares which overlap in the bottom square of the column and the leftmost square of the row (eg. $\delta_{3,6}$ is ).
- (a) The position $\delta'_{s,t}$ looks like $\delta_{s,t}$, but in the play of $\delta'_{s,t}$ whichever player moves first must use the lower left square (rather clearly the best play). Prove that $\delta'_{s,t} \equiv \delta_{s,t}$. Note: There are many ways to do this. One is to prove that $\delta_{s,t} + \delta_{t,s}$ and $\delta'_{s,t} + \delta_{t,s}$ are both type P and apply Lemma 2.16.
- (b) Using (a), find a dyadic position equivalent to $\delta_{s,t}$ whenever at least one of s, t is odd.
- (c) Prove that whenever s, t are positive even integers, there exists an integer c so that $\delta'_{s,t} + *1 + \bullet c$ has type P. Use this to prove that $\delta_{s,t} \equiv *1 + \bullet(-c)$.
- (11) For every nonnegative integer s , find a dyadic position equivalent to the given position in Hackenbush and prove your result:



- (12) For every positive integer s and nonnegative integer t , find a dyadic number equivalent to the given Hackenbush position and prove your result:



- (13) Let $\gamma_{s,t}$ be a position in the game Push (defined in Exercise (6)) with one R in position s and one L in position t with $s < t$. Here we treat position 0 as being pushed off the array and position i as the i^{th} box from the left. Find a dyadic position equivalent to $\gamma_{s,t}$ for every $0 \leq s < t$ and prove your result.
- (14) For every positive integer n , find a dyadic number a_n so that a $3 \times n$ position in Cut-Cake is equivalent to $\bullet a_n$. Prove your formula holds.
- (15) In this problem we investigate the Hackenbush positions α and β with infinitely many edges in the figure below.
- (a) Show that α deserves to be treated as $\bullet \frac{1}{\infty}$ in the following sense: For every dyadic number $a^- < 0$ the position $\bullet(a^-) + \alpha$ is type R, but for every dyadic number $a^+ \geq 0$ the position $\bullet(a^+) + \alpha$ is type L.
- (b) Using Exercise (11), part (c), find a (nondyadic) number b so that β behaves like $\bullet b$ in the following sense: If b^- is a dyadic number with $b^- < b$, then $\bullet(-b^-) + \beta$ has type L, and if b^+ is a dyadic number with $b^+ > b$, then $\bullet(-b^+) + \beta$ has type R.



- (16) In the last section we showed that a 2×2 position in Domineering is equivalent to $\{\bullet 1 \mid \bullet(-1)\}$ and noted that the Simplicity Principle does not apply in this case (since the number on the left is at least that on the right). In this problem we explore strategy (and type) in sums of this kind of position. Consider a position equivalent to

$$\{\bullet a_1 \mid \bullet(-a_1)\} + \cdots + \{\bullet a_m \mid \bullet(-a_m)\} + \bullet b_1 + \cdots + \bullet b_n$$

where $a_1 \geq a_2 \geq \cdots \geq a_m$ are positive integers and $b_1, \dots, b_n$ are integers. Assuming Richard is first to play, describe a winning strategy for either Louise or Richard.

- (17) A set $S \subseteq \mathbb{Q}$ is called
- *closed under negation* if $-x \in S$ whenever $x \in S$,

- *closed under sums* if $x + y \in S$ whenever $x, y \in S$,
- *closed under products* if $xy \in S$ whenever $x, y \in S$.

Note that the dyadic numbers satisfy all three properties.

- Find an infinite list $R_1, R_2, \dots$ of subsets of $\mathbb{Q}$ each closed under negation, sums, and products which satisfy $R_i \not\subseteq R_j$ and $R_j \not\subseteq R_i$ for every i, j with $i \neq j$.
 - Find an infinite list $S_1, S_2, \dots$ of subsets of $\mathbb{Q}$ each closed under negation, sums, and products with $S_1 \subset S_2 \subset S_3 \dots$.
- (18) For every Hackenbush position α , we defined a position $-\alpha$ so that $-\alpha + \alpha$ is type P. For each of the following games, define negation, and prove that your definition satisfies this property:
- Cut-Cake,
 - Push (defined in Exercise (6)),
 - any impartial game.
- (19) In our proof of the Simplicity Principle, we let c be the oldest number in the interval (a_1, b_1) and we assumed that $c = \frac{\ell}{2^k}$ where $\ell, k \geq 1$. Use a similar argument to prove the result when $c \in \mathbb{Z}$.
- (20*) Prove that every Hackenbush position is equivalent to a dyadic position.

Chapter 5

Zero-Sum Matrix Games

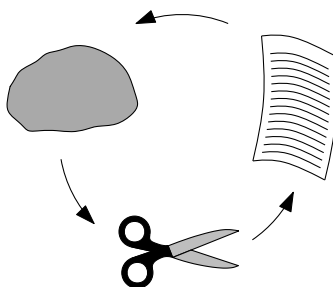


Figure 5.1. Rock-Paper-Scissors

Chapter 5 moves from combinatorial game theory into the world of classical game theory. This is a radical departure! In previous chapters, we studied games in which players alternate moves, each making a sequence of decisions in response to an opponent. Now, we will consider situations in which each player makes a single decision without knowing the opponent's decision. The combination of the two players' choices results in some outcome. Rock-Paper-Scissors is a familiar game of this type—each player secretly chooses to play Rock, Paper, or Scissors. These choices are then simultaneously revealed to determine a winner.

We will use a matrix to model this type of situation. Our two players will be **Rose**, who secretly chooses a **Row**, and **Colin**, who secretly chooses a **Column**. If our players are playing Rock-Paper-Scissors for \$1, then the following matrix would model the game. Here Rose's choice of Rock, Paper, or Scissors corresponds to selecting a row, and Colin's decision corresponds to a choice of column. The entry selected by Rose's row choice and Colin's column choice indicates how much Colin must pay Rose.

Game 5.1 (Rock-Paper-Scissors).

		Colin		
		Rock	Paper	Scissors
Rose	Rock	0	-1	1
	Paper	1	0	-1
	Scissors	-1	1	0

We can use matrices like this to represent a wide variety of decision problems between two players in direct competition (i.e. whatever is good for one player is bad for the other). This dynamic appears frequently in games, as well as in sports and politics and in other real-world situations. Next comes a formal definition for this new kind of game.

Definition 5.2. A *Zero-Sum Matrix Game* is a game played between Rose and Colin according to the following rules. There is a fixed matrix A that is known to both players. Rose secretly chooses one of the rows and Colin secretly chooses one of the columns (neither player knows the other's choice). Then, both players reveal their choices. If Rose chose row i and Colin chose column j , then the (i, j) entry of matrix A is the *payoff* of the game. This payoff indicates how much Colin must pay Rose (of course, if the payoff is negative, then Rose will have to pay Colin). Such games are called zero-sum since whatever one player gains is what the other player loses. Naturally, we will assume that each player wants to maximize his or her money, so Rose wants the payoff to be as high as possible and Colin wants the payoff to be as low as possible.

Here is another example.

Game 5.3 (Two-Finger Morra). This game is played between two players similarly to Rock-Paper-Scissors. On the count of three, each player

plays either one or two fingers with his or her right hand, and simultaneously *guesses* either one or two fingers with the left hand. Each player wants the number of left-hand guess fingers to match the number played by the opponent's right.

If either both players guess correctly or both guess incorrectly, then the game is a tie and nothing is exchanged. If just one player guesses correctly, that player wins from the other an amount equal to the total number of fingers played by both players.

Denote the strategy of playing i and guessing j by pig_j , and the following zero-sum matrix game represents Two-Finger Morra:

	p1g1	p1g2	p2g1	p2g2
p1g1	0	2	-3	0
p1g2	-2	0	0	3
p2g1	3	0	0	-4
p2g2	0	-3	4	0

Exercise (8) will explore this game in detail.

5.1. Dominance

This section will begin to explore play in zero-sum matrix games. How should we compete against a rational opponent? To investigate this question, we introduce a notion of strategy and a concept of dominance for matrix games.

Generally speaking, a strategy is a rule that tells a player what to do at each point where that player has a decision to make. Each player has just one choice to make in a matrix game, so we define a *pure strategy* for Rose to be a choice of row and a *pure strategy* for Colin to be a choice of column. Let's consider strategies in the following matrix game:

$$\text{Rose} \begin{matrix} & \text{Colin} \\ \begin{bmatrix} 2 & 1 \\ 3 & 0 \\ -1 & 0 \end{bmatrix} \end{matrix}$$

We can see that the third row is a rather poor strategy for Rose. No matter what Colin does, Rose's payoff is always higher when she plays row 1 instead of row 3. This highlights the important notion of dominance.

For a zero-sum matrix game A , Rose's strategy of row i *dominates* her strategy of row i' if every entry in row i is greater than or equal to the corresponding entry in row i' . In the above game, both rows 1 and 2 dominate row 3. Additionally, row i *strictly dominates* row i' if every entry in row i is strictly greater than the corresponding entry in row i' . Note in the matrix above that row 1 strictly dominates row 3, but row 2 does not.

Since Colin wants the payoff to be as low as possible, his strategy of column j *dominates* his strategy of column j' if every entry in column j is less than or equal to the corresponding entry in column j' . Again, column j *strictly dominates* column j' if every entry in column j is strictly less than the corresponding entry in column j' . Let's return to the example and see what happens when we eliminate the dominated third row and simplify the game:

$$\begin{array}{cc} & \text{Colin} \\ \text{Rose} & \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix} \end{array}$$

In this smaller game, Colin should play column 2 since it strictly dominates column 1. In light of this, Rose should play row 1 and she will win 1 from Colin on each play of the game. Observe in the original game that Rose's strategy of playing the first row guarantees that she will always win at least 1, while Colin's strategy of playing column 2 guarantees that he will lose at most 1. This approach, called *iterated removal of dominated strategies*, can powerfully clarify a game. Let's continue with a more complicated political example.

Example 5.4 (Linear Voting). Imagine a voting model where there are exactly 7 possible political views, linearly ordered 1, 2, ..., 7 with 1 the most liberal and 7 the most conservative. Suppose that there are exactly 10 voters holding each of these 7 views. Also assume that there are 2 candidates running for office, each of whom must choose to adopt 1 of the 7 views. After the candidates have chosen, then each of the 7 voter blocks will vote for the candidate with views closest to its own, with votes split evenly in case of a tie. If, for example, the first candidate adopts view 2 and the second chooses view 4, then the first candidate will get all of the votes from blocks 1 – 2 and the second candidate will get all of the votes from blocks 4 – 7. The votes from block 3 will be split. The first

candidate will thus have a total of 25 votes, and the second candidate 45 votes. To optimize political capital, each candidate wants to win the maximum number of votes.

We can model this scenario using the zero-sum matrix game below with Rose and Colin as the candidates running for office. Here row (or column) k corresponds to the strategy of adopting view k on the political spectrum and a payoff of n indicates that Rose gets n more votes than Colin does:

$$\begin{bmatrix} 0 & -50 & -40 & -30 & -20 & -10 & 0 \\ 50 & 0 & -30 & -20 & -10 & 0 & 10 \\ 40 & 30 & 0 & -10 & 0 & 10 & 20 \\ 30 & 20 & 10 & 0 & 10 & 20 & 30 \\ 20 & 10 & 0 & -10 & 0 & 30 & 40 \\ 10 & 0 & -10 & -20 & -30 & 0 & 50 \\ 0 & -10 & -20 & -30 & -40 & -50 & 0 \end{bmatrix}.$$

How is this game likely to be played? A bit of investigation reveals that strategy 2 dominates 1 for each player and, similarly, strategy 6 dominates 7. If Rose and Colin are rational, neither of them would ever choose to play either dominated strategy 1 or 7. Eliminating these rows and columns reduces the game to the following 5×5 matrix:

$$\begin{bmatrix} 0 & -30 & -20 & -10 & 0 \\ 30 & 0 & -10 & 0 & 10 \\ 20 & 10 & 0 & 10 & 20 \\ 10 & 0 & -10 & 0 & 30 \\ 0 & -10 & -20 & -30 & 0 \end{bmatrix}.$$

This new, smaller matrix also has dominated strategies. Specifically, the second strategy dominates the first, and the second to last dominates the last. Again, neither rational player will choose a dominated strategy. Crossing those out leaves a 3×3 matrix:

$$\begin{bmatrix} 0 & -10 & 0 \\ 10 & 0 & 10 \\ 0 & -10 & 0 \end{bmatrix}.$$

As before, the first and last strategies for each player are dominated in this matrix. Eliminating these dominated strategies results in the

1×1 matrix $[0]$. It follows that each rational player should play the middle strategy 4 (corresponding to the middle political view) in the original game. If both players do this, the resulting payoff is 0. This example, while artificial, nonetheless illuminates a real strategic dynamic pervasive in politics. For instance, candidates in the United States' presidential election often try to position themselves one way during the primaries to earn a party nomination and then move nearer the center for the general election.

The method of iterated deletion of dominated strategies is an extremely valuable tool in analyzing zero-sum games. Yet, its optimality depends upon both players playing rationally and performing sophisticated logical analysis with the knowledge that the opponent is doing the same. Sometimes this doesn't happen! If, for example, a candidate above chose to play the inferior strategy 2, then an opponent would benefit by playing 3 instead of 4 as directed by the deletion of dominated strategies.

This section ends with a basic theoretical property that holds anytime iterated removal of dominated strategies results in a 1×1 matrix. We say that a zero-sum matrix game A has a *saddle point* at (i, j) if the entry in row i and column j is the smallest in its row and the largest in its column.

Example 5.5. The following zero-sum matrix game has a saddle point at $(2, 3)$:

$$\begin{bmatrix} 1 & 9 & 4 & 0 \\ 5 & 7 & \textcircled{5} & 6 \\ 6 & 1 & 3 & 8 \end{bmatrix}.$$

The term “saddle point” is motivated by the surface of a saddle as in Figure 5.2. Here the center point lies at the intersection of two curves—it is the lowest point on one of the curves and the highest point on the other.

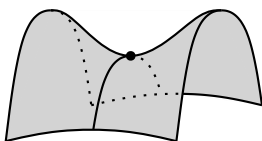


Figure 5.2. A saddle point

Theorem 5.6. *Let A be a zero-sum matrix game. If iterated deletion of dominated strategies reduces A to a 1×1 matrix consisting of the entry in position (i, j) of the original matrix, then A has a saddle point at (i, j) .*

Proof. Given a zero-sum matrix game B , suppose we are interested in the maximum entry of a particular column, say column j . Removing another dominated column has no impact on the maximum entry of our column. Now suppose row i dominates row i' , and we modify B to get the matrix B' by deleting row i' . Since row i dominates row i' in B , the i^{th} entry of column j is greater than or equal to the $(i')^{\text{th}}$ entry. Thus the maximum entry of column j in the matrix B' is equal to the maximum entry of column j in B . This proves that removing dominated rows and columns from a matrix does not alter the maximum entry of any column that was not removed. In particular, if iterated removal of dominated strategies reduces matrix A to a 1×1 matrix consisting of the entry in position (i, j) of A , then this entry is the largest in column j . A similar argument shows that this entry is the smallest in row i . This implies the desired result. $\square$

We have just seen that whenever iterated deletion of dominated strategies reduces a zero-sum matrix game to a 1×1 matrix, the remaining entry corresponds to a saddle point. However, not every saddle point can be found by this method (Exercise (3)).

5.2. Mixed Strategies

Removing dominated strategies can effectively identify an optimal strategy, but what if a matrix game does not include dominated strategies? This section introduces a concept that applies more broadly to arbitrary matrix games. To start, suppose that Rose and Colin are going to play the following game:

$$\begin{array}{cc} & \text{Colin} \\ \text{Rose} & \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \end{array}$$

Rose can always guarantee a payoff of at least 1 by playing the second row. But if Colin recognizes this is her strategy, he can respond by playing the first column so the payoff is no larger than 1. If Rose realizes that Colin

is going to play the first column, then she should play the first row, but then Colin would want to play the first column. This is going in circles! Each player attempts to predict what action the other will take. Yet each player prefers to make unpredictable actions. A natural way to create unpredictability is to introduce a source of randomness. For instance, Colin might flip a coin and choose column 1 if it comes up heads and column 2 if it is tails. Then Colin will randomly play each column with probability $1/2$. Rose might similarly flip two coins and choose row 1 if she gets heads both times and otherwise choose row 2. Then she will play row 1 with probability $1/4$ and row 2 with probability $3/4$:

$$\begin{array}{cc} & \text{Colin} \\ & \begin{array}{cc} 1/2 & 1/2 \end{array} \\ \text{Rose} & \begin{array}{c} 1/4 \\ 3/4 \end{array} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \end{array}$$

Since this notion of a probabilistic strategy is essential for the forthcoming investigation of matrix games, we pause here to introduce some necessary probability theory.

Finite Probability Spaces. A *finite probability space* consists of a finite set S of *outcomes* together with a probability function $p : S \rightarrow [0, 1]$ with the property that $\sum_{x \in S} p(x) = 1$. We view this as a probabilistic process where S is the set of possible outcomes and outcome x occurs with probability $p(x)$.

Example 5.7. Consider an experiment where a coin is tossed twice. The four possible outcomes are given by $S = \{HH, HT, TH, TT\}$. Given a fair coin, each outcome is equally likely, so the probability function p is given by

$$p(HH) = p(HT) = p(TH) = p(TT) = 1/4.$$

There are often natural quantities associated with the outcomes of a probability space. In the above example, for instance, we might be interested in the total number of tails in the two coin tosses. The notion of a random variable captures this idea. Formally, define a *random variable* to be a function $f : S \rightarrow \mathbb{R}$. So $f(x)$ is a quantity associated with the outcome x . In our example, we may define a random variable $f : S \rightarrow \mathbb{R}$

that counts the total number of tails in our experiment. This would be the function given by $f(HH) = 0$, $f(HT) = f(TH) = 1$, and $f(TT) = 2$.

A crucial property for a random variable is its average value. This average value is called the *expected value* and is defined as

$$\mathbb{E}(f) = \sum_{x \in S} p(x)f(x).$$

The expected number of tails in the coin toss experiment is precisely the expected value of the random variable f :

$$\begin{aligned}\mathbb{E}(f) &= p(HH)f(HH) + p(HT)f(HT) + p(TH)f(TH) + p(TT)f(TT) \\ &= (1/4)0 + (1/4)1 + (1/4)1 + (1/4)2 \\ &= 1.\end{aligned}$$

So, if a fair coin is tossed twice, the expected number of tails is 1. While it is certainly possible to see either 0 or 2 tails in two tosses, the average is 1.

Recall the previous 2×2 matrix example, with Rose playing row 1 with probability $1/4$ and row 2 with probability $3/4$ and Colin playing columns 1 and 2, each with probability $1/2$:

$$\begin{array}{cc} & \text{Colin} \\ & \begin{array}{cc} 1/2 & 1/2 \end{array} \\ \text{Rose} & \begin{array}{c} 1/4 \left[\begin{array}{cc} 2 & -1 \end{array} \right] \\ 3/4 \left[\begin{array}{cc} 1 & 2 \end{array} \right] \end{array} \end{array}$$

We have now constructed a probability space where the outcomes are the four possibilities for Rose's and Colin's choices. We can think of these outcomes as cells in the matrix indexed by $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$. The probability the (i, j) entry is the outcome is equal to probability that Rose chooses row i times the probability that Colin chooses column j .¹ Then the probability of outcome $(1, 2)$ is $1/4 \cdot 1/2 = 1/8$. Our matrix associates a quantity, the payoff, with each (i, j) entry. Next we compute the

¹This assumes that Rose's and Colin's probabilistic choices are independent—an assumption we will continue to make.

expected value of this payoff, often called the *expected payoff*:

$$\begin{aligned}
 \text{Expected payoff} &= (1/4)(1/2)2 + (1/4)(1/2)(-1) + (3/4)(1/2)1 + (3/4)(1/2)2 \\
 &= \begin{bmatrix} 1/4 & 3/4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} \\
 &= 5/4.
 \end{aligned}$$

When the players adopt the given probabilistic strategies, the expected payoff of the game is $5/4$. (Note that this is never the payoff from a single play—it is just the average). The computation also reveals that this expected payoff can be computed using a natural matrix product. As we will soon see, this property holds in general.

Mixed Strategies. Some new terminology will formalize this notion of a probabilistic strategy. For an arbitrary $m \times n$ zero-sum matrix game A , define a *mixed strategy* for Rose to be a row vector $\mathbf{p} = [p_1 \ p_2 \ \cdots \ p_m]$ with the property that $p_i \geq 0$ for $1 \leq i \leq m$ and $\sum_{i=1}^m p_i = 1$. When Rose *plays* $\mathbf{p}$, she randomly chooses to play row i with probability p_i .

Analogously for Colin, a *mixed strategy* is a column vector $\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}$,

for which $q_j \geq 0$ for $1 \leq j \leq n$ and $\sum_{j=1}^n q_j = 1$. When Colin *plays* $\mathbf{q}$, he randomly chooses column j with probability q_j . (Again, we assume that Rose's and Colin's probabilistic choices are independent.) Above, for example, Rose played the mixed strategy $[1/4 \ 3/4]$ and Colin played the mixed strategy $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$.

Now consider the mixed strategy $\mathbf{p} = [1 \ 0 \ \cdots \ 0]$ for Rose. In this strategy, Rose plays the first row with probability 1 and all other rows with probability 0. This operates exactly like Rose's pure strategy choice of row 1, so, in this case, we will also call $\mathbf{p}$ a pure strategy. More generally, we define any mixed strategy for Rose or Colin with one entry equal to 1 and all others 0 to be a *pure* strategy. Pure strategies are then just special cases of mixed ones.

Assume that Rose plays $\mathbf{p} = [p_1 \ p_2 \ \cdots \ p_m]$ and Colin plays $\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}$ in the game A . We now have a probability space where the

outcome of entry (i, j) occurs with probability $p_i q_j$. The expected payoff of this game will then be given as follows (here, $a_{i,j}$ denotes the (i, j) entry of the matrix A):

$$\text{Expected payoff} = \sum_{i=1}^m \sum_{j=1}^n p_i a_{i,j} q_j = \mathbf{pAq}.$$

The matrix product $\mathbf{pAq}$ conveniently encodes this important quantity. By possibly taking either $\mathbf{p}$ or $\mathbf{q}$ to be a pure strategy, we obtain the following.

Proposition 5.8. *For a zero-sum matrix game A , the expected payoffs when Rose and Colin play pure or mixed strategies are as follows:*

Expected payoff		Colin plays	
		col j	$\mathbf{q}$
Rose plays	row i	$a_{i,j}$	$(A\mathbf{q})_i$
	$\mathbf{p}$	$(\mathbf{p}A)_j$	$\mathbf{pAq}$

Take a moment here to appreciate this proposition. When Rose plays row i and Colin plays column j , the payoff by definition will be the (i, j) entry of our matrix, $a_{i,j}$. If instead both play the mixed strategies $\mathbf{p}$ and $\mathbf{q}$, the expected payoff (computed above) is $\mathbf{pAq}$. The other cases have a slightly different flavor. When Rose plays mixed strategy $\mathbf{p}$, the product $\mathbf{pA}$ is a row vector and the above proposition tells us what this vector means. Specifically, the j^{th} entry in this row is the expected payoff if Rose plays $\mathbf{p}$ and Colin plays column j . So the row vector $\mathbf{pA}$ encodes all possible expected payoffs for each column Colin might choose, assuming Rose plays $\mathbf{p}$. Similarly, the column vector $A\mathbf{q}$ tells us all of the possible expected payoffs for each row Rose might choose, under the assumption that Colin plays $\mathbf{q}$.

5.3. Von Neumann Solutions

Consider a zero-sum matrix game A and suppose that Rose has decided to play the mixed strategy $\mathbf{p}$. As above, the row vector $\mathbf{p}A$ contains the expected payoff for each possible column which Colin might choose. Rose wants the expected payoff to be large regardless of Colin's choice. Define the *guarantee* of $\mathbf{p}$ to be the minimum entry of the vector $\mathbf{p}A$. So, if the guarantee of $\mathbf{p}$ is 2, then the expected payoff will be at least 2 when Rose plays $\mathbf{p}$ no matter which column Colin chooses.² Colin wants the expected payoffs to be small, so he would like to find a mixed strategy $\mathbf{q}$ so that all entries of $A\mathbf{q}$ are small. Accordingly, we define the *guarantee* of $\mathbf{q}$ to be the maximum entry of $A\mathbf{q}$. So, if the guarantee of $\mathbf{q}$ is 3, then when Colin plays $\mathbf{q}$, the expected payoff will be at most 3 no matter which row Rose plays.

Example 5.9. Given the zero-sum matrix game $A = \begin{bmatrix} 6 & 2 & -3 \\ 3 & 3 & 0 \\ 0 & -3 & 2 \end{bmatrix}$, find the guarantees of Rose's mixed strategy $\mathbf{p} = \begin{bmatrix} 1/6 & 1/2 & 1/3 \end{bmatrix}$ and Colin's mixed strategy $\mathbf{q} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$.

For Rose, $\mathbf{p}A = \begin{bmatrix} 1/6 & 1/2 & 1/3 \end{bmatrix} \begin{bmatrix} 6 & 2 & -3 \\ 3 & 3 & 0 \\ 0 & -3 & 2 \end{bmatrix} = \begin{bmatrix} 5/2 & 5/6 & 1/6 \end{bmatrix}$, so $\mathbf{p}$ has a guarantee of $1/6$. For Colin, $\begin{bmatrix} 6 & 2 & -3 \\ 3 & 3 & 0 \\ 0 & -3 & 2 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 2 \\ -1/3 \end{bmatrix}$, so $\mathbf{q}$ has a guarantee of 2.

To work with guarantees, it will be useful to have an inequality to compare two vectors. For two vectors of the same dimension, say $\mathbf{u} = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$, we define $\mathbf{u} \leq \mathbf{v}$ if $u_i \leq v_i$ for every $1 \leq i \leq n$. In other words, $\mathbf{u} \leq \mathbf{v}$ when each coordinate of $\mathbf{u}$ is less than or equal to the corresponding coordinate in $\mathbf{v}$. Below we see two vectors which can be compared with $\leq$ and two vectors which cannot be

²Note that if $\mathbf{p}$ has a guarantee of v , it is not necessarily true that when Rose plays $\mathbf{p}$, the payoff will always be at least v . What is true is that the *expected* payoff is at least v .

compared with $\leq$:

$$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} \leq \begin{bmatrix} 0 \\ 3 \\ 3 \\ 3 \end{bmatrix}, \quad [1 \ 0] \not\leq [0 \ 1], \quad [0 \ 1] \not\leq [1 \ 0].$$

This vector inequality provides a way to express the guarantees of mixed strategies for Rose and Colin. Namely, a mixed strategy $\mathbf{p}$ for Rose has a guarantee of at least v if and only if $\mathbf{p}A \geq [v \ \cdots \ v]$. Similarly, $\mathbf{q}$ has a guarantee of at most v if and only if $A\mathbf{q} \leq \begin{bmatrix} v \\ \vdots \\ v \end{bmatrix}$.³

Suppose Rose is playing the mixed strategy $\mathbf{p}$ in the zero-sum matrix game A . If $\mathbf{p}$ has a guarantee of r , then all entries of $\mathbf{p}A$ are at least r . So whichever pure strategy Colin chooses, the expected payoff will always be at least r . The next result shows that the same will hold true for any mixed strategy of Colin. In short, if Rose plays a mixed strategy with a guarantee of r , the expected payoff will always be at least r , no matter what Colin does.

Lemma 5.10. *Let $\mathbf{p}$ and $\mathbf{q}$ be mixed strategies for Rose and Colin in the zero-sum matrix game A .*

- (1) *If $\mathbf{p}$ has a guarantee of r , then $\mathbf{p}A\mathbf{q} \geq r$.*
- (2) *If $\mathbf{q}$ has a guarantee of c , then $\mathbf{p}A\mathbf{q} \leq c$.*
- (3) *If $\mathbf{p}$ and $\mathbf{q}$ have guarantees of r and c , then $r \leq c$.*

Proof. Let $\mathbf{p} = [p_1 \ \cdots \ p_m]$ and $\mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$. Now (1) and (2) follow from the inequalities

$$(\mathbf{p}A)\mathbf{q} \geq [r \ \cdots \ r]\mathbf{q} = r(q_1 + \cdots + q_n) = r,$$

$$\mathbf{p}(A\mathbf{q}) \leq \mathbf{p} \begin{bmatrix} c \\ \vdots \\ c \end{bmatrix} = c(p_1 + \cdots + p_m) = c$$

and the last part comes from combining the previous two. □

³Note that this inequality for vectors appeared implicitly with domination. In a zero-sum matrix game A , Rose's strategy of row i dominates row i' if row i is $\geq$ row i' . For Colin, column j dominates column j' if column j is $\leq$ column j' .

Rose and Colin both want to do their best in a zero-sum matrix game. Rose wants a strategy with the highest possible guarantee, while Colin wants one with the lowest possible guarantee. The above lemma says that Rose's guarantee will always be less than or equal to Colin's guarantee. But how close can these guarantees be? Von Neumann's famous Minimax Theorem answers this question. It states that, in fact, Rose's maximum guarantee is always equal to Colin's minimum guarantee. This special number is called the value of the game.

Theorem 5.11 (Von Neumann's Minimax Theorem). *Every zero-sum matrix game A has a unique number v , called the value of A , which is the maximum guarantee of a mixed strategy for Rose and the minimum guarantee of a mixed strategy for Colin.*

This theorem concludes the satisfying fact that every zero-sum matrix game A has a special value v , which may be viewed as the expected payoff under rational play. Rose has a mixed strategy $\mathbf{p}$ to guarantee the expected payoff is at least v and Colin has a mixed strategy $\mathbf{q}$ to guarantee the expected payoff is at most v . This pair of strategies and resulting payoff v provide strong information about the matrix game A . Accordingly, we will call the number v together with two such strategies $\mathbf{p}$ and $\mathbf{q}$ a *von Neumann solution* to the game A . Note that while the value v is unique (every game has just one value) there may be more than one mixed strategy for Rose or Colin with a guarantee of v . So, in general, a zero-sum matrix game may have many von Neumann solutions.

Why is this theorem sometimes called the Minimax Theorem? Colin wants to choose a mixed strategy which has the smallest guarantee. In other words, he is looking for a mixed strategy $\mathbf{q}$ which **minimizes** the **maximum** entry of $A\mathbf{q}$. This is the “minimax” by which the theorem is frequently known. On the other side, Rose wants a mixed strategy with the highest possible guarantee. So she wants a mixed strategy $\mathbf{p}$ which **maximizes** the **minimum** value of $\mathbf{p}A$ —which may be called a “maximin”. Von Neumann's Theorem tells us that Colin's minimax will always be equal to Rose's maximin for matrix game A .

Example 5.12. Consider the matrix game

$$A = \begin{bmatrix} 1 & -1 & 0 & 3 & -1 \\ -2 & 3 & 2 & 1 & 2 \\ 0 & -1 & 4 & 0 & 3 \\ -1 & 5 & 1 & 1 & 2 \\ 3 & 0 & 4 & -1 & 4 \end{bmatrix}$$

together with the mixed strategies

$$\mathbf{p} = \begin{bmatrix} 3/8 & 0 & 0 & 7/24 & 1/3 \end{bmatrix} \quad \text{and} \quad \mathbf{q} = \begin{bmatrix} 11/24 \\ 1/4 \\ 0 \\ 7/24 \\ 0 \end{bmatrix}.$$

A quick calculation reveals that $\mathbf{p}A = \begin{bmatrix} 13/12 & 13/12 & 13/8 & 13/12 & 37/24 \end{bmatrix}$,

so $\mathbf{p}$ has a guarantee of $13/12$. Since $A\mathbf{q} = \begin{bmatrix} 13/12 \\ 1/8 \\ -1/4 \\ 13/12 \\ 13/12 \end{bmatrix}$, then $\mathbf{q}$ also has a

guarantee of $13/12$, so this is the value of the game. It follows that $13/12$ together with the mixed strategies $\mathbf{p}$ and $\mathbf{q}$ form a von Neumann solution for A .

The general problem of finding such von Neumann solutions is quite involved—it is a question central to the subject of linear programming. Appendix B introduces the topic, but a full algorithm for finding von Neumann solutions exceeds the scope of this book. The next chapter does develop techniques to find solutions for small matrix games.

One accessible case for finding a von Neumann solution is when two pure strategies form such a solution. Suppose that A is a zero-sum matrix game and there is a von Neumann solution for A consisting of the number v together with pure strategies for Rose and Colin. If Rose's pure strategy is to play row i , then every entry in row i must be at least v (since this strategy has a guarantee of v). Similarly, if Colin's pure strategy is to play column j , then every entry in column j must be at most v (since this strategy has a guarantee of v). It follows that the (i, j) entry of A is exactly

v and that A has a saddle point at (i, j) . We formalize this observation below (again we use $a_{i,j}$ to denote the (i, j) entry of A).

Observation 5.13. *Let A be a zero-sum matrix game with $a_{i,j} = v$. Then A has a saddle point at (i, j) if and only if v together with the pure strategies of row i and column j form a von Neumann solution.*

Exercises

- (1) If two perfectly rational players play the following zero-sum matrix games, what strategies will they use?

$$(a) \begin{bmatrix} 5 & 4 & 3 \\ 6 & 2 & 0 \end{bmatrix}, \quad (b) \begin{bmatrix} 0 & 6 & 1 \\ 2 & 5 & 0 \\ 6 & 4 & 3 \end{bmatrix}, \quad (c) \begin{bmatrix} 5 & 6 & 4 & 7 \\ 1 & 11 & 3 & 0 \\ 7 & 9 & 5 & 6 \end{bmatrix}.$$

- (2) Colonel Blotto and Count Baloney are opposing commanders in a battle. There are three locations valued 1, 2, 3, and each commander has two military units to deploy (both to the same location or split to different locations). If one commander sends more units to a location than the other, that commander captures the location and wins its value. If both commanders send the same number of units to a location, it is not captured and nothing is won at this location.

- Model this conflict as a zero-sum matrix game.
 - Use the method of eliminating dominated (not necessarily strictly dominated!) strategies to reduce this to a 1×1 game.
 - What strategy would you recommend to these commanders and why?
- (3) Here is a general method for finding all saddle points in a zero-sum matrix game: First, mark all minimum entries in each row. Then mark all maximum entries in each column. The entries with two marks correspond to the saddle points. Use this method to find all saddle points in the following matrices:

$$(a) \begin{bmatrix} 0 & 3 & 8 \\ 5 & 4 & 7 \\ 7 & 2 & 1 \end{bmatrix}, \quad (b) \begin{bmatrix} 5 & 9 & 5 & 7 \\ 2 & 3 & 4 & 10 \\ 5 & 6 & 5 & 8 \\ 3 & 11 & 1 & 6 \end{bmatrix}.$$

- (4) A fair six-sided die numbered $1, \dots, 6$ is tossed and f is the random variable which indicates the number rolled. Compute the following quantities:⁴
- $\mathbb{E}(f)$,
 - $\mathbb{E}(f^2)$,
 - $\mathbb{E}(f^2) - (\mathbb{E}(f))^2$.
- (5) A spinner randomly selects between the letters A, B, C , choosing each with probability $1/3$. Assume the spinner is spun three times in a row giving outcomes such as ABC or BBB .
- What are all the possible outcomes? What is the probability each is selected?
 - Let f be the random variable denoting the total number of times the spinner selects A . For each number $0 \leq i \leq 3$ determine the probability that f has value i .
 - What is the expected value of f ?
- (6) In the zero-sum matrix game $\begin{bmatrix} -1 & 4 \\ 3 & -1 \\ 5 & 2 \end{bmatrix}$, Rose plays the mixed strategy $\begin{bmatrix} 1/6 & 1/3 & 1/2 \end{bmatrix}$ and Colin plays the mixed strategy $\begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$.
- For every $1 \leq i \leq 3$ and $1 \leq j \leq 2$, determine the probability that Rose selects row i and Colin selects column j .
 - What is the expected payoff?
- (7) Consider the zero-sum game $\begin{bmatrix} 3 & 4 & 2 \\ 5 & -2 & 7 \end{bmatrix}$ and the mixed strategies $\mathbf{p} = \begin{bmatrix} 2/3 & 1/3 \end{bmatrix}$ for Rose and $\mathbf{q} = \begin{bmatrix} 1/5 \\ 2/5 \\ 2/5 \end{bmatrix}$ for Colin. Compute the expected payoff when the players use the following strategies:
- Rose plays row 2; Colin plays column 2.
 - Rose plays row 1; Colin plays $\mathbf{q}$.
 - Rose plays $\mathbf{p}$; Colin plays column 3.
 - Rose plays $\mathbf{p}$; Colin plays $\mathbf{q}$.

⁴The number computed in (c) is known as the *variance* of f .

- (8) Show that in Two-Finger Morra, the mixed strategy of playing play-two guess-one with probability $5/12$ and play-one guess-two with probability $7/12$ has a guarantee of 0 for each player. Observe that this yields a von Neumann solution to the game.

- (9) Consider the game $\begin{bmatrix} 2 & -1 & 8 & 3 \\ 1 & 3 & 1 & 2 \\ 5 & 2 & -1 & 1 \end{bmatrix}$. For each mixed strategy below, find the associated guarantee:

$$(a) \begin{bmatrix} 1/4 & 1/4 & 1/2 \end{bmatrix}, \quad (b) \begin{bmatrix} 1/5 & 3/5 & 1/5 \end{bmatrix}, \quad (c) \begin{bmatrix} 1/4 \\ 1/2 \\ 1/4 \\ 0 \end{bmatrix}, \quad (d) \begin{bmatrix} 0 \\ 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}.$$

Can you determine a von Neumann solution from the above?

- (10) The game *Rock-Paper-Scissors-Lizard-Spock* is similar to Rock-Paper-Scissors with each player choosing one of these five options. If both make the same choice, it's a draw. Otherwise, the winner is determined by the rules: Scissors cuts paper covers rock crushes lizard poisons Spock smashes scissors decapitates lizard eats paper disproves Spock vaporizes rock crushes scissors. Model this as a zero-sum matrix game. Guess a solution for Rock-Paper-Scissors-Lizard-Spock and then check that it works.
- (11) Let A and B be $m \times n$ matrices. If $\mathbf{p}$ guarantees Rose $\geq r$ in A and $\mathbf{p}$ guarantees Rose $\geq s$ in B , prove that $\mathbf{p}$ guarantees Rose $\geq r + s$ in $A + B$.
- (12) Let A be a zero-sum matrix game with a von Neumann solution consisting of v , $\mathbf{p}$, and $\mathbf{q}$. Determine and verify von Neumann solutions for the following matrices:
- the matrix A' obtained from A by adding c to every entry,
 - cA for $c \geq 0$,
 - $-A^T$.
- (13) Let A be a zero-sum matrix game with two different saddle points. Prove that these saddle points have the same value.

- (14) Suppose that $\mathbf{p}$ and $\mathbf{p}'$ are mixed strategies for Rose which guarantee her r and r' . Let $0 \leq t \leq 1$.
- Show that $t\mathbf{p} + (1-t)\mathbf{p}'$ is a mixed strategy for Rose.
 - Show that the guarantee of $t\mathbf{p} + (1-t)\mathbf{p}'$ is $\geq tr + (1-t)r'$.
 - Prove that any matrix with at least two different von Neumann solutions has infinitely many von Neumann solutions.
- (15) Let A be a zero-sum matrix game, and let $\mathbf{p}', \mathbf{p}''$ be mixed strategies for Rose and $\mathbf{q}', \mathbf{q}''$ mixed strategies for Colin. Suppose the expected payoff when Rose plays $\mathbf{p}'$ and Colin plays $\mathbf{q}'$ is w' and the expected payoff when Rose plays $\mathbf{p}''$ and Colin plays $\mathbf{q}''$ is w'' . If $w' < w < w''$, show that there exist mixed strategies $\mathbf{p}$ for Rose and $\mathbf{q}$ for Colin so that the expected payoff when Rose plays $\mathbf{p}$ and Colin plays $\mathbf{q}$ is w .
- (16) Let A be an $m \times n$ matrix with all entries nonnegative, and suppose that every column of A has an entry which is at least k . Prove that A has value at least $\frac{k}{n}$. Then give an example to show that this bound cannot be improved.
- (17) Find an infinite set of 2×1 vectors so that no two distinct vectors from your set $\mathbf{u}$ and $\mathbf{u}'$ satisfy $\mathbf{u} \leq \mathbf{u}'$.
- (18) Let f be a random variable which is always nonnegative. Show that for every $a > 0$, the probability that f has value at least a is at most $\frac{\mathbb{E}(f)}{a}$.
- (19) Let A be a zero-sum matrix game, and suppose that A' is obtained from A by deleting a single dominated row or column. Prove that A and A' have the same value.
- (20) Let A be a zero-sum matrix game with von Neumann solution v , together with $\mathbf{p} = [p_1 \ \cdots \ p_m]$ and $\mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$. Prove that $p_i = 0$ whenever $(A\mathbf{q})_i < v$ and $q_j = 0$ whenever $(\mathbf{p}A)_j > v$ (here we use the notation $(\cdot)_i$ to indicate the i^{th} entry of a vector).
- (21) For every matrix M we let $\text{value}(M)$ denote the von Neumann value of M . Our goal in this exercise is to prove that the function $M \rightarrow \text{value}(M)$ is continuous. To prove this, fix a matrix M and let $\epsilon > 0$. Show that there exists a number $\delta > 0$ so that $|\text{value}(M) - \text{value}(M')|$

$< \epsilon$ whenever M' is a matrix with the property that every entry of $M - M'$ has absolute value $< \delta$.

- (22) A *stochastic game* G is a matrix game where each entry of the matrix is either a number a or has the form $a + pG$ where a is a number indicating the immediate payoff and $0 < p < 1$ is a probability. If the entry $a + pG$ is selected by Rose's and Colin's choices, then Colin pays Rose a immediately and with probability p the game is repeated. The total payoff is the sum of the payoffs at each stage. In general when playing a stochastic matrix game, it is possible for players to do different things at different stages. However, we will restrict our attention to having Rose play a single mixed strategy $\mathbf{p}$ and Colin play a mixed strategy $\mathbf{q}$ which they use for every stage of play (it can be shown that Rose and Colin always have optimal strategies of this type).

- (a) Consider the stochastic game $G = \begin{bmatrix} 1 + 1/2G & 3 \\ 7 & -1 + 1/2G \end{bmatrix}$ and suppose that Rose plays the mixed strategy $\mathbf{p} = [1/2 \quad 1/2]$ and Colin plays the mixed strategy $\mathbf{q} = [1/3 \quad 2/3]$. What is the expected immediate payoff? What is the expected probability that the game will be repeated?
- (b) Assume Rose and Colin are still using the mixed strategies $\mathbf{p}$ and $\mathbf{q}$ from the previous part, what is the expected total payoff?
- (c) Generalize the previous result. Let H be an arbitrary stochastic game and let $\mathbf{p}, \mathbf{q}$ be arbitrary mixed strategies for Rose and Colin. If the expected immediate payoff when the players use these strategies is a and the expected probability of repetition is p (i.e. $\mathbf{p}H\mathbf{q} = a + pH$), prove that the expected total payoff is $\frac{a}{1-p}$.
- (23) This exercise continues the investigation of stochastic games begun in the previous exercise. For every stochastic game G and every $x \in \mathbb{R}$, let $G(x)$ be the zero-sum matrix game obtained from G by replacing each entry $a + pG$ by the number $a + px$. The next exercise proves the following result.

Theorem 5.14. *For every stochastic game G , there is a unique real number v such that $G(v)$ has value v . If the mixed strategies $\mathbf{p}^*$ for*

Rose and $\mathbf{q}^*$ for Colin together with the value v form a von Neumann solution to $G(v)$, then whenever Rose plays $\mathbf{p}^*$ in the game G , the expected total payoff is at least v no matter what mixed strategy Colin plays. Similarly, when Colin plays $\mathbf{q}^*$ in G , the expected total payoff is at most v no matter what Rose does.

Consider the stochastic game $G = \begin{bmatrix} 4 + 1/2G & -1 \\ 1 & 2 + 1/2G \end{bmatrix}$.

- (a) Verify that $G(2)$ has value 2 by showing that this number together with the mixed strategies $\mathbf{p} = \begin{bmatrix} 1/4 & 3/4 \end{bmatrix}$ for Rose and $\mathbf{q} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ for Colin form a von Neumann solution to it.
 - (b) Show that whenever Rose plays $\mathbf{p}$ in the original game G , the expected total payoff is at least 2 no matter what mixed strategy Colin plays.
 - (c) Show that whenever Colin plays $\mathbf{q}$ in the original game G , the expected total payoff is at most 2 no matter what mixed strategy Rose plays.
- (24) This exercise proves Theorem 5.14. Let G be a stochastic game.
- (a) Define the function f by $f(x) = \text{value}(G(x)) - x$. It follows from Exercise (21) that f is continuous. Use this to prove that there exists $v \in \mathbb{R}$ for which $v = \text{value}(G(v))$.
 - (b) Assuming $a < b$, prove that $f(a) > f(b)$. (Hint: Let p be the highest probability appearing in the matrix G and show that $\text{value}(G(b)) - \text{value}(G(a)) \leq p(b - a)$.) Deduce that there is a unique number v so that $v = \text{value}(G(v))$.
 - (c) Suppose $v = \text{value}(G(v))$ and let this number together with the mixed strategies $\mathbf{p}^*$ and $\mathbf{q}^*$ for Rose and Colin form a von Neumann solution to $G(v)$. Show that $\mathbf{p}^*$ and $\mathbf{q}^*$ satisfy Theorem 5.14.

Chapter 6

Von Neumann's Minimax Theorem

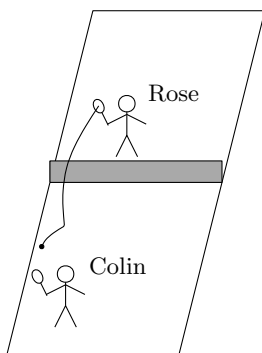


Figure 6.1. Tennis

Chapter 5 introduced von Neumann's famous Minimax Theorem: Every zero-sum matrix game has a value v and pair of mixed strategies $\mathbf{p}$ for Rose and $\mathbf{q}$ for Colin so that both $\mathbf{p}$ and $\mathbf{q}$ have a guarantee of v . This theorem lies at the heart of classical game theory—in fact, it was essential in founding the subject. Still, its significance surpasses the subject of game theory. Von Neumann's result and its generalization to LP-duality

helped to provide the theoretical underpinning for the field of linear programming, a subject central to mathematical optimization and its extensive industrial applications. Indeed, efficient tools developed to solve linear programming problems rank among the most influential extant algorithms. For the interested reader, Appendix B connects the study of zero-sum matrix games to an introduction to linear programming. In particular, this appendix shows how to find von Neumann solutions using a linear program.

Chapter 6 is entirely devoted to von Neumann's Minimax Theorem. We first give a constructive algebraic proof of the theorem for 2×2 games. Then we demonstrate a geometric approach to solve $2 \times n$ and $m \times 2$ matrices. The chapter ends with a full proof of von Neumann's Theorem (although this proof does not explicitly show how to compute solutions).

Let's start with an example from the world of sports.

Example 6.1 (Tennis). Rose and Colin are playing a game of tennis. Rose has come to the net and played a somewhat shallow ball to Colin, who now has the advantage. We assume that Colin will either play a return to Rose's forehand (F) or backhand side (B) (ignoring the possibility of, say, a lob) and Rose must guess by moving to her forehand (F) or backhand side (B). If Rose guesses wrong, she will not get her racket on the ball. Rose, of course, can still win the point if Colin hits it into the net or out of bounds. The following matrix shows the probability that Rose will win the point depending upon the F/B decisions of the two players in this particular scenario. Be aware that this matrix only applies to this very specific situation and not to any other part of the game:

		Colin	
		F	B
Rose	F	80%	10%
	B	20%	50%

Suppose now that Rose works very hard and improves her backhand during the off season. When she plays Colin in the following year and encounters this particular scenario again, the probabilities have changed to the following:

		Colin	
		F	B
Rose	F	80%	10%
	B	20%	60%

Assume that Rose and Colin know each other's game well, so Colin knows Rose's backhand is now better. These are sophisticated players! How will Colin's strategies compare in these two games? Will Rose play more or less to her improved backhand side?

6.1. Equating the Opponent's Results

We will analyze Rose and Colin's strategies in the above tennis example after developing an algebraic technique to find von Neumann solutions for 2×2 games.

2×2 Games. Let's return to the 2×2 zero-sum matrix game example from the last chapter. Consider $A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$ with the mixed strategies $\mathbf{p} = \begin{bmatrix} 1/4 & 3/4 \end{bmatrix}$ for Rose and $\mathbf{q} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ for Colin. Note that Rose's strategy has the following nice property:

$$\mathbf{p}A = \begin{bmatrix} 1/4 & 3/4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5/4 & 5/4 \end{bmatrix}.$$

Regardless of which column Colin chooses, the expected payoff will always be $5/4$. In other words, Rose's strategy equates the expected payoff whether Colin chooses column 1 or column 2. This extremely useful property merits a general definition.

Definition 6.2. For a zero-sum matrix game A , we say that a mixed strategy $\mathbf{p}$ for Rose *equates Colin's results* if all entries of $\mathbf{p}A$ are equal. Similarly, we say that a mixed strategy $\mathbf{q}$ for Colin *equates Rose's results* if all entries of $A\mathbf{q}$ are equal.

In the above example, Colin's mixed strategy does not equate Rose's results since

$$A\mathbf{q} = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 3/2 \end{bmatrix}.$$

Can we find a strategy for Colin that does equate Rose's results? We want $\mathbf{q}' = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ so that both entries of $A\mathbf{q}'$ are equal. In other words, $2q + (-1)(1-q) = q + 2(1-q)$. This simplifies to $3q - 1 = 2 - q$, which has a solution of $q = 3/4$. Note that the vector $\mathbf{q}' = \begin{bmatrix} 3/4 \\ 1/4 \end{bmatrix}$ is indeed a mixed strategy since both entries are at least zero and they sum to 1. For this new vector $\mathbf{q}'$ we have

$$A\mathbf{q}' = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 5/4 \\ 5/4 \end{bmatrix}.$$

We see here that Colin's new strategy $\mathbf{q}'$ also has a guarantee of $5/4$! Thus we have a von Neumann solution for game A —the value $v = 5/4$ together with the mixed strategies $\mathbf{p}$ for Rose and $\mathbf{q}'$ for Colin. In this case, a pair of mixed strategies which equate the opponents' results give a von Neumann solution. This process works for every 2×2 matrix that does not have a dominated row or column. To prove this, we need the following lemma.

Lemma 6.3. *For every 2×2 zero-sum matrix game:*

- (1) *If Rose has no dominant row, Colin has a mixed strategy equating her results.*
- (2) *If Colin has no dominant column, Rose has a mixed strategy equating his results.*

Proof. We will give the argument for (1), assuming Rose has no dominant strategy. Part (2) follows from a similar argument. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a zero-sum matrix game in which Rose does not have a dominant strategy. We want to find a mixed strategy $\mathbf{q} = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ for Colin that equates Rose's results. This gives us the equation $aq + b(1-q) = cq + d(1-q)$, which simplifies to $(a-c+d-b)q = (d-b)$, so $q = (d-b)/((a-c) + (d-b))$. Since Rose has no dominant strategy, it is not possible for $a \geq c$ and $b \geq d$ and also not possible for $a \leq c$ and $b \leq d$. It follows that either $d > b$ and $a > c$ or $d < b$ and $a < c$. In either case, the number q will satisfy $0 \leq q \leq 1$ so $\mathbf{q}$ is a mixed strategy which equates Rose's results, as desired. $\square$

This sets up the following theorem, which gives a simple procedure to find von Neumann solutions in any 2×2 matrix.

Proposition 6.4. *For every 2×2 zero-sum matrix game A , one of the following holds.*

- (1) *Iterated removal of dominated strategies reduces the matrix to a 1×1 matrix $\begin{bmatrix} v \end{bmatrix}$. The number v and the associated pure row and column strategies form a von Neumann solution.*
- (2) *Rose and Colin have mixed strategies $\mathbf{p}$ and $\mathbf{q}$ equating the opponents' results. Then $\mathbf{p}A = \begin{bmatrix} v & v \end{bmatrix}$ and $A\mathbf{q} = \begin{bmatrix} v \\ v \end{bmatrix}$. The number v with $\mathbf{p}$ and $\mathbf{q}$ form a von Neumann solution.*

Proof. If a player has a dominated strategy, then removing it reduces the matrix to the form $\begin{bmatrix} a \\ b \end{bmatrix}$ or $\begin{bmatrix} c & d \end{bmatrix}$. Either way, one of the two numbers in the matrix will be less than or equal to the other. Removing the corresponding dominated strategy leaves a 1×1 matrix. Theorem 5.6 implies that this entry is a saddle point, and then Observation 5.13 implies that (1) holds.

Now assume neither player has a dominant strategy. In this case, the previous lemma tells us that there exists a mixed strategy $\mathbf{p} = \begin{bmatrix} p & 1-p \end{bmatrix}$ for Rose that equates Colin's results and a mixed strategy $\mathbf{q} = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ for Colin that equates Rose's results. So there must exist real numbers u and w so that $\mathbf{p}A = \begin{bmatrix} u & u \end{bmatrix}$ and $A\mathbf{q} = \begin{bmatrix} w \\ w \end{bmatrix}$. Now

$$w = pw + (1-p)w = \mathbf{p} \begin{bmatrix} w \\ w \end{bmatrix} = \mathbf{p}A\mathbf{q} = \begin{bmatrix} u & u \end{bmatrix} \mathbf{q} = qu + (1-q)u = u.$$

So $u = w$ is the value of the matrix, and (2) holds. $\square$

Playing Sports. The world of sports provides a familiar arena for sophisticated competition in zero-sum games. Sporting events generally feature two individuals or teams in direct opposition and such situations can always be modeled with zero-sum games. Expert players and teams practice intensely, and this focused repetition often results in highly

strategic play. While sportscasters commonly deliver blanket statements like “always play to your strengths,” reality is often more subtle than this.

Ideally, rational players will make the easiest way to beat them as difficult as possible. A sophisticated player knows the opponent's strengths and weaknesses and assumes the opponent has reciprocal knowledge. If a player is easy to beat in some way, a strategic opponent will focus on and exploit this. As such, players want the easiest way to beat them to be as difficult as possible.

For an example, we return to the tennis scenario from the start of the chapter. Momentarily suppose that when Rose plays according to her current strategy, it is easier for Colin to win by playing to her backhand than by playing to her forehand. Then Colin will generally play to Rose's backhand, and on average Rose will lose more points than she should. Rose could do better by going to her backhand more often, thus making it more difficult to beat her that way. Indeed, Rose should adopt a strategy of guessing F or B so she will be equally difficult to beat no matter what choice Colin makes. This is exactly the idea of playing to equate the opponent's results, and it is precisely what we see in upper echelons of sporting competition.

Let's look at some numbers in the tennis example. The matrices at the start of the chapter have percentages in the boxes to indicate on average who wins what percentage of points. Say the outcome is listed as 80%. This means the expected number of points for Rose will be $.8(1) + .2(-1)$ since she will win one point with probability .8 and lose one point with probability .2. For each entry in the first matrix, this gives us the following familiar type of zero-sum matrix game:

$$A = \begin{bmatrix} 3/5 & -4/5 \\ -3/5 & 0 \end{bmatrix}.$$

This matrix has no dominated strategy. To solve it we will find a strategy for Rose to equate Colin's results. Since

$$\mathbf{p}A = [p \quad 1-p] \begin{bmatrix} 3/5 & -4/5 \\ -3/5 & 0 \end{bmatrix} = [6/5p - 3/5 \quad -4/5p],$$

Colin's results will be the same when $\frac{6}{5}p - \frac{3}{5} = -\frac{4}{5}p$, or when $p = \frac{3}{10}$. Therefore, Rose should guess forehand 30% of the time and backhand 70% of the time. Assuming Rose does this, the matrix product $\begin{bmatrix} \frac{3}{10} & \frac{7}{10} \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ -\frac{3}{5} & 0 \end{bmatrix} = \begin{bmatrix} -\frac{6}{25} & -\frac{6}{25} \end{bmatrix}$ tells us (by Proposition 6.4) that this game has value $-\frac{6}{25}$. If each player plays rationally, this game will have an expected advantage of $\frac{6}{25}$ of a point to Colin. What happens when Rose improves her backhand? Our sophisticated players are now playing the new game

$$A' = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ -\frac{3}{5} & \frac{1}{5} \end{bmatrix}.$$

This game also has no dominated strategy, so to solve it we will find a strategy for Rose that equates Colin's results. The new strategy $\mathbf{p}' = \begin{bmatrix} p' & 1 - p' \end{bmatrix}$ for Rose gives us

$$\mathbf{p}' A' = \begin{bmatrix} p' & 1 - p' \end{bmatrix} \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ -\frac{3}{5} & \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \frac{6}{5}p' - \frac{3}{5} & -p' + \frac{1}{5} \end{bmatrix}.$$

Equating Colin's results yields $\frac{6}{5}p' - \frac{3}{5} = -p' + \frac{1}{5}$, so $p' = \frac{4}{11}$, which is approximately 36.4%. This means Rose should now go to her forehand about 36.4% of the time and to her backhand about 63.6% of the time. Although her backhand volley has improved, Rose should actually go for her backhand shot less as a result! To find the new value of the game we compute $\mathbf{p}' A' = \begin{bmatrix} -\frac{9}{55} & -\frac{9}{55} \end{bmatrix}$, giving a value of $-\frac{9}{55}$. Overall, Rose's backhand improvement led to a better result for her in this game, which is now only worth an expected advantage of $\frac{9}{55}$ of a point for Colin. The fact that Rose plays her better backhand less often may not be so surprising after all. When Rose's backhand improves, Colin will probably want to play a smaller percentage of shots there. With more shots going toward Rose's forehand, she'll probably want to move that way to take advantage.

Professional athletes appear keenly aware of the strategic nature of the games they play, and in-depth statistical analysis generally shows that they compete in an essentially optimal manner. The key principle highlighted here is always to play such that the easiest way to beat you is as difficult as possible.

6.2. Two-Dimensional Games

This section introduces a graphical technique for finding von Neumann solutions to $2 \times n$ and $m \times 2$ zero-sum matrix games. Let's start with an example.

Example 6.5. Consider the zero-sum matrix game

$$A = \begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & 1 \end{bmatrix}.$$

Suppose that Rose is going to play the mixed strategy $[p \ 1 - p]$. By treating p as a variable, we can investigate expected payoffs in a variety of situations. If Colin plays the first column, the expected payoff is given by the linear function $C_1(p) = p(4) + (1 - p)(-1) = 5p - 1$. A similar calculation shows that when Colin plays the second column or the third column, the expected payoffs are given by $C_2(p) = -5p + 3$ or $C_3(p) = -2p + 1$. Figure 6.2 shows graphs of these linear functions using the horizontal axis for the variable p (which satisfies $0 \leq p \leq 1$) and the vertical axis for the expected payoff.

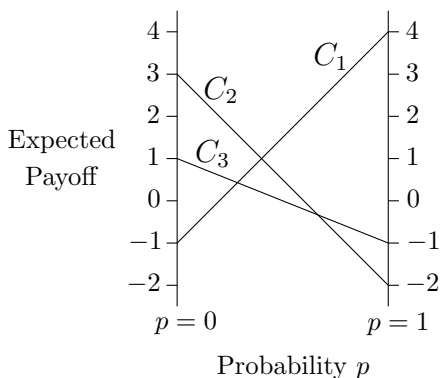


Figure 6.2. Graphing the payoffs

Combining linear functions C_1, C_2, C_3 gives the equation

$$\begin{bmatrix} p & 1 - p \end{bmatrix} A = \begin{bmatrix} C_1(p) & C_2(p) & C_3(p) \end{bmatrix}.$$

For every $0 \leq p \leq 1$, the guarantee of $[p \ 1 - p]$ is the minimum entry of the vector $[p \ 1 - p] A$. Treating p as a variable gives us a *guarantee*

function for Rose, denoted G_R and defined by

$$G_R(p) = \min\{C_1(p), C_2(p), C_3(p)\}.$$

For every $0 \leq p \leq 1$ we have $G_R(p)$ as the guarantee for Rose's mixed strategy $[p \quad 1-p]$. Since G_R is defined by the minimum of the functions C_1, C_2, C_3 , it is straightforward to graph it (see Figure 6.3).

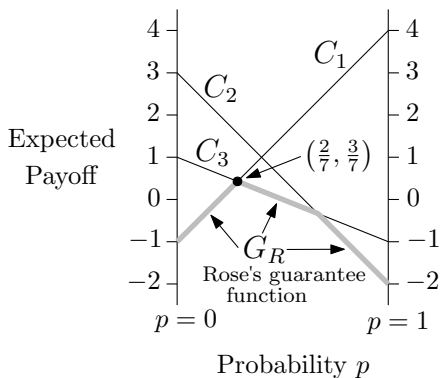


Figure 6.3. Rose's guarantee function

Rose wants to find a mixed strategy with the highest possible guarantee, so Rose wants to maximize $G_R(p)$ over all $0 \leq p \leq 1$. The graph in Figure 6.3 shows that this maximum will be achieved at the point where the lines C_1 and C_3 intersect. If $C_1(p) = C_3(p)$, then $5p - 1 = -2p + 1$ so $7p = 2$ and $p = 2/7$. At $p = 2/7$ we have $C_1(2/7) = C_3(2/7) = 3/7$. Thus the highest point on the graph of G_R has coordinates $(2/7, 3/7)$. In terms of the game, this tells us that the Rose can achieve a maximum guarantee of $3/7$ by the mixed strategy $[2/7 \quad 5/7]$ (i.e. the mixed strategy $[p \quad 1-p]$ for $p = 2/7$).

A full von Neumann solution also includes a mixed strategy for Colin with a guarantee of $3/7$. When Rose plays $[2/7 \quad 5/7]$, the expected payoff is $3/7$ when Colin plays either the first or third column. The payoff is higher if Colin plays the second column. Thus a mixed strategy for Colin with a guarantee of $3/7$ will not use the second column. That is, Colin's strategy

should have the form $\mathbf{q} = \begin{bmatrix} q \\ 0 \\ 1-q \end{bmatrix}$. Following our analysis of 2×2 games,

we now seek a mixed strategy of this form that equates Rose's results. In particular,

$$4q - 1(1 - q) = (-1)q + (1 - q)$$

implies $7q = 2$, so $q = 2/7$ and $\mathbf{q} = \begin{bmatrix} 2/7 \\ 0 \\ 5/7 \end{bmatrix}$. Now calculate

$$A\mathbf{q} = \begin{bmatrix} 4 & -2 & -1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2/7 \\ 0 \\ 5/7 \end{bmatrix} = \begin{bmatrix} 3/7 \\ 3/7 \end{bmatrix}.$$

Thus Colin's strategy $\mathbf{q}$ has a guarantee of $3/7$. We conclude that the value $3/7$ together with the strategies $\mathbf{p} = \begin{bmatrix} 2/7 & 5/7 \end{bmatrix}$ for Rose and $\mathbf{q} = \begin{bmatrix} 2/7 \\ 0 \\ 5/7 \end{bmatrix}$ for Colin form a von Neumann solution for A .

Now let's extend these ideas to more general matrices. Consider a $2 \times n$ zero-sum matrix game given by

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \end{bmatrix}.$$

As before, assume that Rose is playing the mixed strategy $\begin{bmatrix} p & 1 - p \end{bmatrix}$ and view p as a variable with $0 \leq p \leq 1$. If Colin plays according to the pure strategy of choosing column j , then the expected payoff is given by $C_j(p) = pa_{1j} + (1 - p)a_{2j}$. As in the example above, the expected payoff when Colin plays column j is an easily graphed linear function of p . The functions $C_1, \dots, C_n$ give the convenient equation

$$\begin{bmatrix} p & 1 - p \end{bmatrix} A = \begin{bmatrix} C_1(p) & C_2(p) & \cdots & C_n(p) \end{bmatrix}.$$

Based on this, define Rose's *guarantee function* G_R by the rule

$$G_R(p) = \min\{C_1(p), C_2(p), \dots, C_n(p)\}.$$

Now $G_R(p)$ is the guarantee of Rose's mixed strategy $\begin{bmatrix} p & 1 - p \end{bmatrix}$. As before, the graph of G_R is the minimum of the lines $C_1, \dots, C_n$.

Determine a highest point (p^*, v^*) on the graph of G_R (this may be found by computing intersection points of some of the lines $C_1, \dots, C_n$). By construction, Rose's mixed strategy $\begin{bmatrix} p^* & 1 - p^* \end{bmatrix}$ has a guarantee of v^* . We will show that Colin also has a mixed strategy with a guarantee

of v^* , thus giving a von Neumann solution. If there is a column of the matrix with both entries less than or equal to v^* , then this column is a pure strategy with a guarantee of v^* , so we are done. Otherwise, the point (p^*, v^*) will satisfy $0 < p^* < 1$ and will lie at the intersection of two lines C_i and C_j , one with positive slope and the other with negative slope (Exercise (16)). Consider the 2×2 matrix A' obtained by deleting all but columns i and j from the matrix A . Since $C_i(p^*) = C_j(p^*) = v^*$, in this smaller matrix we have

$$\begin{bmatrix} p^* & 1 - p^* \end{bmatrix} A' = \begin{bmatrix} v^* & v^* \end{bmatrix}.$$

So, when playing A' , Rose's mixed strategy $\begin{bmatrix} p^* & 1 - p^* \end{bmatrix}$ equates Colin's results. By our assumptions, one of C_i and C_j has positive slope while the other has negative slope. It follows from this that Rose does not have a dominant strategy in the matrix A' . Therefore, by Lemma 6.3 we may choose a mixed strategy $\begin{bmatrix} q^* \\ 1 - q^* \end{bmatrix}$ for Colin that equates Rose's

results in A' . Now Proposition 6.4 implies that $A' \begin{bmatrix} q^* \\ 1 - q^* \end{bmatrix} = \begin{bmatrix} v^* \\ v^* \end{bmatrix}$. So

v^* together with the mixed strategies $\begin{bmatrix} p^* & 1 - p^* \end{bmatrix}$ and $\begin{bmatrix} q^* \\ 1 - q^* \end{bmatrix}$ form a von Neumann solution in the matrix A' . Now we will use this to obtain a von Neumann solution to A . Define a mixed strategy $\mathbf{q}$ for Colin in the original matrix game A by having Colin play columns i and j with the same probabilities as in our solution to A' and having Colin play all other columns with probability 0 (i.e. we form $\mathbf{q}$ from $\begin{bmatrix} q^* \\ 1 - q^* \end{bmatrix}$ by adding a 0 entry for each column we deleted in going from A to A').

Then $A\mathbf{q} = A' \begin{bmatrix} q^* \\ 1 - q^* \end{bmatrix} = \begin{bmatrix} v^* \\ v^* \end{bmatrix}$, so we have found a von Neumann solution to the original game A consisting of the value v^* and the strategies $\begin{bmatrix} p^* & 1 - p^* \end{bmatrix}$ and $\mathbf{q}$. We summarize this process next.

Procedure 6.6 (Solving $(2 \times n)$ -Dimensional Games).

- (1) Let Rose play $\begin{bmatrix} p & 1 - p \end{bmatrix}$ and graph the expected payoffs for each of Colin's pure strategies $C_1, \dots, C_n$ (each a linear function of p).
- (2) Rose's guarantee function is $G_R(p) = \min\{C_1(p), \dots, C_n(p)\}$.

- (3) Identify a highest point (p^*, v^*) on the graph of G_R .
- (4) The value of the game is v^* .
- (5) Rose's strategy $[p^* \quad 1 - p^*]$ has a guarantee of v^* .
- (6) If Colin has a pure strategy with a guarantee of v^* , this gives a solution. Otherwise, the point (p^*, v^*) lies at the intersection of two lines C_i and C_j , one with positive slope and the other with negative slope. Obtain a strategy for Colin with a guarantee of v^* by playing these two columns i and j to equate Rose's results.

A similar process works for an $m \times 2$ zero-sum matrix game. Suppose Colin plays the mixed strategy $\begin{bmatrix} q \\ 1 - q \end{bmatrix}$ and view q as a variable. If Rose plays the pure strategy of row i , then her expected payoff will be $R_i(q)$ for a linear function R_i . Now Colin's guarantee function is given by the *maximum* of $R_1, \dots, R_m$ (note that we previously defined Rose's guarantee function as a *minimum*). To find the solution, we will determine the *lowest* point on the graph of G_C (whereas before we found the *highest* point on the graph of G_R). This produces the following procedure.

Procedure 6.7 (Solving $(m \times 2)$ -Dimensional Games).

- (1) Let Colin play $\begin{bmatrix} q \\ 1 - q \end{bmatrix}$ and graph the expected payoffs for each of Rose's pure strategies $R_1, \dots, R_m$ (each a linear function of q).
- (2) Colin's guarantee function is $G_C(q) = \max\{R_1(q), \dots, R_m(q)\}$.
- (3) Identify a lowest point (q^*, v^*) on the graph of G_C .
- (4) The value of the game is v^* .
- (5) Colin's strategy $\begin{bmatrix} q^* \\ 1 - q^* \end{bmatrix}$ has a guarantee of v^* .
- (6) If Rose has a pure strategy with a guarantee of v^* , this gives a solution. Otherwise, the point (q^*, v^*) lies at the intersection of two lines R_i and R_j , one with positive slope and the other with negative slope. Obtain a strategy for Rose with a guarantee of v^* by playing rows i and j to equate Colin's results.

6.3. Proof of the Minimax Theorem

This section presents a proof of von Neumann's Theorem. From the many proofs for this significant result, this one was chosen for its emphasis on the essential geometric nature of the problem.

Hyperplanes. Our proof involves the geometry of n -dimensional Euclidean space $\mathbb{R}^n$. We will denote points in $\mathbb{R}^n$ using column vectors. So

if $\mathbf{x} \in \mathbb{R}^n$ is the point with coordinate i equal to x_i , we write $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

If $\mathbf{y} \in \mathbb{R}^n$ is given by $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$, then the *dot product* of $\mathbf{x}$ and $\mathbf{y}$ is defined

to be

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

Fix a particular nonzero point $\mathbf{y}$ and consider dot products of other points with $\mathbf{y}$. For every real number d , define

$$H_d = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = d\}.$$

The set H_0 consists of all $\mathbf{x} \in \mathbb{R}^n$ which have dot product with $\mathbf{y}$ equal to 0 and this forms a subspace of $\mathbb{R}^n$ of dimension $n - 1$. Every other set H_d may be obtained by translating H_0 by a scalar multiple of $\mathbf{y}$. We call any set of the form H_d a *hyperplane* with *normal* $\mathbf{y}$.

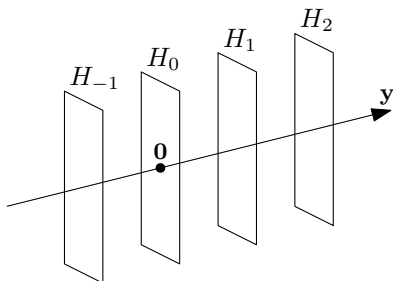


Figure 6.4. Hyperplanes

Figure 6.4 shows the points which have dot products $-1, 0, 1$, and 2 with $\mathbf{y}$. Observe that each of the hyperplanes H_d divides $\mathbb{R}^n$ into those points which have dot product $\geq d$ with $\mathbf{y}$ and those which have dot product $\leq d$ with $\mathbf{y}$.

Convexity. If $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the midpoint between $\mathbf{x}$ and $\mathbf{y}$ is given by the equation $\frac{1}{2}\mathbf{x} + \frac{1}{2}\mathbf{y}$. This is apparent since each coordinate of the midpoint is the average of the corresponding coordinates of $\mathbf{x}$ and $\mathbf{y}$. More generally, the line segment between $\mathbf{x}$ and $\mathbf{y}$, denoted by $\overline{\mathbf{xy}}$, is given by $\overline{\mathbf{xy}} = \{t\mathbf{x} + (1-t)\mathbf{y} \mid 0 \leq t \leq 1\}$.

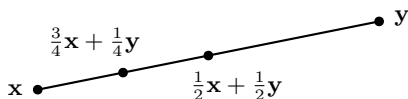


Figure 6.5. A line segment

Recall that a *linear combination* of $\mathbf{x}_1, \dots, \mathbf{x}_k$ is an expression of the form

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_k\mathbf{x}_k,$$

where $c_1, \dots, c_k$ are any real numbers. We call this a *convex combination* if in addition $c_1, \dots, c_k \geq 0$ and $\sum_{i=1}^k c_i = 1$. Define the *convex hull* of $\mathbf{x}_1, \dots, \mathbf{x}_k$ to be the set of all points that may be written as convex combinations of $\mathbf{x}_1, \dots, \mathbf{x}_k$. Note that the convex hull of $\mathbf{x}, \mathbf{y}$ is precisely $\overline{\mathbf{xy}}$. Other convex hulls appear in Figure 6.6.

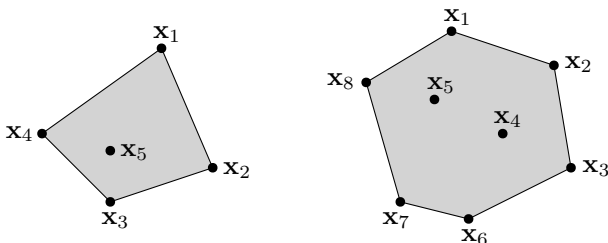


Figure 6.6. Two convex hulls in $\mathbb{R}^2$

For an arbitrary set $S \subseteq \mathbb{R}^n$, we say that S is *convex* if for every $\mathbf{x}, \mathbf{y} \in S$ the entire line segment $\overline{\mathbf{xy}}$ is contained in S . A necessary property of convex hulls is that these sets are indeed convex.

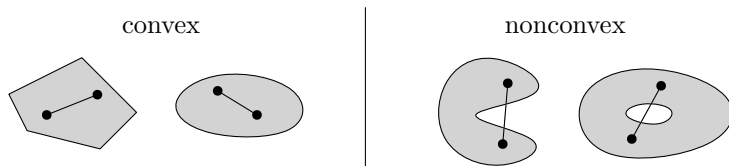


Figure 6.7. Convex and nonconvex sets

Proposition 6.8. *The convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_k$ is a convex set.*

Proof. Let $\mathbf{y}$ and $\mathbf{z}$ be in the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_k$ and consider a point on the line segment $\overline{\mathbf{y}\mathbf{z}}$ given by $t\mathbf{y} + (1-t)\mathbf{z}$ (so $0 \leq t \leq 1$). To complete the proof, it suffices to show that this point is also in the convex hull. By assumption, $\mathbf{y}$ and $\mathbf{z}$ are both convex combinations of $\mathbf{x}_1, \dots, \mathbf{x}_k$, so we may choose $c_1, \dots, c_k \geq 0$ summing to 1 and $d_1, \dots, d_k \geq 0$ summing to 1, so that

$$\mathbf{y} = c_1\mathbf{x}_1 + \dots + c_k\mathbf{x}_k$$

and

$$\mathbf{z} = d_1\mathbf{x}_1 + \dots + d_k\mathbf{x}_k.$$

Now, the point of interest is

$$t\mathbf{y} + (1-t)\mathbf{z} = (tc_1 + (1-t)d_1)\mathbf{x}_1 + \dots + (tc_k + (1-t)d_k)\mathbf{x}_k.$$

All of the coefficients of the $\mathbf{x}_i$ on the right in the above equation are nonnegative, and $\left(t \sum_{i=1}^k c_i\right) + \left((1-t) \sum_{i=1}^k d_i\right) = t + (1-t) = 1$, so this point lies in the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_k$, as desired. $\square$

In fact, Exercise (19) shows that the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_k$ is the unique minimal convex set which contains these points.

The Proof. We will use $\mathbf{0}$ to denote the origin and $\mathbf{1}$ to denote the point with all coordinates equal to 1. We will also call on the transpose to

move between column vectors and row vectors. For example, $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}^T =$

$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$. For convenience, we restate the Minimax Theorem before giving the proof.

Theorem (von Neumann). *For every zero-sum matrix game, the maximum guarantee of a mixed strategy for Rose is equal to the minimum guarantee of a mixed strategy for Colin.*

Proof. Fix a zero-sum matrix game $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$, and let r be the largest number¹ so that Rose has a mixed strategy with a guarantee of r . It follows from Lemma 5.10 that Colin does not have a mixed strategy with a guarantee strictly smaller than r . So, to complete the proof, it suffices to show that Colin has a mixed strategy with a guarantee of at most r . To prove this by contradiction, now suppose that Colin does not have a mixed strategy with a guarantee of at most r .

Modify the matrix A by subtracting r from each entry. Observe that this decreases every payoff by r . After this change, the highest guarantee Rose can achieve is 0, and Colin does not have a mixed strategy with a guarantee which is ≤ 0 . We will see that this leads us to a contradiction.

Define the set

$$\begin{aligned} S &= \{A\mathbf{q} \mid \mathbf{q} \text{ is a mixed strategy for Colin}\} \\ &= \{q_1\mathbf{a}_1 + \cdots + q_n\mathbf{a}_n \mid q_1, \dots, q_n \geq 0 \text{ and } \sum_{i=1}^n q_i = 1\}. \end{aligned}$$

Note that S is precisely the convex hull of the columns of A , so in particular, S is convex. Now define another set as follows:

$$S^+ = \{\mathbf{y} \in \mathbb{R}^n \mid \mathbf{y} \geq \mathbf{x} \text{ for some } \mathbf{x} \in S\}.$$

If the point $\mathbf{0}$ was contained in S^+ , we would have $\mathbf{0} \geq \mathbf{x}$ for some $\mathbf{x} \in S$. However, then there would be a mixed strategy $\mathbf{q}$ for Colin with $A\mathbf{q} = \mathbf{x} \leq \mathbf{0}$, which contradicts the assumption that Colin does not have a mixed strategy with a guarantee of ≤ 0 . Therefore $\mathbf{0}$ is not contained in S^+ . Let $\mathbf{y}$ be a closest point² in S^+ to $\mathbf{0}$ and define the number $d = \mathbf{y} \cdot \mathbf{y} > 0$ and the hyperplane

$$H_d = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = d\}.$$

Next, we proceed with a sequence of three claims.

¹That such a number r exists follows from the continuity of the function $\mathbf{p} \rightarrow \min(\mathbf{p}A)$ and the fact that the set of all mixed strategies $\mathbf{p}$ is compact.

²The fact that there is a closest point follows from the fact that S^+ is closed.

Claim 1. S^+ is convex.

Let $\mathbf{u}^+$ and $\mathbf{w}^+$ be points in S^+ , let $0 \leq t \leq 1$, and consider the point $t\mathbf{u}^+ + (1-t)\mathbf{w}^+$. To prove the claim, it suffices to show that this point is in S^+ . Since $\mathbf{u}^+, \mathbf{w}^+ \in S^+$, we may choose points $\mathbf{u}, \mathbf{w} \in S$ with $\mathbf{u} \leq \mathbf{u}^+$ and $\mathbf{w} \leq \mathbf{w}^+$. It follows from the convexity of S that $t\mathbf{u} + (1-t)\mathbf{w}$ is in S . Now $t\mathbf{u}^+ + (1-t)\mathbf{w}^+ \geq t\mathbf{u} + (1-t)\mathbf{w}$ so $t\mathbf{u}^+ + (1-t)\mathbf{w}^+$ is in S^+ , as desired.

Claim 2. $\mathbf{y} \cdot \mathbf{z} \geq d$ for every $\mathbf{z} \in S^+$.

If there were a point $\mathbf{z} \in S^+$ for which $\mathbf{z} \cdot \mathbf{y} < d$, then $\mathbf{z}$ would lie on the same side of the hyperplane H_d as $\mathbf{0}$, as shown on the left in Figure 6.8. Then, by the convexity of S^+ , the line segment between $\mathbf{z}$ and $\mathbf{y}$ would also be contained in S^+ . However, then $\mathbf{y}$ would not be a closest point in S^+ to $\mathbf{0}$. Thus, the claim holds, and the situation is as on the right in Figure 6.8.

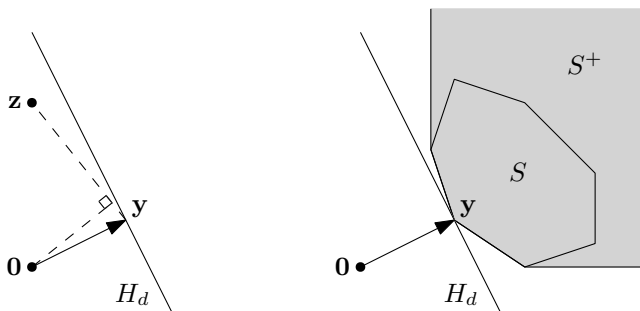


Figure 6.8. The hyperplane H_d

Claim 3. $\mathbf{y} \geq \mathbf{0}$.

Let $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ and let $\mathbf{y}' = \mathbf{y} + \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. The definition of S^+ implies that

$\mathbf{y}' \in S^+$ so Claim 2 gives us $d \leq \mathbf{y} \cdot \mathbf{y}' = \mathbf{y} \cdot \mathbf{y} + y_1 = d + y_1$. It follows that $y_1 \geq 0$. A similar argument for the other coordinates gives us $\mathbf{y} \geq \mathbf{0}$.

With this last claim in place, we can now complete the proof. Define $s = \mathbf{y} \cdot \mathbf{1}$ and note that $s > 0$. Then define $\mathbf{p} = \frac{1}{s}\mathbf{y}$. The vector $\mathbf{p}^\top$ is a mixed strategy for Rose since $\mathbf{p} \geq \mathbf{0}$ and the sum of its entries is $\mathbf{p} \cdot \mathbf{1} = \frac{1}{s}(\mathbf{y} \cdot \mathbf{1}) = 1$. For every point $\mathbf{z} \in S^+$, Claim 2 implies $\mathbf{p} \cdot \mathbf{z} = \frac{1}{s}\mathbf{y} \cdot \mathbf{z} \geq \frac{d}{s}$. So, in particular, every column $\mathbf{a}_i$ of A must satisfy $\mathbf{p} \cdot \mathbf{a}_i \geq \frac{d}{s}$. However, this means that $\mathbf{p}^\top$ is a mixed strategy for Rose with a guarantee of at least $\frac{d}{s}$. This contradicts our initial assumption and completes the proof. $\square$

Exercises

(1) Find von Neumann solutions to the following:

$$(a) \begin{bmatrix} 1 & 2 \\ -2 & 0 \end{bmatrix}, \quad (b) \begin{bmatrix} 0 & 3 \\ 4 & 2 \end{bmatrix}, \quad (c) \begin{bmatrix} 3 & -2 \\ 0 & 1 \end{bmatrix}, \quad (d) \begin{bmatrix} 2 & 7 \\ 5 & -1 \end{bmatrix}.$$

(2) Find von Neumann solutions to the following:

$$(a) \begin{bmatrix} -3 & 5 \\ -1 & 3 \\ 2 & -2 \\ 3 & -6 \end{bmatrix}, \quad (b) \begin{bmatrix} -2 & 5 \\ 1 & 2 \\ 0 & -2 \\ 1 & 3 \end{bmatrix}, \quad (c) \begin{bmatrix} -4 & 2 & 3 & -2 \\ 4 & -1 & -3 & 1 \end{bmatrix}.$$

(3) Use a combination of eliminating dominated strategies and the methods from this chapter to find von Neumann solutions to the following:

$$(a) \begin{bmatrix} 2 & 3 & 1 \\ 3 & -2 & -1 \\ -2 & 4 & -3 \end{bmatrix}, \quad (b) \begin{bmatrix} 4 & 2 & 0 \\ 2 & -1 & 3 \\ 0 & -3 & 1 \end{bmatrix}, \quad (c) \begin{bmatrix} 6 & 2 & 0 \\ 1 & 5 & 9 \\ 0 & 4 & 6 \end{bmatrix}.$$

(4) Find two von Neumann solutions to $\begin{bmatrix} -2 & 0 & 2 \\ 3 & 1 & -1 \end{bmatrix}$.

(5) In Example 6.5, Rose's guarantee function may be expressed as

$$G_R(p) = \begin{cases} 5p - 1 & \text{if } 0 \leq p \leq 2/7, \\ -2p - 1 & \text{if } 2/7 < p \leq 2/3, \\ -5p + 3 & \text{if } 2/3 < p \leq 1. \end{cases}$$

Find similar formulas for the following functions from Exercise (2):

- (a) Colin's guarantee function from (a),
- (b) Rose's guarantee function from (c).

(6) For every $x, y \in \mathbb{R}$ find a von Neumann solution to the following:

$$(a) \begin{bmatrix} 1 & 2 \\ 3 & x \end{bmatrix}, \quad (b) \begin{bmatrix} x & -1 \\ 1 & 4 \end{bmatrix}, \quad (c) \begin{bmatrix} x & 2 \\ 0 & x \end{bmatrix}, \quad (d) \begin{bmatrix} x & 1 \\ y & 5 \end{bmatrix}.$$

(7) *Rock-Paper-Scissors-Dynamite* is like Rock-Paper-Scissors with the added rules that Dynamite ties itself, beats Rock and Paper, but loses to Scissors (which cut the wick of the Dynamite). Express this as a matrix game and find a von Neumann solution.

(8) *Rock-Paper-Scissors-Superman-Kryptonite* is another game like Rock-Paper-Scissors with the following additional rules: Kryptonite beats Superman, Superman ties Superman, Kryptonite ties Kryptonite, and Superman beats all of Rock, Paper, Scissors, all of which beat Kryptonite. Express this as a matrix game and find a von Neumann solution. Hint: Find a strategy that equates the opponents' results to zero and plays Rock, Paper, and Scissors all with the same probability.

(9) The game *Weighted Rock-Paper-Scissors* is similar to Rock-Paper-Scissors but with different payoffs. We may model it using the zero-

sum matrix game $\begin{bmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{bmatrix}$ where $a, b, c > 0$. Find a von

Neumann solution to this game by finding strategies that equate the opponents' results to zero.

(10) In the proof of Lemma 6.3, it was shown that when Rose has no dominant strategy, Colin has a mixed strategy which equates her results. Fill in the missing argument from this lemma by showing that when Colin has no dominant strategy, Rose has a mixed strategy which equates his results.

(11) Let A be an $n \times n$ matrix with $A^T = -A$.

- (a) Prove that A has value 0.
- (b) Prove that A has determinant 0 if n is odd.

(12) Let A be a zero-sum matrix game, let $\mathbf{p}$ be a mixed strategy for Rose which equates Colin's results to w , and let $\mathbf{q}$ be a mixed strategy

for Colin which equates Rose's results to u . That is, we assume

$$\mathbf{p}A = [u \quad \cdots \quad u] \text{ and } A\mathbf{q} = \begin{bmatrix} w \\ \vdots \\ w \end{bmatrix}. \text{ Prove that } u = w \text{ and note}$$

that u together with $\mathbf{p}$ and $\mathbf{q}$ forms a von Neumann solution.

- (13) Let A be an $m \times n$ matrix with the property that whenever the entries of a row are summed, the result is always equal to s , and similarly, whenever the entries of a column are summed, the result is always equal to t . Find a von Neumann solution for A .
- (14) Construct a zero-sum 2×2 matrix game with value 1 for which Rose has a unique mixed strategy with a guarantee of 1, but Colin has infinitely many mixed strategies with a guarantee of 1.
- (15) Let $A = [\mathbf{a}_1 \quad \cdots \quad \mathbf{a}_n]$ be a zero-sum matrix game and suppose that there is a real number $0 \leq q \leq 1$ so that $\mathbf{a}_1 \geq q\mathbf{a}_2 + (1-q)\mathbf{a}_3$. If A' is the matrix obtained from A by deleting the first column, prove that A and A' have the same von Neumann value.
- (16) Prove the following for every $2 \times n$ zero-sum matrix game: Either there exists a von Neumann solution where Colin plays a pure strategy or Rose's guarantee function has a unique maximum attained at p^* where $0 < p^* < 1$.
- (17) A function f is *convex up* if for every $x_1 < x_2$ in the domain and every $0 \leq p \leq 1$ the following equation is satisfied:

$$f(px_1 + (1-p)x_2) \geq pf(x_1) + (1-p)f(x_2).$$

Prove that for every zero-sum $2 \times n$ matrix game, Rose's guarantee function G_R is concave down.

- (18) Let $\mathbf{a}_1, \dots, \mathbf{a}_k \in \mathbb{R}^m$ have convex hull A and let $\mathbf{b}_1, \dots, \mathbf{b}_\ell \in \mathbb{R}^n$ have convex hull B . (Note that all $\mathbf{a}_i, \mathbf{b}_j$ are column vectors.) If W is an $m \times n$ matrix, prove that there exists $\mathbf{a}^* \in A$ and $\mathbf{b}^* \in B$ with the following property: For every $\mathbf{a} \in A$ and $\mathbf{b} \in B$

$$\mathbf{a}^\top W \mathbf{b}^* \leq (\mathbf{a}^*)^\top W \mathbf{b}^* \leq (\mathbf{a}^*)^\top W \mathbf{b}.$$

- (19) Let $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$ with $k \geq 2$.
- (a) Prove that every point $\mathbf{z}$ in the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$ lies on a line segment of the form $\overline{\mathbf{y}\mathbf{x}_k}$ where $\mathbf{y}$ is in the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_{k-1}$.
 - (b) Show that every convex set S containing $\mathbf{x}_1, \dots, \mathbf{x}_k$ will also contain their convex hull. Therefore, the convex hull is the unique minimal convex set containing a given set.
- (20*) One key ingredient in our proof of the Minimax Theorem is a hyperplane separation property. For a particular set $S \subseteq \mathbb{R}^n$ and point $\mathbf{y} \notin S$ we showed that there is a hyperplane with $\mathbf{y}$ strictly on one side and S on the other. This holds more generally for any closed convex subset of $\mathbb{R}^n$. More formally:

Theorem 6.9. *Let $S \subseteq \mathbb{R}^n$ be closed and convex and let $\mathbf{y} \in \mathbb{R}^n$. If $\mathbf{y} \notin S$, there exists $\mathbf{z} \in \mathbb{R}^n$ and $c \in \mathbb{R}$ so that $\mathbf{y} \cdot \mathbf{z} > c$ and $\mathbf{z} \cdot \mathbf{x} \leq c$ for every $\mathbf{x} \in S$.*

For an arbitrary set $C \subseteq \mathbb{R}^n$ define the *polar* of C to be the subset of $\mathbb{R}^n$ given by $C^\circ = \{\mathbf{y} \in \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} \leq 1 \text{ for every } \mathbf{x} \in C\}$.

- (a) The hypercube $Q_n \subseteq \mathbb{R}^n$ is the convex hull of all points of the form $(\pm 1, \pm 1, \dots, \pm 1)$. Find a set of points X so that Q_n° is the convex hull of X .³
 - (b) Prove that every closed convex set $C \subseteq \mathbb{R}^n$ which contains a ball of nonzero radius around the origin satisfies $(C^\circ)^\circ = C$.
- (21*) This exercise gives another proof of the Minimax Theorem. Suppose (for a contradiction) that Theorem 5.11 is false, and choose a counterexample $m \times n$ matrix A for which $m + n$ is minimum. Let r be Rose's highest guarantee and let c be Colin's lowest guarantee for the matrix game A .⁴ Observe that since A is a counterexample, we must have $r < c$. Next choose a mixed strategy $\mathbf{p}$ for Rose with a guarantee of r and a mixed strategy $\mathbf{q}$ for Colin with a guarantee of c . By Exercise (12), at least one of $\mathbf{p}A$ or $A\mathbf{q}$ is not constant, and we assume the former without (significant) loss of generality. By possibly rearranging the order of the columns of A , we may assume that the first ℓ entries of $\mathbf{p}A$ are equal to r and the others are all strictly larger. Define the matrix consisting of the first ℓ columns of A to

³The polytope Q_n° is known as the *cross-polytope*.

⁴The existence of r and c follows from the compactness of the set of mixed strategies.

be A' . Now A' is an $m \times \ell$ matrix with $\ell < n$, so by the minimality of our counterexample, Theorem 5.11 is true for A' . Therefore we may choose a von Neumann solution to A' consisting of a value v' together with strategies $\mathbf{p}'$ and $\mathbf{q}'$.

- (a) Prove that $c \leq v'$.
- (b) Combine $\mathbf{p}$ and $\mathbf{p}'$ to form a mixed strategy for Rose in A with a higher guarantee than r , thus giving a contradiction.

Chapter 7

General Games

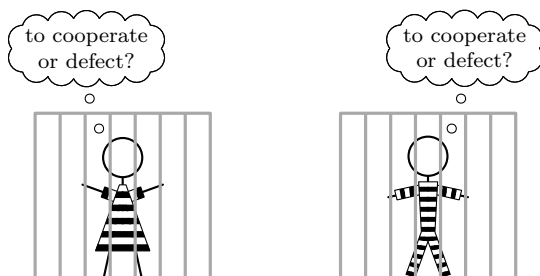


Figure 7.1. The Prisoner's Dilemma

This chapter again dramatically expands our horizons to include more general strategic situations. In all the games so far, two players competed directly, so anything gained by one was lost by the other. Moving forward, games may involve outcomes that are simultaneously good for both players or bad for both players or some combination of the two. These general games are far too rich to hope for some kind of universal theorem that explains optimal play in every situation, but the tools of game theory can nonetheless model and analyze nearly any scenario in which the actions of one player affect the results for another. Precise mathematical representations can illuminate the essence of a strategic situation and, in some cases, strongly indicate best play. Other cases

defy such conclusions, yet the formalism still helps to clarify the situation at hand. One great benefit of studying game theory is developing the ability to abstract key strategic ingredients from a wide variety of interactions.

To represent outcomes that are good for both players or bad for both players requires a richer payoff structure. To allow for this, we now associate each outcome with a pair of numbers (a, b) where a indicates the payoff to Rose and b indicates the payoff to Colin. We assume that Rose and Colin wish to maximize their payoffs (each with no concern for the payoff of the other). In the special case where each outcome is associated with a pair of the form $(a, -a)$, we have a zero-sum game. This more general payoff structure allows us to introduce the concept of a (general) matrix game.

Definition 7.1. A *matrix game* is a game played between Rose and Colin using a fixed matrix A , known to both players, where each entry of A consists of an ordered pair (a, b) where a indicates the payoff to Rose and b the payoff to Colin. To play, Rose secretly chooses a row, Colin secretly chooses a column, and then they reveal their answers. If this row and column selects the entry (a, b) from the matrix A , then Rose gets a *payoff* of a and Colin gets a *payoff* of b .

Let's get things started with a classic.

Game 7.2 (The Prisoner's Dilemma). Rose and Colin have been caught robbing a bank, but the police don't have all the necessary evidence to charge them with the maximum penalty. The police isolate the players and offer each the option to give evidence to convict the other and, in return, receive less jail time. So each player can either cooperate with the other player (C) and stay silent or may defect (D) by turning over evidence. In the matrix of payoffs below $-n$ indicates a prison sentence of n years:

		Colin	
		C	D
Rose	C	$-1, -1$	$-10, 0$
	D	$0, -10$	$-5, -5$

From Rose's perspective, if Colin cooperates, she gets a shorter prison term by defecting, and if Colin defects, she also gets a shorter prison term

by defecting. In other words, no matter what Colin does, Rose does better by defecting. A similar argument shows that Colin should likewise choose defect. Still, if both players defect, each will get a -5 payoff, which is *worse for both players* than the -1 payoffs they could achieve by cooperating. This presents a significant, fundamental, and pervasive dilemma: Rational individual choice leads to inferior outcomes.

7.1. Utility

In the Prisoner’s Dilemma, we assumed that each player wished to minimize his or her own prison sentence with no regard for the other player. This might be a reasonable assumption in some situations but could also be entirely wrong in others. What if Rose and Colin are in love? They might then value years apart in prison very differently than if they are enemies. The next section develops the vital notion of utility, which captures such meaningful variations.

Money vs. Utility. Imagine a scenario where one player, say Rose, considers buying health insurance. Assume for simplicity that the cost of this insurance is \$1,000 per year. With 99% probability, Rose will stay healthy and not need the insurance. With 1% probability, she will have a significant health problem. In that case, if she’s insured, she will pay nothing extra, but if she is uninsured Rose will have to pay \$80,000 on average. The matrix below represents this 1-player game:¹

		Chance	
		99/100 healthy	1/100 problem
Rose	buy ins.	−\$1,000	−\$1,000
	dont buy	\$0	−\$80,000

If Rose buys insurance, her expected payoff will be $-\$1,000$, but if she does not buy insurance, it will be $-\$800$. So not purchasing insurance gives Rose a better expected monetary return than purchasing insurance. Though this may seem surprising at first, the perspective of an insurance provider may clarify why such a payoff structure is likely: To be a viable business, the insurance company must set prices so, on average, income exceeds expenditure for each client.

¹This type of game is known as a *game against the universe*.

This calculation suggests that Rose should *not* buy insurance. But how does Rose truly value the different possible outcomes in this scenario? The matrix shows Rose's payoff of $-\$1,000$ when she buys insurance regardless of whether or not she has a significant health problem. It's unlikely this number reflects how she really values these two very different outcomes. Certainly Rose would much prefer to stay healthy than to have a serious health issue (even with insurance). And if she does fall ill, Rose would be especially relieved to have insurance. Ideally, we could associate with each outcome a number that accurately represents how Rose values the different events. Such numbers, called *utilities*, could further clarify Rose's decision to buy or not to buy insurance. We postpone for now the question of how to determine utilities and focus here on how these values impact the decision. Suppose the following matrix gives Rose's utility for each of the four outcomes:

		Chance	
		$^{99/100}$ healthy	$^{1/100}$ problem
Rose	buy ins.	$-1,000$	$-51,000$
	dont buy	0	$-300,000$

When Rose stays healthy, her utility is 0 when she does not buy insurance and $-1,000$ when she does buy insurance, just as before. If Rose becomes seriously ill, then her utility will be $-51,000$ if she is insured and $-300,000$ if she is not insured. In this latter unfortunate situation, medical bills could financially ruin Rose.

What are Rose's expected payoffs in this new matrix of utilities? If she buys insurance, her expected utility is

$$(^{99/100})(-1,000) + (^{1/100})(-51,000) = -1,500.$$

On the other hand, if Rose doesn't buy insurance, her expected utility is

$$(^{99/100})(0) + (^{1/100})(-300,000) = -3,000.$$

So, in the end, using Rose's utility values alters the game. Purchasing insurance gives Rose a higher expected utility, despite a lower expected monetary payoff.

Even in circumstances where only money is at stake, it is quite possible for rational players to take actions that do not maximize expected monetary payoff. Let's consider a more pleasant situation for Rose, who

will now play a 1-player game starting with a hidden fair coin flip. Rose has the option of either receiving a sure \$1,000,000 or gambling on the result of the coin toss. If Rose gambles and the coin toss is heads, then she gets \$2,200,000, but if the coin came up tails, she gets nothing. The following 1-player game represents the situation:

		Chance	
		1/2 heads	1/2 tails
Rose	sure	\$1,000,000	\$1,000,000
	gamble	\$2,200,000	\$0

Now compute Rose’s expected payoffs for each option. If Rose chooses the sure money, then she gets \$1,000,000. If instead she gambles, her expected payoff is $(1/2)(2,200,000) + (1/2)(0) = \$1,100,000$, which is \$100,000 higher.

The expected dollar reward is greater when Rose gambles, yet many individuals in her position strongly prefer the sure money. A key issue here is the decreasing marginal utility of wealth. In short, a person will value her first million dollars more than her second million dollars. Another factor is Rose’s risk tolerance for gambling with such large amounts of money. Rose might value the security of sure money more than the chance of a higher expected reward. Alternately, Rose might be neutral or even favorable towards risk. The game below accounts for all of these factors and assigns Rose utilities for the various outcomes:

		Chance	
		1/2 heads	1/2 tails
Rose	sure	1,000,000	1,000,000
	gamble	1,600,000	0

If this is accurate, then Rose’s expected utility for gambling is 800,000, less than the 1,000,000 she gets by taking the sure payoff. Once again, we see that Rose will choose differently to maximize her expected utility instead of her expected monetary payoff.

In both examples here, Rose confronted a choice between two options. To get a more meaningful picture of these decision problems, we associated each of Rose’s possible outcomes with her utility—a number representing her value for each outcome relative to the others. Introducing utility radically transformed the nature of the game and generated a

different choice from Rose in both cases. All of this reasoning hinges on the assumption that Rose can meaningfully assign utilities to the various outcomes ... but is this really possible?

Von Neumann and Morgenstern's Lottery. Among the many important contributions to game theory due to von Neumann and Morgenstern is a robust theory of utility. They proved that under certain natural assumptions, rational players can associate outcomes with utilities in such a way that the rational action in any decision problem is to make the choice that maximizes expected utility. Moreover, they explained how to determine these utilities using the idea of a lottery.

To introduce their idea, suppose Rose has two possible outcomes, X and Z , and that she values X with utility 0 and Z with utility 10. Given a lottery resulting in outcome X with probability $\frac{1}{2}$ and outcome Z with probability $\frac{1}{2}$, then Rose's expected utility from this lottery would be $\frac{1}{2}0 + \frac{1}{2}10 = 5$. More generally, in a lottery where Rose gets X with probability p and Z with probability $1 - p$, her expected utility is $(p)0 + (1 - p)10 = 10 - 10p$.

Now introduce a third outcome Y and suppose that Rose ranks the three outcomes $X < Y < Z$, so she prefers Z to Y and she prefers Y to X . Von Neumann and Morgenstern suggested that Rose's utility for Y could be determined by comparing Y to lotteries involving outcome X with probability p and Z with probability $1 - p$ for various values of p . If $p = 1$, then the lottery gives Rose an outcome of X every time—she would certainly prefer Y to that lottery. At the other extreme, when $p = 0$, the lottery gives Rose an outcome of Z every time, so she prefers this lottery to Y . For some number $0 < p < 1$, Rose views the lottery as equivalent to Y . From above, Rose's expected utility in this lottery is $10 - 10p$, so this is the utility assigned to Y . More generally, if there are many outcomes $Y, Y', Y'', \dots$ that Rose prefers over X but less than Z (i.e. $X < Y, Y', Y'', \dots < Z$), then this procedure assigns to each one a utility.

In this process, X need not have utility 0 nor Z utility 10. The same operations work with any utility x for X and any utility z for Z as long as $x < z$. The lottery says how to determine the utilities of all of the outcomes given utilities for X and Z , but how will we find the utilities associated with the least and most favorable outcomes? The answer is

that the utility x for the least favorable outcome X and the utility z for the most favorable outcome Z may be arbitrarily chosen, subject to the constraint $x < z$. In other words, utility is not an absolute measure, but rather a kind of comparative scale similar to temperature metrics. The Celsius scale, for example, declares the freezing point of water to be 0 and the boiling point for water to be 100 and then extends linearly. The Fahrenheit scale sets the freezing point of water at 32 and the boiling point 212. We could likewise build another perfectly reasonable temperature scale by choosing x to be the freezing point and z to be the boiling point. Endpoints on a utility scale work similarly.

Von Neumann and Morgenstern's lottery provides a method to assign utilities that measure how a player compares any collection of outcomes. We can use these utilities instead of basing our analysis on years in prison or dollar reward. This presents a much more nuanced view of players' motivations and facilitates more careful analysis of rational play. In the Prisoner's Dilemma, for example, having players assign a utility to each of the four possible outcomes would create a 2×2 matrix game more meaningfully representing the essence of the situation. Forthcoming chapters feature strategic situations and represent outcomes as payoffs for the players. In these games, we will always assume that the given payoffs accurately reflect each player's utility for a corresponding outcome.

7.2. Matrix Games

This section develops the basic theory of matrix games, a generally applicable model for representing decision problems. We first introduce a handful of dilemmas that help demonstrate the variety of matrix games beyond the Prisoner's Dilemma. Concepts of dominance and equilibria will then help us better understand the play of these games in some cases.

More Dilemmas. To get an idea of what types of decision problems can be represented with matrix games, we will introduce a number of meaningful examples. There is no complete theory advising players how to play an arbitrary matrix game, so we will not be presenting a "solution"

to most of these games here or ever. Our goal is simply to demonstrate the power of the model.

Game 7.3 (Coordination Game). Rose and Colin are test subjects in a psychology experiment. They have been separated, and each player gets to guess either *X* or *Y*. Both players get \$1 if their guesses match and nothing if they do not:

		Colin	
		X	Y
Rose	X	1, 1	0, 0
	Y	0, 0	1, 1

In a game such as this one, *communication* between the players would result in an advantageous outcome. If the players knew the game and were permitted to communicate prior to play, it would be easy for them to agree to make the same choice.

This situation is in contrast to the Prisoner's Dilemma. In that game defect always yields a better payoff than cooperate—a reality unaffected by the existence of communication. Indeed, were advance communication part of the Prisoner's Dilemma, each player would have incentive to try to convince the other one to cooperate, but then to defect in the actual game.

Game 7.4 (Dating Dilemma). Suppose Rose and Colin have started dating and they are on the phone deciding what to do this evening. Rose wants to go to the ball game while Colin prefers to go to the film. Unfortunately, Colin's phone battery dies mid-conversation, so further communication is impossible! Each player must individually decide to go to the Ball game (B) or to the Film (F). The matrix of payoffs is below:

		Colin	
		B	F
Rose	B	2, 1	0, 0
	F	0, 0	1, 2

The players prefer to spend the evening together, so payoffs where the players are in separate places are worst possible for both players. The tricky part of this dilemma is that Rose would prefer to end up with Colin at the Ball game, whereas Colin would rather be with Rose at the Film.

Coordination could help here, but there is still a conflict between the players.

In some situations it may be possible for one of the players to seize the initiative and move first (breaking the usual simultaneity of matrix games). Suppose that Rose committed to going to the ball game and Colin knew of this decision. Then his best move is to attend the ball game, too, giving Rose her favorite outcome. This feature appears in numerous games and, particularly in economics, is called first mover advantage.

Finally, note that although this situation is artificial, scenarios with this type of payoff structure appear commonly in interpersonal dynamics. In a circumstance where the same players might repeatedly encounter this sort of game, the players might arrange to alternate between the two activities. This game will reappear later in our consideration of cooperation.

Game 7.5 (Volunteering Dilemma). Rose and Colin have been a mostly happy couple for some time now, but they disagree about who will do the dishes. Each has the option of either volunteering (V) to do the dishes or staying silent (S). Here is the matrix of payoffs:

		Colin	
		S	V
Rose	S	-10, -10	-2, -2
	V	-2, -2	-1, -1

If neither player volunteers to do the dishes, the payoff is quite bad for both (perhaps they cannot eat dinner). Each player would most like to stay silent and have the other volunteer. Both, though, certainly want to avoid the bad SS outcome. This dynamic is so common that the aboriginal people of Tierra del Fuego have a word for it. *Mamihlapinatapai* (noun): The situation when two people are staring at one another, each hoping the other will volunteer to do something that both want done but neither wants to do.

Game 7.6 (Stag Hunt). Rose and Colin remain a little annoyed with one another over the dishes; each feels a bit uncertain of the trust between them. Now they are headed off to the woods on a hunting trip. Each

player has two strategies—work together and hunt for a stag (S) or go for a rabbit alone (R). Here is the matrix of payoffs:

		Colin	
		S	R
Rose	S	3, 3	0, 2
	R	2, 0	1, 1

Obviously, both players do best here if they cooperate and hunt the Stag. Really the only sticky point is that if one player suspects the other may go for a rabbit, then that player has incentive to choose R, too. Communication is likely to help here, as long as the players trust each other enough to cooperate.

Strategies and Dominance. In the setting of zero-sum matrix games, the concept of dominance sometimes allowed us to determine rational play. This notion naturally extends to (general) matrix games and can be a useful tool for analyzing them, too. In fact, dominance already appeared in the Prisoner's Dilemma—it was the key tool guiding our investigation.

As in zero-sum matrix games, for a matrix game A define a *pure strategy* for Rose to be a choice of row and a *pure strategy* for Colin to be a choice of column. If there are two rows, i and i' , with the property that no matter what column Colin chooses, Rose's payoff from i is greater than or equal to her payoff from i' , then row i *dominates* row i' . Also as before, if, no matter what column Colin chooses, Rose's payoff will be strictly greater when choosing row i than when choosing row i' , then row i *strictly dominates* row i' . We define domination and strict domination analogously for Colin's column strategies.

A rational player maximizing payoff would never play a strictly dominated strategy. In the Prisoner's Dilemma, for instance, the strategy (D) of defecting strictly dominates the strategy (C) of cooperating. Eliminating the dominated strategies leaves (D) as the only rational move. Let's see another example where iteratively deleting dominated strategies is the key to determining rational play.

Game 7.7 (Competing Pubs). Consider two pubs in a tourist town competing for business. On a given night, 200 locals and 200 tourists head to these pubs. For simplicity, assume that each person will order exactly

one drink. Each pub prices drinks at either \$5, \$6, or \$7. The tourists are not discerning, and $\frac{1}{2}$ of them will go to each pub. The locals, on the other hand, know the drink prices, so if one pub is cheaper than the other, that pub will get all of the locals. If prices are equal at both pubs, then the locals will also split $\frac{1}{2}$ and $\frac{1}{2}$. Assuming each pub wants to maximize revenue results in the following matrix game:

		Pub 2			
		\$5		\$6	
Pub 1	\$5	1,000, 1,000	1,500, 600	1,500, 700	
	\$6	600, 1,500	1,200, 1,200	1,800, 700	
	\$7	700, 1,500	700, 1,800	1,400, 1,400	

We see that the \$5 strategy strictly dominates \$7 for both players. Eliminating this row and column leaves a 2×2 matrix where strategy \$5 dominates \$6 for both players. After this dominated row and column are eliminated, the rational solution remains: Both pubs charge \$5 and get a 1,000 payoff. Note that both pubs could get a payoff of \$1,400 if they both set prices at \$7. So, just as in the Prisoner's Dilemma, rational play by the individual pubs results in an inferior outcome.

The iterated removal of strictly dominated strategies led to the above conclusion. Since it is clearly irrational to play a strictly dominated strategy, this approach is a powerful tool in analyzing rational play. What about eliminating strategies which are dominated (but not strictly dominated)? While this type of reduction does help find certain equilibria (as we'll see later), it can lead to some surprising results in non-zero-sum games (Exercise (7)).

Best Responses and Pure Equilibria. In our study of zero-sum matrix games, we imagined fixing a strategy for one player and considering the other player's optimal response to that. Similar methodology contributes to our investigation of non-zero-sum matrix games, and next we introduce some terminology to facilitate this.

Fix a matrix game A and assume that Colin chooses the pure strategy of column j . A *best pure response* to column j is a choice of row for Rose that maximizes her payoff under the assumption that Colin plays column j . Similarly, if i is a row, then a *best pure response* to row i is a

choice of column for Colin that maximizes his payoff under the assumption that Rose plays row i .

In the special case of 2×2 matrices, a graphical technique called a *movement diagram* helps to visualize best responses (an alternative method which applies to larger matrices appears in Exercise (6)). We construct this diagram as follows:

- For each column, draw an arrow from the outcome Rose likes least to the one she likes best (in case of a tie, use a double-headed arrow).
- For each row, draw an arrow from the outcome Colin likes least to the one he likes best (in case of a tie, use a double-headed arrow).

Figure 7.2 shows a simple example.

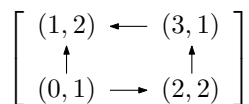


Figure 7.2. A movement diagram

In the Figure 7.2 movement diagram, the upper left cell has both vertical and horizontal arrows pointing toward it. This means that row 1 is a best response to column 1 and that column 1 is a best response to row 1. This implies a certain stability for this pair of strategies. Namely, if each player thinks the other is going to play 1, then each should play 1. Take note that this stability does *not* in any way mean that these strategies are somehow optimal—only that each is a best response to the other. This significant type of stability merits a formal definition: A *pure Nash equilibrium* is a pair of pure strategies, one for Rose and one for Colin, each of which is a best responses to the other.

For identifying pure Nash equilibria and visualizing dynamics of play, movement diagrams can be quite helpful. Figure 7.3 depicts these diagrams for our four dilemmas.

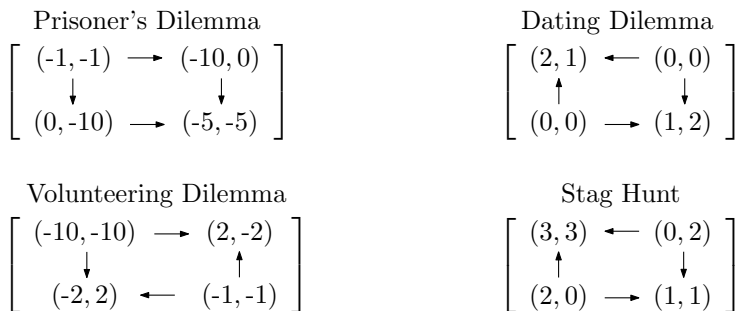


Figure 7.3. Movement diagrams for our dilemmas

7.3. Game Trees

We depicted the play of combinatorial games with game trees in Chapter 1. Game trees, in fact, lend themselves well to model more general multistage strategic interactions. Here, we will add three new features—general payoffs, chance nodes, and information sets—to enhance the power of game trees to model sequential interactions.

General Payoffs. Recall that in Chapter 1 we used W-L-D game trees since the only possible outcomes for each player were a win, a loss, or a draw. We marked each terminal node $+-$, $-+$, or 00 to indicate the winner or that the outcome was a draw. We could view these outcomes as $(1, -1)$, $(-1, 1)$, and $(0, 0)$ and easily extend this to mark each terminal node in a game tree with an ordered pair (a, b) to indicate a payoff of a to Rose and b to Colin. See Figure 7.4 for an example.

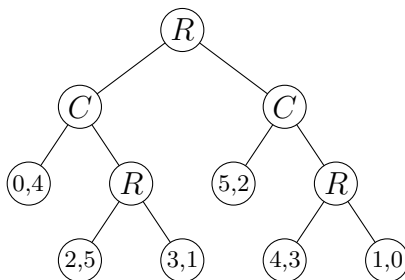


Figure 7.4. A game tree with general payoffs

A *strategy* for a player is as before—a set of decisions indicating which move to make at each node where that player has a decision to make. As with W-L-D game trees, we can work backwards up a general game tree to try to determine how rational players will play. This process appears in Figure 7.5, where each nonterminal node is marked with the payoffs the players will receive if they play rationally from that point down.

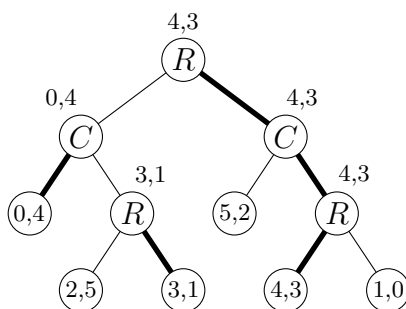


Figure 7.5. Rational play

This process works well unless there are payoffs (a, b) and (a', b') that agree in exactly one coordinate and can break down the working backwards procedure (see Exercise (9)).

Randomness. Next we will enhance our game trees by incorporating randomness of the type appearing when players roll dice, toss a coin, or call on another source of randomness to make a decision. To enable game trees to model such situations, we introduce a new type of nonterminal node called a *chance* node (indicated in Figure 7.6 by *Ch*). Suppose a game reaches a chance node N and the possible next nodes from N are $N_1, N_2, \dots, N_k$. Then the game will randomly move from N to one of the nodes $N_1, \dots, N_k$ according to given probabilities. More precisely, each N_i comes with a probability p_i (used to label the edge from N to N_i). If node N is reached, the game will then move to node N_i with probability p_i . Naturally, the numbers $p_1, \dots, p_k$ will all be nonnegative and will sum to 1. Just as the game tree is known to both players, these probabilities are also common information.

Game 7.8 (Coin Toss). In this game, Rose calls either “heads” or “tails” and then a fair coin is tossed. If Rose correctly predicts the coin toss, the outcome is $(1, -1)$ and otherwise it is $(-1, 1)$. Figure 7.6 depicts a game tree for this game. Here the node marked R is one where Rose has a choice; the nodes marked Ch are chance nodes.

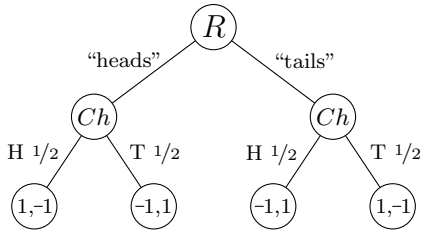


Figure 7.6. A game tree for Coin Toss

Incomplete Information. Finally, we would like to use game trees in scenarios in which the players do not have full information about the state of the game. Consider a Blackjack position, in which each player starts with one card face up and another face down. A Blackjack position in a game between Rose and Colin might appear as in Figure 7.7 to an outside spectator.

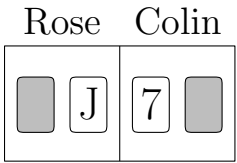


Figure 7.7. Spectator’s view

Rose may peek at her face down card, so she knows both of her cards and one of Colin’s. Colin knows both of his cards, but only one of Rose’s. When we refer to the position of a game, we will always assume that a position includes *all* information, even that which may be hidden from the players. For instance, the position corresponding to the spectator’s view from Figure 7.7 might be the one depicted in Figure 7.8.

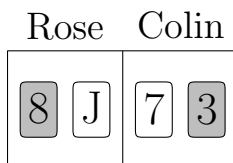


Figure 7.8. The position

Now consider the situation from Rose's perspective. What does she know about the position? Since Rose can see both face up cards and can also peek to see her face down card, she can distinguish between many positions. But since she can't see Colin's face down card, there are also many positions between which she cannot distinguish. Figure 7.9 shows a number of possible Blackjack positions. Positions indistinguishable to Rose are connected with a dotted line.

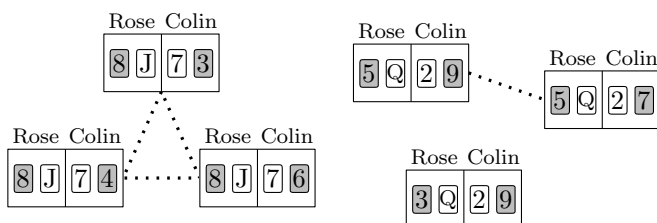


Figure 7.9. Some positions from Rose's perspective

In general, the position of a game always includes all relevant information about the state of the game. Each player may or may not be able to distinguish between different positions of the game. A set consisting of all of the positions that appear the same to Rose is called an *information set* for Rose.² In the Blackjack example, one information set for Rose consists of all positions in which she has a face down 8 and a face up J and Colin has a face up 7. Information sets for Colin are defined analogously.

Now we will add dotted lines to a game tree to represent an information set. If two nodes in a game tree are labeled for the same player,

²This standard terminology is admittedly somewhat confusing since an information set indicates a lack of information.

who cannot distinguish between those two positions in the game, indicate this by a dotted line connecting these two nodes of the tree. Note that in this case, the available moves from these two nodes should look the same to the player choosing between them. Next we will see how to use information sets to model a game with coins which plays a little like Poker or Blackjack.

Game 7.9 (Coin Poker). At the start of play, Rose and Colin each put one chip in the pot as ante and each player tosses a coin. Rose sees the result of her toss, but not Colin's, and vice versa. It is then Rose's turn to play, and she may either fold, ending the game and giving Colin the pot, or bet and place 2 more chips in the pot. If Rose bets, then it is Colin's turn to play and he may either fold, giving Rose the pot, or he may call and place 2 chips in the pot. In this latter case, both coin tosses are revealed. If both players have the same coin toss, the pot is split between them. Otherwise, the player who tossed heads wins the entire pot.

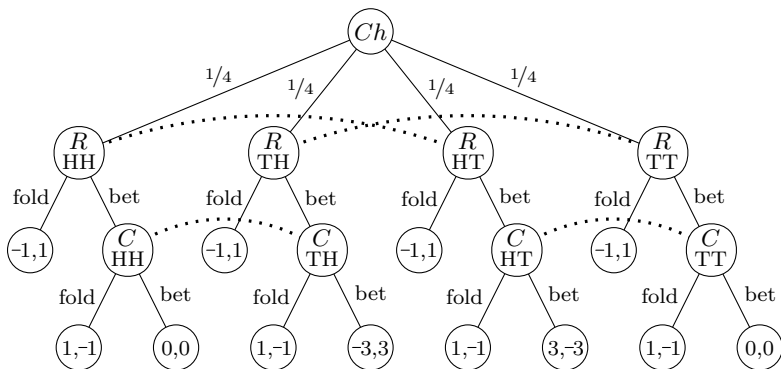


Figure 7.10. A game tree for Coin Poker

In this example, Rose has two information sets. The first consists of the two nodes where she flipped heads and the second contains the two nodes where she flipped tails, so call these information sets H^* and T^* . Colin similarly has two information sets, one where he flipped heads and one where he flipped tails, so call these $*H$ and $*T$.

What about strategy in this game tree? If Rose has T and Colin has T , Rose would probably like to bet, but if Rose has T and Colin has H , she

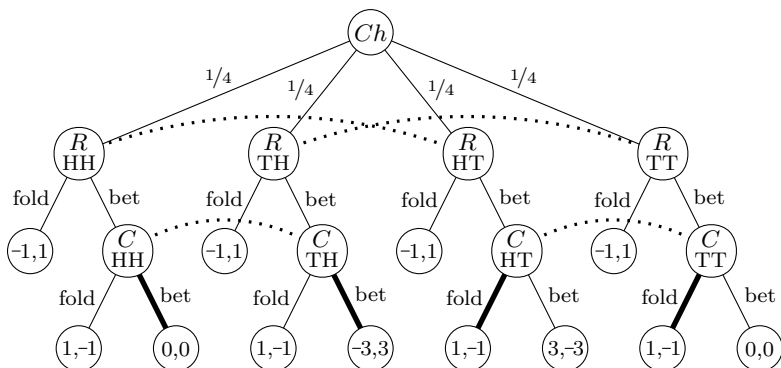


Figure 7.11. Colin's strategy H

would probably prefer to fold. Rose unfortunately cannot distinguish between these two positions. Rose's strategy must determine her move using only the information she has. In other words, a strategy for Rose must make the same choice in both of the above circumstances. Expanding our earlier definition, we now define a *strategy* for a player in a game tree to be a choice of action at each information set where that player has a decision. In our example, Rose has information sets H^* and T^* corresponding to the result of her coin toss. In each information set, she can choose to either bet or fold, so Rose has a total of four strategies. She can Always bet (A), Never bet (N), bet only with Heads (H) or bet only with Tails (T). Although Colin's position in this game is quite different, he also has the choice to bet or fold in each of his information sets, so we will also use A, N, H, and T to indicate his strategies.

Adding randomness and incomplete information markedly expands game trees' capacity to model a wide variety of situations with sequential decisions. Unfortunately, these same improvements mean the recursive working backwards procedure in general no longer applies to determine rational play.

7.4. Trees vs. Matrices

To tap the modeling power of game trees, we need a new method to study them in general. In fact, every game tree can be represented by a matrix

game, and conversely any matrix game can be represented by a game tree. So these two different types of games are really just two different ways to visualize the same game. The matrix representation is generally known as the normal or strategic form (since it crystallizes the strategies), while the tree representation is called the extensive form (since it is based on the extended sequence of possible decisions).

Trees to Matrices. Let's first convert the Coin Poker game tree into a matrix.

As we saw, in the game Coin Poker, Rose and Colin each have four strategies, denoted A, N, H, and T. Suppose that Rose adopts the strategy of always betting (A) and Colin adopts the strategy of betting only with heads (H). Then the outcome depends only on the chance node and we will have one of the following possibilities:

probability	coins (RC)	payoffs
$1/4$	TT	$(1, -1)$
$1/4$	TH	$(-3, 3)$
$1/4$	HT	$(1, -1)$
$1/4$	HH	$(0, 0)$

In this case, the expected payoff for Rose is given by $1/4(1) + 1/4(-3) + 1/4(1) + 1/4(0) = -1/4$ and the expected payoff for Colin is $1/4(-1) + 1/4(3) + 1/4(-1) + 1/4(0) = 1/4$. More compactly, we can compute the expected payoffs for both players simultaneously as

$$1/4(1, -1) + 1/4(-3, 3) + 1/4(1, -1) + 1/4(0, 0) = (-1/4, 1/4).$$

Similarly, we can compute for any possible choice of strategies for Rose and Colin all of the expected payoffs, which always have the form $(a, -a)$. We can now take a rather sophisticated *strategic* view of this game. Instead of having Rose and Colin sequentially play the game, imagine they each simply choose a strategy ahead of time. We know the expected payoff for every possible strategy pair, so we can represent the game tree as a matrix game. The Coin Poker game tree results in the

following zero-sum matrix game:

		Colin			
		A	T	H	N
Rose	A	0	$\frac{5}{4}$	$-\frac{1}{4}$	1
	T	$-\frac{5}{4}$	$-\frac{1}{4}$	-1	0
	H	$\frac{1}{4}$	$\frac{1}{2}$	$-\frac{1}{4}$	0
	N	-1	-1	-1	-1

A quick analysis of this matrix reveals strategy A strictly dominates T for Rose. Without the row corresponding to T, Colin's strategy H dominates all of his other strategies. This leaves a one-column game in which Rose has two dominant strategies, H and A. Thus, we have found two saddle points to this game consisting of the pure strategy H for Colin and either the pure strategy H or A for Rose. We conclude that the value $-\frac{1}{4}$ and the pure strategies associated with either saddle point together form a von Neumann solution.

This process generalizes naturally to an arbitrary game tree.

Procedure 7.10 (Tree-to-Matrix). Consider an arbitrary game tree with strategies $S_1, S_2, \dots, S_m$ for Rose and $T_1, T_2, \dots, T_n$ for Colin. For each choice of a strategy S_i for Rose and a strategy T_j for Colin, we know how every nonchance node in the game tree will act. Therefore, each possible outcome will occur with some probability, and we can compute the expected payoffs for this pair of strategies S_i and T_j . Define a matrix A by setting the (i, j) entry to be the pair of numbers (a, b) where a and b are the expected payoffs for Rose and Colin playing strategy S_i opposite T_j . The matrix A is called the *strategic form* of the game.

Let's do one more example of this tree-to-matrix process with a game that is not zero sum.

Example 7.11. In Figure 7.12, Rose and Colin each have just a single information set, and in each of these information sets each player has just two options. Say Rose has strategies W and X and Colin has strategies Y and Z . Let's compute the expected payoffs when Rose plays W and Colin plays Y . With probability $\frac{1}{3}$, the initial chance node chooses left. Since Rose chooses W , a second chance node gives expected payoffs of

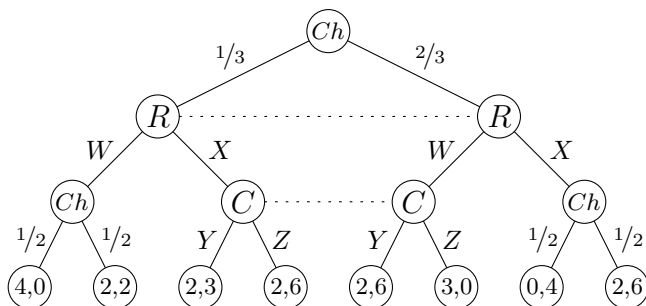


Figure 7.12. A game tree

$\frac{1}{2}(4, 0) + \frac{1}{2}(2, 2) = (3, 1)$. With probability $\frac{2}{3}$, the initial chance node chooses right. If Rose plays W and Colin Y , then the game ends at a terminal node with payoffs $(2, 6)$. Altogether, we find that expected payoffs when Rose plays W and Colin plays Y are

$$\frac{1}{3}\left(\frac{1}{2}(4, 0) + \frac{1}{2}(2, 2)\right) + \frac{2}{3}(2, 6) = (7/3, 13/3).$$

We may likewise compute the expected payoffs for all possible choices of strategies. Combining these gives us the following strategic form matrix:

		Colin	
		Y	Z
Rose	W	$\left[\frac{7}{3}, \frac{13}{3}\right]$	$(3, \frac{1}{3})$
	X	$\left[\frac{4}{3}, \frac{13}{3}\right]$	$(\frac{4}{3}, \frac{16}{3})$

In this matrix game, Rose's strategy of W strictly dominates X . Eliminating this dominated row leaves the matrix $\left[\left(\frac{7}{3}, \frac{13}{3}\right), (3, \frac{1}{3})\right]$ in which Colin has a dominant strategy of Y . So, assuming our players are rational, Rose will play W and Colin will play Y .

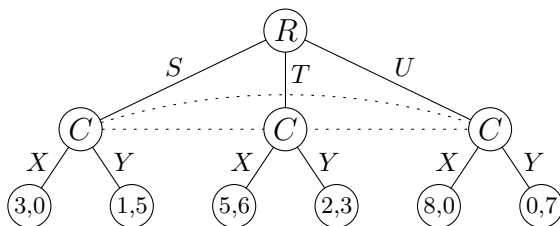
This matrix form suppresses all of the details concerning who can move where and when, and instead focuses on comparing how a strategy for Rose matches up with a strategy for Colin.

Matrix-to-Tree. Just as every game tree can be turned into a matrix game, so every matrix game can be turned into a game tree. To introduce our matrix-to-tree process, we will again consider an example.

Example 7.12. Consider the following 3×2 matrix game:

		Colin	
		X	Y
Rose	S	3, 0	1, 5
	T	5, 6	2, 3
	U	8, 0	0, 7

We usually imagine Rose and Colin revealing their choices simultaneously in a matrix game, whereas in a game tree their decisions are necessarily sequential. Nevertheless, it is straightforward to model a matrix game with a tree. We will have Rose make her decision and move first and have Colin make his decision and move second. Since Colin is not permitted to have knowledge of Rose's choice, we put all of his decision nodes in one large information set. For our example game, the corresponding game tree follows:



In this game tree, Rose makes the first choice of S , T , or U , and then Colin chooses either X or Y , with no knowledge of Rose's choice. This is just the same as the play of the original matrix game. The general process is a straightforward extension of this example.

Procedure 7.13 (Matrix-to-Tree). Let M be a matrix game with m rows and n columns. Construct a game tree where the root node offers Rose the choices $1, 2, \dots, m$ each corresponding to a row. Every node directly below the root node will be a decision node for Colin with the choices $1, 2, \dots, n$ each corresponding to a column, and we place all of these nodes in the same information set. Below each of these decision nodes for Colin is a terminal node, and the terminal node is given payoffs from the (i, j) entry of M if Rose selected row i and Colin selected row j . This tree is called the *extensive form* of M .

We have now demonstrated that the expressive power of matrix games is theoretically equivalent to that of game trees. Both variants capture all features one might hope for in modeling most decision problems between two players. Going forward we will focus primarily on matrix games since these are structurally simpler and more convenient to work with mathematically.

Exercises

- (1) Find all pure Nash equilibria in the dilemmas from Section 7.2
- (2) Draw the movement diagram, find all dominated strategies, and find all pure Nash equilibria for each matrix game below:

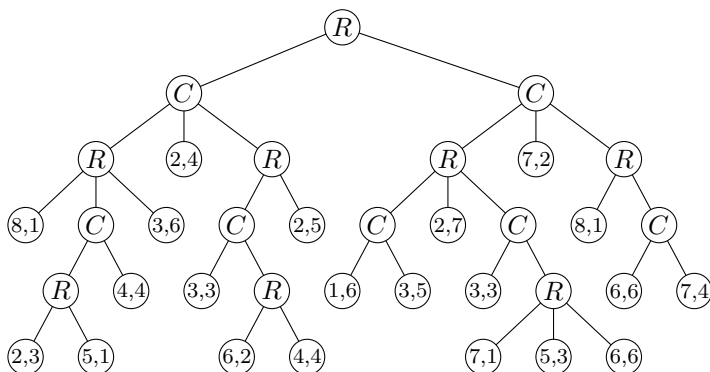
$$(a) \begin{bmatrix} (1, 1) & (2, 3) \\ (3, 2) & (1, 1) \end{bmatrix}, \quad (b) \begin{bmatrix} (2, 5) & (0, 4) \\ (4, 1) & (1, 3) \end{bmatrix}, \quad (c) \begin{bmatrix} (2, 0) & (1, 1) \\ (1, 1) & (2, 0) \end{bmatrix}.$$

- (3) If two rational players play the following matrix games, what will be the outcome?

$$(a) \begin{bmatrix} (0, 3) & (3, 5) & (6, 4) \\ (1, 1) & (1, 5) & (2, 0) \\ (5, 6) & (2, 1) & (3, 7) \end{bmatrix}, \quad (b) \begin{bmatrix} (3, 2) & (2, 5) & (-1, 1) \\ (2, 3) & (1, 0) & (5, 2) \\ (0, 0) & (3, 1) & (3, -1) \end{bmatrix}.$$

- (4) Consider the following outcomes: You get \$0, you get \$100, next Saturday is a beautiful day, you get a perfect score on your next homework assignment. Use von Neumann and Morgenstern's lottery method to assign personal utilities to these events.

- (5) If two rational players play the following game, what will be the outcome?



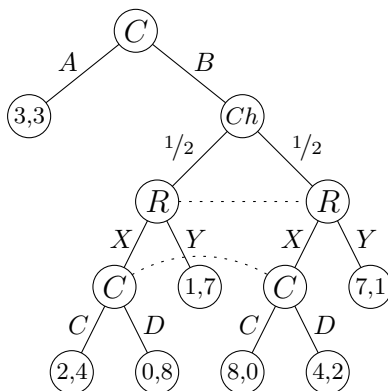
- (6) Here is a method for finding pure Nash equilibria in matrix games: For each row i , mark all entries for which Colin's payoff is maximum (i.e. Colin's best responses to i). Then, for each column j , mark all entries for which Rose's payoff is maximum (i.e. Rose's best responses to j). The entries receiving two marks correspond to the pure Nash equilibria. Find all pure Nash equilibria in each matrix:

$$(a) \begin{bmatrix} (4, 1) & (2, 2) & (10, 1) \\ (2, 2) & (3, 4) & (1, 0) \\ (5, 6) & (1, 3) & (0, 8) \end{bmatrix}, \quad (b) \begin{bmatrix} (5, 4) & (2, 0) & (3, 3) \\ (1, 3) & (3, 4) & (0, 5) \\ (0, 1) & (4, 5) & (2, 1) \\ (3, 2) & (1, 1) & (6, 3) \end{bmatrix}.$$

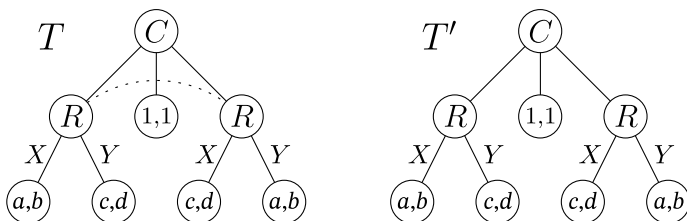
- (7) Note that the 1×2 matrix $\begin{bmatrix} (2, 1) & (3, 1) \end{bmatrix}$ can be reduced either to $\begin{bmatrix} (2, 1) \end{bmatrix}$ or to $\begin{bmatrix} (3, 1) \end{bmatrix}$ by eliminating dominated strategies (since each of Colin's pure strategies dominates the other). Use this idea to construct a 2×2 matrix which can be reduced by the method of eliminating dominated strategies to three different 1×1 matrices.
- (8) Use the Matrix-to-Tree Procedure (Procedure 7.10) to construct a game tree for the Dating Dilemma.
- (9) Explain why the strategy of working backwards up the tree to determine rational strategies does not work when there are payoffs of the form (a, b) and (a, b') with $b \neq b'$. Give an example to illustrate your point.

- (10) *Coin Jack* is a game played between Rose and Colin. First, both players put one chip in the pot as ante. Then Rose flips two coins with the first one revealed to both players and the second revealed only to her. Rose can either fold, giving the pot to Colin and ending the game, or she can bet by putting one more chip in the pot. If she bets, then Colin tosses a coin and keeps the result secret from Rose. Then Colin can choose to fold, giving the pot to Rose and ending the game, or he can bet by placing one more chip in the pot. If Colin bets, then all coin tosses are revealed. Colin wins the pot if he has at least as many heads with his one coin as Rose has in total, and otherwise Rose wins the pot.
- (a) Model this game with a game tree.
 - (b) How many strategies does Rose have?
 - (c) How many strategies does Colin have?
 - (d) Suppose that Rose bets when the revealed coin is heads and folds when it is tails, and suppose that Colin bets only when he has heads. What would be the expected payoffs?
- (11) *AKQ* is a game played between Rose and Colin using a 3-card deck consisting of an Ace, a King, and a Queen (ranked in this order). At the start each player puts one chip in the pot as ante. Then each player is dealt a card from the deck. Rose goes first and can choose to either fold or bet. If she folds, her ante goes to Colin and the game ends. If she bets, then she places two more chips in the pot and it is Colin's turn to play. He can either fold, giving his ante to Rose and ending the game, or he can bet and place two more chips in the pot. If he bets, then the cards are revealed and the player with the higher card wins the pot.
- (a) Model this game with a game tree.
 - (b) How many strategies does Rose have?
 - (c) How many strategies does Colin have?
 - (d) If Rose always bets and Colin bets with an Ace and folds with anything else, what would be the expected payoffs?
- (12) We say that a game tree is *zero-sum* if every terminal node has payoffs of the form $(a, -a)$. Let T be a game tree and let A be the strategic form. For each of the following statements, provide either a proof or a counterexample.

- (a) If T is zero-sum, then A is zero-sum.
 (b) If A is zero-sum, then T is zero-sum.
- (13) Find the strategic form of the following game tree:



- (14) Let M be a matrix game which can be reduced to its (i, j) entry by repeatedly deleting dominated strategies. Prove that the strategies i for Rose and j for Colin form a pure Nash equilibrium.
- (15) Consider the game trees T and T' in the figure below.
- (a) For the tree T , find a probabilistic strategy for Colin which guarantees him an expected payoff of $(b + d)/2$.
- (b) Find values for the variables a, b, c, d so that under rational play, Rose has a lower expected payoff in T' than in T (so the extra information she has in T' is bad for her!):



- (16) There are n outcomes which Rose ranks $X_1 < \dots < X_n$. We will apply the lottery method to assign utilities to $X_1, \dots, X_n$ twice (using different scales). Assume that for each $1 < i < n$, Rose considers X_i to be equivalent to a lottery where she gets X_1 with probability p_i and X_n with probability $(1 - p_i)$. For the first assignment of utilities, give X_1 the value x_1 , give X_n the value x_n (with $x_1 < x_n$), and use the

lottery method to assign every other X_i a number x_i . For the second assignment, give X_1 the value x'_1 , give X_n the value x'_n (with $x'_1 < x'_n$), and use the lottery method to assign every other X_i a number x'_i . Find a linear function $f(x) = cx + d$ so that $f(x_i) = x'_i$ for every $1 \leq i \leq n$.

- (17) Let A and B be matrix games of the same dimension. We say that A and B are *equivalent* and write $A \equiv B$ if there exist linear functions $f_1(x) = c_1x + d_1$ and $f_2(x) = c_2x + d_2$ with $c_1, c_2 > 0$ so that A can be transformed into B by applying f_1 to each entry in Rose's payoff matrix and applying f_2 to each entry in Colin's payoff matrix. Our theory of utility suggests that players should behave the same in equivalent matrix games.

(a) Show that $\equiv$ is an equivalence relation by proving the following:

- (i) $A \equiv A$.
- (ii) If $A \equiv B$, then $B \equiv A$.
- (iii) If $A \equiv B$ and $B \equiv C$, then $A \equiv C$.

(b) Show that the following matrix game is equivalent to a zero-sum game:

$$\begin{bmatrix} (5, -4) & (-1, -1) & (-5, 1) \\ (-3, 0) & (1, -2) & (5, -4) \end{bmatrix}.$$

(c) Use Procedure 6.6 to determine rational play in the above game.

- (18) Consider a game tree which is zero-sum (i.e. all payoffs have the form $(a, -a)$), has only trivial information sets (i.e. whenever Rose or Colin has a choice, they are certain which node they are at), but may have chance nodes. Generalize Zermelo's Theorem by proving that there exists a number v together with a strategy for Rose and a strategy for Colin, so that Rose's strategy guarantees her an expected payoff of $\geq v$ no matter what Colin does and Colin's strategy guarantees him an expected payoff of $\geq -v$ no matter what Rose does.
- (19) Goliath Mart owns N stores numbered $1, \dots, N$. At present, none of the stores are unionized, but on week i , the employees at store i vote on whether or not to form a union. If they do not form a union, everything stays as is and both Goliath Mart and the employees get payoffs of 0. If the employees unionize, Goliath Mart will either accept the union or close the store. If Goliath Mart accepts the union at store i , the employees at this store receive better compensation

for a +2 payoff; however, the profits of Goliath Mart shrink by 1 million, which we treat as a -1 payoff. If store i is closed, the Goliath Mart loses all profits from that store for a payoff of -3 , but all store i employees would lose their jobs, which we treat as a payoff of -10 .

- (a) Use a game tree to model this game for $N = 2$.
 - (b) If the players are perfectly rational, what will happen?
 - (c) Could Goliath Mart benefit by developing a reputation as an irrational player?
- (20) Game theory can be utilized to analyze problems in philosophy. Model each of the following philosophical questions as a matrix game. Discuss the strengths and limitations of this model: In what ways does a matrix representation clarify the problem? How is the question perhaps ill-suited to a matrix? Do the insights or solutions suggested by the matrix support what seems to you to be a reasonable resolution of the problem?
- (a) For Pascal's Wager?
 - (b) For Newcomb's Paradox?

Chapter 8

Nash Equilibrium and Applications

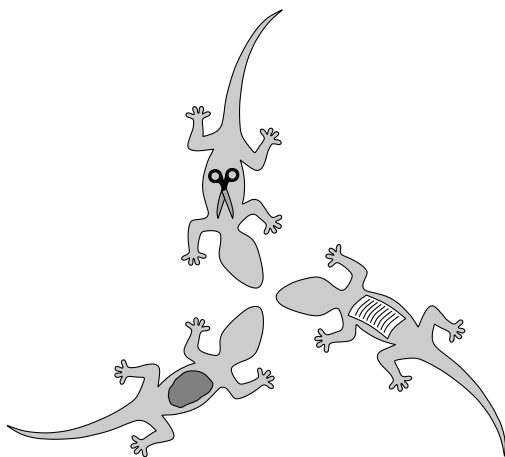


Figure 8.1. Lizards playing Rock-Paper-Scissors

Matrix games can represent a wide variety of decision problems between two players, but how should rational players play in such a general context? This chapter develops Nash's Equilibrium Theorem, one of the great triumphs of classical game theory. The proof indicates how rational players in an environment of repeated play will move towards a certain type of equilibrium point. In the many instances where a matrix game is played repeatedly, Nash's Theorem has great predictive power in addition to its mathematical beauty. This chapter introduces Nash's

Theorem and explores applications in evolutionary biology and business. Chapter 9 presents a proof of Nash's Equilibrium Theorem.

To get things started, try playing the following game a few times in succession.

Game 8.1 (A Number Game). This is a game between player 1 and player 2. Player 1 secretly selects an integer $0 \leq n_1 \leq 100$ and player 2 secretly selects an integer $0 \leq n_2 \leq 100$. Then the players' choices are revealed, and player 1 gets a payoff of $(100 - n_1 - n_2)n_1$ while player 2 gets a payoff of $(100 - n_1 - n_2)n_2$.

See Section 8.3 for a discussion of rational play in this game.

8.1. Nash Equilibrium

We have seen that a pure Nash equilibrium consists of two pure strategies, each of which is a best pure response to the other. This is a nice concept, but limited as many matrices do not have pure equilibria. In the more general setting of mixed strategies, Nash's Equilibrium Theorem proves that a mixed strategy equilibrium always exists.

Mixed Strategies. In a matrix game A , each entry is an ordered pair of real numbers (x, y) indicating payoffs of x to Rose and y to Colin. Considering only the first coordinates of these entries (i.e. just the payoffs to Rose) gives a matrix R that we call *Rose's payoff matrix*. Similarly, the second coordinates form a matrix C , called *Colin's payoff matrix*.

Example 8.2.

$$A = \begin{bmatrix} (1, 4) & (2, 0) & (3, 3) \\ (3, 0) & (1, 2) & (2, 1) \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 0 & 3 \\ 0 & 2 & 1 \end{bmatrix}.$$

Matrix Game Rose's Payoff Matrix Colin's Payoff Matrix

The notion of a mixed strategy from the context of zero-sum games generalizes to matrix games. Namely, if A is an $m \times n$ matrix game, then a *mixed strategy* for Rose is a row vector $\mathbf{p} = [p_1 \ \cdots \ p_m]$ where $p_1, \dots, p_m \geq 0$ and $\sum_{i=1}^m p_i = 1$. Similarly, a *mixed strategy* for Colin will

be a column vector $\mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$ for which $q_1, \dots, q_n \geq 0$ and $\sum_{j=1}^n q_j = 1$.

To *play* strategy $\mathbf{p}$, Rose randomly chooses a row, selecting row i with probability p_i , and to *play* $\mathbf{q}$, Colin randomly chooses a column, selecting column j with probability q_j .

As before, we want to compute the players' expected payoffs. Assume that Rose plays $\mathbf{p}$ and Colin plays $\mathbf{q}$. To compute Rose's expected payoff, consider her payoff matrix R . Rose gets a payoff corresponding to the (i, j) entry of this matrix with probability $p_i q_j$. It follows that Rose's expected payoff is given by $\mathbf{pRq} = \sum_{i=1}^m \sum_{j=1}^n p_i r_{i,j} q_j$. Colin's payoffs are given by his payoff matrix C , so Colin's expected payoff in this case is $\mathbf{pCq}$.

Example 8.2 (Continued). Assume Rose plays $\mathbf{p} = \begin{bmatrix} 1/4 & 3/4 \end{bmatrix}$ and Colin plays $\mathbf{q} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$. Then

$$\begin{array}{l} \text{Rose's expected} \\ \text{payoff} \end{array} = \mathbf{pRq} = \begin{bmatrix} 1/4 & 3/4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = 2,$$

$$\begin{array}{l} \text{Colin's expected} \\ \text{payoff} \end{array} = \mathbf{pCq} = \begin{bmatrix} 1/4 & 3/4 \end{bmatrix} \begin{bmatrix} 4 & 0 & 3 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix} = 4/3.$$

Best Response. How can we determine best responses (both pure and mixed) for our players? First, we will introduce a new term. Suppose that $\mathbf{p}$ is a mixed strategy for Rose and the i^{th} entry of $\mathbf{p}$ is 0. In this case, Rose will never select row i when she plays $\mathbf{p}$. On the other hand, if the i^{th} entry of $\mathbf{p}$ is positive, then she will select row i some of the time. We say that $\mathbf{p}$ *calls on* row i if $p_i > 0$. Similarly, we say that Colin's mixed strategy $\mathbf{q}$ *calls on* column j if $q_j > 0$. Equipped with this new term, let's think about best responses in a particular example.

Example 8.3. Consider the following 3×2 matrix game A together with Rose's payoff matrix R :

$$A = \begin{bmatrix} (3, 2) & (-1, 1) \\ (3, 1) & (0, 3) \\ (1, 4) & (2, 6) \end{bmatrix}, \quad R = \begin{bmatrix} 3 & -1 \\ 3 & 0 \\ 1 & 2 \end{bmatrix}.$$

We will suppose throughout this example that Colin will always play the mixed strategy $\mathbf{q} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$. Then vector $R\mathbf{q}$ below indicates Rose's expected payoff for each possible pure strategy she could select:

$$R\mathbf{q} = \begin{bmatrix} 1 \\ 3/2 \\ 3/2 \end{bmatrix}.$$

When Rose plays a pure strategy of the first row, her expected payoff is 1. If she plays either the second or the third row, then her expected payoff is $3/2$. Thus, the pure strategies that maximize Rose's expected payoff are rows 2 and 3.

Now suppose that Rose is going to play the mixed strategy $\mathbf{p} = [p_1 \ p_2 \ p_3]$. What mixed strategies will give her the highest possible expected payoffs? Could she possibly do better with a mixed strategy than with a pure one? Rose's expected payoff when she plays $\mathbf{p}$ is

$$\mathbf{p}R\mathbf{q} = [p_1 \ p_2 \ p_3] \begin{bmatrix} 1 \\ 3/2 \\ 3/2 \end{bmatrix} = p_1 + (3/2)p_2 + (3/2)p_3.$$

To maximize Rose's expected payoff, we maximize the quantity on the right-hand side of the previous equation over all possible mixed strategies for Rose. In order for $\mathbf{p} = [p_1 \ p_2 \ p_3]$ to be a mixed strategy, we must have $p_1, p_2, p_3 \geq 0$ and $p_1 + p_2 + p_3 = 1$. It follows that

$$p_1 + (3/2)p_2 + (3/2)p_3 \leq 3/2(p_1 + p_2 + p_3) = 3/2.$$

So, we see that Rose cannot get an expected payoff that is greater than $3/2$. We already know that Rose's pure strategies of playing row 2 or row 3 achieve this highest possible expected payoff of $3/2$. Thus, Rose cannot get a higher expected payoff using a mixed strategy than she could with a pure strategy. Are there any mixed strategies different from the pure strategies of row two and row three that achieve this best possible expected payoff? To see the answer, return to the inequality from above: $p_1 + (3/2)p_2 + (3/2)p_3 \leq (3/2)p_1 + (3/2)p_2 + (3/2)p_3$. Note that when p_1 is not zero (i.e. $p_1 > 0$) the right-hand side is strictly larger. This tells us that Rose's expected payoff will be $< 3/2$. On the other hand, when $p_1 = 0$ the left- and right-hand sides will be equal, so Rose's expected payoff will be exactly $3/2$. We conclude that the mixed strategies that

give Rose the highest possible expected payoff are precisely those of the form $\mathbf{p} = [0 \ p_2 \ p_3]$ (note that this also includes the pure strategies of rows two and three). Therefore, Rose's optimal mixed strategies are precisely those that only call on rows that are best pure responses.

With this guiding example, we now extend the notion of best responses to include the use of mixed strategies. Consider a matrix game A and suppose that Colin is going to play a mixed strategy $\mathbf{q}$. We say that a pure strategy, row i , is a *best pure response* to $\mathbf{q}$ if this row gives Rose the maximum expected payoff over all of her pure strategies. Similarly, we say that a mixed strategy $\mathbf{p}$ for Rose is a *best response* if it gives her the maximum expected payoff over all mixed strategies. We similarly define best pure responses and best responses for Colin. The following proposition says how to find all best responses to a mixed strategy for either Colin or Rose.

Proposition 8.4. *Consider a matrix game with payoff matrices R, C .*

Rose's best responses to Colin's mixed strategy $\mathbf{q}$	
Pure	Every row i for which the i^{th} entry of $R\mathbf{q}$ is max
Mixed	Every $\mathbf{p}$ which only calls on best pure responses to $\mathbf{q}$

Colin's best responses to Rose's mixed strategy $\mathbf{p}$	
Pure	Every column j for which the j^{th} entry of $\mathbf{p}C$ is max
Mixed	Every $\mathbf{q}$ which only calls on best pure responses to $\mathbf{p}$

Proof. We prove the result only for Rose's best responses since Colin's case follows by a similar argument. When Colin plays $\mathbf{q}$ and Rose plays row i , her expected payoff will be $(R\mathbf{q})_i$ (i.e. the i^{th} entry of the vector $R\mathbf{q}$). So, Rose's best pure responses are precisely those rows i for which the i^{th} entry of $R\mathbf{q}$ is a maximum entry.

Next, define M to be the maximum entry in the vector $R\mathbf{q}$. If Rose plays the mixed strategy $\mathbf{p} = [p_1 \ \cdots \ p_m]$ and Colin plays $\mathbf{q}$, then her expected payoff will be

$$\mathbf{p}R\mathbf{q} = [p_1 \ \cdots \ p_m](R\mathbf{q}) = \sum_{i=1}^m p_i(R\mathbf{q})_i \leq \sum_{i=1}^m p_i M = M.$$

We see that Rose cannot do better than an expected payoff of M with a mixed strategy. If her mixed strategy $\mathbf{p}$ calls on a row i that is not a best pure response to $\mathbf{q}$, then the i^{th} entry of $R\mathbf{q}$ is strictly smaller than M so $p_i(R\mathbf{q})_i < p_i M$. The above equation shows that $\mathbf{p}$ will give Rose an expected payoff less than M . On the other hand, if every row called on by $\mathbf{p}$ is a best pure response to $\mathbf{q}$, then $p_i(R\mathbf{q})_i = p_i M$ will hold for every $1 \leq i \leq m$ and Rose will get her best possible expected payoff of M . $\square$

Nash Equilibrium. The single most important theorem in classical game theory pairs best responses to produce what's called a Nash equilibrium. For a matrix game A , a pair of mixed strategies $\mathbf{p}$ for Rose and $\mathbf{q}$ for Colin form a *Nash equilibrium* if $\mathbf{p}$ is a best response to $\mathbf{q}$ and $\mathbf{q}$ is a best response to $\mathbf{p}$.

Theorem 8.5 (Nash). *Every matrix game has a Nash equilibrium.*

We prove this important theorem in Chapter 9. For now, let's explore an example.

Example 8.6. Consider the following 5×5 matrix game A together with the payoff matrices R for Rose and C for Colin:

$$A = \begin{bmatrix} (1, 4) & (-1, 1) & (3, 7) & (2, -3) & (0, 2) \\ (5, 3) & (1, -1) & (1, 0) & (3, 1) & (2, 4) \\ (0, -1) & (2, 3) & (-1, 2) & (1, 0) & (4, 1) \\ (1, 3) & (3, 7) & (2, -4) & (-2, 6) & (6, 0) \\ (3, 2) & (0, -1) & (1, 0) & (4, 3) & (-2, 5) \end{bmatrix},$$

$$R = \begin{bmatrix} 1 & -1 & 3 & 2 & 0 \\ 5 & 1 & 1 & 3 & 2 \\ 0 & 2 & -1 & 1 & 4 \\ 1 & 3 & 2 & -2 & 6 \\ 3 & 0 & 1 & 4 & -2 \end{bmatrix}, \quad C = \begin{bmatrix} 4 & 1 & 7 & -3 & 2 \\ 3 & -1 & 0 & 1 & 4 \\ -1 & 3 & 2 & 0 & 1 \\ 3 & 7 & -4 & 6 & 0 \\ 2 & -1 & 0 & 3 & 5 \end{bmatrix}.$$

Consider the strategies $\mathbf{p} = \begin{bmatrix} 0 & 9/17 & 3/17 & 5/17 & 0 \end{bmatrix}$ for Rose and

$$\mathbf{q} = \begin{bmatrix} 2/31 \\ 0 \\ 0 \\ 12/31 \\ 17/31 \end{bmatrix} \text{ for Colin. If Colin plays } \mathbf{q}, \text{ then } R\mathbf{q} = \begin{bmatrix} 26/31 \\ 80/31 \\ 80/31 \\ 80/31 \\ 20/31 \end{bmatrix}. \text{ This gives}$$

Rose's expected payoff for each possible row she might select. We see that rows two, three, and four are Rose's best pure responses to $\mathbf{q}$ —they give her an expected payoff of $80/31$. More generally, her best responses to $\mathbf{q}$ will be all mixed strategies that only call on rows two, three, and four (i.e. mixed strategies of the form $[0 \ p_2 \ p_3 \ p_4 \ 0]$). In particular, her strategy $\mathbf{p}$ is indeed a best response to $\mathbf{q}$.

On the other side, assume that Rose is going to play $\mathbf{p}$. Then the vector $\mathbf{pC} = [39/17 \ 35/17 \ -14/17 \ 39/17 \ 39/17]$ indicates Colin's expected payoff for each possible column he might choose. We see that columns one, four, and five are Colin's best pure responses to $\mathbf{p}$ and these give him an expected payoff of $39/17$. More generally, his best responses to $\mathbf{p}$ will be all mixed strategies that only call on columns one, four, and five

(i.e. all mixed strategies of the form $\begin{bmatrix} q_1 \\ 0 \\ 0 \\ q_4 \\ q_5 \end{bmatrix}$). Thus, Colin's strategy $\mathbf{q}$ is a best response to $\mathbf{p}$.

Since Rose's strategy $\mathbf{p}$ is a best response to Colin's strategy $\mathbf{q}$ and $\mathbf{q}$ is also a best response to $\mathbf{p}$, this pair of strategies forms a Nash equilibrium. This equilibrium does *not* indicate exactly how the players should necessarily play the matrix game A . In fact, this particular matrix game has four more Nash equilibria! Though a Nash equilibrium does not prescribe how to play the game, it does indicate a kind of stability. If Rose and Colin are going to play this game repeatedly and they have settled into the mixed strategies of Rose playing $\mathbf{p}$ and Colin playing $\mathbf{q}$, then neither player has any incentive to change. In this way, a Nash equilibrium is a stable point.

Computing Equilibria in 2×2 Games. The complicated problem of computing Nash equilibria in matrix games is well beyond the scope of this book. We nonetheless *do* have the tools to find Nash equilibria in 2×2 games. Recall that in the study of zero-sum matrix games we used a technique called equating the opponent's results to find von Neumann solutions. Here, we will use a natural generalization of this in the context of general matrix games.

Definition 8.7. Let A be a 2×2 matrix game with payoff matrices R for Rose and C for Colin. We say that a mixed strategy $\mathbf{p}$ for Rose *equates Colin's results* if both entries of $\mathbf{p}C$ are equal (so Colin will get the same payoff no matter what he does). Similarly, we say that a mixed strategy $\mathbf{q}$ for Colin *equates Rose's results* if both entries of $R\mathbf{q}$ are equal (so Rose will get the same payoff no matter what she does).

Theorem 8.8. *In every 2×2 matrix game, one of the following holds:*

- (1) *Iterated removal of dominated strategies reduces the matrix to 1×1 . This row and column form a pure Nash equilibrium.*
- (2) *Rose and Colin both have mixed strategies that equate the other player's results and these form a Nash equilibrium.*

Proof. Suppose first that one player has a dominant strategy. Without (significant) loss of generality, assume that Rose has a dominant strategy of row i . Choose column j to be a best pure response for Colin when Rose plays row i . Now the pure strategies of row i and column j form a pure strategy Nash equilibrium.

Next suppose that neither player has a dominant strategy. Then Rose does not have a dominant strategy in the zero-sum game R , so by Lemma 6.3, Colin has a mixed strategy $\mathbf{q}$ that equates Rose's results in R (so both entries of $R\mathbf{q}$ are equal). Similarly, Colin has no dominant strategy in the zero-sum game C , so by Lemma 6.3, Rose has a mixed strategy $\mathbf{p}$ that equates Colin's results in C (so both entries of $\mathbf{p}C$ are equal). It follows that $\mathbf{p}$ and $\mathbf{q}$ are best responses to one another and thus form a Nash equilibrium. $\square$

Example 8.9. Find a Nash equilibrium in the following 2×2 matrix game:

$$A = \begin{bmatrix} (1, -2) & (0, 2) \\ (-1, 3) & (3, -1) \end{bmatrix}.$$

The payoff matrices are $R = \begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix}$ for Rose and $C = \begin{bmatrix} -2 & 2 \\ 3 & -1 \end{bmatrix}$ for Colin. Neither player has a dominant strategy, so by the above theorem, each player has a mixed strategy that equates the other's results and together these form a Nash equilibrium. First let us find a strategy

$\mathbf{p} = [p \quad 1-p]$ for Rose that equates Colin's results. Equating Colin's first and second column payoffs gives us

$$p(-2) + (1-p)3 = p(2) + (1-p)(-1).$$

Thus $8p = 4$ and $p = 1/2$, so $\mathbf{p} = [1/2 \quad 1/2]$. Now we will look for a strategy $\mathbf{q} = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ for Colin that equates Rose's results. Equating Rose's first and second row payoffs yields

$$q(1) + (1-q)0 = q(-1) + (1-q)3.$$

This gives us $5q = 3$ so $q = 3/5$ and $\mathbf{q} = \begin{bmatrix} 3/5 \\ 2/5 \end{bmatrix}$. Since the strategies $\mathbf{p}$ and $\mathbf{q}$ both equate the other player's payoffs, they are best responses to one another. Therefore, $\mathbf{p} = [1/2 \quad 1/2]$ and $\mathbf{q} = \begin{bmatrix} 3/5 \\ 2/5 \end{bmatrix}$ form a Nash equilibrium.

8.2. Evolutionary Biology

This section adopts a perspective from modern evolutionary biology and views evolution as a game. Charles Darwin first articulated evolution as a competition in which success corresponds to abundant offspring in future generations. If a genetic mutation develops that gives rise to a variation beneficial to survival, this new variant will outcompete the original and eventually replace it (perhaps over the course of many generations). Consider this so-called "survival of the fittest" as nature adopting a dominant strategy in the evolutionary competition. However, this simple notion of dominance cannot fully describe the rich interactions evident in nature. In many cases, mutation leads to a new variant that neither dominates nor is dominated by the original. This new variant might have advantages in some situations and disadvantages in others. A game-theoretic approach to the competition of evolution facilitates a more nuanced view of these dynamics. It also equips us to predict genetic balances in populations.

Hawk vs. Dove. We begin this section with a famous game in evolutionary biology, Hawk vs. Dove, that introduced game theory to the study of evolutionary biology in a landmark paper of John Maynard Smith and George Price.

Example 8.10 (Hawk vs. Dove). Imagine that members of a certain species engage in pairwise competitions for a scarce resource. Suppose for simplicity that the species has two genetic variants, call them *hawk* and *dove*. The hawks of the species are very aggressive and will always fight for the resource. In contrast, the doves are passive and will wait around to see if the opponent gives up and goes away. Many variations of this game appear in nature. For instance, male dung beetles, *Onthophagus acuminatus*, come in two varieties; some have a large pair of front horns, while others have very small (or possibly even nonexistent) horns. In other species, this type of distinction might be less physically obvious and instead might express itself as a predisposition toward aggressive or passive behavior.

Next assign some utilities to these outcomes to enable calculations. Assume that the resource is worth 10. If two hawks engage, they will fight, and the winner earns the resource for +10, and the loser scores -20 for suffering defeat. On average, the expected payoff when two hawks engage is -5 for each. When a hawk and a dove compete, the hawk takes the resource for 10 and the dove gets 0. When two doves compete, they waste time posturing, which costs each -1 . One of the doves will eventually give up and the other will get the +10 resource, so the expected payoff for each dove will be 4. We can express this in a familiar matrix game as follows:

	Hawk	Dove
Hawk	$-5, -5$	$10, 0$
Dove	$0, 10$	$4, 4$

Suppose that the population is presently split so the probability that an individual is a hawk is p . Now consider the situation from the perspective of an individual I who might be a hawk or a dove. When I competes, the probability that I will face a hawk is p and the probability I will encounter a dove is $1 - p$. From the perspective of I , this gives the following 1-player game:

		Chance	
		p Hawk	$1 - p$ Dove
I	Hawk	-5	10
	Dove	0	4

Now we can compute I 's expected payoff if I is a hawk or if I is a dove. If I is a hawk, then I 's expected payoff will be $p(-5) + (1-p)10 = 10 - 15p$. On the other hand, if I is a dove, the expected payoff is $p(0) + (1-p)4 = 4 - 4p$. These two payoffs will be equal when $10 - 15p = 4 - 4p$, so $p = 6/11$. When $p > 6/11$, the payoff is higher if I is a dove, while if $p < 6/11$, the payoff is higher if I is a hawk. Figure 8.2 illustrates this situation with a number line.

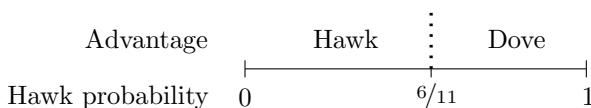


Figure 8.2. Advantage in Hawk-Dove

Imagine for a moment that the species is presently divided such that the probability that an individual is a hawk is larger than 0 but smaller than $6/11$. Since our computations show that the hawks have an advantage (i.e. a hawk will receive a higher expected payoff than a dove in our pairwise competitions), we can expect that the average hawk will be fitter and healthier than the average dove. The average hawk will therefore have more healthy offspring than the average dove. Consequently, over time the division of the species will shift so that hawks comprise a higher percentage of the population. That is, if $0 < p < 6/11$ initially, we expect to see p increase over time. Conversely, if the species is presently divided so that the probability that an individual is a hawk is greater than $6/11$ but smaller than 1, then doves will have the advantage. In this case, the doves will be fitter and have more offspring. That is, if $6/11 < p < 1$, to start, then we expect over time to see p decrease.

We have now argued that the population distribution should eventually settle so the probability that an individual is a hawk is $6/11$. Notice that the key to the calculation was finding a strategy for the column player that equated the row player's results. Thanks to the symmetry between the row and column players in our original matrix, these calculations give a Nash equilibrium consisting of the strategies $\left[\frac{6}{11} \quad \frac{5}{11} \right]$ and $\left[\frac{6}{11} \right]$.

Symmetry and Nash Equilibria. The Hawk-Dove game exhibits a type of symmetry between the row and column players common to this type of evolutionary game. Namely, both the row and column players have the same strategies and corresponding payoffs. Define a square matrix game A to be *symmetric* if the payoff matrices R, C for Rose and Colin satisfy $R^T = C$. In other words, a matrix game A is symmetric if Rose's payoff in entry (i, j) is always the same as Colin's payoff for the entry (j, i) . So, both the Hawk-Dove game and the Prisoner's Dilemma are symmetric. For a symmetric matrix game A we say that a column vector $\mathbf{d}$ is a *symmetric Nash equilibrium* if the pair of strategies $\mathbf{d}^T$ for Rose and $\mathbf{d}$ for Colin form a Nash equilibrium. A slight modification of the proof of Nash equilibrium yields the following result.

Theorem 8.11. *Every symmetric matrix game has a symmetric Nash equilibrium.*

Whenever pure strategies correspond to genetic variants in an evolutionary game, we should always expect the population distribution to form a Nash equilibrium $\mathbf{d}$. Otherwise, there would be a pure strategy with a better expected payoff than $\mathbf{d}$. The corresponding genetic variant would then be more successful than the average, and over time the percentage of individuals with this variant would increase. So, our population distribution should form a symmetric Nash equilibrium, and the above theorem conveniently guarantees that such an equilibrium must always exist. In the Hawk-Dove game, for example, $\begin{bmatrix} 6/11 \\ 5/11 \end{bmatrix}$ is a symmetric Nash equilibrium.

Example 8.12 (Common Side-Blotched Lizard). The males of this type of lizard have either orange, blue, or yellow coloration on their throats, and each of the three types has a different mating strategy. The orange-throated males are strongest, and their strategy is to control large amounts of territory with many rocks since the female lizards enjoy sunning themselves on these rocks. The blue-throated males are of medium strength and they generally control a smaller amount of territory. In competition, the stronger orange lizards beat blue ones. The yellow-throated males are the smallest and weakest, but they have a very similar color to that of the female. Instead of strategically controlling territory, the camouflaged yellow-throated male sneaks into the territory

of another lizard and mates with females there. Since the orange lizard controls so much territory, yellow can take advantage and beat orange. However, the blue-throated male can defend his smaller territory against yellow. Altogether, then, orange lizards beat blue, blue lizards beat yellow, and yellow lizards beat orange. For our purposes, the following matrix game adequately models the situation:

	Orange	Blue	Yellow
Orange	0, 0	1, -1	-1, 1
Blue	-1, 1	0, 0	1, -1
Yellow	1, -1	-1, 1	0, 0

This zero-sum game is identical to Rock-Paper-Scissors! These lizards are essentially playing a kind of evolutionary version of this familiar game. Thanks to the symmetry between the three strategies here, it

is natural to suspect that $\mathbf{d} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$ will be a symmetric Nash equilibrium. To check this, let R be the payoff matrix for the row player and

consider the vector $R\mathbf{d} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. Every pure strategy is a best response to

$\mathbf{d}$, so Proposition 8.4 implies that $\mathbf{d}^\top$ is a best response to $\mathbf{d}$. So, $\mathbf{d}$ is a symmetric Nash equilibrium.

Evolutionary Stability. We have argued that populations should settle into a symmetric Nash equilibrium based on evolutionary considerations. From an evolutionary standpoint, however, not all equilibria are the same. Some are less stable than others, and these less stable equilibria tend not to be selected. The Nash equilibrium in the Hawk-Dove game exhibits strong stability and the population will tend to settle on a distribution in which $6/11$ of the individuals are hawks. Next, we consider a game with a different type of equilibrium.

Example 8.13. Recall the Coordination game from Chapter 7:

	X	Y
X	1, 1	0, 0
Y	0, 0	1, 1

Let's consider this as an evolutionary game. So, our species will have two variants, X and Y , and in pairwise competitions, X 's will fare better when opposite other X 's and similarly Y 's will do better opposite other Y 's. There are two pure Nash equilibria here consisting of X, X and Y, Y and these correspond to one of the two variants dying off and the other taking over. Yet, there is also another symmetric Nash equilibrium in this game given by the strategy $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ that results in a population distribution evenly split between the X and Y variants.

As in the Hawk-Dove game, imagine that an individual is an X with probability p . Does variant X or Y have the advantage? The perspective of a single individual yields the following 1-player game:

		Chance	
		p X	$1 - p$ Y
I	X	1	0
	Y	0	1

Here individual X gets an expected payoff of p , while Y gets an expected payoff of $1 - p$ in this game. Hence X 's will do better when $p > 1/2$, and Y 's will do better when $p < 1/2$, and both will do equally well when $p = 1/2$. We can again illustrate this advantage (Figure 8.3) on a number line as in the Hawk-Dove game.

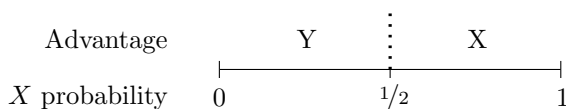


Figure 8.3. Advantage in the Coordination game

Suppose that the present population distribution has a higher percentage of X 's than Y 's. In this case, the X individuals will be more successful than the Y 's on average. As a result, we expect that the percentage of type X will increase even more. This will result over time in a population with a higher and higher percentage of X 's. Eventually the population will be 100% type X . If, however, the population were distributed with a higher percentage of Y 's, then the Y 's would be more successful than the X 's. Then the population would shift to have an even

smaller percentage of X 's and, in time, the population would be 100% type Y . Although $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ is indeed a symmetric Nash equilibrium, it is not very stable in the sense that any small fluctuation is likely to lead, over time, to one of the pure Nash equilibria consisting of all X or all Y .

So the Nash equilibrium in the Hawk-Dove game is stable in a way that the $\begin{bmatrix} 1/2 & 1/2 \end{bmatrix}$ equilibrium in the Coordination game is not. How can we formalize this notion of stability in general? Maynard Smith famously answered this question with the following definition of a stable equilibrium of the type that should be favored in an evolutionary setting. In particular, an equilibrium of this type could not be disrupted by an alternative strategy.

Definition 8.14. Let A be a symmetric matrix game with payoff matrix R for the row player. A strategy $\mathbf{d}$ is an *evolutionarily stable strategy* if it satisfies the following:

- (1) $\mathbf{d}$ is a symmetric Nash equilibrium.
- (2) For every pure strategy $\mathbf{p}$ that is a best response to $\mathbf{d}$, we have $\mathbf{d}^\top R \mathbf{p} > \mathbf{p}^\top R \mathbf{p}$.

In order for a symmetric Nash equilibrium $\mathbf{d}$ to be an evolutionarily stable strategy, any pure strategy alternative that is a best response to $\mathbf{d}$ must have the property that $\mathbf{d}$ is a better response to this alternative than it is to itself. Why are these strategies stable? Imagine that the population consists of variants $X_1, \dots, X_n$ and that the present distribution is

given by the evolutionarily stable strategy $\mathbf{d} = \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix}$. Now suppose that,

due to random fluctuations, the X_i population increases slightly. If X_i is not a best response to $\mathbf{d}$, then in pairwise competitions, these X_i individuals will do worse than the rest of the population on average. This will cause the X_i population to decrease, returning toward distribution $\mathbf{d}$. Next suppose that X_i is a best response to $\mathbf{d}$ and let $\mathbf{p}$ be the pure strategy of playing X_i (i.e. so $\mathbf{p}$ is a column vector with a 1 in the i^{th} position and 0 elsewhere). Since X_i is a best response to $\mathbf{d}$, it would not be surprising to see a small increase in the percentage of individuals of type X_i . However, if the original distribution $\mathbf{d}$ satisfies $\mathbf{d}^\top R \mathbf{p} > \mathbf{p}^\top R \mathbf{p}$,

then the average player gets a better payoff competing against X_i than X_i gets competing against itself. In this new population distribution with a slightly higher percentage of variant X_i , the X_i players are once again at a disadvantage. This will cause their numbers to decline, returning the population to distribution $\mathbf{d}$.

Let's verify this definition in some familiar games. We identified symmetric Nash equilibria of $\mathbf{d} = \begin{bmatrix} 6/11 \\ 5/11 \end{bmatrix}$ in Hawk-Dove, $\mathbf{d} = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}$ in our Lizard game, and $\mathbf{d} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ in the Coordination game. In a symmetric Nash equilibrium, $\mathbf{d}^\top$ must be a best response to $\mathbf{d}$, so by Proposition 8.4, any pure strategy that is called on by $\mathbf{d}^\top$ must be a best pure response to $\mathbf{d}$. So, in all three of our games, every pure strategy is a best response to $\mathbf{d}$. A little calculation shows that, in the given games, every pure strategy $\mathbf{p}$ satisfies the following:

$$\begin{array}{ll} \text{Hawk-Dove} & \mathbf{d}^\top R \mathbf{p} > \mathbf{p}^\top R \mathbf{p}, \\ \text{Lizard} & \mathbf{d}^\top R \mathbf{p} = \mathbf{p}^\top R \mathbf{p}, \\ \text{Coordination} & \mathbf{d}^\top R \mathbf{p} < \mathbf{p}^\top R \mathbf{p}. \end{array}$$

The first fact implies that $\mathbf{d} = \begin{bmatrix} 6/11 \\ 5/11 \end{bmatrix}$ is an evolutionarily stable strategy in Hawk-Dove. This corresponds to the strong stability property uncovered in our earlier discussion. In the Coordination game, the symmetric Nash equilibrium $\mathbf{d} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$ is not an evolutionarily stable strategy. This is an unstable equilibrium. Finally, in the Lizard game, our symmetric Nash equilibrium is not evolutionarily stable, but it is right on the borderline. Accordingly, the male population in this species of lizards does not maintain a precise distribution between orange, blue, and yellow, instead it fluctuates a little over time around the equilibrium.

8.3. Cournot Duopoly

Game-theoretic situations appear frequently in the world of economics, where actions of one party often impact the outcome for another party.

This section presents a famous model concerning two companies in competition (a duopoly).

Setup. The demand curve communicates one particularly important relationship in economic theory, that between the price of a commodity and the amount the public will purchase. For convenience, we will set our timescale at 1 year, so if (p, q) is a point on the demand curve, then at price p , the public will demand q units of the commodity over the course of 1 year. Barring exceptional circumstances, the higher the price the lower the amount the public will demand. Thus, the demand curve generally has a negative slope as depicted as in Figure 8.4.¹

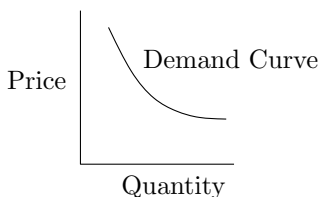


Figure 8.4. The demand curve

Going forward, we assume that the demand curve is given by the following simple linear equation:

$$P = a - bQ.$$

Here a and b are constants, P denotes the price, and Q denotes the quantity of the commodity demanded.

Although real-world demand curves are generally not linear, linear functions frequently can approximate important parts of the demand curve. Moreover, since the analysis here is robust under changes to our assumption on the demand curve, the conclusions we will reach apply quite broadly. The model here is carefully chosen to be rich enough to reveal the essential features of interest, but simple enough to allow for easy computations.

¹Although it may initially seem backwards, the standard convention is to write Quantity on the horizontal axis and Price on the vertical. The motivation for this will be discovered later in this section.

Another simplifying assumption relates to production costs. Namely, we assume that producing each unit has a cost of c . So, the total cost for a company producing Q units is given by the equation

$$\text{Cost} = cQ.$$

Monopoly. Start by considering a monopoly—just one company producing the commodity. This company makes only one decision: the quantity to produce. Given this quantity, the demand curve sets the price. So, if the company decides to produce Q units, the price will be $P = a - bQ$, which gives the following:

$$\begin{aligned}\text{Cost} &= cQ, \\ \text{Revenue} &= PQ = (a - bQ)Q.\end{aligned}$$

The profit U is equal to revenue minus cost, so

$$U = (a - bQ)Q - cQ = -bQ^2 + (a - c)Q.$$

The company naturally wants to choose Q to maximize profit. The profit is given by a quadratic function of Q for which the graph is a downward-opening parabola. Accordingly, the value of Q that will maximize profit will be the unique point where the derivative is equal to zero. So, to find the quantity that maximizes profit, set $\frac{dU}{dQ} = 0$ and solve for Q :

$$0 = \frac{dU}{dQ} = -2bQ + a - c.$$

To maximize profit, the company should produce exactly $Q^* = \frac{a-c}{2b}$ units. With this level of production, yearly profit will be

$$\begin{aligned}U^* &= -b(Q^*)^2 + (a - c)Q^* \\ &= -\frac{(a-c)^2}{4b} + \frac{(a-c)^2}{2b} \\ &= \frac{(a-c)^2}{4b}.\end{aligned}$$

A monopoly poses no game-theoretic question. This is just an optimization problem! Let's add a second company and see how the dynamics change.

Duopoly. Now introduce a second company. Assume that Company 1 and Company 2 compete in the same marketplace and produce equivalent versions of the commodity (i.e. interchangeable goods) so the public will not see any significant distinction between the two brands.

As before, each company has just a single decision to make: how many units to produce. Assume that Company 1 produces Q_1 units while Company 2 Produces Q_2 units. In this case, the total number of units produced will be $Q_1 + Q_2$. According to the above demand curve, the price of the commodity can be computed by

$$P = a - b(Q_1 + Q_2).$$

Assume that each company is interested in maximizing its profit and pause for a moment to recognize all the ingredients for a game. Company 1 and Company 2 both choose a number of units to produce, Q_1 and Q_2 , respectively. Based on these choices, each company will make a certain profit, which we view as its payoff. Notice that the actions of one company affect the price and therefore profit of the other company. Now we have a game! If maximum production were capped at, say, N , then we could express this as a matrix game. Each player must choose a number between 0 and N and then payoffs correspond to pairs of choices.

To analyze rational competition in this situation, we need profit functions (the payoffs in our game) for both companies. The revenue for each company is given by the price times the quantity sold, so

$$\text{Revenue for Company } i = PQ_i = (a - bQ_1 - bQ_2)Q_i.$$

Assume that each company has a production cost of c per unit. So, if Company i produces Q_i units, then

$$\text{Cost for Company } i = cQ_i.$$

What about profit? The profit for each company is given by revenue minus costs. So, the profit for Company i , denoted U_i , is given by

$$U_1 = (a - bQ_1 - bQ_2)Q_1 - cQ_1 = -bQ_1^2 + (a - c - bQ_2)Q_1,$$

$$U_2 = (a - bQ_1 - bQ_2)Q_2 - cQ_2 = -bQ_2^2 + (a - c - bQ_1)Q_2.$$

First consider this situation from the perspective of Company 1. Suppose Company 1 knows (based on previous actions, or industry information, etc.) that Company 2 will be producing Q_2 units. How many units should Company 1 produce? In game-theoretic terminology, what is the best response for Company 1 when Company 2 plays Q_2 ? If we treat Q_2 as fixed, then the profit function U_1 is a function of the single variable Q_1 . Since U_1 is a quadratic function of Q_1 and its graph is a downward-opening parabola, the maximum profit will be achieved at the unique point where the derivative $\frac{dU_1}{dQ_1}$ is equal to zero. This gives us the equation

$$0 = \frac{dU_1}{dQ_1} = -2bQ_1 + a - bQ_2 - c.$$

So, the quantity Company 1 should produce to maximize profit is

$$Q_1^* = \frac{a-bQ_2-c}{2b}.$$

In game-theoretic terminology, when Company 2 produces Q_2 , the best response for Company 1 is to produce Q_1^* units determined by the above equation. A similar analysis concludes that when Company 1 produces Q_1 units, the best response for Company 2 is to produce Q_2^* units where

$$Q_2^* = \frac{a-bQ_1-c}{2b}.$$

Suppose these two companies have a long history of competing in this marketplace. They would each know a lot about the other company's actions, and we would expect these rational players to use strategies that are best responses to one another. In other words, we would expect these two strategies to form a Nash equilibrium. Suppose exactly this: The production quantities Q_1^* and Q_2^* form a Nash equilibrium.

If Q_1^* is a best response to Q_2^* , then from above, we have

$$Q_1^* = \frac{a-bQ_2^*-c}{2b}.$$

Similarly, if Q_2^* is a best response to Q_1^* , then we have

$$Q_2^* = \frac{a-bQ_1^*-c}{2b}.$$

Given these two equations in two unknowns, we can solve for the unknowns. Multiply the first equation through by $4b$ and $4bQ_1^* = 2a - 2bQ_2^* - 2c$. Multiply the second by $2b$, so $2bQ_2^* = a - bQ_1^* - c$. Now substitute to find $4bQ_1^* = 2a - (a - bQ_1^* - c) - 2c$. Thus $Q_1^* = \frac{a-c}{3b}$ and, similarly, $Q_2^* = \frac{a-c}{3b}$. So, in other words, we have just computed

that the only Nash equilibrium in this game is given by the strategy pair $Q_1^* = Q_2^* = \frac{a-c}{3b}$.

Since this is the *only* Nash equilibrium in the game, it is the anticipated behavior from two well-informed, rational companies. How does this compare with the monopoly situation? With just one company, $\frac{a-c}{2b}$ was the optimal quantity to produce. In the duopoly case, at the Nash equilibrium, each company produces $\frac{a-c}{3b}$, which means the total number of goods produced is $\frac{2(a-c)}{3b}$. So, the total number of goods produced is greater for the duopoly. As a result, the price will be lower when there are two companies competing than when there is just one.

The profit for Company 1 at the Nash equilibrium is

$$U_1^* = (a - c)Q_1^* - b(Q_1^* + Q_2^*)Q_1^* = \frac{(a - c)^2}{3b} - \frac{2(a - c)^2}{9b} = \frac{(a - c)^2}{9b}.$$

Likewise, Company 2 will have profits of $U_2^* = \frac{(a-c)^2}{9b}$. So, the combined profits of the two companies will be $\frac{2(a-c)^2}{9b}$, which is less than the monopoly profit of $\frac{(a-c)^2}{4b}$.

But if the two companies worked together, they could operate just like a monopoly. Each company, for instance, could produce half the optimal monopoly quantity, i.e. $\frac{a-c}{4b}$, sell at the monopoly price, and earn half the monopoly profit, $\frac{(a-c)^2}{8b}$. This is more profit than at the duopoly Nash equilibrium! Here is the striking reality: If both companies compete rationally, each profits strictly less than if they work together and limit production.

This dynamic parallels the Prisoner's Dilemma. Rational individual competition results in an outcome less desirable than what could be achieved through cooperation. This strategic dynamic appears frequently in business competition. In fact, antitrust laws exist to prevent companies from signing agreements to cooperate in this manner. The above example demonstrates why such agreements can improve profits for companies and how preventing this action can benefit consumers with lower prices.

In closing, note that the game at the start of this chapter is equivalent to the duopoly problem in the case when $a = 100$, $b = 1$, and $c = 0$. Specifically, in the game each player chooses a quantity Q_i and

the first player hopes to maximize a profit of $(100 - Q_1 - Q_2)Q_1$ while the second hopes to maximize a profit of $(100 - Q_1 - Q_2)Q_2$. These values are artificial, but analysis reveals the central feature. The only Nash equilibrium occurs when each player chooses $33\frac{1}{3}$, but both players get a higher payoff if each chooses 25.

Exercises

(1) For the matrix game $\begin{bmatrix} (3, 2) & (7, 1) & (4, 5) \\ (1, 3) & (2, 6) & (3, 3) \end{bmatrix}$, what are the payoff matrices for Rose and Colin?

(2) Consider the matrix game $\begin{bmatrix} (2, 4) & (4, 2) \\ (2, 8) & (6, 0) \\ (8, 2) & (4, 4) \end{bmatrix}$ and the mixed strategies

$\mathbf{p} = \begin{bmatrix} 1/2 & 0 & 1/2 \end{bmatrix}$ for Rose and $\mathbf{q} = \begin{bmatrix} 1/4 \\ 3/4 \end{bmatrix}$ for Colin. Find the expected payoffs for both players in each case below:

- (a) Rose plays row 1 and Colin plays column 2.
- (b) Rose plays row 2 and Colin plays $\mathbf{q}$.
- (c) Rose plays $\mathbf{p}$ and Colin plays column 1.
- (d) Rose plays $\mathbf{p}$ and Colin plays $\mathbf{q}$.

(3) Find all Nash equilibria for the dilemmas from Section 7.2.

(4) Find a Nash equilibrium for each matrix game below:

$$(a) \begin{bmatrix} (3, 2) & (0, 4) \\ (1, 6) & (5, 2) \end{bmatrix}, \quad (b) \begin{bmatrix} (5, 8) & (2, 3) \\ (2, 5) & (1, 2) \end{bmatrix}, \quad (c) \begin{bmatrix} (9, 1) & (2, 4) \\ (5, 8) & (7, 3) \end{bmatrix}.$$

(5) Use a combination of eliminating dominated strategies and the methods from this chapter to find a Nash equilibrium:

$$(a) \begin{bmatrix} (5, 1) & (3, 3) & (2, 9) \\ (3, 4) & (6, 6) & (3, 5) \\ (2, 5) & (2, 4) & (8, 6) \end{bmatrix}, \quad (b) \begin{bmatrix} (1, 9) & (5, 5) & (3, 6) \\ (9, 3) & (6, 2) & (2, 5) \\ (2, 8) & (1, 2) & (6, 3) \\ (7, 6) & (2, 7) & (1, 4) \end{bmatrix}.$$

- (6) Rose is playing $\mathbf{p} = \begin{bmatrix} 2/5 & 0 & 0 & 3/5 \end{bmatrix}$ and Colin is playing $\mathbf{q} = \begin{bmatrix} 1/3 \\ 2/3 \\ 0 \end{bmatrix}$

in the matrix game $\begin{bmatrix} (1, 2) & (4, 5) & (2, 3) \\ (5, 1) & (2, 2) & (0, 4) \\ (6, 1) & (0, 2) & (5, 0) \\ (7, 2) & (1, 0) & (3, 1) \end{bmatrix}$.

- (a) What are Rose's best responses to $\mathbf{q}$?
 (b) What are Colin's best responses to $\mathbf{p}$?
 (c) Do $\mathbf{p}$ and $\mathbf{q}$ form a Nash equilibrium?
 (7) For every $x, y, z \in \mathbb{R}$, find a Nash equilibrium:

(a) $\begin{bmatrix} (4, x) & (2, 0) \\ (1, 2) & (y, 3) \end{bmatrix}$, (b) $\begin{bmatrix} (5, x) & (1, -1) \\ (y, 0) & (3, 1) \end{bmatrix}$, (c) $\begin{bmatrix} (x, y) & (1, 3) \\ (0, 0) & (z, 6) \end{bmatrix}$.

- (8) Male dung beetles come in two types, some have horns (H) and some have no horns (N). Suppose we are considering pairwise competition for a resource. Two horned beetles will battle each other for this resource, earning an average payoff of -20 each. If a horned beetle encounters a nonhorned beetle, the horned beetle will win the resource, earning 40 , while the nonhorned beetle will get -5 . Finally, if two nonhorned beetles compete, they will split the resource for a payoff of 20 each. Based on this simple model, what percentage of male beetles would you expect to be horned?
- (9) In baseball, the batter has an advantage when he has the opposite handedness from the pitcher (e.g. batter is left-handed and pitcher is right-handed). Suppose that the batting average in Major League Baseball according to handedness is given by the following matrix:

		Pitcher	
		L	R
Batter	L	.255	.275
	R	.275	.265

So, on average when a right-handed batter faces a right-handed pitcher, the batter gets a hit 26.5% of the time. If we focus purely on this statistic, what percentage of pitchers and what percentage of hitters would you expect to be left-handed?

- (10) Show that every symmetric 2×2 matrix game has a symmetric Nash equilibrium.
- (11) This problem concerns Nash equilibria and domination.
- Show that whenever $\mathbf{p}$ and $\mathbf{q}$ form a Nash equilibrium in a matrix game, neither $\mathbf{p}$ nor $\mathbf{q}$ calls on a pure strategy which is strictly dominated.
 - Construct a matrix game A with a Nash equilibrium consisting of $\mathbf{p}$ and $\mathbf{q}$ for which both $\mathbf{p}$ and $\mathbf{q}$ call on pure strategies which are (not strictly) dominated.
- (12) Find a 2×2 matrix game A which has a Nash equilibrium $\mathbf{p}, \mathbf{q}$ where $\mathbf{p}$ is pure but $\mathbf{q}$ is not pure. Show that A also has a Nash equilibrium consisting of two pure strategies.
- (13) Consider the Hawk-Dove game, but suppose that instead of the species divided between hawks and doves, we have two variants: aggressive and passive. The aggressive variant uses the hawk strategy with probability $2/3$ and the dove strategy with probability $1/3$, while the passive variant uses the dove strategy with probability $2/3$ and the hawk strategy with probability $1/3$. How will the species be divided between the aggressive and passive variants?
- (14) This problem explores Fisher's Principle, a famous game-theoretic explanation for the roughly 50/50 split of males/females seen in many species (even in species with different reproductive dynamics). Consider three generations of a species, and suppose that in all three generations the percentage of females is higher than the percentage of males. Let f_2, m_2 be the number of females and males in the second generation (so $f_2 > m_2$) and let n_3 be the total number of individuals in the third generation.
- Determine the expected number of children for a male and for a female in the second generation.
 - Would an individual in the first generation expect more grandchildren from a male or female child?
 - Suppose that our population is made up of two variants, ϕ and μ , where ϕ individuals are biased toward having more female children and μ individuals are biased toward having more male children. Which variant in the first generation is likely to be most

successful? How might you expect this to change the male/female balance of the overall population over time?

- (15) In this problem Company 1 and Company 2 are competing in a duopoly as in Section 8.3. As before, Company i can choose what quantity Q_i it wishes to produce, and the production cost for company i , C_i , is a function of Q_i . Find an equilibrium point in each variation below.
- (a) The cost functions are $C_1 = 40Q_1$ and $C_2 = 60Q_2$ and the price is given by $P = 200 - Q_1 - Q_2$.
 - (b) The cost functions are $C_i = cQ_i$ for $i = 1, 2$ and the price is $P = a - b(Q_1 + Q_2)^2$.
 - (c) The cost functions are $C_i = cQ_i$ for $i = 1, 2$, and the prices are different. Company 1 sells at the price $P_1 = a - b(2Q_1 + Q_2)$ while Company 2 sells at the price $P_2 = a - b(Q_1 + 2Q_2)$.
- (16) Two hotels on the same road compete for customers. We model the road as the line segment $[0, 1]$ and assume that Hotel i is at position x_i where $0 < x_1 < x_2 < 1$. Each hotel sets a room rate and we will assume that Hotel i charges price P_i . A customer at a location x will have to pay a transportation cost of $c|x - x_i|$ to get to Hotel i so the total expense for such a consumer to stay at Hotel i will be $P_i + c|x - x_i|$. Assume that every consumer chooses the hotel only to minimize total expense.
- (a) Find a number $0 \leq x^* \leq 1$ so that every consumer at a location $x < x^*$ will go to Hotel 1 and every consumer at a location $x > x^*$ will go to Hotel 2.
 - (b) If an interval of length ℓ goes to Hotel i , its revenue is $P_i \ell$. Use x^* to determine the revenues for each hotel.
 - (c) Assume each hotel wants to maximize revenue and find a Nash equilibrium for the prices P_1 and P_2 .

- (17) Prove that $\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$ is the unique symmetric Nash equilibrium for the Lizard Game.

- (18) Let A be a symmetric matrix game with payoff matrix R for the row player, and assume that R is invertible.
- (a) Assuming there exists a symmetric Nash equilibrium which calls on every pure strategy, show that it must be unique, and find a formula for it.
 - (b) When does such an equilibrium exist?
 - (c) Again assuming such an equilibrium exists, determine when it is an evolutionarily stable strategy.
- (19) Show that Nash's Equilibrium Theorem implies von Neumann's Minimax Theorem.
- (20) Rose and Colin are competing in an auction where the prize is \$100. Each player bids a whole dollar amount which is at least \$1 and at most \$99. Both players lose the money they bid, but if one player bids a higher number than the other, this player gets the \$100 prize. In the case of a tie, neither player gets the prize. Find a symmetric Nash equilibrium for this game which calls on every possible pure strategy.

Chapter 9

Nash's Equilibrium Theorem

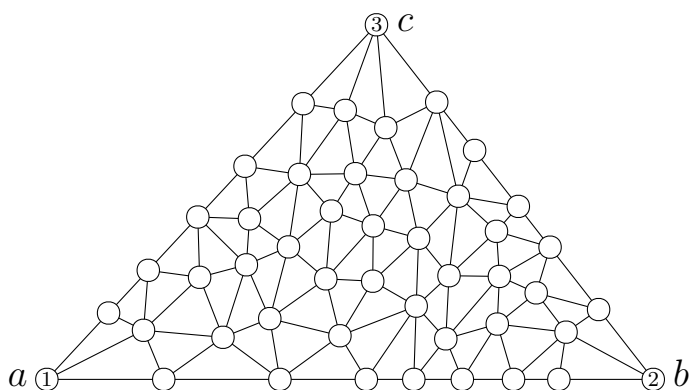


Figure 9.1. Triangle Solitaire

This chapter presents a proof of Nash's Equilibrium Theorem. The intricate proof of this deep result even involves the game of Triangle Solitaire shown in Figure 9.1! For accessibility, the argument here focuses on the special case of 2×2 games. Each result cleanly generalizes to higher dimensions and all those details appear in Appendix C.

We now know how to calculate a Nash equilibrium in a 2×2 matrix game, and it's easy to hope that higher-dimensional cases could be

handled similarly, perhaps with more complicated algebraic expressions. This is simply not the case. Analogously, the familiar quadratic formula finds roots of a second-degree polynomial and there are similar formulae for finding roots of polynomials of degrees three and four. However, there is no such formula for a fifth-degree polynomial ... the situation is too complex to admit simple algebraic solutions. It is nevertheless still possible to reason and prove things about the roots of these polynomials.¹ Likewise, there are explicit formulae for finding Nash equilibria in small cases, but such conveniences do not exist for higher dimensions. Hence, we take a less direct approach to proving the existence of a Nash equilibrium. Figure 9.2 outlines the major pieces of the argument.

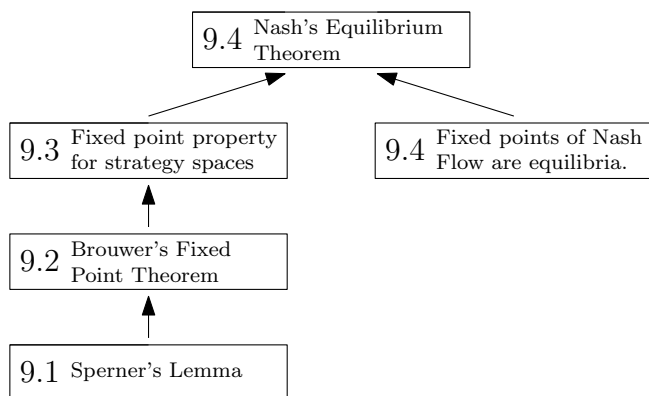


Figure 9.2. Steps in proving Nash's Theorem

We begin with the following one-player game.

Game 9.1 (Triangle Solitaire). This game uses the triangle in Figure 9.1.² To play, write a 1, 2, or 3 in each of the empty circles according to the rule that every circle on the side between the corners labeled i and j must get either an i or a j . So, for instance, every circle on the bottom of the big triangle must be filled with either a 1 or a 2, but those

¹For instance, there is an easy proof that a polynomial of degree 5 must have at least one root. This follows from the fact that a polynomial $f(x)$ of degree 5 either has $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$ or has $\lim_{x \rightarrow \infty} f(x) = -\infty$ and $\lim_{x \rightarrow -\infty} f(x) = \infty$. In either case, we can deduce that f has both positive and negative values, so by the Intermediate Value Theorem it must have a root.

²A larger version of this game board can be found at the end of the book and also online at www.ams.org/bookpages/stml-80.

in the interior can be labeled 1, 2, or 3. The goal in this game is to minimize the number of little triangles with three differently labeled corners. How well can you do?

9.1. Sperner's Lemma

This section builds to a proof of a beautiful general result due to Emanuel Sperner. Despite its significance, this result is traditionally called a lemma. We begin with a one-dimensional version that involves a subdivided line segment. The two-dimensional version calls on the triangle in Figure 9.1.

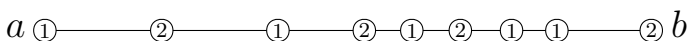


Figure 9.3. Sperner's Lemma in 1D

Lemma 9.2 (Sperner 1D). *Let $\overline{ab}$ be a line segment that is subdivided into edges by adding some new vertices. Assume that each vertex is labeled according to the following rules:*

- *a is labeled 1 and b is labeled 2.*
- *Every other vertex is labeled 1 or 2.*

Then there exist an odd number of edges with endpoints of both numbers.

Proof. Imagine starting at a and walking along the line segment to b . Starting at a vertex labeled 1 and walking to one labeled 2 means switching numbers an odd number of times. Thus, there are an odd number of edges with ends of different labels. $\square$

This one-dimensional result figures into the following proof that the game of Triangle Solitaire will always have at least one small triangle with all three labels.

Lemma 9.3 (Sperner 2D). *Let $\triangle abc$ be subdivided into small triangles by adding new vertices and edges. Assume that each vertex is given a label according to the following rules:*

- *The labels on a, b, c are (respectively) 1, 2, 3.*
- *Every vertex on $\overline{ab}$ is labeled 1 or 2.*

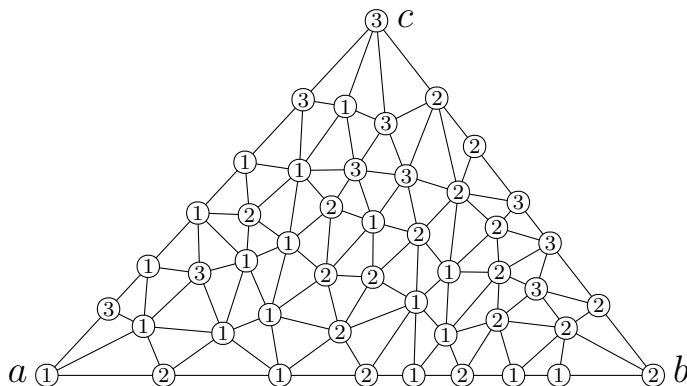


Figure 9.4. Sperner in 2D

- Every vertex on $\overline{bc}$ is labeled 2 or 3.
- Every vertex on $\overline{ac}$ is labeled 1 or 3.
- Every vertex inside $\triangle abc$ is labeled 1, 2, or 3.

Then there are an odd number of small triangles with vertices of all three labels.

Proof. Imagine Figure 9.4 as a floor plan for a house, so each little triangle is a room with three walls. Then add a door along every wall (edge) that has one end labeled 1 and the other labeled 2. Now consider the possibilities. A room with at least one door must have one vertex of label 1 and one of label 2. If the third vertex has label 3, then the room has just one door. Otherwise, the third vertex is labeled either 1 or 2, and, in either case, the room will have exactly two doors. No room has more than two doors, so a person walking through the house must walk only forward or backwards along a pathway. Figure 9.5 highlights the pathways in this labeling.

Some rooms have no doors and some pathways form cycles—these are irrelevant. Focus on the pathways that have a beginning and an end. The first and last door on any such pathway must either be a door to the outside (along $\overline{ab}$) or a door into a room with just one door. Since each such pathway has two ends, the total number of doors to the outside plus the number of rooms with exactly one door is even. The number of

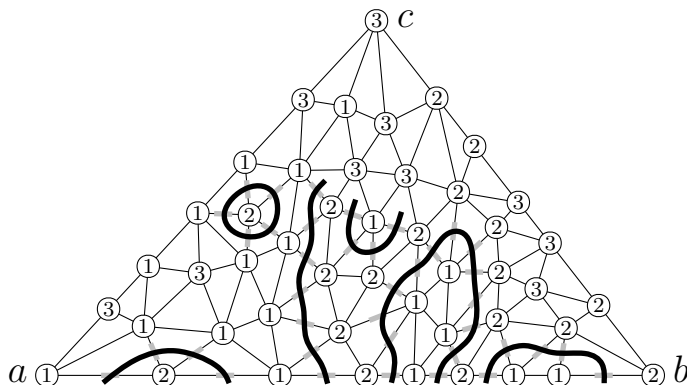


Figure 9.5. Pathways

doors to the outside must be odd by Sperner 1D. Therefore, the number of rooms with exactly one door must also be odd. In particular, there is at least one room with just one door. This is a small triangle with vertices of all three labels! $\square$

Higher Dimensions. To extend Sperner's Lemma to higher dimensions requires sets in $\mathbb{R}^n$ that behave like line segments in $\mathbb{R}^1$ and triangles in $\mathbb{R}^2$. The notions of hyperplane and convex hull that we saw with von Neumann's Theorem in Chapter 6 recur here. One way to describe a triangle is as the convex hull of 3 points in $\mathbb{R}^2$ that do not lie on a common line. This idea generalizes to the following definition of an n -dimensional simplex.

Definition 9.4. An n -simplex is the convex hull of $n + 1$ points in $\mathbb{R}^n$ that do not lie on a common hyperplane.

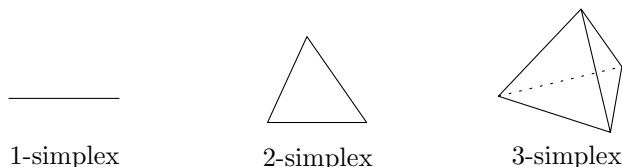


Figure 9.6. Small-dimensional simplexes

Observe that the 1D Sperner Lemma involved a 1-simplex that was subdivided and had vertices labeled 1 and 2. The 2D Sperner Lemma involved a 2-simplex that was subdivided with vertices labeled 1, 2, and 3. More generally, the n -dimensional Sperner Lemma uses an n -simplex that has vertices labeled $1, 2, \dots, n + 1$.

Lemma 9.5 (Sperner). *Consider an n -simplex given as the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_{n+1}$ that has been subdivided into small simplexes. Suppose that each vertex is given a label from $1, 2, \dots, n + 1$ satisfying the following rule:*

- *If a vertex has label i , then it does not lie in the convex hull of the points $\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}_{i+1}, \dots, \mathbf{x}_{n+1}$.*

Then the number of small simplexes with all $n + 1$ labels is odd.

The proof of this n -dimensional result is a generalization of the proof for the 2D version and appears explicitly in Appendix C.

9.2. Brouwer's Fixed Point Theorem

Some subsets of $\mathbb{R}^n$ exhibit a special property called the fixed point property. Perhaps surprisingly, Sperner's Lemma features prominently in the proof of a topological result called Brouwer's Fixed Point Theorem concerning this characteristic. This result is essential in proving Nash's Equilibrium Theorem.

Fixed Points. This section focuses on subsets $X \subseteq \mathbb{R}^n$ and functions of the form $f : X \rightarrow X$. Define a *fixed point* of such a function f to be a point $x \in X$ for which $f(x) = x$ (i.e. it is a point that is fixed by the function).

Examples 9.6.

- (1) The function $f_1 : [0, 1] \rightarrow [0, 1]$ given by $f_1(x) = 1 - x$ has $1/2$ as a fixed point since $f_1(1/2) = 1/2$.
- (2) The function $f_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $f_2(x, y) = (y + 1, x - 1)$ has $(1, 0)$ as a fixed point since $f_2(1, 0) = (1, 0)$.
- (3) The function $f_3 : \mathbb{R} \rightarrow \mathbb{R}$ given by $f_3(x) = e^x$ has no fixed point since there is no solution to the equation $x = e^x$.

Moving forward, it will be useful to know if *all* continuous functions on a particular set have a fixed point.

Definition 9.7. We say that a set $X \subseteq \mathbb{R}^n$ has the *fixed point property* if every continuous function $f : X \rightarrow X$ has a fixed point.

Note that this definition only concerns continuous functions. To prove that a set X does *not* have the fixed point property, simply find one continuous function from X to itself that has no fixed points.

Examples 9.8.

- (1) Consider a circle $C \subseteq \mathbb{R}^2$. Now, choose an angle $0 < \theta < 2\pi$ and define a function $f : C \rightarrow C$ by the rule that f rotates each point around the circle by an angle of θ . Since this is a continuous function with no fixed point, we conclude that C does not have the fixed point property.
- (2) Consider the set consisting of the entire real number line $\mathbb{R}$. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x + 1$ is a continuous function with no fixed point. Therefore, $\mathbb{R}$ does not have the fixed point property.

It is considerably more difficult to prove that a set X does have the fixed point property since this means showing that *every* continuous function $f : X \rightarrow X$ has a fixed point. In the case when X is a closed interval, we have good tools to solve this problem.

Theorem 9.9 (Brouwer 1D). *The closed interval $[0, 1]$ has the fixed point property.*

Proof. We must show that every continuous function from the interval $[0, 1]$ to itself has a fixed point. Let $f : [0, 1] \rightarrow [0, 1]$ be such a function and define the new function $g : [0, 1] \rightarrow \mathbb{R}$ by the rule $g(x) = x - f(x)$. Now g is also a continuous function and $g(0) = -f(0) \leq 0$ while $g(1) = 1 - f(1) \geq 0$. It follows from the Intermediate Value Theorem that there exists a point c in $[0, 1]$ so that $g(c) = 0$. This point c satisfies $0 = g(c) = c - f(c)$, so it is a fixed point of the function f . Since this holds for every continuous function $f : [0, 1] \rightarrow [0, 1]$, we have established that $[0, 1]$ has the fixed point property, as desired. $\square$

Next, we will prove that a certain triangle also has the fixed point property.

Triangles. Define Δ_2 to be the solid planar triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$. More formally,

$$\Delta_2 = \{(x, y) \in \mathbb{R}^2 \mid x, y \geq 0 \text{ and } x + y \leq 1\}.$$

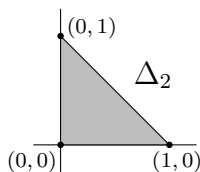


Figure 9.7. The triangle Δ_2

To prove that Δ_2 has the fixed point property, we need to consider continuous functions $f : \Delta_2 \rightarrow \Delta_2$. While it's common to visualize functions $f : \mathbb{R} \rightarrow \mathbb{R}$ using a graph, this technique is not so helpful for picturing a function $f : \Delta_2 \rightarrow \Delta_2$. Instead, draw Δ_2 and draw a collection of arrows to indicate the output of particular points under f (an arrow indicates that the initial point (x, y) is mapped to the terminal point $f(x, y)$). Since only continuous functions concern us, these arrows give a good idea of how f acts on nearby points, too.

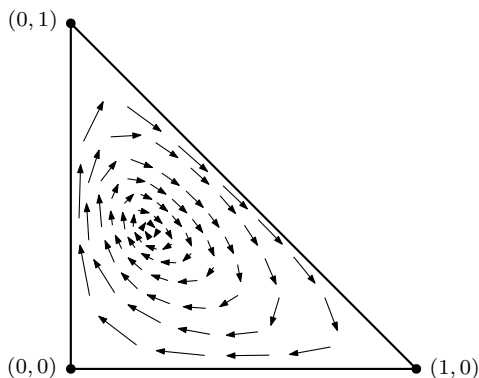


Figure 9.8. A continuous function $f : \Delta_2 \rightarrow \Delta_2$

Let (x, y) be a point in the triangle Δ_2 and suppose that $f(x, y) = (x', y')$. Define $(x', y') - (x, y) = (x' - x, y' - y)$ to be the *direction of* (x, y) . Observe that in Figure 9.8, each arrow indicates the direction of its initial point.

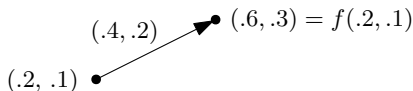


Figure 9.9. The direction of $(.2, .1)$ is $(.4, .2)$.

Now for an unusual move. Instead of considering all possible directions for a point, divide the directions into three groups: West, Southeast, and Northeast. This divides the points in the triangle into three sets that give a very rough indication of where the points go when we apply the function f . Formally, assign each point in Δ_2 a label 1 (for West), 2 (for Southeast), and 3 (for Northeast) according to the following rule: If $(x', y') = f(x, y)$, then

$$(x, y) \text{ has label } \begin{cases} 1 & \text{if } x' < x, \\ 2 & \text{if } x' \geq x \text{ and } y' < y, \\ 3 & \text{if } x' \geq x \text{ and } y' \geq y. \end{cases}$$

Figure 9.10 helps visualize these directions. In words, when the direction associated with a point has an angle of θ , the label will be a 1 if $\frac{\pi}{2} < \theta \leq \frac{3\pi}{2}$, a 2 if $\frac{3\pi}{2} < \theta < 2\pi$, and a 3 if $0 \leq \theta \leq \frac{\pi}{2}$.

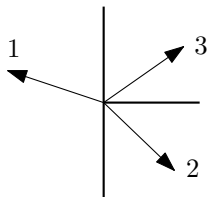


Figure 9.10. Assigning labels to directions

Consider a point (x, y) that is not a fixed point, and suppose it has a direction with angle $\frac{\pi}{4}$. By the above labeling scheme, this point will have label 3. Because f is continuous, every point near (x, y) will have a similar direction. So all points sufficiently close to (x, y) also have label 3.

Next suppose that (x, y) has direction with angle $\frac{\pi}{2}$ and is thus assigned label 3. This point would be on the boundary between points labeled 1 and 3, so all points sufficiently close would be labeled 1 or 3. Similarly if (x, y) has direction with angle $-\frac{\pi}{2}$, all points sufficiently close would be labeled 1 or 2, and if the direction has angle 0, all points sufficiently close would be labeled 2 or 3. The only way for (x, y) to have points of all three labels arbitrarily close to it is for (x, y) to be a fixed point.

The following proof utilizes exactly this feature of the labeling. For any continuous function f , we ignore everything except the associated labeling of the points. We show that there exists a point (x, y) with points of all three labels arbitrarily close to it, and from this we deduce that f has (x, y) as a fixed point.

Theorem 9.10 (Brouwer 2D). *The triangle Δ_2 has the fixed point property.*

Proof. Let $f : \Delta_2 \rightarrow \Delta_2$ be continuous, and label the points of Δ_2 in accordance with the above description.

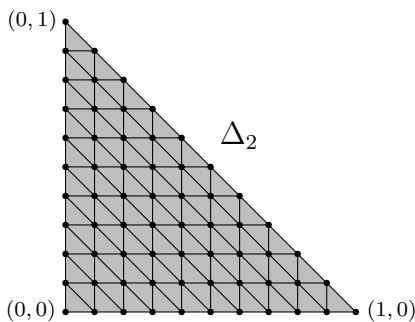


Figure 9.11. A subdivision of Δ_2

Claim. *For every $\ell \geq 0$ either there is a fixed point or a point in Δ_2 with distance $\leq \frac{1}{2^\ell}$ to points of all three labels.*

Subdivide the triangle Δ_2 into small triangles using a fine mesh (as in Figure 9.11) so that the center of each small triangle is a distance at most $\frac{1}{2^\ell}$ from each of its three vertices. Now consider all of the vertices of this subdivision. If one of them is a fixed point, then there is nothing left

to prove, so assume no vertex is a fixed point. Consider the 1, 2, 3 labeling of these vertices and observe that this labeling satisfies the assumptions of Sperner 2D (with $a = (1, 0)$, $b = (0, 1)$, and $c = (0, 0)$). The point $(0, 0)$, for instance, must get a label of 3, the point $(1, 0)$ must get a label of 1, and the points on the line segment between these two all have the form $(x, 0)$, so they will be labeled 1 or 3. It follows that there is a small triangle with vertices of all three labels. By construction, the center of that triangle is a distance $\leq \frac{1}{2^\ell}$ from points of all three labels.

By applying the above claim for $\ell = 1, 2, 3, \dots$ we either find a fixed point (thus completing the proof) or we generate a sequence of points in the triangle $(x_1, y_1), (x_2, y_2), \dots$ so that (x_i, y_i) is a distance $< 1/2^i$ from points of all three labels. It follows³ that there is a particular point (x^*, y^*) in the triangle that has points of all three labels arbitrarily close to it, so (x^*, y^*) is a fixed point. $\square$

Higher Dimensions. To generalize this to higher dimensions we introduce some special n -simplexes defined as

$$\Delta_n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1, \dots, x_n \geq 0 \text{ and } x_1 + \dots + x_n \leq 1\}.$$

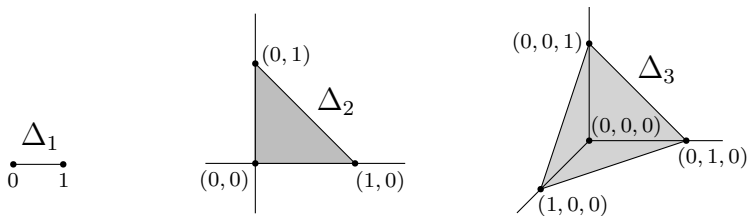


Figure 9.12. More simplexes

So Δ_1 is the line segment $[0, 1]$ and Theorem 9.9 showed that it has the fixed point property. Theorem 9.10 proved that Δ_2 also has the fixed point property. The Brouwer Fixed Point Theorem asserts that this holds true in general.

Theorem 9.11 (Brouwer). *The simplex Δ_n has the fixed point property for every $n \geq 1$.*

³Since Δ_2 is compact, this sequence has a convergent subsequence.

The proof of this more general result is a straightforward extension of the proof of Theorem 9.10 that calls upon the general Sperner Lemma. See Appendix C for details.

9.3. Strategy Spaces

So far in this chapter, we have no hint whatsoever of a matrix game! This is about to change: We will use certain subsets of $\mathbb{R}^n$ to describe all possible pairs of mixed strategies for Rose and Colin and then apply the Brouwer Fixed Point Theorem to prove that these subsets have the fixed point property. This constitutes the main step in the anticipated proof of Nash's Equilibrium Theorem.

2×2 matrices. Suppose that Rose and Colin are playing a 2×2 matrix game A . Recall that a mixed strategy for Rose is a vector of the form $\mathbf{p} = [p \quad 1 - p]$ where $0 \leq p \leq 1$. Although there are two entries in this vector, the single real number p entirely determines $\mathbf{p}$. Likewise, a mixed strategy for Colin is a vector $\mathbf{q} = \begin{bmatrix} q \\ 1 - q \end{bmatrix}$, but just the single number q is enough to describe this strategy completely. So we may identify Rose's mixed strategies with numbers $0 \leq p \leq 1$ and Colin's mixed strategies with numbers $0 \leq q \leq 1$.

How can we simultaneously describe a mixed strategy for Rose and one for Colin? In the case of a 2×2 matrix game as above, a pair of real numbers (p, q) with $0 \leq p \leq 1$ and $0 \leq q \leq 1$ is enough. This set of ordered pairs makes up what is formally called a strategy space.

Definition 9.12. The *strategy space* of a 2×2 matrix game is

$$S_{2,2} = \{(p, q) \in \mathbb{R}^2 \mid 0 \leq p \leq 1 \text{ and } 0 \leq q \leq 1\}.$$

The strategy space $S_{2,2}$ is the familiar set of points that make up the unit square. Associating each point (p, q) in $S_{2,2}$ with the pair of mixed strategies $[p \quad 1 - p]$ for Rose and $\begin{bmatrix} q \\ 1 - q \end{bmatrix}$ for Colin equips us with a nice geometric interpretation of all possible pairs of strategies.

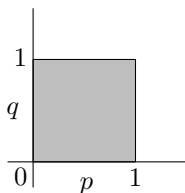


Figure 9.13. The strategy space of a 2×2 game

Strategy spaces will reappear with Nash flow in the following section. For now, recognize that the strategy space $S_{2,2}$ is a square that encodes pairs of mixed strategies for the players. The desired result here is that the strategy space of every matrix game has the fixed point property. We will need another concept from the world of topology to achieve this.

Topological Equivalence. Thanks to the Brouwer Fixed Point Theorem, we know that the triangle Δ_2 has the fixed point property. To deduce from this the fact that square $S_{2,2}$ also has the fixed point property will involve showing that Δ_2 and $S_{2,2}$ have a certain kind of equivalence—topological equivalence—and that any topologically equivalent sets either both have the fixed point property or neither does.

Central to this key notion of equivalence is the definition of a bijection. A function $f : X \rightarrow Y$ is a bijection if every $y \in Y$ is the image of exactly one point $x \in X$. So, a bijection gives a correspondence that pairs up the points between X and Y . Assuming f is a bijection, define an inverse function $f^{-1} : Y \rightarrow X$ by the rule that $f^{-1}(y) = x$ where x is the unique point in X for which $f(x) = y$. Bijections are precisely those functions that have inverses.

Definition 9.13. We say that two sets $X, Y \subseteq \mathbb{R}^n$ are *topologically equivalent* if there is a bijection $g : X \rightarrow Y$ with the property that both g and g^{-1} are continuous.

Example 9.14. Consider the intervals $[0, 1]$ and $[0, 2]$ in $\mathbb{R}$. The function $g : [0, 1] \rightarrow [0, 2]$ given by $g(x) = 2x$ is a bijection between $[0, 1]$ and $[0, 2]$. Note that $g^{-1} : [0, 2] \rightarrow [0, 1]$ is given by $g^{-1}(x) = \frac{1}{2}x$. Since both g and g^{-1} are continuous, $[0, 1]$ and $[0, 2]$ are topologically equivalent.

Notably, the following proposition shows that whenever one of two topologically equivalent sets has the fixed point property, the other does,

too. For example, Theorem 9.9 proved that the interval $[0, 1]$ has the fixed point property, so the topological equivalence from the previous example means that $[0, 2]$ also has the fixed point property.

Proposition 9.15. *If X and Y are topologically equivalent and X has the fixed point property, then Y has the fixed point property.*

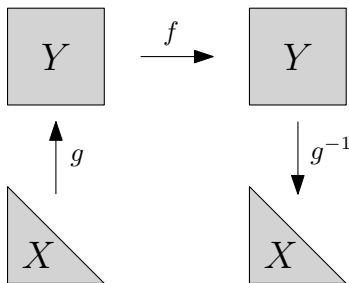


Figure 9.14. Inheriting the fixed point property

Proof. Assume that X and Y are topologically equivalent and also that X has the fixed point property. To prove that Y has the fixed point property, let $f : Y \rightarrow Y$ be an arbitrary continuous function. Since X and Y are topologically equivalent, there exists a continuous bijection $g : X \rightarrow Y$ so that g^{-1} is also continuous. Now combine the functions g, f , and g^{-1} as in Figure 9.14. More precisely, construct a new function from X to itself given by the rule

$$x \rightarrow g^{-1}(f(g(x))).$$

Since this function is continuous and X has the fixed point property, it must have a fixed point. So, we may choose a point $x \in X$ for which $x = g^{-1}(f(g(x)))$. Next apply the function g to both sides of this equation to get

$$g(x) = f(g(x)).$$

Now set $y = g(x)$ and observe that $y \in Y$ and $f(y) = y$. Thus f has a fixed point and, since f was an arbitrary continuous function, we conclude that Y has the fixed point property. $\square$

With these tools in hand, we are now ready to prove the main theorem for this section.

Corollary 9.16. *The strategy space $S_{2,2}$ has the fixed point property.*

Proof. With Theorem 9.10 and the above proposition, we can prove that $S_{2,2}$ has the fixed point property by showing that $S_{2,2}$ is topologically equivalent to Δ_2 . This follows from the continuous functions $g : \Delta_2 \rightarrow S_{2,2}$ and $g^{-1} : S_{2,2} \rightarrow \Delta_2$ defined by

$$\begin{aligned} g(x, y) &= \left(x + \frac{x+y-|x-y|}{2}, y + \frac{x+y-|x-y|}{2} \right), \\ g^{-1}(x, y) &= \left(x - \frac{x+y-|x-y|}{4}, y - \frac{x+y-|x-y|}{4} \right). \end{aligned} \quad \square$$

There is nothing particularly special about this triangle and square. It is tangential to our investigations here, but, in fact, every solid polygon and every circle plus its interior are topologically equivalent to one another and to many more shapes as well.⁴

Higher Dimensions. In a 2×2 matrix game, a mixed strategy for Rose has the form $[p \quad 1-p]$ so it can be described with a single real number p . More generally, in an $m \times n$ matrix game, Rose's mixed strategies will have the form $\mathbf{p} = [p_1 \quad \cdots \quad p_m]$. Here again there is some redundancy. Since $p_1 + \cdots + p_m = 1$, the last coordinate of Rose's mixed strategy can be deduced from the earlier ones: $p_m = 1 - (p_1 + \cdots + p_{m-1})$. Therefore, each possible mixed strategy for Rose can be associated with a point $(p_1, \dots, p_{m-1})$ where $p_1, \dots, p_{m-1} \geq 0$ and $p_1 + \cdots + p_{m-1} \leq 1$. We will adopt this convenient description of Rose's mixed strategies. Cor-

respondingly, each mixed strategy $\mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$ for Colin can be associated with the point $(q_1, \dots, q_{n-1})$, which will satisfy $q_1, \dots, q_{n-1} \geq 0$ and $q_1 + \cdots + q_{n-1} \leq 1$.

With this interpretation, define the *strategy space* of a general $m \times n$ matrix game A to be

$$\begin{aligned} S_{m,n} &= \{(p_1, \dots, p_{m-1}, q_1, \dots, q_{n-1}) \in \mathbb{R}^{m+n-2} \mid \\ &\quad p_1, \dots, p_{m-1}, q_1, \dots, q_{n-1} \geq 0, \sum_{i=1}^{m-1} p_i \leq 1, \text{ and } \sum_{j=1}^{n-1} q_j \leq 1\}. \end{aligned}$$

Just as in the 2×2 case, each point in the strategy space corresponds to a pair of strategies, one for Rose and one for Colin. In the same way that

⁴More generally, any two closed convex sets in $\mathbb{R}^n$ with nonzero volume are topologically equivalent.

we used the 2D version of the Brouwer Fixed Point Theorem to prove that $S_{2,2}$ has the fixed point property, the general version of Brouwer's Theorem can be used to prove that $S_{m,n}$ has the fixed point property. Full details of this similar argument appear in Appendix C.

Lemma 9.17. *For every pair of positive integers m, n , the strategy space $S_{m,n}$ has the fixed point property.*

9.4. Nash Flow and the Proof

This section finally concludes our proof of Nash's Equilibrium Theorem. We have already done the hard work of proving that the strategy space has the fixed point property. What remains is to introduce a continuous function called Nash flow on the strategy space and use the fixed point property to locate a Nash equilibrium.

Nash Flow. Imagine that Rose and Colin are playing a game of Ping-Pong. Rose notices that she is somewhat more successful today when she hits to Colin's backhand rather than his forehand. It would be silly for Rose to respond by hitting every single ball to Colin's backhand—he would quickly realize and exploit her strategy. It would make more sense for Rose instead to adjust and play a slightly higher percentage of her shots to his backhand.

This situation has a very natural and important game-theoretic analogue. Suppose that Rose and Colin are playing a 2×2 matrix game, with Rose using the strategy $\mathbf{p} = [p \quad 1 - p]$ and Colin using the strategy $\mathbf{q} = \begin{bmatrix} q \\ 1 - q \end{bmatrix}$. If Rose observes that she does better playing the second row than playing the first against strategy $\mathbf{q}$, she might decide to modify her strategy $\mathbf{p}$ to play row 2 more often. As in the Ping-Pong game above, a subtle adaptation makes more sense than an abrupt change to playing the second row 100% of the time. Nash flow exhibits this idea of making a small adjustment to improve.

For a 2×2 matrix game A , Nash flow is a function, denoted f_A , that maps a point (p, q) in the strategy space to another point in the strategy space $f_A(p, q) = (p', q')$. Formally, then, $f_A : S_{2,2} \rightarrow S_{2,2}$. We think of this as updating Rose's and Colins' strategies $[p \quad 1 - p]$ and $\begin{bmatrix} q \\ 1 - q \end{bmatrix}$

to new strategies $[p' \ 1 - p']$ and $\begin{bmatrix} q' \\ 1 - q' \end{bmatrix}$. Each player modifies his or her initial strategy to do better against the other player.

Example 9.18. Figure 9.15 depicts the Nash flow function f_A for the matrix $A = \begin{bmatrix} (1, 2) & (2, 3) \\ (0, 3) & (3, 1) \end{bmatrix}$.⁵

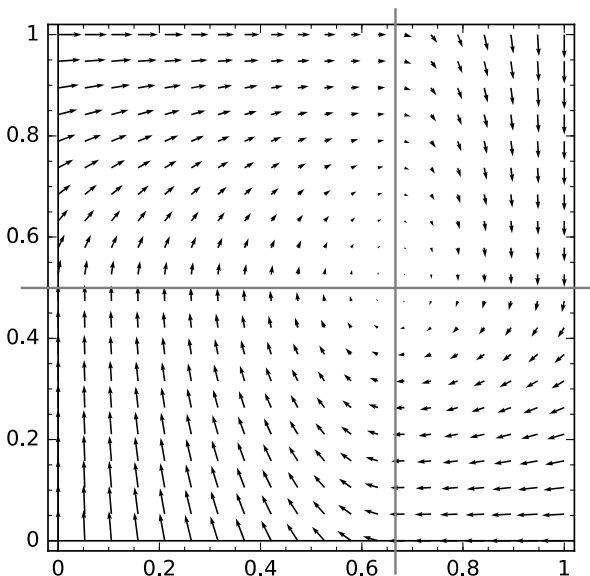


Figure 9.15. Nash flow for a 2×2 matrix

In Figure 9.15, arrows point to the right in the region where $q > 1/2$. This corresponds to Rose increasing the probability that she plays the first row. Rose indeed gets a better payoff if she plays the first row instead of the second row when $q > 1/2$. On the other hand, in the region where $q < 1/2$, the vectors are all directed to the left, indicating Rose's adjustment to decrease the probability that she chooses the first row. When $q < 1/2$, Rose in fact gets a better payoff playing the second row instead of the first. Near the point $(2/3, 1/2)$, the arrows shrink to length zero. This

⁵This plot was created using the computing program Sage. See www.ams.org/bookpages/stml-80 for a link to the code.

suggests that Rose has little incentive to modify her strategy in the region. Indeed $(2/3, 1/2)$ is the unique fixed point of the Nash flow f_A , corresponding to the fact that $p = 2/3$ and $q = 1/2$ form the unique Nash equilibrium of A , so neither player has any incentive to change from this point.

Definition 9.19 (Nash Flow). The Nash flow for a 2×2 matrix game A is a continuous function $f_A : S_{2,2} \rightarrow S_{2,2}$. As usual, let R and C denote Rose's and Colin's payoff matrices and let (p, q) be a point in the strategy space. The precise definition of the function will describe how to compute the new point $f_A(p, q) = (p', q')$.

Point (p, q) in the strategy space is associated with the strategies $\mathbf{p} = \begin{bmatrix} p & 1-p \end{bmatrix}$ for Rose and $\mathbf{q} = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ for Colin. Rose's (expected) payoffs for playing row 1 or 2 against Colin's strategy $\mathbf{q}$ are given by

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = R\mathbf{q}.$$

So a_1 is Rose's payoff if she plays row 1 against Colin's $\mathbf{q}$ strategy, and a_2 is her payoff if she plays row 2. Similarly, if Rose plays $\mathbf{p}$, Colin's payoffs for playing either column 1 or column 2 are

$$\begin{bmatrix} b_1 & b_2 \end{bmatrix} = \mathbf{p}C.$$

To articulate the players' updated strategies, first define the *plus function* as follows:

$$(x)^+ = \begin{cases} x & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

Now, at last, we can define Nash flow for the point (p, q) . Let $f_A(p, q) = (p', q')$ where

$$p' = \frac{p + (a_1 - a_2)^+}{1 + |a_1 - a_2|}, \quad q' = \frac{q + (b_1 - b_2)^+}{1 + |b_1 - b_2|}.$$

Let's investigate the situation. The behavior of p' and q' are very similar, so this analysis focuses on p' . First note that the denominator in the expression for p' is positive and the numerator is nonnegative, so $p' \geq 0$. Since this denominator is always at least as large as the numerator, $p' \leq 1$. Therefore $0 \leq p' \leq 1$, so p' does correspond to a strategy for Rose.

If $a_1 = a_2$, then $p' = p$ and Rose will not alter her strategy. She gets the same payoff playing row 1 or row 2 so she has no incentive to change. Next suppose Rose does better playing row 2 than row 1, so $a_2 > a_1$. In this case the formula says $p' = \frac{p}{1+t}$ where $t = |a_1 - a_2| > 0$. If $p = 0$, then Rose is already playing the pure strategy of row 2 and $p' = 0$, so she will not change. On the other hand, if $p > 0$, then $p' < p$ so Rose's new strategy will have her playing the first row with lower probability and the second with higher probability. In the $a_2 < a_1$ case, Rose does better playing the first row than the second. Here, the formula simplifies to $p' = \frac{p+t}{1+t}$ where $t = a_1 - a_2 > 0$. If $p = 1$, then Rose is already playing the pure strategy of choosing row 1 and $p' = p$ so she will not change. Otherwise, $p < 1$ and $p' > p$, so Rose's new strategy will have her play the first row with lower probability and the second with higher, just as desired.

In sum, this modification scheme gives Rose a sensible response. If row 1 and row 2 give her equal payoffs, she does not change strategy. If row 1 gives Rose a better payoff than row 2, then she modifies her strategy to play row 1 more frequently (if possible). Similarly, if row 2 gives a better payoff than row 1, then Rose alters her strategy to play row 2 more frequently (if possible). Rose's new strategy p' will be exactly the same as her original p if and only if p is a best response to q . A similar analysis for Colin results in the following key property.

Lemma 9.20. *For every 2×2 matrix A , the fixed points (p, q) of Nash flow f_A are precisely those points for which $\mathbf{p} = \begin{bmatrix} p & 1-p \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} q \\ 1-q \end{bmatrix}$ form a Nash equilibrium of A .*

Nash flow can be very helpful for getting a sense of the strategy space for a particular game. Let's reconsider this function for another one of the dilemmas from Chapter 7.

Example 9.21. Figure 9.16 depicts the Nash flow associated with the Dating Dilemma game given by the matrix $\begin{bmatrix} (2, 1) & (0, 0) \\ (0, 0) & (1, 2) \end{bmatrix}$. This game has two pure Nash equilibria corresponding to the case when the players go to the same venue. These outcomes correspond to the fixed points

$(0, 0)$ and $(1, 1)$ in the Nash flow. There is an additional Nash equilibrium when Rose plays $\begin{bmatrix} 2/3 & 1/3 \end{bmatrix}$ and Colin plays $\begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}$.

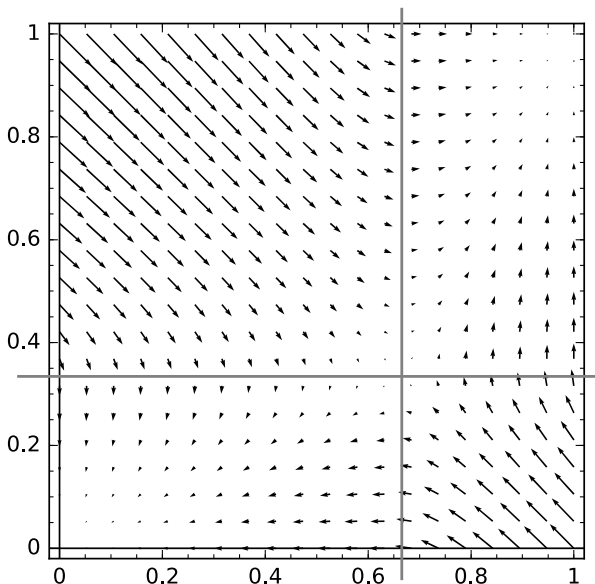


Figure 9.16. Nash Flow for the Dating Dilemma

In closing, we note the considerable value of Nash flow for studying stability properties of Nash equilibria (as in the discussion of evolution in Chapter 8). It's evident in the Dating Dilemma that the two pure strategy Nash equilibria are very stable—all points in the strategy space near either of these points are directed toward it. The mixed strategy equilibrium is less stable since some nearby points are directed away from it.

The Proof. We are finally ready to prove Nash's Equilibrium Theorem. This proof is, in fact, a fairly straightforward consequence of Corollary 9.16 and Lemma 9.20.

Theorem. Every 2×2 matrix game A has a Nash equilibrium.

Proof. The existence of a Nash equilibrium for the 2×2 game A follows from the statements below:

- (1) Nash flow is a continuous function $f_A : S_{2,2} \rightarrow S_{2,2}$ with the property that every fixed point corresponds to a Nash equilibrium of A (Lemma 9.20).
- (2) The strategy space $S_{2,2}$ has the fixed point property (Corollary 9.16), and thus f_A has a fixed point. $\square$

Higher Dimensions. Just as we defined for every 2×2 matrix game A the Nash flow $f_A : S_{2,2} \rightarrow S_{2,2}$, it is possible to define a (continuous) Nash flow function for an arbitrary $m \times n$ matrix game. The details appear explicitly in Appendix C, but the key is the following generalization of Lemma 9.20.

Lemma 9.22. *For every $m \times n$ matrix A , the fixed points of Nash flow $f_A : S_{m,n} \rightarrow S_{m,n}$ are precisely those points that correspond to a Nash equilibrium of A .*

With this in place, the full proof of the Nash Equilibrium Theorem for an arbitrary $m \times n$ matrix game follows from the same reasoning as in the 2×2 case.

Theorem (Nash). *Every matrix game has a Nash equilibrium.*

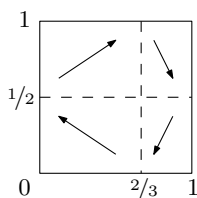
Proof. Let A be an $m \times n$ matrix game with strategy space $S_{m,n}$. The proof that A has a Nash equilibrium follows from the following two properties:

- (1) Nash flow is a continuous function $f_A : S_{m,n} \rightarrow S_{m,n}$ with the property that every fixed point corresponds to a Nash equilibrium of A (Lemma 9.22).
- (2) The strategy space $S_{m,n}$ has the fixed point property (Lemma 9.17), so f_A has a fixed point. $\square$

Exercises

- (1) Find all fixed points for the functions below:
 - (a) $f : [0, 2] \rightarrow [0, 2]$ defined by $f(x) = 1 + x^2/4$,
 - (b) $g : [-1, 2] \rightarrow [-1, 2]$ defined by $g(x) = x^2/3 + 2x/3 - 1/4$,
 - (c) $h : [0, 4] \rightarrow [0, 4]$ defined by $h(x) = x^2 - 3x + 3$.
- (2) Find all fixed points for the functions below:
 - (a) $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $f(x, y) = (y - 1, x + y)$,
 - (b) $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $g(x, y) = (2 - y, -2x + 1)$,
 - (c) $h : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $h(x, y) = (xy - 1, 3x + 2)$.
- (3) Define the square $S = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1\}$. Sketch (as in Figure 9.8) each function from S to itself:
 - (a) $f : S \rightarrow S$ given by $f(x, y) = (x^2, y^2)$,
 - (b) $g : S \rightarrow S$ given by $g(x, y) = (x/2, 1 - y/2)$,
 - (c) $h : S \rightarrow S$ given by $h(x, y) = (1 - y, x)$.
- (4) For $A = \begin{bmatrix} (0, 5) & (3, 3) \\ (6, 2) & (1, 4) \end{bmatrix}$, evaluate the Nash flow function f_A at each point:
 - (a) $f_A(1/3, 1/3)$,
 - (b) $f_A(1/2, 1/4)$,
 - (c) $f_A(0, t)$ for $0 \leq t \leq 1$,
 - (d) $f_A(t, t)$ for $0 \leq t \leq 1$.
- (5) This problem concerns functions that may be discontinuous.
 - (a) Construct a function $f : [0, 1] \rightarrow [0, 1]$ with no fixed point.
 - (b) For every positive integer k , construct a function $f : [0, 1] \rightarrow [0, 1]$ which has exactly k fixed points.
- (6) Find a continuous function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with the following given property:
 - (a) f has no fixed points.
 - (b) f has exactly one fixed point.
 - (c) f is not the identity but has infinitely many fixed points.
- (7) Consider the matrix $A = \begin{bmatrix} (1, 2) & (2, 3) \\ (0, 3) & (3, 1) \end{bmatrix}$ from Example 9.18. When Rose plays the mixed strategy $[p \quad 1 - p]$, Colin gets a higher payoff playing the first column when $p < 2/3$ and a higher payoff playing

the second column when $p > 2/3$. This corresponds to the fact that the Nash flow function goes upward to the left of the line $p = 2/3$ and downward to the right. By considering Rose's payoffs we can determine that the Nash flow function will go rightward above the line $q = 1/2$ and leftward below the line $q = 1/2$. Based on this we obtain a rough plot of Nash flow shown in the figure below (compare with Figure 9.15).

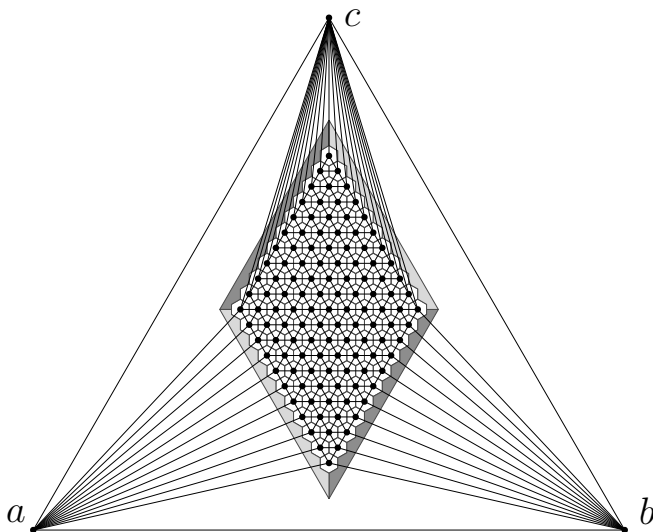


Find rough plots of Nash flow for the following matrix games:

$$(a) \begin{bmatrix} (3, 1) & (0, 6) \\ (1, 4) & (5, 3) \end{bmatrix}, \quad (b) \begin{bmatrix} (1, 3) & (3, 8) \\ (9, 3) & (2, 2) \end{bmatrix}, \quad (c) \begin{bmatrix} (1, 6) & (3, 4) \\ (5, 2) & (4, 3) \end{bmatrix}.$$

- (8) Prove that the given set does not have the fixed point property.
 - (a) $\{x \in \mathbb{R} \mid 0 < x < 1\}$,
 - (b) $\{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \geq 0\}$,
 - (c) $\{(x, y) \in \mathbb{R}^2 \mid 0 < x < 1 \text{ and } 0 < y < 1\}$,
 - (d) $\{(x, y) \in \mathbb{R}^2 \mid 0 < x^2 + y^2 < 1\}$.
- (9) Construct continuous functions with the following properties:
 - (a) $f : [0, \infty) \rightarrow [0, \infty)$ has 0 as its unique fixed point.
 - (b) $g : [0, 1] \rightarrow [0, 1]$ had 0, 1 as its only fixed points.
 - (c) $h : \mathbb{R} \rightarrow \mathbb{R}$ has exactly k fixed points.
- (10) Let $a < b$ and consider the closed interval $[a, b]$.
 - (a) Prove that $[a, b]$ has the fixed point property by generalizing the proof of Theorem 9.9.
 - (b) Prove that $[a, b]$ is topologically equivalent to $[0, 1]$ by generalizing Example 9.14.
- (11) This exercise uses Sperner's Lemma to prove that Hex cannot end in a draw. Suppose that Rose is marking cells with $*$ and trying to connect the lower left side with the upper right side, while Colin is marking cells with $\circ$ and trying to connect the lower right side with the upper left. Assume that they continue play until all cells

are occupied. Now construct a subdivided triangle on top of the Hex board as shown in the figure below:



Next we label vertices of the subdivided triangle. Give a label 1, b label 2, c label 3, and label every other vertex v according to the following:

- v gets label 1 if its cell contains $*$ and this cell can be joined to the lower left side by a connected path of cells labeled $*$.
- v gets label 2 if its cell contains $\circ$ and this cell can be joined to the lower right side by a connected path of cells labeled $\circ$.
- v gets label 3 if neither of the above conditions apply.

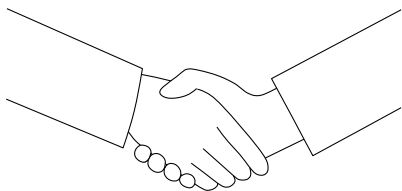
Prove that either Rose or Colin won the game.

- (12) If A is an $n \times n$ matrix and $\mathbf{b} \in \mathbb{R}^n$, the function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $f(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ is an *affine transformation*.
- (a) Show that the composition of two affine transformations is an affine transformation.
 - (b) Determine when the affine transformation $f(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ is invertible, and find a formula for the inverse when it exists.
 - (c) Let $\mathbf{e}_i \in \mathbb{R}^n$ be the vector with a 1 in coordinate i and 0 elsewhere. If $\mathbf{x}_1, \dots, \mathbf{x}_n$ are linearly independent, find an invertible matrix A so that $A\mathbf{x}_i = \mathbf{e}_i$ for every $1 \leq i \leq n$.

- (d) If $\mathbf{x}_0, \dots, \mathbf{x}_n \in \mathbb{R}^n$ do not lie on a common hyperplane, show that there exists an invertible affine transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ so that $f(\mathbf{x}_0) = \mathbf{0}$ and $f(\mathbf{x}_i) = \mathbf{e}_i$ for $1 \leq i \leq n$.
- (e) Prove that every simplex in $\mathbb{R}^n$ is topologically equivalent to Δ_n .
- (f) Prove that any two simplices in $\mathbb{R}^n$ are topologically equivalent.
- (13) Let A be a symmetric 2×2 matrix game with payoff matrix $R = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ for Rose (so Colin's payoff matrix is $C = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$). In this exercise we will follow the paradigm from this chapter to prove that A has a symmetric Nash equilibrium (i.e. we will prove Theorem 8.11 for the special case of 2×2 matrices).
- (a) Show that for every $0 \leq t \leq 1$ the Nash flow function satisfies $f_A(t, t) = (u, u)$ for some u .
- (b) Define a function $g : [0, 1] \rightarrow [0, 1]$ by the rule $g(t) = u$ if $f_A(t, t) = (u, u)$. Use a theorem from this chapter to show that g has a fixed point, and then deduce from this that A has a symmetric Nash equilibrium.
- (14*) Let T be the boundary of the triangle Δ_2 . Prove that there does not exist a continuous function $f : \Delta_2 \rightarrow T$ with the property that every $\mathbf{x} \in T$ is a fixed point of f . Hint: Use Sperner's Lemma.

Chapter 10

Cooperation



Many games model decision problems where experience suggests that cooperation may be beneficial, yet we haven't seen any cooperation so far in our analysis of games. Could game theory be useful in determining how and when rational players cooperate? This chapter expands our horizons beyond the one-time play of a matrix game to explore this question. First, we change perspective: Instead of *playing* a matrix game, imagine that two players *negotiate* about their actions prior to play. This setting gives rise to the concept of a negotiation set and establishes context for a beautiful theorem about selecting a fair outcome. In the end, the chapter returns to game play embellished by a new type of game with many stages. The well-known Folk Theorem indicates a mechanism by which cooperation may appear in such an environment.

Imagine a decision problem for Rose and Colin. Assume that each player has two choices. In combination, these will select one of four possible outcomes (denoted W, X, Y, Z):

		Colin	
		W	X
Rose	Y		
	Z		

As usual, each player assigns a utility to indicate a value for each of these four outcomes relative to the others. Assume the following 2×2 matrix game results:

		Colin	
		4, 2	1, 4
Rose	0, 1		
	3, 0		

If probabilistic actions are involved, outcome X may occur with probability $\frac{1}{3}$ and outcome Z with probability $\frac{2}{3}$. In this case, Rose's and Colin's expected payoffs are given by $\frac{1}{3}(1, 4) + \frac{2}{3}(3, 0) = (\frac{7}{3}, \frac{4}{3})$. It is natural to view this possibility of having X occur with probability $\frac{1}{3}$ and Z occur with probability $\frac{2}{3}$ as a type of "outcome" in its own right and to denote it by $\frac{1}{3}X + \frac{2}{3}Z$. We will use the term *mixed outcome* to describe such situations.

How should Rose and Colin play this game without any communication? What if Rose and Colin sit down and negotiate in advance? Could they reach a more favorable mixed outcome through cooperation?

10.1. The Negotiation Set

Perhaps surprisingly, a geometric framework can assist Rose and Colin's negotiations in a matrix game. To set a baseline for the negotiations, we first investigate how Rose and Colin might expect to fare in a game with no communication.

Security Levels. Consider the game from the introduction from Rose's perspective. If she plays a mixed strategy $\mathbf{p} = [p \quad 1-p]$, then we can compute her expected payoff for each possible choice Colin makes.

Rose's payoff matrix is $R = \begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix}$, so when she plays mixed strategy

$\mathbf{p} = [p \quad 1 - p]$, the vector of interest is given by

$$\mathbf{p}R = [p \quad 1 - p] \begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix} = [4p \quad 3 - 2p].$$

Thus, if Rose plays the mixed strategy $\mathbf{p}$, her expected payoff is $4p$ when Colin chooses the first column and $3 - 2p$ when he chooses the second. A conservative plan for Rose would be to choose a mixed strategy so that the smallest entry of $[4p \quad 3 - 2p]$ is as large as possible. This strategy gives Rose the highest guaranteed expected payoff. For this particular game, this can be achieved when $4p = 3 - 2p$, so $p = \frac{1}{2}$. If Rose does play the mixed strategy $[\frac{1}{2} \quad \frac{1}{2}]$, then she gets an expected payoff of 2 no matter what Colin does. This may well look familiar—right now Rose is just playing as she would in the zero-sum matrix game R . She is completely ignoring Colin's payoffs. This may not be particularly strategic play for Rose, but it does at least offer her a baseline guarantee.

Definition 10.1. For an arbitrary matrix game, Rose's *security level* is the maximum expected payoff she can guarantee herself. Similarly, Colin's *security level* is the maximum expected payoff he can guarantee himself.

The above example generalizes to yield the following theorem that says how to calculate Rose's security level for an arbitrary matrix game.

Theorem 10.2. *In a matrix game with Rose's payoff matrix R , Rose's security level is the von Neumann value of R .*

Proof. Let v together with the strategies $\mathbf{p}$ for Rose and $\mathbf{q}$ for Colin be a von Neumann solution to the zero-sum matrix game R . If Rose plays $\mathbf{p}$, then $\mathbf{p}R \geq [v \quad \dots \quad v]$ so Rose will be guaranteed an (expected) payoff of at least v no matter what Colin does. On the other hand, if Colin plays

$\mathbf{q}$, then $R\mathbf{q} \leq \begin{bmatrix} v \\ \vdots \\ v \end{bmatrix}$, so Rose will have an (expected) payoff of at most v .

Thus, v is Rose's security level. □

Computing Colin's security level is similar, but with a slight twist (or rather a flip). To find Colin's security level, consider his payoff matrix C . His security level will be the highest possible guarantee he can

achieve using a mixed strategy $\mathbf{q}$. In other words, we seek $\mathbf{q}$ so that the minimum entry of $C\mathbf{q}$ is as large as possible. This is nearly the same as the usual zero-sum game format, except that Colin, the column player, is now looking to maximize (instead of minimize) the payoff. To adjust for this, transpose the matrix to interchange the roles of the players.

Theorem 10.3. *In a matrix game with Colin's payoff matrix C , Colin's security level is the von Neumann value of $C^\top$.*

Proof. Exercise (12). □

In the 2×2 example game from the introduction, Colin's payoff matrix is given by $C = \begin{bmatrix} 2 & 4 \\ 1 & 0 \end{bmatrix}$. So $C^\top = \begin{bmatrix} 2 & 1 \\ 4 & 0 \end{bmatrix}$. In the zero-sum matrix game $C^\top$, the second column dominates the first. Removing the dominated column leaves a smaller matrix in which the first row dominates the second. Thus $C^\top$ has a von Neumann value of 1 and Colin has a security level of 1 in this game.

At first glance, it might seem sensible for Rose (or Colin) to play so as to guarantee his or her security level. But this may be a bad idea! Matrix game A might contain a strictly dominated column—a poor choice for Colin. Assuming Colin plays rationally, he never chooses this dominated column, so there is no reason for Rose to worry about him playing it. Accounting for this may permit Rose a better guarantee. More generally, rational play of matrix games against rational players usually involves careful consideration of the payoffs for both players.

The Payoff Polygon. We have now determined what Rose and Colin can guarantee themselves independently. What if they communicate? Imagine that, instead of playing the game immediately, Rose and Colin first negotiate about what might be a fair or reasonable outcome. For clarity, suppose they are bound to act in accordance with the agreement resulting from their discussion. Consequently, concepts developed here apply whenever there exists some social, legal, or moral mechanism in place to prevent default on an agreement.

Let's return to the introductory game with the four outcomes W, X, Y, Z and the corresponding payoffs as shown below:

		Colin		
Rose	W	(4, 2)	X	(1, 4)
	Y	(0, 1)	Z	(3, 0)

The payoff associated with each outcome is an ordered pair (x, y) , so we can visualize these payoffs as points in the plane, as shown in Figure 10.1. In this setting, Rose prefers the selected point to be as far to the right as possible, while Colin wants it to be as high up as possible.

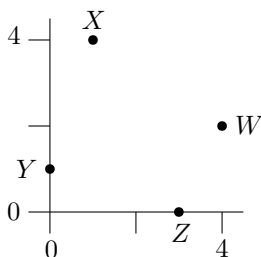


Figure 10.1. Payoffs

Consider the mixed outcome $\frac{1}{2}W + \frac{1}{2}Z$. To achieve this, half of the time Rose would choose row 1 and Colin would choose column 1, and the other half of the time Rose would choose row 2 and Colin would choose column 2. So this particular mixed outcome is one that could only be achieved by players who are cooperating in a kind of negotiated setting. If the players do coordinate to achieve the mixed outcome $\frac{1}{2}W + \frac{1}{2}Z$, then their expected payoffs will be $\frac{1}{2}(4, 2) + \frac{1}{2}(3, 0) = (\frac{7}{2}, 1)$. Geometrically, this is the midpoint between W and Z in Figure 10.1. More generally, the players may agree to select outcomes W, X, Y, Z with probabilities p_W, p_X, p_Y, p_Z whenever p_W, p_X, p_Y, p_Z are nonnegative numbers that sum to 1. The corresponding expected payoffs will then be given by $p_W(4, 2) + p_X(1, 4) + p_Y(0, 1) + p_Z(3, 0)$. Geometrically, this is a convex combination of the four points associated with W, X, Y, Z . In general, then, the set of all possible expected payoffs of mixed outcomes is precisely the convex hull of the points $(4, 2), (1, 4), (0, 1)$, and $(3, 0)$. This is the payoff polygon for the matrix game in Figure 10.2.

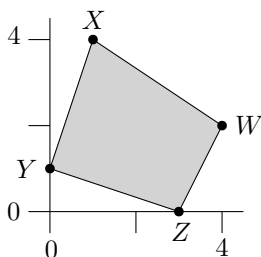


Figure 10.2. The payoff polygon

Next, generalize this to an arbitrary matrix game.

Definition 10.4. The *payoff polygon* of a matrix game A is the convex hull of the ordered pairs appearing as entries of A .

The payoff polygon provides a useful geometric viewpoint for comparing various mixed outcomes. To get a sense of these polygons, let's revisit the dilemmas from Chapter 7 and consider the associated payoff polygons:

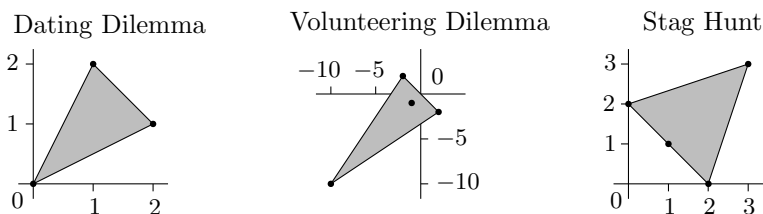


Figure 10.3. Payoff polygons for some dilemmas

In terms of modeling real-world situations, the mixed outcomes here may or may not make sense. In the Dating Dilemma, for example, if Rose and Colin will go on many dates, it is certainly reasonable for them to go to a ball game (B) half the time and to a film (F) half the time. This gives the mixed outcome $\frac{1}{2}B + \frac{1}{2}F$, which would correspond to the point $\frac{1}{2}(2, 1) + \frac{1}{2}(1, 2) = (\frac{3}{2}, \frac{3}{2})$ in the payoff polygon. There are other circumstances where the game in question will not be repeated. Perhaps Rose and Colin are planning a wedding. Rose wants to get married in France, while Colin wants to get married in Hawaii. This decision game could

generate a similar payoff polygon, but the situation is different. Likely, Rose and Colin will not repeat this same game. They still may flip a fair coin to make the decision and this is a perfectly reasonable process if they both agree. However, it is certainly possible that Rose or Colin might be unwilling to let a coin flip decide this important question. In that case, they have a dilemma indeed!

The Negotiation Set. The payoff polygon depicts all possible payoffs associated with the mixed outcomes. It is a little inconvenient to have to consider *all* of these points for the purposes of negotiation. Is there a sensible way to focus on just some reasonable points in the payoff polygon?

At the start of this section, we considered playing a matrix game and we defined the security levels for Rose and Colin. Recall that the players' security levels are the highest possible expected payoff they can guarantee themselves. If our players instead negotiate, it would then be irrational for either player to agree to a mixed outcome that gives him or her an expected payoff less than his or her security level.

Optimality offers another consideration. It would be irrational for Rose and Colin to agree on a mixed outcome with expected payoff (x, y) if there were another point (x', y') in the payoff polygon with $(x, y) < (x', y')$ (i.e. $x < x'$ and $y < y'$). Why would they agree to (x, y) if another point is strictly better for both players? More generally, if there is a point (x'', y'') with $(x, y) \leq (x'', y'')$ (so each player does at least as well at this new point) and $(x, y) \neq (x'', y'')$ (so at least one player does strictly better), then the players should favor the point (x'', y'') over (x, y) . Based simply on optimality, then, negotiations should ignore points in the payoff polygon (x, y) if it is possible to increase one or the other coordinate (or both) and stay in the polygon. Taking all of this into account generates the following definition.

Definition 10.5. The *negotiation set* is the set of all points (x, y) in the payoff polygon satisfying the following properties:

- (1) x is at least Rose's security level and y is at least Colin's security level.
- (2) There is no point (x', y') in the payoff polygon with $(x, y) \leq (x', y')$ and $(x, y) \neq (x', y')$.

Von Neumann and Morgenstern introduced the negotiation set to focus attention on the reasonable negotiated outcomes. This well-accepted notion serves as a helpful guide in negotiations. For the earlier 2×2 example, we already computed Rose's security level to be 2 and Colin's as 1, so the negotiation set is as in Figure 10.4

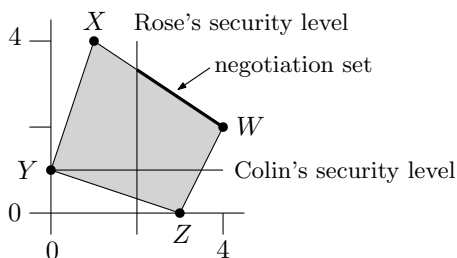


Figure 10.4. The negotiation set

Now let us return to the dilemmas from Chapter 7 to consider the associated payoff polygons and negotiation sets as shown in Figure 10.5.

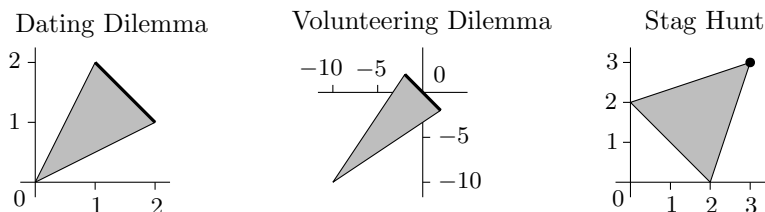


Figure 10.5. Negotiation sets for some dilemmas

In Stag Hunt, the negotiation set consists of just a single point. Von Neumann and Morgenstern would argue in this case that this point is the only reasonable outcome for the players. In other games such as the

Dating Dilemma and the Volunteering Dilemma, the negotiation set is larger and serves just to focus attention on the sensible outcomes for a negotiation.

10.2. Nash Arbitration

A negotiation set, as developed above, contains all reasonable possibilities for a negotiated outcome. Our goal in this section is to prove a beautiful theorem that selects a particular point from the negotiation set that is arguably a “fair” outcome. First, we abstract further from the setting of games to an even more general framework for negotiations.

In the previous section, we considered a matrix game A and associated it with a payoff polygon, security levels, and a negotiation set. Now define the *status quo* point associated with A to be the point (x_0, y_0) at the intersection of the lines corresponding to Rose’s and Colin’s security levels. As suggested by the terminology, view the status quo point as a starting point in the negotiation. Note that the status quo point indicates the security levels for both Rose and Colin. Given a payoff polygon and its status quo point, we can quickly determine the associated security levels, and thus the negotiation set.

General Negotiations. In fact, these concepts of payoff polygon, status quo point, and negotiation set apply even in places where there is no game! Indeed, many actual negotiations that are not based directly on games can be modeled using these ideas, and this can be a useful construct for sharpening our understanding of such situations. Imagine, for example, a dispute at a company between labor and management. Labor would like to see salaries increase by \$500 per year, while management would prefer they decrease by \$500 per year. Labor would like to increase pension payments by \$100 per month, while management would prefer to decrease them by \$100 per month. Finally, labor would like to increase the afternoon breaktime by 20 minutes, while management would like to decrease it by 20 minutes. (It is not necessary for each such possibility to be added or subtracted by the same amount; see for instance Exercise (7).)

Assume that labor and management have linear utility scales for each of these three possibilities, where the value of the present arrange-

ment is zero and the utilities associated with each possible increase or decrease are shown below:

Payoffs (labor, management)			
	Salary (S)	Pension (P)	Breaktime (B)
Increase (+)	4, -6	8, -3	2, -1
Decrease (-)	-4, 6	-8, 3	-2, 1

For instance, labor values an increase in salary, denoted S^+ , at 4, whereas management values this outcome at -6. On the other hand, the corresponding decrease in salary, denoted S^- , is valued at -4 by labor and at 6 by management.

More generally, imagine all of these three quantities moving either up or down, and assume the utility that labor and management associate with an outcome is given by the sum of the three quantities. For example, an outcome of $S^+P^-B^+$ gives labor a payoff of $4 - 8 + 2 = -2$ and management a payoff of $-6 + 3 - 1 = -4$.

Just as with matrix games, we can plot the utilities for labor and management associated with these outcomes. The convex hull of these points forms the payoff polygon, and we will regard the origin as the status quo point since the utility for both labor and management of the present situation is 0. Using this status quo point, we can then determine the negotiation set. All of this appears in Figure 10.6.

Consider the point corresponding to $\frac{3}{8}S^-P^+B^+ + \frac{5}{8}S^-P^+B^-$ (i.e. the point on the line segment between $S^-P^+B^+$ and $S^-P^+B^-$ that is $\frac{3}{8}$ of the way from $S^-P^+B^+$ to $S^-P^+B^-$). Using the interpretation from the previous section, we could think of this point as corresponding to an outcome where $\frac{3}{8}$ of the time we choose $S^-P^+B^-$ and the other $\frac{5}{8}$ of the time we choose $S^-P^+B^+$. This would mean decreasing salaries by \$500 per year, increasing pensions by \$100 per month, and $\frac{3}{8}$ of the time decreasing the break by 20 minutes and the other $\frac{5}{8}$ of the time increasing the break by 20 minutes.

However, breaktime is an inherently continuous quantity. So instead of choosing this strange sounding combination, we could just in-

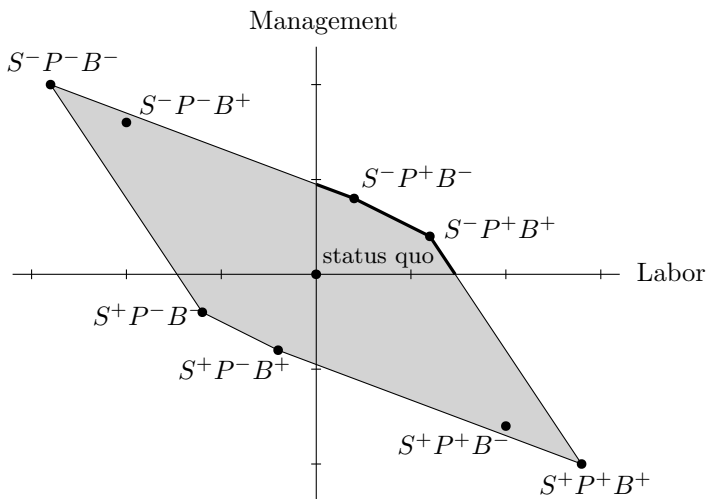


Figure 10.6. The payoff polygon for our labor-management dispute

crease the breaktime by 5 minutes (yielding the same average since $\frac{3}{8}(-20) + \frac{5}{8}(20) = 5$). Indeed, all salary, pension, and breaktime quantities are inherently continuous. As a result, we can associate each point in our polygon with a particular change in salary, pension, and breaktime. The payoff polygon thus produces a detailed encoding of the utilities that the two parties associate with any possible adjustment to salaries, pensions, and breaktime within the given bounds.

Arbitration. Von Neumann and Morgenstern focused on the negotiation set as a set of reasonable outcomes. Could there be a way of choosing a particular point in this negotiation set that is somehow “fair”? This brings us to the concept of an arbitration scheme.

Formally speaking, we’d like to find a rule that, given a convex polygon P and a status quo point (x_0, y_0) , will select a *solution point* (x, y) from the polygon. We define such a rule to be an *arbitration scheme*. Note that we will always assume that the polygon contains at least one point (x, y) with $x \geq x_0$ and $y \geq y_0$ —otherwise there would be no negotiation set.



Figure 10.7. An arbitration scheme

Next we introduce a particular arbitration scheme.

Definition 10.6 (Nash arbitration). If there is a point (x, y) in the polygon with $x > x_0$ and $y > y_0$, then choose such a point (x, y) for which $(x - x_0)(y - y_0)$ is maximum. Otherwise, the negotiation set has size 1 (Exercise (11)) and the solution point (x, y) is the unique point in the negotiation set.

Before considering any questions of fairness, we revisit the labor-management negotiation. What point in the payoff polygon does Nash arbitration select? The optimization required to find the solution to Nash arbitration is fairly straightforward to compute. In this example, the status quo point is $(0, 0)$, so the function to maximize is xy (more generally it is $(x - x_0)(y - y_0)$ for status quo point (x_0, y_0)). The maximum will be one of the points in the negotiation set, so it will lie on one of three line segments. To find the maximum over all three, consider each one individually, and then take the best. Start with the middle line segment. This line segment is given by the equation $y = -\frac{1}{2}x + 5$ with the bounds $2 \leq x \leq 6$. So, to maximize xy on this line segment we want to maximize the function $f(x) = x(-\frac{1}{2}x + 5)$ for $2 \leq x \leq 6$. The maximum will either be at an endpoint (i.e. $x = 2$ or $x = 6$) or at an interior point $2 < x < 6$ for which $f'(x) = 0$. Checking these points reveals that the maximum is attained when $x = 5$, and in fact this turns out to be the solution point to Nash arbitration. This corresponds to decreasing salaries by \$500 per year, increasing pensions by \$100 per month, and increasing breaktime by 10 minutes per day.

Nash arbitration is a procedure that will always select a point from the payoff polygon, so it is indeed a valid arbitration scheme. But is the solution point determined by this scheme really fair? How can we even make sense of this question from a rigorous mathematical perspective? Nash provided a particularly attractive answer to these questions. His answer involves the following four axioms, always satisfied by his arbitration scheme (Exercise (16)).

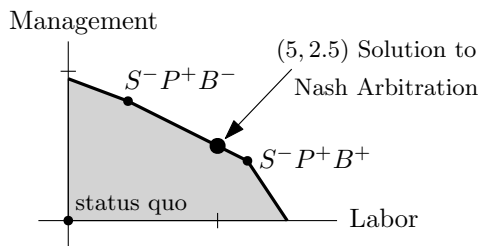


Figure 10.8. Nash arbitration for our labor-management dispute

Definition 10.7 (Nash's Axioms).

- (1) Rationality. The solution point is in the negotiation set.
- (2) Linear Invariance. If either Rose's or Colin's utility is transformed by a linear function with positive slope, the solution point is also transformed by applying this same function to that player's payoff.
- (3) Symmetry. If the payoff polygon is symmetric about the line of slope 1 through (x_0, y_0) , then the solution point (x, y) is on this line.
- (4) Independence of Irrelevant Alternatives. Suppose (x, y) is the solution point when the arbitration scheme is applied to polygon P and status quo point (x_0, y_0) . Let Q be another polygon completely contained in P also containing (x, y) . In this case, (x, y) is also the solution point when the arbitration scheme is applied to Q with status quo point (x_0, y_0) .

Von Neumann and Morgenstern suggested the first axiom here when they introduced the concept of negotiation set, and it is quite sensible and well accepted. The second axiom is also quite reasonable. For instance, if one player doubled his utilities, we would expect the new solution point to double in this coordinate. Similarly, if one player were to add a fixed value of t to each of her utilities, this should shift the solution point by adding t to the appropriate coordinate. The third axiom also exhibits a natural fairness property. In a game like the Dating Dilemma or

the Volunteering Dilemma where the positions of the players are symmetric, neither player should be favored, so the solution point should lie on the line of slope 1 through (x_0, y_0) . The last axiom also seems quite sensible. After all, if (x, y) is the best solution for the polygon P and status quo point (x_0, y_0) and we modify the situation by shrinking the polygon in such a way that we still keep the point (x, y) , we should expect the solution point for this new polygon (still using the status quo point (x_0, y_0)) to be the same.

If we accept these reasonable axioms as fair properties, then we should focus our attention on arbitration schemes that satisfy them. Shockingly, Nash proved that his arbitration scheme is the only one that can satisfy all four! So, if we agree that a fair arbitration scheme should satisfy these seemingly innocuous axioms, then Nash arbitration is the only scheme that can be fair.

Theorem 10.8 (Nash). *The only arbitration scheme satisfying all four of Nash's axioms is Nash arbitration.*

Proof. Denote Nash arbitration by $\mathcal{N}$ and let $\mathcal{A}$ be another arbitration scheme that satisfies all of Nash's axioms. To prove the theorem, it suffices to show that, for an arbitrary polygon P with status quo point (x_0, y_0) , the solution point selected by $\mathcal{N}$ and by $\mathcal{A}$ is the same.

Begin by modifying the polygon P . Subtract x_0 from each of Rose's payoffs and y_0 from each of Colin's to shift the status quo point to $(0, 0)$. If $\mathcal{N}$ and $\mathcal{A}$ select the same solution point for this new polygon, then axiom (2) implies that they also select the same solution point from the original.

If there is no point (x, y) in the payoff polygon with $x > 0$ and $y > 0$, the negotiation set has size one, and the first axiom implies that this point is selected by both $\mathcal{N}$ and $\mathcal{A}$. Assume, then, that there is a point in P with $x > 0$ and $y > 0$. Let (x^*, y^*) be the point in P with $x^* > 0$ and $y^* > 0$ that maximizes x^*y^* . Now modify P by dividing all x -coordinates by x^* and all y -coordinates by y^* . Again, by the second axiom, it suffices to show that for this new polygon, the solution point selected by $\mathcal{N}$ and $\mathcal{A}$ is the same. However, now we know that for this new polygon $\mathcal{N}$ will select $(1, 1)$.

We claim that the entire polygon P now lies on or below the line given by the equation $x + y = 2$. Suppose (for a contradiction) that there is a point (x_1, y_1) in P for which $x_1 + y_1 > 2$. Now consider the line segment between $(1, 1)$ and (x_1, y_1) , and zoom in on the region of the plane near the point $(1, 1)$ as shown in Figure 10.9.

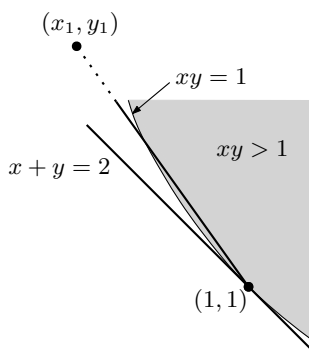


Figure 10.9. The line segment $(1, 1)$ to (x_1, y_1) enters the region $xy > 1$.

Here we see a branch of the hyperbola given by the equation $xy = 1$ passing through $(1, 1)$. At this point, the line tangent to the hyperbola is the line with equation $x + y = 2$. All points that lie above this branch of our hyperbola satisfy $xy > 1$ and this region is shaded in Figure 10.9. Now, the line segment between $(1, 1)$ and (x_1, y_1) must enter the region $xy > 1$ and it follows that the line segment between $(1, 1)$ and (x_1, y_1) contains a point (x, y) with $xy > 1$. By the convexity of P , this entire line segment is in P , so P contains a point (x, y) with $xy > 1$. This is a contradiction, so polygon P must lie on or below the line given by $x + y = 2$ as claimed.

Now use the fact that polygon P lies on or below the line $x + y = 2$ to choose a certain triangle that contains P . That is, choose a triangle T so that one vertex of T lies on the line $x = y$, the opposite side lies on the line $x + y = 2$ in such a way that T is symmetric about the line $x = y$, and T contains P .

Axiom (3) implies that when arbitration scheme $\mathcal{A}$ is applied to the triangle T with status quo point $(0, 0)$ it must select the point $(1, 1)$. Now axiom (4) implies that $\mathcal{A}$ must also select this point when applied to P

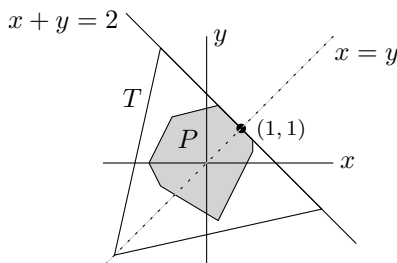


Figure 10.10. A triangle T which contains P

with status quo point $(0, 0)$. Since this is the same point selected by $\mathcal{N}$, this completes the proof. $\square$

If, in the example above, labor and management agree to abide by an arbitration scheme satisfying Nash's axioms, then they should accept the Nash arbitration output as a fair solution. However, not everyone agrees with Nash's axioms! More specifically, some theorists consider the fourth axiom to be somewhat questionable and, consequently, there is no universal agreement on what the fair outcome of this type of arbitration should be. Kalai-Smorodinsky arbitration, for example, uses a different axiom in place of Nash's fourth and it has been shown that there is a unique arbitration scheme satisfying these axioms.

10.3. Repeated Games and the Folk Theorem

So far in considering cooperation, we have relied on an external agent to force the players to honor the solution they agreed to in negotiations. Applying Nash arbitration to the Prisoner's Dilemma, for example, yields a solution point that requires both players to cooperate. But if nothing binds the players to actually cooperate when they play, rationality dictates that they should defect. Is there any setting in which rational players might cooperate without an external force?

Reducing a decision problem to a matrix game removes the possibility of future interaction—a potentially significant factor. While a matrix game is concerned with only immediate payoffs, real interactions are rarely final and this sometimes motivates individuals to make different

choices. This section enriches the types of games to account for the possibility of repeated play. In this more complicated setting, cooperation may be rational even when it does not give the best immediate return.

Since the forthcoming games are extremely complex in terms of possible strategies, we will not make any attempt to give general solutions or even guidelines for how to play. We will instead focus on simply demonstrating that there exist pairs of strategies that cooperate in equilibrium. The main theorem from this section, known as the Folk Theorem, pushes this phenomena to its extreme. This famous theorem says that, in some sense, any plausible payoffs can be achieved by a pair of strategies in equilibrium. The Folk Theorem applies broadly in the social sciences as it provides a way in which cooperation may emerge without any external agent.

Repeated Games. Concepts from this section apply in very broad settings, but we begin with the 2×2 matrix game PD that is a variant of the Prisoner's Dilemma:

	C	D
C	2, 2	0, 3
D	3, 0	1, 1

The matrix game PD

While the numbers in this matrix game are different from those in the standard Prisoner's Dilemma, this is strategically just the same. Pure strategies for each player are still cooperate (C) and defect (D), and, as before, Defect is strictly dominant for both players. However (as in the usual Prisoner's Dilemma), when both players defect, the payoffs of (1, 1) are worse than the (2, 2) with cooperation.

We previously discussed the possibility of playing a matrix game over and over again (especially in the context of probabilistic strategies), but now we consider repeated play in a very different manner. Namely, we are going to introduce a new type of game that itself has many stages of play. Here is a first example.

Game 10.9 (Probabilistic Repeated Prisoner's Dilemma). Begin by playing one round of the above game of Prisoner's Dilemma and recording

the scores. Then each player flips a coin, and if both flip tails, the game ends. Otherwise, play another round of the game (again recording the scores) and then toss coins to see if the game continues. Keep playing until one of the coin tosses results in a pair of tails. The goal (as always) is to maximize total payoff without considering the other player's payoff.

At first blush it might seem that this game should be essentially similar to the one-time matrix game PD, but it's not. This is a far more complicated realm. The key difference in this repeated game is that the choices of each player in later rounds are permitted to depend on what the other player did in earlier rounds. As a result, this repeated game becomes far more strategically complex than just a one-time play of PD. Unlike the one-time play of PD—with just a single pure Nash equilibrium (where both players defect)—we will prove that the probabilistic repeated version has a rich variety of pure Nash equilibria (many of which feature cooperation). With this example in hand, consider the general repeated matrix game central to this section.

Definition 10.10. For a matrix game A and a number $0 < \delta < 1$, define the game $\text{Repeat}(A, \delta)$ as follows. On each round, Rose and Colin play the matrix game A . Then they toss a biased coin to see if the game continues. With probability δ the coin comes up heads, and the game continues to another round. With probability $1 - \delta$ the coin comes up tails and the game ends. The payoff for each player is the sum of that player's payoffs over all of the rounds. Note that the particular game we introduced at the start of this section is equivalent to $\text{Repeat}(\text{PD}, \frac{3}{4})$.

There is another interpretation of the game $\text{Repeat}(A, \delta)$ that appears in economics. Instead of flipping a coin and possibly ending the game, imagine that the players are going to play the game A infinitely many times, but the payoff for each player will be determined by the payoff for the first round plus δ times the payoff on the second round, plus δ^2 times the payoff on the third round, and so on. To see that these two interpretations are essentially equivalent, consider a player who gets a payoff of 2 in each round. This new interpretation gives a total payoff of $2 + 2\delta + 2\delta^2 + \dots$. In the original probabilistically repeated interpretation, our player will get 2 on the first round, with expected payoff for the second round of 2δ (i.e. the payoff of 2 times the probability that the

game continues to the second round), and the expected payoff for the third round will be $2\delta^2$, and so on. This new interpretation is known as “discounting the future” and is based on the simple economic property that one would rather receive a dollar today than receive a dollar tomorrow. In a business-related repeated-play scenario, then, the players should value a payoff of $\$n$ in an early round more than a payoff of $\$n$ in a later round.

Strategies and Equilibria in Repeated Games. What about the notion of strategy in a repeated matrix game? Generally, a (pure) strategy in a game (e.g. a combinatorial game, game tree, or matrix game) is a plan that tells a player what to do each time there is a decision to make. Since a repeated matrix game has many rounds, a strategy must tell a player what to do on each one. This strategy may now also take into account what the other player has done on earlier rounds. This brings us to the following formalization.

Definition 10.11. A *pure strategy* in the game $\text{Repeat}(A, \delta)$ is a rule that indicates what choice to make on the k^{th} round, depending on what has happened in rounds $1, \dots, k - 1$.

All of the following are pure strategies in the game $\text{Repeat}(\text{PD}, \delta)$:

Strategy	Rule
Always Defect	Defect on every round.
Always Cooperate	Cooperate on every round.
Grim Trigger	Cooperate on every round until the other player defects. Then defect on all future rounds.
Silly Prime	Cooperate unless the total number of times the other player has defected is a prime.

Of course, rational play dictates that each player should defect in a one-time game of Prisoner’s Dilemma and a pair of defect strategies gives the only possible Nash equilibrium. What happens in repeated games? With the definition below of Nash equilibrium extended to the setting of repeated games, we will look to find cooperative equilibria in this new setting.

If S and T are pure strategies in $\text{Repeat}(A, \delta)$, we say that S is a *best response* to T if S gives a highest possible payoff over all possible strategies played against T . We say that the pair S, T forms a *pure Nash equilibrium* if S is a best response to T and T is a best response to S . In order to investigate equilibria and best responses, the formula below for the sum of a geometric series is frequently useful.

Proposition 10.12. $t + t\delta + t\delta^2 + \dots = \frac{t}{1-\delta}$ for every $0 < \delta < 1$.

Proof. $(1-\delta)(t + t\delta + t\delta^2 + \dots) = t + (t\delta - t\delta) + (t\delta^2 - t\delta^2) + \dots = t.$ ¹ $\square$

So, whenever a player receives a payoff of t on each round in $\text{Repeat}(A, \delta)$, her total payoff is $\frac{t}{1-\delta}$. With this, it is straightforward to find the payoffs when the strategies Always Cooperate and Always Defect play against either themselves or the other:

Strategies			Payoffs
Always Cooperate	vs.	Always Cooperate	$(\frac{2}{1-\delta}, \frac{2}{1-\delta})$
Always Defect	vs.	Always Defect	$(\frac{1}{1-\delta}, \frac{1}{1-\delta})$
Always Defect	vs.	Always Cooperate	$(\frac{3}{1-\delta}, 0)$

This table demonstrates that Always Cooperate is not a best response either to itself or to Always Defect. So, Always Cooperate does not form a pure Nash equilibrium with either itself or Always Defect. On the other hand, it is straightforward to verify that Always Defect is a best response to itself. Indeed, if one player were going to defect every round no matter what, then the other player would get the highest payoff by also defecting on every round. So, two strategies of Always Defect do form a pure Nash equilibrium. In this repeated game, the strategies Always Defect and Always Cooperate act much like the strategies Defect and Cooperate in the one-round game.

Grim Trigger. The strategy Grim Trigger effectively has two phases. In the first phase, Grim Trigger is cooperative and plays C each time. If the other player ever fails to cooperate, it turns into a retaliatory phase and plays D every time. Suppose that a player plays against Grim Trigger. What are the possible payoffs? If the player cooperates on each round,

¹This proof uses the fact that the sequence $t, t\delta, t\delta^2, \dots$ is absolutely convergent.

then Grim Trigger will stay in its cooperative phase and will also cooperate each round. Thus, the payoff will be $\frac{2}{1-\delta}$. Now suppose the player chooses a strategy that cooperates for the first k rounds and then defects on round $k + 1$. The Grim Trigger strategy will then be cooperative for rounds $1, \dots, k + 1$ and then turn retaliatory and defect from round $k + 2$ onward. In light of this, there is no point to cooperating on round $k + 2$ or afterward. In other words, the best payoff against Grim Trigger is either that obtained by cooperating every round or by cooperating for k rounds and then defecting afterward. The following calculation shows the payoff for this latter strategy:

$$\begin{aligned}
 \text{Payoff} &= 2 + 2\delta + \dots + 2\delta^{k-1} + 3\delta^k + \delta^{k+1} + \delta^{k+2} + \dots \\
 &= (2 + 2\delta + 2\delta^2 + \dots) + \delta^k - (\delta^{k+1} + \delta^{k+2} + \dots) \\
 &= \frac{2}{1-\delta} + \delta^k - \frac{\delta^{k+1}}{1-\delta} \\
 &= \frac{2}{1-\delta} + \delta^k \left(1 - \frac{\delta}{1-\delta}\right).
 \end{aligned}$$

When $\delta > \frac{1}{2}$, the quantity $1 - \frac{\delta}{1-\delta}$ is negative and a player is best off cooperating every round. On the other hand, when $\delta < \frac{1}{2}$, this quantity is positive, and a player earns the best payoff when δ^k is as large as possible. Since $0 < \delta < 1$, this is achieved at $k = 0$. It follows that when $\delta < \frac{1}{2}$, always defecting results in the best payoff against Grim Trigger. In the boundary case when $\delta = \frac{1}{2}$, then $1 - \frac{\delta}{1-\delta} = 0$ and the above calculation shows that cooperating every time or cooperating for k rounds and defecting thereafter (for every k) all yield the same (best possible) payoff.

What happens if two copies of Grim Trigger play against one another? By definition, each of these strategies will cooperate each round, so both players will get a payoff of $\frac{2}{1-\delta}$. In the case when $\delta \geq \frac{1}{2}$, cooperating every round gives the highest possible payoff vs. Grim Trigger as shown above. So, in this case, Grim Trigger is a best response to Grim Trigger. Finally, a pure Nash equilibrium consisting of cooperating strategies!

Let us pause to recount the key features of Grim Trigger that brought about this equilibrium. In a way, Grim Trigger insists that the other player adopt the plan of cooperating on each round. Someone playing against Grim Trigger who deviates from this plan on round k can get a

higher payoff on round k , but on all future rounds, $k + 1, k + 2, \dots$, Grim Trigger will punish the player entering its retaliatory phase, and then the player gets poor payoffs. So long as $\delta > \frac{1}{2}$, future earnings from cooperating with Grim Trigger exceed the one-time payoff from defecting, so a player is better off cooperating.

The Folk Theorem. The version of the Folk Theorem presented here is an elementary but instructive one. In fact, there is not just one Folk Theorem, but rather a body of related results known by this name. These theorems all assert (under various assumptions) that any “reasonable” payoffs can result from two strategies that form a pure Nash equilibrium in a repeated game. In particular, these theorems demonstrate a way in which cooperation may emerge in equilibrium strategies (without any appeal to an external agent) for a wide range of repeated games.

The proof below is essentially a sophisticated generalization of the key principle behind the Grim Trigger equilibrium above. This theorem requires one extra bit of terminology. We need a modified notion of the payoff in a repeated matrix game that facilitates meaningful comparison of payoffs in the games $\text{Repeat}(A, \delta)$ and $\text{Repeat}(A, \delta')$ when $\delta \neq \delta'$. Define the *normalized payoff* of $\text{Repeat}(A, \delta)$ to be the (usual) payoff multiplied by $1 - \delta$. Note that a player who earns 2 in each round of the game $\text{Repeat}(A, \delta)$ will then get a normalized payoff of $(1 - \delta) \frac{2}{1 - \delta} = 2$, independent of the value of δ .

Theorem 10.13 (Folk). *Let A be a matrix game with status quo point (x_0, y_0) and let (x, y) be a point in the payoff polygon with $x > x_0$ and $y > y_0$. Then for all δ sufficiently close to 1, $\text{Repeat}(A, \delta)$ has a pair of strategies with normalized payoffs (x, y) that form a pure Nash equilibrium.*

Proof. We will construct a pair of coordinated probabilistic strategies that form a pure Nash equilibrium and that have (x, y) as the expected payoffs for the players (it is possible to remove this randomization, but we omit this detail). The constructed strategies, $\mathcal{R}$ for Rose and $\mathcal{C}$ for Colin, will have two phases: a cooperative phase and a retaliatory phase. Just like Grim Trigger, $\mathcal{R}$ and $\mathcal{C}$ begin in a cooperative phase and remain there until the other player deviates from a certain predetermined course. Once one player deviates the other will move to the retaliatory phase and play to punish in every future round.

Since (x, y) is a point in the payoff polygon, we can express (x, y) as a convex combination of entries of A . Therefore, there exist nonnegative numbers $p_1, \dots, p_\ell$ that satisfy $p_1 + \dots + p_\ell = 1$ together with a selection of rows $r_1, \dots, r_\ell$ and columns $c_1, \dots, c_\ell$ that satisfy the following equation (here $a_{i,j}$ denotes the (i, j) entry of A):

$$(x, y) = \sum_{i=1}^{\ell} p_i a_{r_i, c_i}.$$

Since the numbers $p_1, \dots, p_\ell$ are nonnegative and $p_1 + \dots + p_\ell = 1$, these are probabilities we can use to define the cooperative phase of the strategies. On each round of play while $\mathcal{R}$ and $\mathcal{C}$ are in the cooperative phase, the players choose a random index $1 \leq i \leq \ell$ according to the rule that i is selected with probability p_i . If the index i has been selected, then the predetermined course will be for Rose to play row r_i and for Colin to play the column c_i . Assuming the players follow this course, the above equation tells us that the expected payoff on each round will be exactly (x, y) . It follows that the expected normalized payoffs for the play of $\text{Repeat}(A, \delta)$ will also be (x, y) .

Both strategies $\mathcal{R}$ for Rose and $\mathcal{C}$ for Colin will stay in the cooperative phase so long as the other player follows the above predetermined course. However, if Colin were to deviate from this course, Rose's strategy $\mathcal{R}$ would go into retaliatory phase. In this phase Rose plays according to a mixed strategy $\mathbf{p}$, chosen so that Colin's expected payoff on each round will be at most his security level y_0 . Similarly, if Rose were to deviate from the course, Colin's strategy $\mathcal{C}$ would go into retaliatory phase in which Colin plays according to a mixed strategy $\mathbf{q}$ to limit Rose's expected payoff on each round to her security level x_0 .

At this point, we have constructed a strategy $\mathcal{R}$ for Rose and a strategy $\mathcal{C}$ for Colin. When both players adopt these strategies, both will stay in the cooperative phase and the resulting normalized payoffs will be (x, y) . It remains to prove that these strategies form a pure Nash equilibrium when δ is sufficiently close to 1. So, to complete the proof, we need to show that for δ sufficiently close to 1, $\mathcal{C}$ is a best response to $\mathcal{R}$ and vice versa.

Assume that Colin plays according to $\mathcal{C}$ and that Rose plays according to $\mathcal{R}$ up until round k , when she deviates. We will prove that when δ

is high enough, Rose gets a worse payoff than if she continued to follow the strategy $\mathcal{R}$. On round k , Rose will surprise Colin and she may get a very high payoff, but she cannot get a higher payoff than M , which we define to be the maximum value in Rose's payoff matrix. For round $k + 1$ and beyond, $\mathcal{C}$ will be in the retaliatory phase, so on each such round Rose will get an expected payoff of at most x_0 . We can compute Rose's expected payoffs for rounds k and beyond as follows:

$$\delta^{k-1}M + \delta^k x_0 + \delta^{k+1}x_0 + \dots = \delta^{k-1} \left(M + \frac{\delta x_0}{1 - \delta} \right).$$

On the other hand, if Rose were to continue cooperating, her expected payoffs from rounds k and higher would be

$$\delta^{k-1}x + \delta^k x + \dots = \delta^{k-1} \left(\frac{x}{1 - \delta} \right).$$

Continuing to follow $\mathcal{R}$ gives Rose a better payoff whenever $\frac{x - \delta x_0}{1 - \delta} > M$. Since $x > x_0$, the limit as δ approaches 1 from below $\frac{x - \delta x_0}{1 - \delta}$ is ∞ . So, if $0 < \delta < 1$ is sufficiently close to 1, Rose's payoff will be better when she continues to follow $\mathcal{R}$ than when she deviates from it. In other words, for all sufficiently high values of δ , the strategy $\mathcal{R}$ is a best response to $\mathcal{C}$. A similar argument for $\mathcal{C}$ shows that these strategies form a pure Nash equilibrium, as desired. $\square$

Axelrod's Olympiad. The Folk Theorem demonstrates the existence of cooperative equilibria in a wide variety of repeated games. However, the strategies, like the Grim Trigger, constructed to prove this theorem would be rather unreasonable in most circumstances. For instance, if one abandoned any friendship as soon as a friend committed even a small annoyance, very soon no friends would remain—a sad circumstance, indeed! Furthermore, the mere existence of cooperating strategies in equilibrium doesn't really indicate that cooperation is a good idea—only that it may be rational as a response in certain settings. Is there a way to argue that cooperation really is advantageous in an appropriate setting?

In response to this question, Robert Axelrod devised a tournament where players compete in games of Repeated Prisoner's Dilemma. In

his competition, the competitors are strategies for playing Repeated Prisoner's Dilemma, denoted $S_1, \dots, S_k$. These strategies are viewed as genetic variants of a species that exhibits different strategies in pairwise interactions. The first generation is evenly divided among the different variants $S_1, \dots, S_k$. Now the expected payoff is computed for each strategy S_i competing against a randomly chosen opponent. Strategies that did well in the first round make up a higher percentage of the population in the second round and strategies that did worse make up a smaller percentage (as one would expect in a true evolutionary competition). This competition continues for many stages until the population distribution has stabilized. The strategy making up the highest percentage of the final population wins.

Axelrod invited a number of social scientists to submit computer programs designed to play against one another in a repeated game of Prisoner's Dilemma. Each program chose whether to cooperate or defect each round based on what happened during the previous rounds. So, in other words, each program executed a strategy S_i for this Repeated Prisoner's Dilemma game. Axelrod had these programs compete in an olympiad structured as the evolutionary competition described above. The champion was the following delightfully simple submission by Anatol Rapoport.

Strategy 10.14 (Tit-for-Tat). Cooperate on the first round. On each future round, do what the other player did on the last round.

After publicizing his findings, Axelrod ran his olympiad again. The second competition featured some strategies specifically designed to beat Tit-for-Tat, but Tit-for-Tat won yet again! (The strategies coded to beat Tit-for-Tat tended to do poorly when squared off against other strategies.) Tit-for-Tat has many nice properties, but its key feature is that it encourages cooperation. It will retaliate if the other player defects, so as not to be exploited. However, after retaliating, it will be forgiving in the sense that it will cooperate again.

Tit-for-Tat should not be viewed as *the* best strategy for repeated play of Prisoner's Dilemma. Even in Axelrod's evolutionary competitions, many other strategies performed nearly as well. Nonetheless, Axelrod's Olympiads and numerous subsequent related competitions all

reach one very robust conclusion: In evolutionary play of Repeated Prisoner's Dilemma (and similar games), the strategies that emerge successful are highly cooperative. Although decision-making is considerably more nuanced than any of our models, Axelrod's competition still does capture some of its essence, and already we can see here the great value of cooperation.

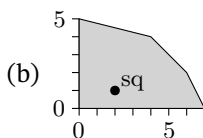
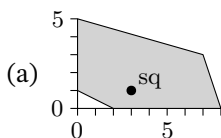
Exercises

(1) For each matrix game below do the following:

- (i) Find the payoff polygon.
- (ii) Find the security level for each player.
- (iii) Find the negotiation set.

$$(a) \begin{bmatrix} (0, 0) & (3, 3) \\ (1, 5) & (4, 0) \end{bmatrix}, \quad (b) \begin{bmatrix} (0, -1) & (2, 7) \\ (6, 6) & (8, 1) \end{bmatrix}, \quad (c) \begin{bmatrix} (5, 4) & (4, 5) \\ (0, 7) & (8, 0) \end{bmatrix}.$$

(2) Determine the negotiation set from each figure below:

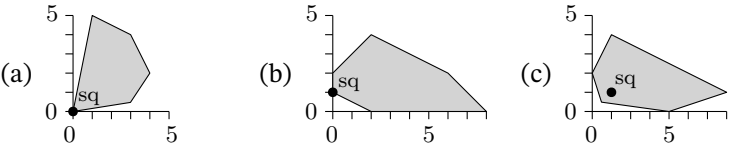


(3) Given a 2×2 matrix game A constructed using the entries $(-1, -1)$, $(6, 1)$, $(2, 8)$, and $(5, 5)$.

- (a) What is the payoff polygon for A ?
- (b) Arrange the entries so that A has security level 2 for Rose and $\frac{41}{13}$ for Colin.
- (c) Arrange the entries so that A has a negotiation set of size 1.

(4) Use the axioms to find the solution point to Nash arbitration applied to the dilemmas with payoff polygon from Figure 10.3.

(5) Find the solution point to Nash arbitration:



(6) Find the solution point to Nash arbitration on $\left[\begin{matrix} (0, y) & (5, 5) \\ (0, 0) & (x, 0) \end{matrix} \right]$.

(7) A company has informally agreed to hire a job applicant, and a representative for the company is negotiating with the applicant about the terms of the job offer. The usual starting salary is \$75,000, but this can be either increased or decreased by up to \$3,200. There is a possible one-time signing bonus of an amount between \$0 and \$12,000. Finally, the usual number of vacation days for a starting employee is 10, but this can be negotiated upward by up to 5 days. So, each aspect of the job offer, salary (S), vacation (V), and bonus (B), has a maximum value, which we indicate by a +, and a minimum value, which we indicated by a −. The following table indicates the utility that the applicant and the company assign to the max and min in each case:

Payoffs (applicant, company)			
	Salary (S)	Vacation (V)	Bonus (B)
Maximum (+)	4, −6	3, −2	2, −1
Minimum (−)	−4, 6	0, 0	0, 0

There are 8 pure outcomes such as $S^+V^-B^+$ where each of the three aspects S, V, B is either at the maximum or minimum value. Each such outcome is associated with payoffs for the applicant and company (e.g. $S^+V^-B^+$ is associated with the payoffs $(4 + 0 + 2, -6 + 0 - 1) = (6, -7)$).

- (a) Plot the payoff polygon associated with the 8 pure outcomes.
- (b) Find the solution point to Nash arbitration with status quo point $(0, 0)$.
- (c) Find the salary, bonus, and number of vacation days that correspond to the solution point given by Nash arbitration.

- (8) This exercise concerns two new arbitration schemes defined below. As usual, the input to an arbitration scheme is a polygon P together with a status quo point (x_0, y_0) and the output is a point in P called the solution point.

Definition 10.15 (Kalai-Smorodinsky Arbitration). Let $x_{\max}$ and $y_{\max}$ be the maximum x -coordinate and y -coordinate of a point in the negotiation set. Define the solution point to be the intersection between the negotiation set and the line segment between (x_0, y_0) and $(x_{\max}, y_{\max})$.

Definition 10.16 (Egalitarian Arbitration). For the solution point, choose a point (x, y) in the negotiation set which maximizes $x + y$. If there is more than one such point, choose the one which minimizes $|x - y|$.

- (a) Apply Kalai-Smorodinsky arbitration to (b) and (c) from Exercise (5) and find the solution points.
 (b) Apply egalitarian arbitration to (a) and (b) from Exercise (5) and find the solution points.
- (9) Consider the matrix game A shown below:

	X	Y
X	3, 3	2, 4
Y	4, 2	1, 1

Here are three strategies for playing $\text{Repeat}(A, \delta)$:

Strategy	Rule
Always X	Play X on every round.
Always Y	Play Y on every round.
XY Trigger	Play X on every round until the other player plays Y . Then play Y on every future round.

- (a) Compute the expected payoffs when the players play $\text{Repeat}(A, \frac{2}{3})$ using the following strategies:
 (i) Both players play Always X .
 (ii) Both players play Always Y .
 (iii) Rose plays Always X and Colin plays Always Y .
- (b) Do any strategy pairs from part (a) form a pure Nash equilibrium?

- (c) If the players play $\text{Repeat}(A, \delta)$ and both play XY Trigger, what are the expected payoffs?
- (d) For what values of δ do two copies of XY Trigger form a pure Nash equilibrium?
- (10) *Alternating Trigger* is a strategy for playing the Repeated Prisoner's Dilemma game PD. Just like Grim Trigger, this strategy has a cooperative phase and a retaliatory phase. During the cooperative phase, Alternating Trigger follows the alternating sequence C, D, C, D, C, ..., and this continues as long as the other player plays exactly the same sequence. If the other player ever deviates from this sequence, then Alternating Trigger goes into a retaliatory phase and uses D for every future round.
- (a) If Rose and Colin play $\text{Repeat}(PD, \delta)$ and both use the strategy Alternating Trigger, what are the expected payoffs?
- (b) For what values of δ does a pair of Alternating Trigger strategies form a Nash equilibrium?
- (11) Let P be a convex payoff polygon with status quo point (x_0, y_0) and assume (as usual) that P contains a point (x_1, y_1) with $(x_1, y_1) \geq (x_0, y_0)$. Prove that either there is a point (x, y) in P with $(x, y) > (x_0, y_0)$ or the negotiation set has size 1.
- (12) Prove Theorem 10.3.
- (13) Consider the matrix game $A = \begin{bmatrix} (1, 1) & (1, 0) \\ (0, 0) & (0, 1) \end{bmatrix}$ and observe that its payoff polygon P is the unit square. In particular, the point $(\frac{1}{2}, 1)$ is in P since we may write it as $\frac{1}{2}(0, 1) + \frac{1}{2}(1, 1)$. Our players could coordinate to get expected payoffs $(\frac{1}{2}, 1)$ by flipping a (fair) coin, and in the case of heads, Rose plays row 1 and Colin plays column 1, while in the case of tails, Rose plays row 2 and Colin plays column 2. Note that it is necessary for the players to coordinate their choices to achieve this result. In this exercise we will consider the set S of expected payoffs which could be achieved without this coordination. More formally, we define S to be the set of all points (x, y) which satisfy $(x, y) = [p \quad 1-p] A \begin{bmatrix} q \\ 1-q \end{bmatrix}$ for some $0 \leq p \leq 1$ and $0 \leq q \leq 1$. Plot S in the plane.

- (14) Let P be a payoff polygon with status quo point (x_0, y_0) and assume that the negotiation set of P contains a line segment given by the equation $y = -cx + d$ where $a \leq x \leq b$. Suppose that the point on this line segment which maximizes $(x - x_0)(y - y_0)$ is (x^*, y^*) where $a < x^* < b$. Prove that (x^*, y^*) is the solution point to Nash arbitration.
- (15) Let $c > 0$ and let (x_0, y_0) be a point which lies below the line $y = -cx + d$. Prove that the point on this line which maximizes $(x - x_0)(y - y_0)$ also lies on the line through (x_0, y_0) with slope c .
- (16) Prove that Nash arbitration satisfies all four of Nash's axioms.
- (17) For a positive integer n and a matrix game A , the game $n \times A$ is a repeated game where A will be played exactly n times in succession. The payoff for each player in $n \times A$ is the sum of their payoffs over the n rounds of play. Suppose two perfectly rational players play the game $n \times PD$. Determine what they will do and prove it.
- (18) The arbitration schemes below are defined in Exercise (8). For each of them, either prove it satisfies each of Nash's four axioms or give a counterexample showing it does not:
- egalitarian arbitration,
 - Kalai-Smorodinsky arbitration.
- (19) There are others beyond Nash's four axioms that one might like an arbitration scheme to satisfy. Here is one such axiom.
- (5) Increasing Opportunity. Suppose (x, y) is the solution point for the polygon P with status quo point (x_0, y_0) . If P' is a polygon containing P , then the solution point (x', y') for P' with status quo point (x_0, y_0) satisfies $x' \geq x$ and $y' \geq y$.
- Show that it is impossible for any arbitration scheme to satisfy axiom (1) (from Nash) and axiom (5) above. Hint: Make three polygons (all with the same status quo point) so that the first has the single point (x_1, y_1) as its negotiation set, the second has the single point (x_2, y_2) as its negotiation set, and the third has the line segment between (x_1, y_1) and (x_2, y_2) as its negotiation set.
- (20*) Consider the following axiom for an arbitration scheme:
- Suppose (x, y) is the solution point for the polygon P with status quo point (x_0, y_0) . Let P' be a polygon containing P which has

the same values for both $x_{\max}$ and $y_{\max}$ as P . If (x', y') is the solution point for P' with status quo point (x_0, y_0) , then $x' \geq x$ and $y' \geq y$.

Prove that Kalai-Smorodinsky arbitration (defined in Exercise (8)) is the unique arbitration scheme which satisfies Nash's axioms (1), (2), (3) and axiom (6) above.

Chapter 11

n -Player Games

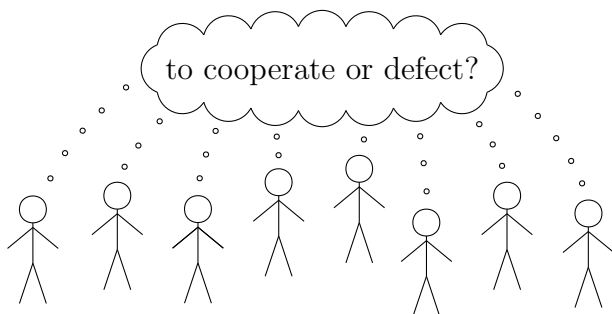


Figure 11.1. Tragedy of the Commons

The previous six chapters introduced the classical theory of 2-player games, a powerful framework for representing decision problems between two parties. Rich theoretical ideas like the Nash Equilibrium Theorem and Nash arbitration help identify rational play and even suggest fair negotiated outcomes. But what if there are more than two players? Welcome to the world of n -player games, involving circumstances that feature many individuals making decisions, all of which may affect the outcome for everyone. The tools of classical n -player game theory expressively model the dynamics of a collection of rational players. Many ideas

from the context of 2-player games—notably, the Nash Equilibrium Theorem—still apply in this more general setting, but we will also encounter some new complexities. Let's begin with another classic.

Game 11.1 (Tragedy of the Commons). This is a game played between 100 farmers who share a field for grazing cattle. Each farmer can decide either to put one or two cows in the field. Each farmer gets +50 for each cow she has on the field. However, the field only sustainably supports 100 cows, and each cow beyond that depletes the field's nutrients. This costs everyone. Every farmer gets -1 for each extra cow.

Say that a farmer who puts one cow in the field is cooperating (C) and a farmer who puts two cows in the field is defecting (D). To analyze this, consider the perspective of farmer i , and suppose that k of the other farmers have chosen to defect. The 2×1 matrix below depicts the payoff for each possible choice of farmer i :

		k other D's	
farmer i	C	50 $- k$	
	D	100 $- (k + 1)$	

Farmer i achieves maximum payoff by defecting and putting two cows in the field. This holds true for every possible value of k , so farmer i gets the highest payoff by defecting no matter what the other farmers do. Thus (D) is the rational play for every farmer. Still, if everyone defects, every farmer will get a payoff of $100 - 100 = 0$, obviously worse than the payoff of 50 every farmer obtains when everyone cooperates. As in the Prisoner's Dilemma, we are confronted here with another situation in which rational individual play leads to an inferior outcome.

The heart of this dilemma is the fact that the penalty for a farmer putting an extra cow in the field is a cost shared among all the farmers, while only the owner of the extra cow earns the benefit. Although this is clearly a toy example, the dynamic present here appears in a variety of more meaningful situations:

- People have the option of cooperating by disposing trash responsibly or defecting by littering. Each person who litters gets a bonus for getting rid of trash. However, for every player who litters, there is a small penalty to everyone for the accumulated trash.

- Fishing companies can cooperate with sustainable fishing or defect by fishing in an unsustainable way. Unsustainable fishing yields a higher profit for a company. However, the fish population declines with each company that fishes unsustainably, thus penalizing every company.
- Nations can cooperate by limiting CO_2 emissions or defect by continuing as is. A country that defects will have greater productivity. However, each country that defects contributes to the global accumulation of CO_2 , thus causing a penalty to everyone.

In such situations, we prefer to have all players cooperate, but rationality dictates they should all defect. To arrange for cooperation requires that we *change the game* by adding incentives or penalties to alter the payoffs associated with the different outcomes. How might constructs such as social norms and laws be used to alter the payoffs in each of the three instances above?

11.1. Matrix Games

Our investigations of 2-player games centered on matrix games. Higher-dimensional matrix games likewise figure vitally in the study of n -player games. Just as in the study of 2-player matrix games, n -player matrix games can be used to model a wide variety of decision problems.

Basic Definitions. We begin with 3-player matrix games. Rose and Colin are still playing, as before, and now a new player, called Larry, will join them. Represent this game not by a two-dimensional matrix with just rows and columns, but instead by a three-dimensional matrix with rows, columns, and *layers*. As before, Rose will choose a row of the matrix and Colin will choose a column; now Larry will choose a layer. These three choices select a matrix entry of the form (x, y, z) , indicating payoffs of x to Rose, y to Colin, and z to Larry.

Example 11.2. Figure 11.2 shows a 3-player matrix game between Rose, Colin, and Larry. So, for instance, if Rose chooses A , Colin chooses C , and Larry chooses F , the payoffs will be $(1, 4, -1)$.

		Larry <i>E</i>		Larry <i>F</i>	
		Colin		Colin	
		<i>C</i>	<i>D</i>	<i>C</i>	<i>D</i>
Rose	<i>A</i>	2, -1, 3	1, 1, -2	1, 4, -1	3, 1, 1
	<i>B</i>	0, 3, 2	2, 0, 1	-1, 1, 0	1, 1, 1

Figure 11.2. A 3-player matrix game

Although it becomes challenging to draw matrix games between large numbers of players, these matrices are nonetheless helpful constructs. To play a matrix game with players 1, 2, ..., n requires an n -dimensional matrix, say with dimensions $d_1 \times d_2 \times \cdots \times d_n$. Each entry of the matrix will have the form $(x_1, x_2, \dots, x_n)$ indicating a payoff of x_i to player i in the event the players' choices select this entry.

For instance, a matrix can model the 100-player Tragedy of the Commons. Each of the 100 players has just two options (cooperate or defect), so the matrix game has dimensions $\underbrace{2 \times 2 \times \cdots \times 2}_{100}$. Every cell of this matrix corresponds to a choice of cooperate or defect for each of the players, and cell entries are of the form $(x_1, \dots, x_{100})$, indicating a payoff of x_i to player i .

Dilemmas. Such n -player matrix games provide an insightful mathematical model for situations involving many players. We have already articulated one dilemma, the Tragedy of the Commons, as a 100-player version of the Prisoner's Dilemma. Below are 100-player versions of two more dilemmas from Chapter 7, each of which can be represented by a $\underbrace{2 \times 2 \times \cdots \times 2}_{100}$ matrix. These two examples give only a taste of the rich variety of phenomena this framework can model.

Game 11.3 (General Volunteering Dilemma). This is a game played between 100 roommates. The bathroom needs to be cleaned, and everyone must either cooperate by offering to clean it (C) or defect (D) by not offering. If everyone defects, then the bathroom stays dirty and results in a penalty of -10 to each player. If at least one person cooperates by volunteering to clean the bathroom, then the players who defect get a payoff

of 2 for the clean bathroom. If k players cooperate, then they split the work and each gets a payoff of $2 - 20/k$:

		Other Roommates	
		$k \geq 1$ play C	all play D
Roommate i	C	$2 - 20/(k+1)$	-18
	D	2	-10

We note that this General Volunteering Dilemma is a simple generalization of the 2-player Volunteering Dilemma of Game 7.5. As there, each player here wants to defect while at least one other player cooperates—a familiar strategic dynamic.

Game 11.4 (Investing Dilemma). This is a game played among 100 investors. Each player has \$1 and can choose either to invest (I) or hold (H) this money. If at least 90% of the players choose to invest, then each investor earns \$1. Otherwise, each investor loses \$1. The matrix below indicates the outcome for player i depending on the other players:

		Other Investors	
		≤ 88 play I	≥ 89 play I
Investor i	I	-1	1
	H	0	0

This game is essentially a many-player version of Stag Hunt. Each investor benefits if enough other players also invest. Yet there may be some uncertainty about whether the other players can all be trusted to invest. Like Stag Hunt, this game has two pure Nash equilibria, one where every player plays I and the other where each chooses H. Everyone prefers the first equilibrium, but potential mistrust makes for a tricky dynamic.

Dominance. In both zero-sum and general 2-player matrix games, the notion of dominance enhanced our investigation. There is an analogous concept for n -player matrix games and it will prove similarly useful. Suppose player i has two pure strategies s and s' with the property that no matter what every other player does, her payoff will always be at least as good when she selects s as when she selects s' . In this case, we say that strategy s *dominates* strategy s' . Analogously, if, no matter what every other player does, player i does strictly better when playing s than when playing s' , we say that s *strictly dominates* s' .

Example 11.5. In the matrix game from Figure 11.3, Larry's strategy E dominates strategy F . Eliminating the dominated strategy F reduces this to a 2×2 game in which Rose's strategy B dominates strategy A . So, assuming the players are rational, Larry will play E , Rose will play B , and then Colin will play C .

		Larry E		Larry F	
		Colin		Colin	
Rose	A	1, -1, 3	2, -1, 1	3, -1, 2	-1, 2, 0
	B	2, 0, 3	3, -1, 2	1, -1, 0	2, 2, 1

Figure 11.3. Dominance in a 3-player matrix game

In the Tragedy of the Commons, each farmer achieved a higher pay-off from defecting rather than cooperating no matter what the other players did. This corresponds to the property that in the associated matrix game the strategy of defect strictly dominates that of cooperate for every player. Next we will introduce an amusing n -player game to analyze the use of dominance.

Game 11.6 (The $\frac{2}{3}$ of the Average Game). This is an n -player game in which each player secretly chooses an integer between 1 and 100 and then the choices are revealed. The person or persons who selected the number(s) closest to $\frac{2}{3}$ of the average number evenly divide a pot of money.

What are rational players likely to do in this game? Since each player is choosing a number between 1 and 100, the highest possible average is 100. Therefore, $\frac{2}{3}$ of the average will always be less than 67. It follows that the strategy of playing 67 dominates all higher numbers. If we eliminate strategies 68 and above for all of the players, then no player will guess a number larger than 67, so the average will necessarily be at most 67. But then $\frac{2}{3}$ of the average will be less than 45, so the strategy of playing 45 dominates all of the higher strategies. This trend continues and iterated removal of dominated strategies further reduces the game until every player chooses 1.

This analysis suggests that a collection of perfectly rational players who share the common knowledge that the other players are rational will all choose 1. Note that the method of iteratively removing strictly dominated strategies relies heavily on the rationality of the other players and the extent to which this is known. When the $\frac{2}{3}$ of the Average game is actually played, it generally takes some time before the players converge to choosing small numbers.

11.2. Coalitions

One new consideration in n -player games is that of coalitions. For instance, in a 3-player game, two of the players might decide to team up against the third. This section explores n -player games with an eye toward such coalitions and it culminates in an entirely new way of thinking about n -player games.

For simplicity and concreteness, assume for the remainder of this chapter that all payoffs are actual dollar amounts and that each player receives one unit of utility from each dollar. The important side effect of this is that players in a coalition may decide to distribute their winnings differently than the actual result from play (i.e. they may arrange side payments). For instance, if Rose and Colin form a coalition against Larry in a 3-player game and the payoffs were $(5, 1, -2)$, Rose and Colin might agree to split the $5 + 1 = 6$ dollars so they each get 3. Transferable utility is the technical term for this significant assumption. Additionally, assume that the players can communicate and cooperate with one another. These assumptions lead to a mathematically rich and interesting realm of game theory (albeit one that is not universally applicable).

Coalitions in Matrix Games. Before introducing any general ideas, let's begin by considering coalitions in a particular case.

Example 11.7. Consider the following 3-player matrix game:

		Larry E			
		Colin		Larry F	
		C	D	C	D
Rose	A	2, -1, 3	1, 1, -2	1, 4, -1	3, 1, 1
	B	0, 3, 2	2, 0, 1	-1, 1, 0	1, 1, 1

Suppose first that Rose and Colin decide to form a coalition against Larry. Rose's pure strategies are A and B while Colin has the pure strategies C and D . Acting as a coalition, they can decide what Rose will do and what Colin will do. This gives the Rose-Colin coalition four pure strategies: AC , AD , BC , and BD . This Rose-Colin coalition can also play a mixed strategy. For instance, it could play AC with probability $1/2$ and BD with probability $1/2$. Note that this strategy requires Rose and Colin to coordinate their actions, so they would indeed need to be working as a team.

Assuming this coalition has formed, the game reduces to the following 2-player matrix game between Rose-Colin and Larry. Here the payoff to Rose-Colin indicates the sum of the payoffs in the original game for Rose and Colin. So, for instance, in the original game, Rose playing A and Colin playing D and Larry playing F gives a payoff of 3 to Rose and 1 to Colin. In the new game, then, when the Rose-Colin coalition plays AD and Larry plays F , the payoff to Rose-Colin is $3 + 1 = 4$:

		Larry	
		E	F
Rose-Colin	AC	1, 3	5, -1
	AD	2, -2	4, 1
	BC	3, 2	0, 0
	BD	2, 1	2, 1

What is the maximum total payoff the Rose-Colin coalition can guarantee? This is precisely the Rose-Colin security level in the above 2-player game. It follows from Theorem 10.2 that the Rose-Colin security level is the von Neumann value of the zero-sum matrix given by the Rose-Colin payoffs as shown in Figure 11.4. To compute this, we can invoke Procedure 6.7. In this case, Larry has a guarantee function G_L and the value of the game is given by the lowest point on the graph of this function. As shown in the figure, this value is $12/5$, so this is the maximum total expected payoff the Rose-Colin coalition can guarantee.

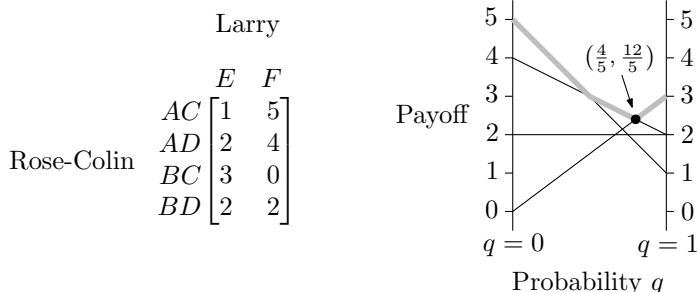


Figure 11.4. Rose-Colin vs. Larry

What about Larry? What is the maximum value he can guarantee himself working alone? The worst case scenario for Larry is when Rose and Colin form a coalition and play against him. Thus Larry's best guarantee in this 3-player game is his security level in the 2-player game Larry vs. Rose-Colin. In Figure 11.5 we calculate this to be $1/7$ using Procedure 6.7.

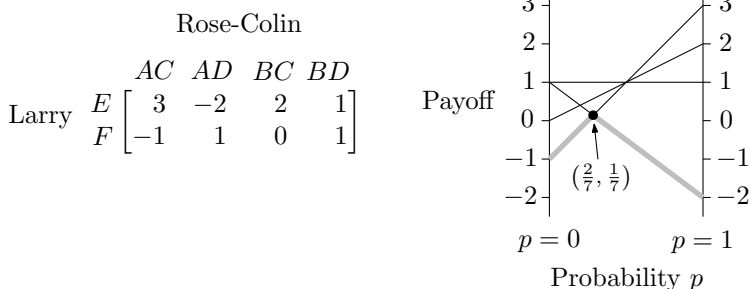


Figure 11.5. Rose-Colin playing to limit Larry

Coalitional Form. The above example suggests an important general concept, namely the total payoff a coalition can guarantee itself (independent of the actions of the other players). More formally, for any matrix game M introduce a function, denoted f_M , called the *coalitional form* of M . The coalitional form f_M is defined by the rule that for every subset S of players, $f_M(S)$ denotes the maximum total expected payoff the players in the set S can guarantee themselves when working as a coalition. Let's return to our 3-player game and determine its coalitional form.

Example 11.7 (Continued). Let's call the matrix from this example M and determine its coalitional form, f_M . The coalitional form is defined for every subset S of players, even the empty set. However, the empty set doesn't ever win any money so $f_M(\emptyset) = 0$. Another extreme is the case when all of the players work together. In this case, the three players can decide to play any combination of the three strategies. They could, for instance, decide that Rose will play B , Colin will play C , and Larry will play E , getting payoffs of $(0, 3, 2)$. In fact, $0 + 3 + 2 = 5$ is the highest sum the three can achieve, so $f_M(\{\text{Rose, Colin, Larry}\}) = 5$. Since it is a bit cumbersome to write $f(\{\dots\})$ when applying f to a nonempty set, we will simplify the notation. Going forward, we will drop the inner set of braces (e.g. write $f_M(\text{Rose, Colin, Larry}) = 5$). Previous considerations of the Rose-Colin coalition determined that, working together, the best guarantee for Rose and Colin is $12/5$, so $f_M(\text{Rose, Colin}) = 12/5$. We've likewise seen that Larry's best guarantee working alone is $1/7$, so $f_M(\text{Larry}) = 1/7$. A similar analysis for the other coalitions produces the full coalitional form for this game:

$$\begin{aligned} f_M(\emptyset) &= 0, & f_M(\text{Rose, Colin, Larry}) &= 5, \\ f_M(\text{Rose}) &= 1, & f_M(\text{Colin, Larry}) &= 13/5, \\ f_M(\text{Colin}) &= 3/5, & f_M(\text{Rose, Larry}) &= 17/7, \\ f_M(\text{Larry}) &= 1/7, & f_M(\text{Rose, Colin}) &= 12/5. \end{aligned}$$

Next, we look beyond this particular example to consider the general case. Suppose that M is an n -player matrix game and we are interested in determining the maximum expected payoff that a certain subset S of players can guarantee itself (i.e. we want to determine $f_M(S)$). Assume further that S is not empty and is also not equal to the set of all players. Define T to be the set of all players not in S and note that T is also not empty and not equal to the set of all players. The worst case for coalition S is if all of the players in T decided to work together in a coalition against coalition S . This results in a 2-player game where the S -coalition plays against the T -coalition. The pure strategies for the S -coalition correspond to any combination of pure strategies for the players in S . The pure strategies for the T -coalition consist of any combination of pure strategies for the players in T . The maximum amount that S can guarantee itself is precisely the security level for S in this 2-player game. Using Theorem 10.2 on these 2-player games yields the following process.

Procedure 11.8 (Finding the Coalitional Form). Consider an n -player matrix game M for players $\{1, \dots, n\}$. The coalitional form of M , denoted f_M , is given by the following rule:

- (1) $f_M(\emptyset) = 0$.
- (2) $f_M(1, \dots, n)$ is equal to the maximum sum of payoffs of the players over all cells of the matrix.
- (3) When $S \subseteq \{1, \dots, n\}$ has at least one, but not all, of the players, let T be the set of players not in S and define a zero-sum 2-player matrix game M_S as follows. Each row of M_S corresponds to a pure strategy for S (i.e. a choice of pure strategy for each player in S) and each column of M_S corresponds to a pure strategy for T (i.e. a choice of pure strategy for each player in T). The entry in position i, j of M_S is the total payoff that S achieves when playing strategy i opposite strategy j for T . Now $f_M(S)$ is the value of M_S .

Note that the above procedure generalizes the above example. To determine the maximum guarantee for Rose-Colin, we computed the value of the zero-sum matrix game where Rose-Colin plays against Larry and each entry is the total payoff to Rose-Colin. Similarly, to compute Larry's maximum guarantee, we computed the value of the zero-sum matrix game where Larry plays against Rose-Colin and each entry is Larry's payoff.

Coalitional Games. The coalitional form f_M associated with a matrix game M gives us a very different way of thinking about the game. This representation ignores the strategic considerations and instead only tells us about the strength of the coalitions. We now distance ourselves from even having a matrix and view a coalitional form as a game in and of itself.

Definition 11.9. We define a *coalitional game* to consist of a set of players $\{1, 2, \dots, n\}$ and a function f that assigns each subset $S \subseteq \{1, 2, \dots, n\}$ a real number $f(S)$ indicating the total payoff that S can obtain as a coalition. Assume that f always satisfies the following:

- (1) $f(\emptyset) = 0$.
- (2) $f(S \cup T) \geq f(S) + f(T)$ whenever S and T are disjoint.

The first condition just enforces the rule that the empty set cannot guarantee a positive payoff. The second condition is natural since whenever S and T are disjoint sets of players, the total that these sets of players can obtain when working together should always be at least the sum of what S and T can obtain working independently. Let's begin our exploration of coalitional games with a simple but instructive example.

Game 11.10 (Divide the Dollar). This is a game played between three players 1, 2, 3. They have one dollar to divide among the three players, and if any two out of the three agree on a division, then that is the outcome. We represent this coalitional game with a function called d (for Divide and Dollar):

$$\begin{aligned} 0 &= d(\emptyset) = d(1) = d(2) = d(3), \\ 1 &= d(1, 2) = d(1, 3) = d(2, 3) = d(1, 2, 3). \end{aligned}$$

Imagine for a moment how the players might negotiate a split of this dollar. Player 1 might suggest a division of $(1/3, 1/3, 1/3)$ since this gives each player the same amount. However, player 2 might suggest to player 3 a two-way split, so each would get $1/2$ by forming a coalition and agreeing upon the division $(0, 1/2, 1/2)$. But then player 1 could offer player 3 an even better split, suggesting $(1/3, 0, 2/3)$. Player 2 might respond by offering to form a coalition with player 1 and divide the dollar as $(1/2, 1/2, 0)$. As you can probably see, this could go on forever!

Just as there is no all-encompassing theory to explain how two rational players should play a general matrix game, there does not exist a complete theory to resolve an arbitrary coalitional game. Many useful concepts have nonetheless been developed to study and help understand coalitional games. The remainder of this chapter introduces a few of these ideas.

Consider a coalitional game f for the players $1, \dots, n$. As in Chapter 10, imagine that the players are going to negotiate about the play of this game. Since the payoffs are transferrable dollars, it will be possible for the players to exchange money after the game is played. If the players are going to agree to divide up the winnings so that player i gets x_i dollars, what properties should we expect of the payoff vector $(x_1, \dots, x_n)$?

Certainly every individual player deserves a payoff at least as large as his or her guarantee in the game. So we should restrict our attention to those payoffs $(x_1, \dots, x_n)$ that satisfy $x_i \geq f(i)$ for every player i . There is also a global optimality condition for the entire group. Namely, the total payoff should be equal to $f(1, \dots, n)$ (i.e. the maximum that the players can achieve working together). If the payoffs $(x_1, \dots, x_n)$ were to add up to less, then it would be possible for the players all to form a coalition and (possibly after exchanging money) every player could get a higher payoff. This motivates the following important definition.

Definition 11.11. If f is a coalitional game for the players $1, \dots, n$, then a vector of payoffs $(x_1, \dots, x_n)$ with x_i going to player i is an *imputation* if it satisfies the following properties:

- (1) Individual Rationality. $x_i \geq f(i)$ for every $1 \leq i \leq n$.
- (2) Group Rationality. $x_1 + x_2 + \dots + x_n = f(1, 2, \dots, n)$.

Von Neumann and Morgenstern introduced the concept of an imputation as a reasonable vector of payoffs in a coalitional game. This well-accepted notion will be our focus going forward in our investigation of these games. Let's examine the set of imputations for Divide the Dollar.

The set of imputations for Divide the Dollar consists of all triples (x_1, x_2, x_3) with $x_1, x_2, x_3 \geq 0$ and $x_1 + x_2 + x_3 = 1$. This set is a triangle in $\mathbb{R}^3$ with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ as depicted in Figure 11.6.

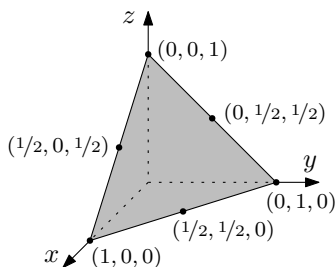


Figure 11.6. A triangle in $\mathbb{R}^3$

To analyze a negotiation between two players, Chapter 10 introduced the concept of a negotiation set. In the special case when the negotiation

set consisted of a single point, the problem of selecting a reasonable and fair negotiated outcome was easy—just pick that point. In the setting of n -player coalitional games, certain games are similarly uninteresting from the standpoint of negotiations since they have just a single imputation. Next we introduce a term for these games.

Definition 11.12. A coalitional game f for the players $1, \dots, n$ is called *inessential* if

$$f(1, 2, \dots, n) = f(1) + f(2) + \dots + f(n),$$

and otherwise we call f *essential*.

Lemma 11.13. A coalitional game f for the players $1, \dots, n$ is inessential if and only if it has exactly one imputation (in this case the unique imputation must be $(f(1), \dots, f(n))$).

Proof. If f is inessential, then $(f(1), f(2), \dots, f(n))$ is an imputation. Furthermore, it must be the only one since any other imputation must have coordinate i at least $f(i)$ and must sum to $f(1, \dots, n) = f(1) + \dots + f(n)$. Next suppose f is essential and define $m = f(1, \dots, n) - (f(1) + \dots + f(n))$, noting that $m > 0$. For every player j , we can form an imputation $(x_1, \dots, x_n)$ by the following rule:

$$x_i = \begin{cases} f(i) & \text{if } i \neq j, \\ f(j) + m & \text{if } i = j. \end{cases}$$

This gives us $n \geq 2$ imputations, thus completing the proof. $\square$

At this point in our study of coalitional games, we have focused on the set of imputations and have defined inessential games that have only a single imputation. Ideally, we would like a way to select a single imputation, or perhaps a small set of imputations, that would be sensible negotiated outcomes for essential games. We use the term “solution concept” for a rule that selects one or more imputations and we now introduce two solution concepts based on the following notion of dominance.

Domination. The initial Divide the Dollar analysis ran in circles as players formed coalitions and then broke them to form different coalitions over and over and over. Some terminology here will explain precisely what was going on there and, in fact, this revisits a many-player variant of the now-familiar notion of dominance.

Consider a coalitional game f for the players $1, \dots, n$ and imagine that someone has proposed the imputation $(x_1, \dots, x_n)$ as a way to split the total payoff. Suppose that there exists a subset of players S and another imputation $(y_1, \dots, y_n)$ with the following properties:

- $y_i > x_i$ for all $i \in S$.
- $f(S) \geq \sum_{i \in S} y_i$.

In this case, the players in S could form a coalition in the game and then split their total payoff so that each player in S gets a higher payoff than they receive in the imputation $(x_1, \dots, x_n)$. Consequently, these players have a legitimate objection to the proposed division $(x_1, \dots, x_n)$. In this case, we say that the imputation $(y_1, \dots, y_n)$ *S-dominates* $(x_1, \dots, x_n)$.

With this notion of dominance, we can better understand the repeated cycle of one Divide the Dollar proposal being rejected in favor of another. Player 1 first suggested the imputation $(1/3, 1/3, 1/3)$, but this was $\{2, 3\}$ -dominated by the imputation $(0, 1/2, 1/2)$, which was $\{1, 3\}$ -dominated by $(1/3, 0, 2/3)$. This, in turn was $\{1, 2\}$ -dominated by $(1/2, 1/2, 0)$. Because of domination, the players move from one proposal to the next. This suggests some value in finding imputations that are not dominated by any others.

Definition 11.14. The *core* of a coalitional game f for the players $1, \dots, n$ is the set of all imputations $(x_1, \dots, x_n)$ with the property that $\sum_{i \in S} x_i \geq f(S)$ for every set S of players. In other words, an imputation $(x_1, \dots, x_n)$ is in the core if every set of players receives a total payoff at least as large as they could guarantee themselves by working as a coalition.

In fact, as proved in Exercise (12), the core of f is precisely the set of all imputations that are not dominated by any other imputation. Since domination is such a fundamental principle, the core is a natural and important set of imputations. There do exist coalitional games with an empty core, so in these cases the core isn't particularly useful. In fact, it's straightforward to verify that the core of Divide the Dollar has nothing in it (Exercise (4)).

Next, we give another solution concept for coalitional games based on domination.

Definition 11.15. A *stable set* is a set J of imputations with the following properties:

- (1) No imputation in J dominates another imputation in J .
- (2) Every imputation not in J is dominated by one in J .

We claim that $J = \{(1/2, 1/2, 0), (1/2, 0, 1/2), (0, 1/2, 1/2)\}$ is a stable set in Divide the Dollar. It's easy to check that none of the imputations in J dominates another, so the first condition is satisfied. For the second condition, consider an arbitrary imputation (x_1, x_2, x_3) and suppose (without loss of generality) that $x_1 \leq x_2 \leq x_3$. If $x_1 \leq x_2 < 1/2$, then this imputation is $\{1, 2\}$ -dominated by $(1/2, 1/2, 0)$. Otherwise we must have $(x_1, x_2, x_3) = (0, 1/2, 1/2)$, but then our imputation is already in J .

In fact, Divide the Dollar has many other stable sets (see Exercise (15)), so a stable set should not be viewed as somehow optimal. Nonetheless, these sets certainly have some nice features. Since every imputation is dominated by one in the stable set, it is reasonable to argue that we may restrict our attention to a stable set without significant loss. Also, since no imputation in a stable set dominates another, we avoid running in circles by restricting our attention to a stable set.

Cores and stable sets are solution concepts based on the fundamental notion of domination. Some games, like Divide the Dollar, unfortunately have no core. Still other coalitional games have no stable sets. So, on the downside, these solution concepts are somewhat limited in their applicability, but on the significant upside they are valuable when they do apply. The next section returns to questions of fairness and introduces another solution concept called Shapley Value—something that always exists and is unique!

11.3. Shapley Value

Nash arbitration appeared in Chapter 10 as a rule that can be used to select an arguably fair point in a payoff polygon. Here, we introduce a roughly analogous idea for coalitional games. The new setting is an n -player coalitional game f . We would like to determine an imputation $(x_1, \dots, x_n)$ that, in some sense, fairly represents the value player i contributes to the coalitions.

Define a *valuation scheme* to be a rule that takes as input a coalitional game f and outputs an imputation $(x_1, \dots, x_n)$ for this game. A valuation scheme may be regarded as a special type of function of the form shown in Figure 11.7.

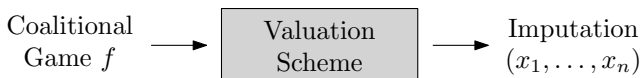


Figure 11.7. A valuation scheme

Introducing Shapley Value. Central to this section is the following valuation scheme due to Lloyd Shapley.

Definition 11.16 (Shapley Value). Given an n -player coalitional game, we will give a rule to determine an imputation $(x_1, \dots, x_n)$. To compute the x_i values, consider all $n!$ ways to order the n players. For each ordering, imagine growing a larger and larger coalition by starting with the empty set and adding one player at a time according to this order, until all n players are part of the coalition. Determine how much value player i adds to this growing coalition as follows: If S is the set of players before i in the ordering, then player i adds a value of $f(S \cup \{i\}) - f(S)$ to the growing coalition. The payoff x_i assigned to player i will be the average value this player adds over all possible orderings.

Example 11.17. Consider a coalitional game with players $\{1, 2, 3\}$ given by the following function f :

$$\begin{aligned}
 f(1) &= 2, & f(1, 2) &= 8, \\
 f(\emptyset) &= 0, & f(2) &= 3, & f(1, 3) &= 7, & f(1, 2, 3) &= 10, \\
 f(3) &= 1, & f(2, 3) &= 6.
 \end{aligned}$$

To compute the Shapley Value, consider all $3! = 6$ orderings of the players. For each ordering, determine the value that each player adds to the growing coalition. As an example, for the ordering 123 the first player adds $f(1) = 2$ to the growing coalition. The second player contributes $f(1, 2) - f(1) = 8 - 2 = 6$, and the third contributes $f(1, 2, 3) - f(1, 2) =$

$10 - 8 = 2$. The table below indicates the value each player adds for each possible ordering:

Ordering	Value added by		
	Player 1	Player 2	Player 3
123	2	6	2
132	2	3	5
213	5	3	2
231	4	3	3
312	6	3	1
321	4	5	1
Total	23	23	14
Average	$23/6$	$23/6$	$14/6$

The Shapley Value is given by the averages in the above table so it is $(23/6, 23/6, 7/3)$.

A quick check reveals that $(23/6, 23/6, 7/3)$ is indeed an imputation for this game f since the entries sum to $10 = f(1, 2, 3)$ and each player i gets a payoff of at least $f(i)$. Next, we give a proof that this holds in general.

Proposition 11.18. *Shapley Value is a valuation scheme.*

Proof. Suppose that Shapley Value applied to f gives $(x_1, \dots, x_n)$. We need to prove that $(x_1, \dots, x_n)$ is an imputation, so we must show that it satisfies both the Individual Rationality and Group Rationality conditions.

For Individual Rationality, let i be an arbitrary player and show that $x_i \geq f(i)$. To do so, consider an arbitrary ordering of the players, and consider how much player i adds by joining the growing coalition. If S is the set of players before i , then the value added by player i is $f(S \cup \{i\}) - f(S)$. The definition of a coalitional game implies that $f(S \cup \{i\}) - f(S) \geq f(i)$. Thus, for every possible ordering, player i contributes at least $f(i)$ to the value of the growing coalition. Since x_i is an average of these values, it follows that $x_i \geq f(i)$, as desired.

To prove Group Rationality, we need to show that $(x_1, \dots, x_n)$ satisfies $x_1 + \dots + x_n = f(1, \dots, n)$. To do this, first consider the ordering of the players $1, 2, \dots, n$ and think about how much this ordering will add

to the total payoffs for the players. In this case, the sum of the values added by all of the players to the coalition is

$$\begin{array}{r}
 f(1) \\
 f(1, 2) - f(1) \\
 f(1, 2, 3) - f(1, 2) \\
 \vdots \\
 + \quad f(1, \dots, n) - f(1, \dots, n-1) \\
 \hline
 f(1, \dots, n)
 \end{array}$$

So, for this ordering, the total value added by the players is $f(1, \dots, n)$. A similar argument shows that the total value added for every ordering will be $f(1, \dots, n)$. Therefore $x_1 + \dots + x_n$, which is the sum of the average values added by the players, will equal $f(1, \dots, n)$, as desired. $\square$

Shapley's Axioms. We have now defined Shapley Value and proven that it is indeed a valuation scheme. That said, finding a valuation scheme isn't really anything special. Another (somewhat silly) valuation scheme is this: For every n -player coalitional game f give every player $i \neq n$ the minimum value $x_i = f(i)$, and give everything else to player n by setting $x_n = f(1, \dots, n) - f(1) - f(2) - \dots - f(n-1)$. This valuation scheme certainly doesn't seem like a particularly fair one, but how can we argue that Shapley Value is fair and this one isn't?

Just as Nash introduced fairness axioms for an arbitration scheme, Shapley likewise introduced certain axioms that articulate appealing properties of a valuation scheme. Similar to Nash, Shapley then proved that Shapley Value is the unique valuation scheme that satisfies all of these axioms. So, if you accept that a fair valuation scheme should obey Shapley's axioms, then Shapley Value is the only value scheme which can be fair

Definition 11.19 (Shapley's Axioms). These are axioms for an n -player valuation scheme $\mathcal{V}$. We will assume that applying $\mathcal{V}$ to the coalitional game f gives the imputation $(x_1, \dots, x_n)$.

- (1) Irrelevance. If there is a player i who adds nothing to any coalition (i.e. $f(S) = f(S \cup \{i\})$ for every $S \subseteq \{1, \dots, n\}$), then $x_i = 0$.

- (2) Symmetry. If there are two players i, j so that every set $S \subseteq \{1, \dots, n\}$ with $i, j \notin S$ satisfies $f(S \cup \{i\}) = f(S \cup \{j\})$ (i.e. players i and j are symmetric), then $x_i = x_j$.
- (3) Invariance Under Sums. Suppose $\mathcal{V}$ assigns the coalitional games f' and f'' the imputations $(x'_1, \dots, x'_n)$ and $(x''_1, \dots, x''_n)$. If $f = f' + f''$, then $(x_1, \dots, x_n) = (x'_1, \dots, x'_n) + (x''_1, \dots, x''_n)$.

The first axiom seems clearly fair. After all, why would a player who never contributes anything deserve a positive payoff? The second is a natural fairness property similar to Nash's symmetry axiom: If two players are symmetric, they should get equal payoffs. The last axiom also looks reasonable. If player i gets a payoff of x_i for the coalitional game f and a payoff of x'_i for the coalitional game f' , then $x_i + x'_i$ looks like a sensible payoff for $f + f'$. However, taken together, these axioms are very powerful as evidenced by the following theorem of Shapley, the centerpiece of this section.

Theorem 11.20 (Shapley). *The only n -player valuation scheme satisfying the above axioms is Shapley Value.*

To prove Shapley's Theorem, we need to do two things. First, we must prove that Shapley Value does indeed obey all three of Shapley's axioms. Then we need to show that any other valuation scheme that obeys these axioms must be the same as Shapley Value. The lemma below accomplishes the first task.

Lemma 11.21. *Shapley Value obeys Shapley's axioms.*

Proof. The fact that Shapley Value obeys the first axiom is an immediate consequence of its definition. The third axiom can also be deduced straight from the definition. For the second axiom, consider a coalitional game where players i and j satisfy the symmetry condition that $f(S \cup \{i\}) = f(S \cup \{j\})$ for every set S not including i or j . Now, for every possible ordering σ of the players, another ordering σ' comes from interchanging the positions of players i and j . It follows from the symmetry property that the value i adds to the growing coalition for σ will be equal to the value j adds to the growing coalition for σ' and vice-versa. It follows that Shapley Value will assign i and j the same payoffs. $\square$

Proof Preliminaries. The key to proving Shapley's Theorem is showing that every coalition game can be expressed as a combination of certain special coalition games. We are now ready to introduce these special coalition games. For a nonempty set $T \subseteq \{1, \dots, n\}$, define the coalitional game h_T by the rule that for every subset of players $S \subseteq \{1, \dots, n\}$

$$h_T(S) = \begin{cases} 1 & \text{if } T \subseteq S, \\ 0 & \text{otherwise.} \end{cases}$$

So, the function h_T assigns a coalition a value of 1 if this coalition contains every member of T , and otherwise it assigns this coalition a value of 0. A straightforward check reveals that h_T is a coalitional game for every nonempty set T , and more generally, for any number $c \geq 0$ the game $c \cdot h_T$ is a coalitional game. The next lemma gives a key property of these games.

Lemma 11.22. *Let $\mathcal{V}$ be a valuation scheme that satisfies Shapley's axioms (1) and (2). Let $T \subseteq \{1, \dots, n\}$ be nonempty and let $c \geq 0$. The imputation $(x_1, \dots, x_n)$ that $\mathcal{V}$ assigns to $c \cdot h_T$ is given by the rule*

$$x_i = \begin{cases} \frac{c}{|T|} & \text{if } i \in T, \\ 0 & \text{if } i \notin T. \end{cases}$$

Proof. If $i \notin T$, then player i adds nothing to any coalition (i.e. $f(S) = f(S \cup \{i\})$ holds for all S). In this case, the first axiom implies that $x_i = 0$. If i and j are two players in T , then i and j will obey the symmetry condition in axiom (2), so $x_i = x_j$. Consequently, every player not in T gets 0, while every player in T gets the same value. To find this common value, note that, because we have a valuation scheme, $x_1 + \dots + x_n = c \cdot h_T(1, \dots, n) = c$. It follows that every $i \in T$ will satisfy $x_i = \frac{c}{|T|}$, as desired. $\square$

We now have all the necessary ingredients for the proof of Shapley's Theorem. A short detour back to Divide the Dollar will help illuminate the main idea in the forthcoming argument. Recall that Divide the Dollar is a coalitional game with players $\{1, 2, 3\}$ and function d given by the rule

$$d(S) = \begin{cases} 0 & \text{if } |S| \leq 1, \\ 1 & \text{if } |S| \geq 2. \end{cases}$$

Suppose that we have a valuation scheme $\mathcal{V}$ that satisfies all of Shapley's axioms. What happens when we apply this valuation to the above coalitional game d ? The central idea is the following equation that expresses d in terms of the special functions h_T :

$$d = h_{\{1,2\}} + h_{\{1,3\}} + h_{\{2,3\}} - 2h_{\{1,2,3\}}.$$

This equation is straightforward to verify (for instance on the input $\{1, 2, 3\}$ the left-hand side evaluates to 1 and the right-hand side evaluates to $1 + 1 + 1 - 2 = 1$). Adding $2h_{\{1,2,3\}}$ to both sides of this equation gives us

$$d + 2h_{\{1,2,3\}} = h_{\{1,2\}} + h_{\{1,3\}} + h_{\{2,3\}}$$

(an equation without any negative terms). By assumption, valuation scheme $\mathcal{V}$ satisfies the first two axioms, so the previous lemma shows that $\mathcal{V}$ must assign $2h_{\{1,2,3\}}$ the imputation $(\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$ and must assign $h_{\{1,2\}}$, $h_{\{1,3\}}$, and $h_{\{2,3\}}$ the imputations $(\frac{1}{2}, \frac{1}{2}, 0)$, $(\frac{1}{2}, 0, \frac{1}{2})$, and $(0, \frac{1}{2}, \frac{1}{2})$, respectively. If $\mathcal{V}$ assigns the Divide the Dollar game d the imputation (x_1, x_2, x_3) , then, since all terms in the previous equation are nonnegative, we may apply axiom (3) to it to obtain

$$(x_1, x_2, x_3) + (\frac{2}{3}, \frac{2}{3}, \frac{2}{3}) = (\frac{1}{2}, \frac{1}{2}, 0) + (\frac{1}{2}, 0, \frac{1}{2}) + (0, \frac{1}{2}, \frac{1}{2}).$$

From this, we conclude that $(x_1, x_2, x_3) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. We have thus demonstrated that any valuation scheme $\mathcal{V}$ that satisfies all three of Shapley's axioms must assign the imputation $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ to Divide the Dollar, just as Shapley Value does. The proof of Shapley's Theorem is a generalization of this argument.

Proof of Shapley's Theorem. We have already shown that Shapley Value obeys Shapley's axioms. To complete the proof, we now show that Shapley Value is the only scheme satisfying all three axioms. To do this, let $\mathcal{V}$ be an arbitrary valuation scheme satisfying all three of the axioms. We will show that $\mathcal{V}$ and Shapley Value assign the same imputation to an arbitrary n -player coalition game f . The key to the proof is the following claim.

Claim. *The coalitional game f may be expressed as a linear combination of h_T games.*

To prove the claim, we proceed in steps. At each step, we construct a new function that maps subsets of $\{1, \dots, n\}$ to numbers. For the first step, define for each player i the number $c_{\{i\}} = f(i)$. Now define a new function f_1 (using f) by the rule

$$f_1 = f - c_{\{1\}}h_{\{1\}} - c_{\{2\}}h_{\{2\}} - \dots - c_{\{n\}}h_{\{n\}}.$$

The key property of f_1 is that it assigns the value 0 to any set of size ≤ 1 (this follows immediately from its definition). For the next step, define for any two distinct players, i and j , the value $c_{\{i,j\}} = f_1(i, j)$. Then define the function f_2 from f_1 by

$$f_2 = f_1 - c_{\{1,2\}}h_{\{1,2\}} - c_{\{1,3\}}h_{\{1,3\}} - \dots - c_{\{n-1,n\}}h_{\{n-1,n\}}.$$

Observe that f_2 assigns 0 to any set of size ≤ 2 . Continuing this process results in a function f_n that evaluates to 0 on every set of size $\leq n$; i.e. f_n is the zero function. Together, these equations yield an expression for the original function f as a linear combination of h_T games, thus completing the proof of the claim. In symbols, this is

$$(11.1) \quad f = \sum_{T \subseteq \{1, \dots, n\}} c_T h_T.$$

This equation is not quite in the right form to apply the axioms. If there is a term on the right in equation (11.1) with a negative coefficient, say $-10h_T$, we add $10c_T$ to both sides of the equation to get rid of this negative sign. Doing this for all such terms gives an equation of the form

$$(11.2) \quad f + \sum_{T \subseteq \{1, \dots, n\}} a_T h_T = \sum_{T \subseteq \{1, \dots, n\}} b_T h_T,$$

where a_T and b_T are nonnegative for every T . It follows from Lemma 11.22 that for every $T \subseteq \{1, \dots, n\}$ both $\mathcal{V}$ and Shapley Value assign the same imputation to the game $a_T h_T$. Both valuation schemes similarly assign the same imputation to $b_T h_T$. Since both $\mathcal{V}$ and Shapley Value satisfy the third axiom (and every sum of coalitional games is a coalitional game), this means that they both assign the same imputation to the coalitional game $\sum_{T \subseteq \{1, \dots, n\}} a_T h_T$ and similarly both assign the same imputation to the coalitional game $\sum_{T \subseteq \{1, \dots, n\}} b_T h_T$. By applying the third axiom to equation (11.2), we deduce that both $\mathcal{V}$ and Shapley Value assign the same imputation to f , as desired. $\square$

This establishes Shapley's important theorem. Although this theorem is rightly held in high regard, it is nonetheless not the case that Shapley Value is considered to be *the* fair valuation scheme (just as Nash arbitration is not regarded as *the* fair arbitration scheme). There are many other meaningful valuation schemes, so Shapley Value should be viewed as an important part of a bigger picture.

Using Shapley Value. Shapley Value is a valuation scheme with widespread application, even frequently in places where there is no obvious game. For example, consider a voting scenario in which the voters do not have equal power.

Example 11.23. In the United Nations Security Council, there are 5 permanent members and 10 temporary members. In order to pass a (substantive) resolution, all 5 of the permanent members must vote in favor and at least 4 of the temporary members must vote in favor.

Coalitional games can model voting situations like this. For a given subset S of voters, the key question is whether or not S , working as a coalition, has the strength to pass a resolution (i.e. if every member in S votes for the resolution and everyone not in S votes against, does it pass?). For every set S of voters, define

$$f(S) = \begin{cases} 1 & \text{if the coalition } S \text{ can pass a resolution,} \\ 0 & \text{otherwise.} \end{cases}$$

This function f is a special type of coalitional game that conveniently encodes this voting situation. Since this situation is both common and relevant, we give this type of game a general definition.

Definition 11.24. A coalitional game f for the players $1, \dots, n$ is called a *voting game* if $f(1, \dots, n) = 1$ and every $S \subseteq \{1, \dots, n\}$ has $f(S) = 0$ or $f(S) = 1$.

Voting games provide a sensible model for a variety of voting scenarios. Suppose we represent an n -player voting scenario with a voting game f , and the Shapley Value of f is $(x_1, \dots, x_n)$. What does this Shapley Value mean? In this case, there is no payoff for the voters to divide. In-

stead the numbers $x_1, \dots, x_n$ give a good measure of the relative strengths of the different voters. This application of Shapley Value is known as the Shapley-Shubik Index.

Example 11.23 (Continued). To compute the Shapley-Shubik Index for the United Nations Security Council, treat this as a voting game and compute the Shapley Value for each member. For an ordering of the players, the value of the growing coalition will stay at 0 until the point where a voter is added who gives the coalition enough strength to pass a resolution, and at that point the value jumps to 1. So there is just one voter, called the *swing voter*, who adds value to the growing coalition, specifically a value of 1. Let T be a temporary member and consider how many times T will be the swing voter over all orderings. For this to happen, all of the permanent members must be before T in the ordering, and exactly 3 of the temporary members must be in before T . (So, T must be in the 9th position.) The total number of ways this happens is $\ell = \binom{3}{3}8!6! = 2438553600$.¹ The value assigned to T by the Shapley-Shubik Index will be the total number of times T is the swing voter, divided by the total number of orderings ($15! = 1307674368000$), which is approximately .00186. To compute the value for a permanent member, use the symmetry of the game to note that any two permanent members get the same value. Since the sum of the values for all members is 1, an easy computation shows that every permanent member will have a value of approximately .19627. Therefore, every permanent member has a little more than 100 times the voting power of each temporary member.

The Shapley-Shubik Index gives a meaningful measure of the relative strengths of different voters. Alternate approaches to measuring voting strength—such as the Banzhaf Index—also provide similar information. So, the Shapley-Shubik Index is a significant contribution to voting theory, but not the final word on measuring voting power.

¹This is because there are $\binom{3}{3}$ ways to choose which temporary members will be before T , and once this choice has been made, there will be $8!$ ways to order the members in front of T and $6!$ ways to order the members behind T .

Exercises

- (1) How will rational players play the following game?

		Larry E	
		Colin	
Rose	A	2, -1, 3	-2, 5, 2
	B	3, 3, -2	8, -2, 0

		Larry F	
		Colin	
Rose	A	3, -1, 5	2, 1, 3
	B	1, 5, 0	-1, 4, 1

- (2) Denote the following 3-player matrix game by M :

		Larry E	
		Colin	
Rose	A	6, 4, -3	3, 6, -1
	B	-1, 1, 3	0, -2, 4

		Larry F	
		Colin	
Rose	A	1, -4, 6	-1, 4, 3
	B	2, 1, 2	3, 0, 0

For the coalitional form f_M , find the following evaluations:

- | | |
|--|---|
| (a) $f_M(\emptyset)$, | (d) $f_M(\text{Rose}, \text{Colin})$, |
| (b) $f_M(\text{Rose}, \text{Colin}, \text{Larry})$, | (e) $f_M(\text{Rose})$, |
| (c) $f_M(\text{Larry})$, | (f) $f_M(\text{Colin}, \text{Larry})$. |

- (3) For each coalitional game, plot the set of imputations

$$(a) \quad \begin{aligned} f(1) &= 1, & f(1, 2) &= 3, \\ f(\emptyset) &= 0, & f(2) &= 1, & f(1, 3) &= 3, & f(1, 2, 3) &= 5, \\ f(3) &= 1, & f(2, 3) &= 3. \end{aligned}$$

$$(b) \quad \begin{aligned} g(1) &= 1, & g(1, 2) &= 4, \\ g(\emptyset) &= 0, & g(2) &= 2, & g(1, 3) &= 5, & g(1, 2, 3) &= 10, \\ g(3) &= 3, & g(2, 3) &= 6. \end{aligned}$$

- (4) Show that the core of Divide the Dollar is empty.

- (5) Prove that the core of the following coalitional game f contains just the imputation $(1, 1, 1)$:

$$\begin{aligned} f(1) &= 0, & f(1, 2) &= 2, \\ f(\emptyset) &= 0, & f(2) &= 0, & f(1, 3) &= 2, & f(1, 2, 3) &= 3, \\ f(3) &= 0, & f(2, 3) &= 2. \end{aligned}$$

- (6) A jazz trio consists of a guitarist G , a saxophone player S , and a drummer D . They will earn \$400 for performing together at a pub, and the drummer suggests they divide this money evenly. The guitarist objects to this division because she can earn \$150 for playing on her own at a cafe. Form a coalitional game f by assigning each subset of this trio the earnings from playing together (without the others). Assume f is as follows:

$$\begin{aligned} f(G) &= 150, & f(G, S) &= 250, \\ f(\emptyset) &= 0, & f(S) &= 100, & f(G, D) &= 300, & f(G, S, D) &= 400, \\ f(D) &= 0, & f(S, D) &= 250. \end{aligned}$$

Assume the trio performs at the pub and use Shapley Value to find a fair division of the \$400.

- (7) Consider a committee with $2n$ members and 1 chairperson. To pass a resolution requires a strict majority of the $2n$ voting members. If the vote results in a tie, then the chairperson casts a deciding vote. Determine the Shapley-Shubik Index for this voting game. Why does the chairperson have voting power equal to that of every other member?
- (8) *Odd-Person-Out* is a 3-player game where each player has a coin which he or she can choose to play as Heads or Tails. The choices of the players are revealed simultaneously. If all three make the same choice, no money is exchanged. Otherwise, the player who made the minority choice must pay each of the other players \$1.
- Model this game as a 3-player matrix game.
 - Determine the coalitional form.
 - A *pure Nash equilibrium* in an n -player game is a choice for each player i of a pure strategy S_i with the property that no player can improve his payoff by switching his choice (assuming all other players do not switch). What are the pure Nash equilibria in this game?

- (9) This exercise offers a generalization of Cournot Duopoly to allow for more players. Assume there are k companies numbered $1, \dots, k$ producing a commodity and that company i is producing Q_i units. As before, the cost to company i will be cQ_i and we assume that the demand curve is given by the equation for price: $P = a - b(Q_1 + \dots + Q_k)$. Find a Nash equilibrium.
- (10) Let f be an inessential voting game for the players $1, \dots, n$. Prove that there is a player j with the property that $f(S) = 1$ if $j \in S$ and otherwise $f(S) = 0$. Note: Here j may be regarded as a dictator since a resolution passes if and only if j votes for it.
- (11) Let f be a coalitional game for the players $1, 2, \dots, n$.
- Prove that the set of all imputations of f is convex.
 - Prove that the core of f is convex.
- (12) Let f be a coalitional game for the players $1, \dots, n$. Prove that the core of f is precisely the set of all imputations which are not dominated by any other imputation.
- (13) Here is a simple valuation scheme. Let f be a coalitional game for the players $1, \dots, n$. Put $m = f(1, \dots, n) - (f(1) + \dots + f(n))$ and then output the imputation $(x_1, \dots, x_n)$ defined by the rule $x_i = f(i) + \frac{m}{n}$. For each of Shapley's axioms, either prove that this valuation scheme satisfies the axiom or give a counterexample showing it does not.
- (14) Construct an essential coalitional game for which the core is a stable set.
- (15) In the game Divide the Dollar, define J_x to be the set of all imputations in which player 1 gets a payoff of x . Prove that J_x is a stable set for every $0 \leq x < \frac{1}{2}$.
- (16) Let f be an essential voting game for the players $1, \dots, n$ and let S be a minimal set of players with $f(S) = 1$ (so every $S' \subset S$ has $f(S') = 0$). Define J to be the set of all imputations $(x_1, \dots, x_n)$ such that $x_i = 0$ for every $i \notin S$. Prove that J is a stable set.
- (17) Two coalitional games f, g for the players $1, \dots, n$ are *equivalent*, denoted $f \equiv g$, if there exist $a > 0$ and $b_1, \dots, b_n \in \mathbb{R}$ so that the following holds for every $S \subseteq \{1, \dots, n\}$:

$$g(S) = af(S) + \sum_{i \in S} b_i.$$

In this case, we may view the game g as formed by scaling the game f by a constant factor of a and then giving each player i a side-payment of b_i . Prove that $\equiv$ is an equivalence relation by showing that it satisfies the following properties whenever f, g, h are coalitional games on the same set of players:

- (a) $f \equiv f$.
 - (b) If $f \equiv g$, then $g \equiv f$.
 - (c) If $f \equiv g$ and $g \equiv h$, then $f \equiv h$.
- (18) An n -player matrix game A is called *zero-sum* if every entry $(x_1, \dots, x_n)$ of A satisfies $x_1 + \dots + x_n = 0$. A coalitional game f for the players $1, \dots, n$ is called *zero-sum* if it satisfies the property $f(S) + f(\{1, \dots, n\} \setminus S) = 0$ for every $S \subseteq \{1, \dots, n\}$.
- (a) Prove that whenever A is a zero-sum matrix game, its coalitional form f_A is zero-sum.
 - (b) Show that the converse is false: Find a matrix game B which is not zero-sum so that the coalitional form f_B is zero-sum.
- (19) A coalitional game f for the players $1, \dots, n$ is called *constant-sum* if there is a number m so that $f(S) + f(\{1, \dots, n\} \setminus S) = m$ holds for every $S \subseteq \{1, \dots, n\}$. Prove that every essential constant-sum coalitional game f has an empty core.
- (20) There is a strange market for gloves. All the pairs of gloves are the same and each trader has only one glove to sell. There are ℓ traders, each of whom has a left glove, and r traders, each of whom has a right glove. We model this as a coalitional game f , where for every set S of traders, $f(S)$ is the total number of pairs of gloves they can form.
- (a) If $\ell > r$, determine the core of f .
 - (b) If $\ell = r$, determine the core of f .
- (21*) In this exercise we introduce another valuation scheme. Let f be a coalitional game for the players $1, \dots, n$ and let $\mathbf{x} = (x_1, \dots, x_n)$ be an imputation. For a subset of players $S \subseteq \{1, \dots, n\}$ we define the *excess of S relative to $\mathbf{x}$* to be

$$e_{\mathbf{x}}(S) = \left(\sum_{i \in S} x_i \right) - f(S).$$

The excess of S is the total additional amount that the players in S receive in the imputation $\mathbf{x}$ minus what they could guarantee themselves. So, a coalition S will be happier if $\mathbf{x}$ gives them greater excess. Order the subsets of $\{1, \dots, n\}$ by $S_1, \dots, S_N$ so that $e_{\mathbf{x}}(S_1) \leq \dots \leq e_{\mathbf{x}}(S_N)$ and define the *excess vector* of $\mathbf{x}$ by

$$\mathbf{e}_{\mathbf{x}} = [e_{\mathbf{x}}(S_1) \quad e_{\mathbf{x}}(S_2) \quad \dots \quad e_{\mathbf{x}}(S_N)].$$

Choose an imputation $\mathbf{x}$ so that the first entry of the excess vector is maximum. If there is more than one imputation achieving this maximum, then among these, choose one for which the second entry is maximum. If there is more than one imputation achieving this, then among all such imputations, choose one for which the third entry is maximum, and so on. Prove that this process selects a unique imputation (thus giving a valuation scheme).

Chapter 12

Preferences and Society

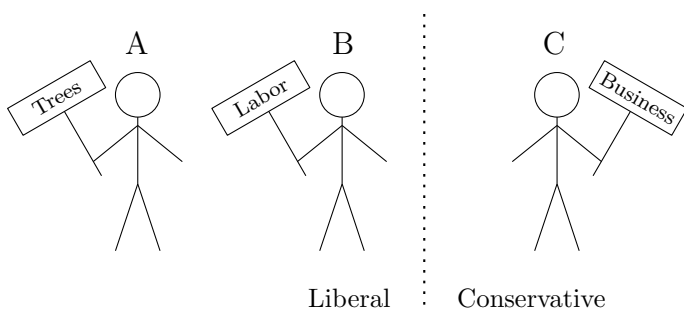


Figure 12.1. Politicians

This chapter explores situations concerning collections of opinionated people. We want to arrive at a collective decision resulting in an outcome that in some way reasonably reflects the preferences of these individuals. The three sections in this chapter are independent, yet each features a mathematical theorem about reaching a collective decision of this type. In Section 12.1, we use Sperner's Lemma to show how a group of people can fairly divide a cake; in Section 12.2, we introduce the Gale-Shapley algorithm for finding stable pairings; and in Section 12.3, we prove Arrow's famous Impossibility Theorem about voting systems.

Before we investigate these significant theorems, let us consider voting—one particularly relevant situation where individual preferences are combined to reach a group decision. In an election, each individual voter has a preferred ranking of the various candidates. A good voting system should act by somehow combining these individual preferences to select a particular candidate.

Example 12.1 (Voting Scenario). Assume that there are three candidates called A , B , and C . Suppose that A is very liberal, B is liberal, and C is conservative. Assume further that the electorate is divided in terms of preferences as follows:

15% Very Liberal	40% Liberal	45% Conservative
A	B	C
B	A	B
C	C	A

So, the very liberal voters prefer A to B to C , the liberal voters prefer B to A to C , and the conservative voters prefer C to B to A .

In standard plurality voting, each voter selects a favorite candidate, and the candidate with the most votes wins. In the above scenario, C will be elected with 45% of the vote if each voter casts a vote for his or her sincere favorite. However, a rational voter would view an election as a type of game and thus play strategically. If the Very Liberal voters chose to vote for the liberal candidate B instead of their sincere favorite A , then candidate B would win the election with 55% of the vote. The Very Liberal voters would prefer this outcome of candidate B to the outcome of candidate A , so they are motivated to vote insincerely for candidate B . This type of action is called strategic voting—a key part of voting theory.

As we have just seen, plurality voting encourages sophisticated voters to take strategic action—something we would like to avoid if possible. Instead, we prefer a voting system that would eliminate the incentive to vote insincerely and then combine the preference information fairly to elect a candidate.

In fact, there are many other voting systems that have been devised to improve on standard plurality voting. One of these alternative voting systems is called *instant runoff*. In instant runoff voting, each voter declares a preferred linear ordering of the candidates. Then the following

rule is repeatedly applied to elect a candidate: If there is a candidate who receives a majority of first place votes, then that candidate is elected. Otherwise, the candidate with the fewest first place votes is eliminated, and everyone's rankings are reconsidered with that candidate removed. If we ran an instant runoff election in the above scenario and each voter expressed sincere preferences, then candidate *A* would be eliminated first (getting only 15% of the first place votes). With candidate *A* removed, then candidate *B* would get 55% of the first place votes to win the runoff between *B* and *C*.

Although instant runoff voting has some advantages over simple plurality, there are still inherent difficulties with this system. In fact, in some sense, there do not exist ostensibly fair voting systems for elections with at least three candidates. Kenneth Arrow first proved this surprising and meaningful fact, and his famous Impossibility Theorem appears at the end of this chapter. We will first explore two positive instances where theory provides an appealing collective outcome to a problem involving individual preferences.

12.1. Fair Division

This section takes on the problem of dividing assets. Imagine that a collection of people wish to divide a common good among themselves and want to find a way of doing this that is “fair”. For fun, suppose that the players are going to divide a cake. (This theory will apply more broadly with any continuously divisible assets.)



Figure 12.2. Cake

Of course, it is quite possible that not all players value each piece of cake the same. Some of the players may be especially fond of the frosting, while other players dislike it. Perhaps there is a player keen to have a piece with a certain bit of the text from the cake. To model such variations, assume that each player i has a personal utility function u_i that indicates how player i values various pieces. More precisely, assume that u_i assigns each possible piece of cake P a nonnegative real number $u_i(P)$ that indicates the utility player i associates with piece P . We assume (normalizing if necessary) that each player assigns the entire cake a utility of 1. The key features of these utility functions are:¹

- u_i is continuous.
- For every division of the cake into n pieces $P_1, \dots, P_n$ we have $u_i(P_1) + \dots + u_i(P_n) = 1$.

Equitable Division. At this point, we have a rich framework offering an effective model for numerous division problems. But how can we tell if a division is fair? Is there a way to make this precise? If there are n players involved, it would be natural for each player to desire a piece of cake he or she values at $\geq 1/n$. We call this notion equitable and a formal definition follows.

Definition 12.2. Assume players $1, 2, \dots, n$ have utility functions $u_1, \dots, u_n$. If the cake is divided into pieces $P_1, \dots, P_n$ with piece P_i going to player i , then we call this division *equitable* if $u_i(P_i) \geq 1/n$ for every player i .

So far, we have been thinking about taking a particular cake and dividing it among the players. Ideally, though, we prefer a mechanism to do this in general. More precisely, what we would like is a procedure that would take inputs from the players and use that information to give us a division of the cake. Next, a nifty procedure of this type that requires the players to be quick on their feet!

Definition 12.3 (Moving Knife). This is a system for dividing a cake among n players. One person is elected to be the *cutter* and wield the

¹In a more formal and sophisticated setting, we would take the cake C to be a Lebesgue measurable subset of $\mathbb{R}^3$ and treat each u_i as a measure on C that is absolutely continuous with respect to the Lebesgue measure and satisfies $u_i(C) = 1$. Of course, each piece is assumed to be Lebesgue measurable.

knife. The cutter begins with the knife to the left of the cake and slowly moves it over the cake to the right. At any point, any player (including the cutter) can call “stop” and the cutter will make a cut. The player who called stop gets the piece to the left of the knife, and then the game continues among the remaining players in the same manner (when there is only one player left, she gets the entire remaining portion).

Moving Knife produces an equitable division if all of the players are careful. This is the content of our next theorem.

Theorem 12.4. *In Moving Knife, each player i has a strategy guaranteeing that the received piece P_i satisfies $u_i(P_i) \geq 1/n$. In particular, if every player plays according to this strategy, the resulting division will be equitable.*

Proof. Here is the strategy for player i : As soon as she values the portion of the cake to the left of the knife at $1/n$, she calls “stop”. We claim the piece P_i that player i gets will satisfy $u_i(P_i) \geq 1/n$. If at any point during the course of the procedure, player i calls “stop”, then by the definition of this strategy we have $u_i(P_i) = 1/n$. On the other hand, if player i never calls stop, then all $n - 1$ of the other players will have called “stop” and for every other player j , the received piece P_j will satisfy $u_i(P_j) \leq 1/n$. So, in this case, the total value of the pieces that i did not get will be at most $(n-1)/n$, so it must be that $u_i(P_i) \geq 1/n$, as desired. $\square$

We have now entered the fascinating subject of mechanism design. Roughly speaking, this is the challenge of trying to design mechanisms or procedures that have desirable guarantees even when they are used by sophisticated players (who may play strategically and even form coalitions). Note that for Moving Knife, even if all of the players $2, \dots, n$ form a coalition, the previous theorem tells us that player 1 can still guarantee herself a piece P_1 for which $u_1(P_1) \geq 1/n$.

Envy-Free Division. Equitable divisions are appealing, but it is not necessarily the case that an equitable division would make everyone happy. Even with an equitable division, one player might envy another.

Example 12.5. Three children numbered 1, 2, 3 have divided a cake into pieces 1, 2, 3 with child i getting piece i . The following table indicates how each child values each of the three pieces of cake:

Utility		Piece		
		1	2	3
Child	1	4/12	3/12	5/12
	2	1/3	1/3	1/3
	3	1/5	2/5	2/5

Note that for each child, the total sum of the values assigned to the three pieces is 1 (as it should be for any division). It is straightforward to check that this division with child i getting piece i is an equitable one. Indeed child 1 values piece 1 at $4/12 = 1/3$, child 2 values piece 2 at $1/3$, and child 3 values piece 3 at $2/5 > 1/3$. However, it is also the case that child 1 values the piece child 3 received at $5/12$, which is greater than the value ($4/12$) child 1 assigned to piece 1. Despite this equitable division, child 1 still envies child 3.

This example suggests a property even stronger than equitable. An appealing division would be one in which every player values her own piece greater than or equal to the value of any other players' piece. This is the concept of an envy-free division, formalized below.

Definition 12.6. Let players $1, \dots, n$ have utility functions $u_1, \dots, u_n$. If the cake is divided into pieces $P_1, \dots, P_n$ with piece P_i going to player i , then we call this division *envy-free* if $u_i(P_i) \geq u_i(P_j)$ for every pair of players i, j (i.e. player i does not envy player j).

So, in an envy-free division, every player is getting a piece that he considers to be one of the best. To see that every envy-free division is also an equitable one, note that for an envy-free division, player i has $u_i(P_j) \leq u_i(P_i)$ for every $1 \leq j \leq n$, which implies

$$1 = u_i(P_1) + \dots + u_i(P_n) \leq n(u_i(P_i)).$$

So envy-free is indeed a stronger property than equitable. Are there division schemes that produce desirable envy-free divisions? In general, this is an extremely complicated question, but there is a familiar procedure that gives envy-free divisions when there are just two players (i.e. $n = 2$).

Definition 12.7 (Split and Choose). This is a procedure for dividing a cake among two players. One player is called the *splitter* and the other is the *chooser*. The splitter divides the cake into two pieces. The chooser selects one piece and the other piece goes to the splitter.

Though there is a lack of symmetry between the roles of the splitter and the chooser, both players can guarantee themselves a portion they value at least as highly as that of the other player.

Theorem 12.8. *In Split and Choose, each player has a strategy that guarantees a piece valued at least as highly as that of the other player. In particular, if both players use such strategies, the resulting division will be envy-free.*

Proof. For the splitter to guarantee a piece he values at least as high as that of the chooser, the splitter should divide the cake into two pieces that he considers to be of equal value. For the chooser to guarantee a piece she values at least as highly as that of the splitter, she should select a piece she considers to have the highest value. $\square$

It is possible to iterate Split and Choose so it applies to more than two players, but the resulting procedure will not be envy-free (See Exercise (3)). Finding procedures for many players that come with envy-free guarantees is extremely difficult.

Envy-Free Division for Many Players. These methods of dividing cake also apply to other circumstances of dividing a continuous asset. However, many of our procedures also apply in some unexpected situations. For instance, suppose that three people, Amir, Bao, and Carol, have decided to rent a 3-bedroom house together. The common areas will be shared, but each person will get his or her own bedroom. If the bedrooms are identical, then it would be natural to split the rent evenly. It is, however, likely that the bedrooms differ in terms of floor space, storage, windows, and other features. In light of this, the players may want to divide the rent unevenly according to the bedrooms. An envy-free division would be ideal here. More precisely, we could hope to assign each of Amir, Bao, and Carol a rent payment and a bedroom so that the sum of the three rent payments is the total rent and no player would want to switch positions with another.

This section proves a theoretical result that implies the existence of an envy-free division of cake. Our proof gives an efficient procedure to find a nearly envy-free division, and then the existence of a true envy-free division follows from a limit argument. In fact, this procedure to find a nearly envy-free division is already quite useful. For instance, an online algorithm² based on the work of Francis Su uses this procedure to divide rent. The algorithm asks the players a sequence of questions about preferred rooms and rent rates. From this information, it suggests a division that is close to envy-free (up to a small amount of money). The players can continue to get divisions of the rent that are closer and closer to envy-free until they are within some negligible amount of money.

Theorem 12.9. *Given a cake and a set of players $1, \dots, n$ with utility functions $u_1, \dots, u_n$, there exists an envy-free division of the cake.*

Proof. This proof is based on Sperner's Lemma. We will describe the proof for $n = 3$ players, Amir, Bao, and Carol, but the proof generalizes immediately to larger numbers of players. Start by giving the cake coordinates in the usual manner by placing it on a real number line so that the interval $[0, 1]$ is a diameter of the cake. Now, only consider divisions of the cake that correspond to two vertical cuts as shown in Figure 12.3.



Figure 12.3. Dividing the cake

Any two such cuts split the interval $[0, 1]$ into three intervals with lengths x, y, z as shown in Figure 12.3. Note that the possibilities for

²See www.ams.org/bookpages/stml-80 for a link.

(x, y, z) are given by the set

$$T = \{(x, y, z) \mid x, y, z \geq 0 \text{ and } x + y + z = 1\}.$$

The set T is a triangle in $\mathbb{R}^3$ as seen in Figure 12.4.

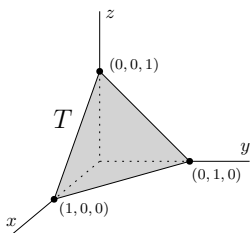


Figure 12.4. A triangle in $\mathbb{R}^3$

At present, each point in the triangle T is associated with a division of the cake. Now take the triangle T and subdivide it into small equilateral triangles in the manner shown in Figure 12.5 (but possibly a much finer subdivision). Assign letters A , B , or C to the vertices of this triangulation in such a way that every small triangle has all three letters at its three corners. The pattern in Figure 12.5 can easily be extended to larger triangulations of the same type.

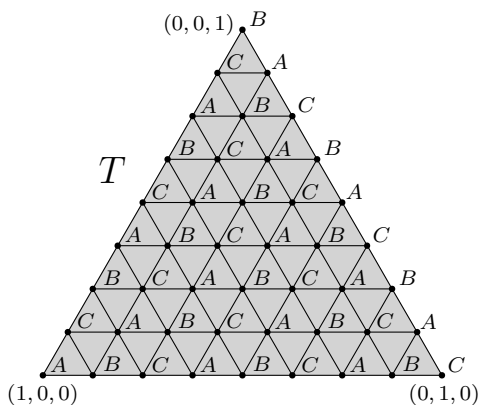


Figure 12.5. Choices for Amir, Bao, and Carol

Now, for each vertex with the letter A , this point in T corresponds to a division of the cake into three pieces, and we let Amir choose his

favorite among these, either piece 1, 2, or 3. So, in other words, Amir will assign this vertex a label from $\{1, 2, 3\}$. Similarly, at every vertex with the letter B (or letter C) we will have Bao (or Carol) choose a favorite piece in the corresponding cake division.

Each vertex in the triangulation now has a label in $\{1, 2, 3\}$. We claim that this labeling satisfies the hypothesis from Sperner's Lemma. To check this, first consider an extreme point, say $(1, 0, 0)$. Although we do not know which player gets to make a choice at this vertex, we know that every player will choose the first piece since the others are empty—and we assume every player prefers some cake to no cake! Thus $(1, 0, 0)$ is labeled 1. Similarly, $(0, 1, 0)$ gets label 2 and $(0, 0, 1)$ gets label 3. Next consider a vertex that lies along the line segment between $(1, 0, 0)$ and $(0, 1, 0)$. Any point on this line segment has z -coordinate equal to 0. It follows that whichever player makes the choice at this vertex will choose either piece 1 or piece 2 (since piece 3 is empty). So, the only labels to appear on such vertices are 1 and 2. A similar argument for the intervals between $(0, 1, 0)$ and $(0, 0, 1)$ and between $(1, 0, 0)$ and $(0, 0, 1)$ shows that this labeling does indeed satisfy the hypothesis for Sperner's Lemma.

By Sperner's Lemma there is a small triangle that has all three labels 1, 2, 3. By this labeling, the vertices must have all three letters A , B , and C as shown in Figure 12.6.

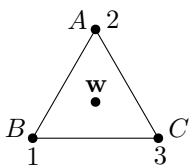


Figure 12.6. An approximately good division

Let $\mathbf{w}$ be the center point of this triangle. Now, $\mathbf{w}$ corresponds to a division of the cake that is very close to the divisions at the three corners. Further, at each corner point, a different player wanted a different piece. In Figure 12.6, for instance, Amir wants piece 2 at his corner, Bao wants piece 1 at his corner, and Carol wants piece 3 at her corner. So, if we were to divide the cake as specified by $\mathbf{w}$ and give Amir piece 2, Bao piece

1 and Carol piece 3, that would be close to an envy-free division point. Repeating the process with a sequence of finer and finer triangulations, will generate a sequence of these approximately good division points and there will be a subsequence of these which converge to a true envy-free division point. $\square$

Fair division is a subtle and complex problem with some clearly meaningful applications. Fortunately, the mathematical tools introduced here can be used to give very robust and satisfying divisions (not just of cake and rent!). Indeed, this is an instance where some highly sophisticated mathematical ideas generate a beautiful solution to a significant problem.

12.2. Stable Marriages

In this section, we investigate a problem involving the formation of stable marriages in a collection of women and men, each with preferences among the possible partners. Although this problem may initially appear to be merely an amusing bit of recreational mathematics, the methods here in fact apply more broadly since they can be used to find desirable allocations or matchings among various different entities. Perhaps most notably the Gale-Shapley algorithm presented here has been used around the world to match medical students with hospital internships.

The setup for this problem involves a set of k women and a set of k men, and we are interested in pairing up each man with one woman. Each woman and each man has certain partner preferences. To model this, assume that each woman w has a preferred linear ordering of the men denoted by $>_w$. So, if woman w prefers man m to man m' , then write $m >_w m'$. Similarly, assume that every man m has a preferred linear ordering of the women and denote this by $>_m$. So if man m prefers woman w to woman w' , express this as $w >_m w'$.

Define a *total marriage* to be a partition of these $2k$ individuals into k couples, where each couple consists of one woman and one man. We will be interested in finding a total marriage which has a certain stability property relative to the preferences.

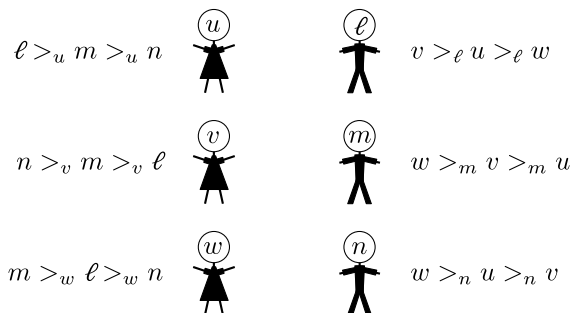


Figure 12.7. Women and men with individual preferences

Consider the example from Figure 12.7 and suppose that we have decided to pair v with m and w with n as shown in Figure 12.8. Observe that in this situation, w prefers m to her present partner, and similarly m prefers w to his present partner. As you can readily appreciate, this is an unstable arrangement, which may cause the couples to split. In this case, we say that $\{w, m\}$ is an *instability*.

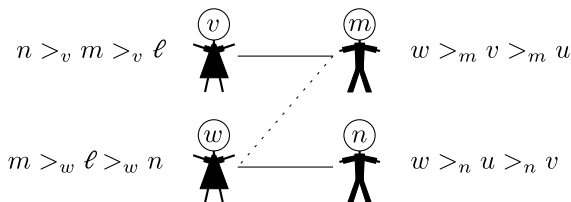


Figure 12.8. An unstable coupling

If no such instability exists, define a total marriage to be *stable*. In words, it is stable if there does not exist a woman w and a man m who each prefer one another over the present partner. Figure 12.9 indicates a stable total marriage for our example population. To verify this, check that each woman and man who are not paired together as a couple do not form an instability (i.e. at least one of them prefers a present partner to the other). We will not carry out this verification, but we encourage the reader to do a spot check.

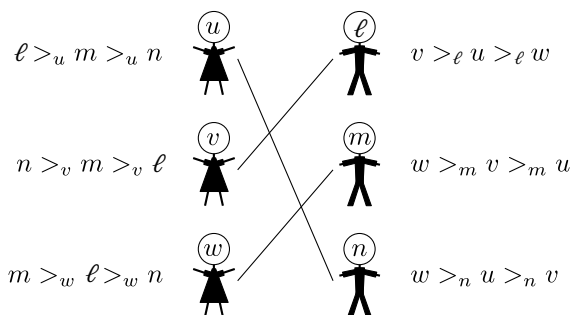


Figure 12.9. A stable total marriage

The main result from this section is a beautiful theorem that proves a stable total marriage always exists. The proof is based on a clever algorithm that always outputs a stable total marriage!

Algorithm 12.10 (Gale-Shapley).

Input: k women and k men with individual preferences.

Output: A set of couples.

Procedure: In each round, every man proposes to the woman he most prefers among those who have not yet rejected him. If every woman receives at most one proposal, stop and output the corresponding couples. Otherwise, every woman who receives more than one proposal says “maybe” to the man she prefers among her proposers and rejects the others.

To use the Gale-Shapley algorithm for pairing medical students with hospitals, begin by having each student rank the hospitals in preferred order and each hospital rank the students in preferred order. Unlike the couplings of men and women above, hospitals in general will accept many students. To model this variation, if some hospital H wants t students, make t copies of H in the problem (each with the same preferences) and have each copy look for a single student. Even after making these duplications, there may be more students than spots at the hospitals, but this is not a problem. In fact, the Gale-Shapley algorithm still works with different numbers of men and women (Exercise (7)).

Theorem 12.11. *The Gale-Shapley algorithm outputs a stable total marriage.*

Proof. To prove this theorem, we need to show a number of things. First, we need to prove that the algorithm will terminate, then we need to show that it outputs a total marriage, and finally we have to prove that this total marriage is stable.

Check first that the algorithm terminates and does not run forever. To see this, note that on any round where the algorithm does not terminate, at least one man gets a rejection. Since men do not propose to women who have already rejected them, each man can only be rejected k times. Our total population has just k men, so the total number of rejections cannot be more than k^2 . It follows that the Gale-Shapley algorithm does indeed terminate.

The algorithm does terminate, so it outputs a set of couples, call it M . For the next two parts of the proof, we rely on the following key claim. The proof of this claim follows immediately from the definition of the algorithm.

Claim. *Every woman w is married to the man m she most prefers among those who propose to her in the algorithm. In particular, if w has at least one proposal, then w will be married at the end.*

Next, prove that our set of couples M is a total marriage (i.e. every man and every woman appear in some couple in M). Suppose (for a contradiction) that there is a man m who is not coupled with any woman in M . In this case, m must not have proposed on the final round, and this means that m must have been rejected by every woman. However, it then follows from the claim that every woman must be married since every woman must receive at least one proposal. Since there are the same number of men and women, this is a contradiction. We conclude that the output set of couples M is a total marriage.

We finally prove that our total marriage M is stable. Suppose (for a contradiction) that it is not, and consider an instability consisting of the couples $\{m, w\}$ and $\{m', w'\}$ where w, w' are women and m, m' are men, for which $m' >_w m$ and $w >_{m'} w'$ (i.e. w and m' prefer one another over the partners they are coupled with). It follows from the definition of the algorithm that m' must have proposed to w at some step (since m' would propose to w before w'). Applying the claim to w , we see that w must be

married to either m' or a person she prefers to m' , thus contradicting our assumptions. It follows that M is a stable total marriage, as claimed. $\square$

Although there is an inherent symmetry between the roles of the men and the women in the problem, we can see that the Gale-Shapley algorithm treats the men and the women very differently. Indeed, the men move downward among their preferences (i.e. they start with their favorite woman, then their second favorite, and so on) while the women move upward (i.e. they keep a man only until someone preferable comes along). Intriguingly, this framework is beneficial for the men and detrimental to the women. Next, we give some further terminology to make this statement precise.

The Gale-Shapley algorithm produces a particular stable total marriage. However, in general there may be many different stable total marriages. Define $\mathcal{M}$ to be the set consisting of all stable total marriages. Now, a given man or woman might be interested in knowing potential partners in the many marriages in $\mathcal{M}$. Let us say that a woman w is *attainable* by a man m (and similarly, m is *attainable* by w) if there exists a stable marriage in $\mathcal{M}$ in which w, m form a couple.

Remarkably, the Gale-Shapley algorithm produces a stable total marriage which is best possible among stable total marriages for every man and worst possible for every woman!

Theorem 12.12. *Let M^* be the stable total marriage output by the Gale-Shapley algorithm, and suppose that the man m and woman w form a couple in M^* . Then both of the following hold:*

- (1) *Among all women attainable by m , he most prefers w .*
- (2) *Among all men attainable by w , she least prefers m .*

Proof. We prove the first part; the second is Exercise (6). Let M be a stable total marriage with couples $\{m_1, w_1\}, \{m_2, w_2\}, \dots, \{m_k, w_k\}$ where $m_1, \dots, m_k$ are men and $w_1, \dots, w_k$ are women. We will prove that in the course of the Gale-Shapley algorithm, no man m_i will be rejected by the woman w_i . Therefore, no man will prefer his partner in M to his partner in M^* , thus yielding the desired result.

Suppose (for a contradiction) that this is false, and consider the first round of the Gale-Shapley algorithm on which it fails. So we may assume that on this round, woman w_i rejects man m_i , but no such rejection has happened in earlier rounds. Since w_i rejects m_i , she must receive a proposal from a man m_j she prefers to m_i . By assumption, m_j has not yet been rejected by w_j , so, because he has proposed to w_i , it must be that m_j prefers w_i to w_j . However, we have now discovered that $\{m_j, w_i\}$ is an instability to the total marriage M . This contradicts the assumption that M is stable. We conclude that no m_i will be rejected by w_i . $\square$

Once again in this section, we have seen a useful decision-making tool arise from a pretty piece of mathematical reasoning. For some potential applications, it would be beneficial to have a procedure for finding stable marriages which is less biased toward either the men or women. Finding such procedures remains an interesting theoretical challenge.

12.3. Arrow's Impossibility Theorem

This section proves the famous Arrow Impossibility Theorem about voting systems. As we saw at the beginning of this chapter, voting is a natural game-theoretic environment and an immensely important one from a practical standpoint. Although it is going a bit too far to suggest that this theorem shows that no practically fair voting system can exist, this theorem does reveal a fundamental difficulty in the design of good voting systems.

The proof of this theorem requires rigorous definitions for a number of concepts. For our purposes, the *candidates* will always be given capital letters, and we will focus a great deal of attention on the special case when there are three candidates A, B, C since this is where the essence of the trouble lies. The *voters* will always be numbered $1, 2, \dots, n$. We will assume that each voter has a preferred linear ordering of the candidates. Together we define the collection of these preferences for each possible voter as a *profile*. For instance, the following figure shows a profile in

the case of three candidates A, B, C and 10 voters:

<i>Voter</i>	1	2	3	4	5	6	7	8	9	10
<i>Rank</i>	C	A	C	B	B	A	B	A	C	C
	B	B	A	C	A	C	A	B	B	B
	A	C	B	A	C	B	C	C	A	A

A *voting system* is a rule that takes a profile (i.e. the preferences of each voter) as input and then outputs a particular ordering (or ranking) of the candidates.

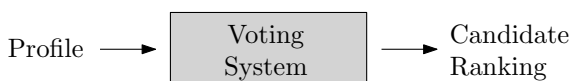


Figure 12.10. A voting system

We are interested in finding a voting system that will give us an output ranking of candidates fairly representing the aggregate preferences of the individuals. For instance, in the case when there are just two candidates, say A and B , and we have an odd number of voters, a sensible voting system would be one that outputs the particular ordering of A and B selected by the majority of the voters. This voting system likely strikes everyone as fair and representative.

What if there are three or more candidates? Can we still design a voting system that seems fair? Investigating questions like this lead Kenneth Arrow to his impossibility theorem. Before this result, let us pause to comment on the applicability of voting systems as we have formally defined them. The example at the start of this chapter together with a bit of consideration should motivate the idea of having voters vote not just for their favorite candidate, but instead input a list of the candidates in their preferred order. From a practical standpoint, though, we usually want to elect just one of the candidates, and not rank all of them. So, we could imagine having a rule that takes a profile and just selects one of the candidates. However, given such a rule, it's easy to use it to rank all the candidates. Just run the process once to select the winner, then remove that winner from the list of candidates and repeat. Realistically, then, any such system of electing a candidate can be turned into a voting system anyway.

Just as Nash and Shapley introduced reasonable and apparently fair axioms for arbitration and valuation schemes, Arrow likewise introduced three desirable axioms for voting systems.

Definition 12.13 (Arrow's Axioms).

- (1) Nondictatorship. The output ranking is not determined just by a single voter.
- (2) Unanimity. If every individual prefers candidate X to candidate Y , the output ranking prefers X to Y .
- (3) Independence of Irrelevant Alternatives. The society's preference between two candidates X and Y should depend only on how each voter compares X and Y .

The first two of these axioms are sensible and seem incontrovertibly fair. The last axiom also seems fair and reasonable. Why should the output ranking comparing X and Y depend on how the voters think about another candidate Z ? Shockingly, Arrow proved the following result, showing that these three natural axioms are impossible to satisfy simultaneously!

Theorem 12.14 (Arrow). *No voting system can satisfy all of the above axioms when there are at least three candidates.*

We described a voting system as a rule that takes a profile as input and then outputs a linear ordering of the candidates. We can also think of this as a function that maps the set of profiles to the set of linear orderings of the candidates. It won't be necessary for us to name this function since there will only be one voting system under consideration. However, it will be helpful to introduce some notation for these inputs and outputs. We will denote profiles using letters such as $\mathcal{P}$ and $\mathcal{Q}$, possibly with subscripts. We will use $>_{\mathcal{P}}$ to denote the linear ordering of the candidates which is output by our voting system when profile $\mathcal{P}$ is the input. So, for instance, if when profile $\mathcal{P}$ is input to the voting system, the output ranking has A first, B second, and C third, then we will write $A >_{\mathcal{P}} B$ and $B >_{\mathcal{P}} C$ (and also $A >_{\mathcal{P}} C$).

One way to define a voting system that would work for any number of candidates is just to have the voting system output the particular ranking selected by the first voter. More generally, we could just select

a single voter i and have the voting system always output the ranking i selects. Any such voting system is called a *dictatorship*, and Arrow's first axiom declares that we want a voting system that is not one of those.

Arrow's second axiom is also quite sensible, as the formalism easily expresses. A voting system satisfies Arrow's second axiom if whenever we input a profile $\mathcal{P}$ in which every single voter prefers candidate X to candidate Y , the output ranking $>_{\mathcal{P}}$ must satisfy $X >_{\mathcal{P}} Y$.

The third axiom compares the outputs from different profiles. Say that profiles $\mathcal{P}$ and $\mathcal{Q}$ *agree in comparing X and Y* if for every voter i , either i prefers X to Y in both profiles $\mathcal{P}$ and $\mathcal{Q}$ or i prefers Y to X in both $\mathcal{P}$ and $\mathcal{Q}$. With this terminology, the last axiom implies that whenever $\mathcal{P}$ and $\mathcal{Q}$ agree in comparing X and Y , then either both output rankings from these profiles prefer X to Y or they both prefer Y to X . Next we restate Arrow's axioms using this terminology and then prove his theorem.

Definition (Arrow's Axioms, version 2)

- (1) Nondictatorship. The voting system is not a dictatorship.
- (2) Unanimity. If in profile $\mathcal{P}$ every voter prefers X to Y , then $X >_{\mathcal{P}} Y$.
- (3) Independence. If the profiles $\mathcal{P}$ and $\mathcal{Q}$ agree in comparing X and Y , the output rankings $>_{\mathcal{P}}$ and $>_{\mathcal{Q}}$ either both rank X above Y or both rank Y above X .

Proof of Theorem 12.14. Initially assume that there are just three candidates A , B , and C and number the voters $1, 2, \dots, n$ (as usual). We will assume that we have a voting system that obeys both unanimity and independence (axioms (2) and (3)) and then show that such a system must be a dictatorship (thus violating axiom (1)). To achieve this, we will consider a handful of different profiles $\mathcal{P}$ and use axioms (2) and (3) to determine the behavior of the ranking $>_{\mathcal{P}}$ output by our voting system. We begin with a very simple profile. Define $\mathcal{P}_0$ to be the profile in which every voter ranks $A > B > C$. It then follows from unanimity that the output

ranking will be $A >_{\mathcal{P}_0} B >_{\mathcal{P}_0} C$:

Profile $\mathcal{P}_0$						Output
<i>Voter</i>	1	2	3	$\cdots$	n	
	A	A	A		A	$>_{\mathcal{P}_0}$
	A	A	A		A	A
<i>Rank</i>	B	B	B	$\cdots$	B	B
	C	C	C		C	C

Next imagine modifying $\mathcal{P}_0$ one voter at a time using the ordering of the voters by having each voter move C from last to first in her ranking until the first point at which this affects the output ranking. Of course, if we go all the way to the point where every voter ranks $C > A$, then the output ranking must also rank $C > A$ (by unanimity) so at some point in the process, this output ranking will have to change. Let $\mathcal{P}_1$ be the last profile in this sequence which still has output ranking $A > B > C$. Assume that in profile $\mathcal{P}_1$ the first $m - 1$ voters have preferences $C > A > B$ and from voter m onward all others have preferences $A > B > C$. So, our assumptions imply

Profile $\mathcal{P}_1$								Output
<i>Voter</i>	1	$\cdots$	$m - 1$	m	$m + 1$	$\cdots$	n	
	C		C	A	A		A	$>_{\mathcal{P}_1}$
	C		C	A	A		A	A
<i>Rank</i>	A	$\cdots$	A	B	B	$\cdots$	B	B
	B		B	C	C		C	C

Now consider modifying $\mathcal{P}_1$ to make a new profile we call $\mathcal{Q}_1$ by having every voter interchange A and B in their rankings. Since $\mathcal{P}_1$ and $\mathcal{Q}_1$ agree in comparing C and A and also agree in comparing C and B , it follows from independence that $>_{\mathcal{Q}_1}$ must rank C last. Furthermore, by unanimity we must have $B >_{\mathcal{Q}_1} A$. This gives us the following:

Profile $\mathcal{Q}_1$								Output
<i>Voter</i>	1	$\cdots$	$m - 1$	m	$m + 1$	$\cdots$	n	
	C		C	B	B		B	$>_{\mathcal{Q}_1}$
	C		C	B	B		B	B
<i>Rank</i>	B	$\cdots$	B	A	A	$\cdots$	A	A
	A		A	C	C		C	C

Now define $\mathcal{P}_2$ to be the profile obtained from $\mathcal{P}_1$ by having voter m switch his preferences to $C > A > B$. Note that by our choice of m , the output ranking for this new profile will not be $A >_{\mathcal{P}_2} B >_{\mathcal{P}_2} C$. We claim that $C >_{\mathcal{P}_2} A >_{\mathcal{P}_2} B$ as shown in the following diagram:

Profile $\mathcal{P}_2$								Output
Voter	1	...	$m-1$	m	$m+1$	...	n	$>_{\mathcal{P}_2}$
Rank	C		C	C	A	A		C
	A	...	A	A	B	...	B	A
	B		B	B	C	C		B

In order to prove this claim, we will also consider the profile $\mathcal{Q}_2$ that is obtained from $\mathcal{Q}_1$ by having voter m switch his preferences to $C > B > A$. We also claim that $C >_{\mathcal{Q}_2} B >_{\mathcal{Q}_2} A$ as shown in the following diagram:

Profile $\mathcal{Q}_2$								Output
Voter	1	...	$m-1$	m	$m+1$	...	n	$>_{\mathcal{Q}_2}$
Rank	C		C	C	B	B		C
	B	...	B	B	A	...	A	B
	A		A	A	C	C		A

To prove these claims, first note that the profiles $\mathcal{P}_2$ and $\mathcal{Q}_2$ agree in comparing C and A and also agree in comparing C and B . It follows from this and the independence axiom that if $A >_{\mathcal{P}_2} C >_{\mathcal{P}_2} B$, then we have $A >_{\mathcal{Q}_2} C >_{\mathcal{Q}_2} B$, but then $\mathcal{Q}_2$ outputs a ranking with A above B , contradicting unanimity. Therefore the ranking $A >_{\mathcal{P}_2} C >_{\mathcal{P}_2} B$ is impossible. We must have $A >_{\mathcal{P}_2} B$ by unanimity, and, by assumption, we do not have $A >_{\mathcal{P}_2} B >_{\mathcal{P}_2} C$ (since we assumed $\mathcal{P}_2$ was the first not to have this output). It follows that we must have $C >_{\mathcal{P}_2} A >_{\mathcal{P}_2} B$ as claimed. Now the independence axiom implies that $\mathcal{Q}_2$ must also rank C above A and B , so by unanimity we conclude that $C >_{\mathcal{Q}_2} B >_{\mathcal{Q}_2} A$, as claimed.

Let $\mathcal{P}_3$ be the profile obtained from $\mathcal{P}_1$ or $\mathcal{P}_2$ by changing the preferences of voter m to $A > C > B$. Now $\mathcal{P}_3$ agrees with $\mathcal{P}_1$ in comparing A and C so by independence we must have $A >_{\mathcal{P}_3} C$. Similarly, $\mathcal{P}_3$ agrees with $\mathcal{P}_2$ in comparing C and B so we must have $C >_{\mathcal{P}_3} B$. This gives us

the following:

Profile $\mathcal{P}_3$								Output
<i>Voter</i>	1	...	$m-1$	m	$m+1$	...	n	$>_{\mathcal{P}_3}$
<i>Rank</i>	C		C	A	A		A	A
	A	...	A	C	B	...	B	C
	B		B	B	C		C	B

Similarly, we define $\mathcal{Q}_3$ to be the profile obtained from $\mathcal{Q}_1$ or $\mathcal{Q}_2$ by changing the preferences of voter m to $B > C > A$. Similarly to the above analysis for $\mathcal{P}_3$, we find that $\mathcal{Q}_3$ agrees with $\mathcal{Q}_1$ in comparing B and C so $B >_{\mathcal{Q}_3} C$; also $\mathcal{Q}_3$ agrees with $\mathcal{Q}_2$ in comparing A and C so $C >_{\mathcal{Q}_3} A$. This gives us the following:

Profile $\mathcal{Q}_3$								Output
<i>Voter</i>	1	...	$m-1$	m	$m+1$	...	n	$>_{\mathcal{Q}_3}$
<i>Rank</i>	C		C	B	B		B	B
	B	...	B	C	A	...	A	C
	A		A	A	C		C	A

So far, we haven't seen anything too shocking. However, the profiles $\mathcal{P}_3$ and $\mathcal{Q}_3$ have a peculiar property that we can exploit to reach some surprising conclusions. The key feature is that all voters except m have C in an extreme position but the output has C in the middle. Suppose we begin from the profile $\mathcal{P}_3$ and have any subset of voters not including m interchange the ranks of A and B . This new profile will agree with $\mathcal{P}_3$ in comparing C and A and will also agree in comparing C and B . Therefore, all of the new profiles $\mathcal{P}_3^*$ we can obtain in this manner will have the same output ranking of $A > C > B$. In particular, you may note that we can get an unfair-looking outcome by having all voters except m switch

to rank B above A . Then according to our voting system, even though every voter except m ranks B above A , the output ranking has A above B !

Profiles $\mathcal{P}_3^*$								Output
Voter	1	...	$m-1$	m	$m+1$	...	n	
	C		C	A	*		*	$>_{\mathcal{P}_3^*}$
Rank	*	...	*	C	*	...	*	A
	*		*	B	C		C	C
								B

We see a similar thing starting from $\mathcal{Q}_3$. If we modify this profile by having any set of voters other than m switch the rankings of A and B , the resulting profile will still agree with $\mathcal{Q}_3$ in comparing both C and A and comparing C and B so all of these new profiles $\mathcal{Q}_3^*$ will still output a ranking of $B > C > A$:

Profiles $\mathcal{Q}_3^*$								Output
Voter	1	...	$m-1$	m	$m+1$	...	n	
	C		C	B	*		*	$>_{\mathcal{Q}_3^*}$
Rank	*	...	*	C	*	...	*	B
	*		*	A	C		C	C
								A

Now consider an arbitrary profile $\mathcal{R}$. If voter m ranks $A > B$ in this profile, then $\mathcal{R}$ will agree with one of our $\mathcal{P}_3^*$ profiles in comparing A and B . Therefore, by our independence axiom we must have $A >_{\mathcal{R}} B$. Conversely, if voter m ranks $B > A$ in $\mathcal{R}$, then $\mathcal{R}$ will agree with one of our $\mathcal{Q}_3^*$ profiles in comparing A and B . In this case, independence implies that $B >_{\mathcal{R}} A$. In other words, we have just proven that the output ranking will have $A > B$ if and only if this was the preference of voter m . We describe this by saying that m is a *local dictator* for A, B .

By similar reasoning, we find that there is a voter m' that is a local dictator for B, C and voter m'' is a local dictator for A, C . If m, m', m'' are not all equal, then we may assume (without loss of generality) that $m'' \neq m$ and $m'' \neq m'$. In this case, we can find a profile where m and m' rank the candidates $A > B > C$ and voter m'' ranks $C > A$. However, by the local dictator property, our output ranking for this profile must then rank $A > B$ and $B > C$ and $C > A$, which is impossible. It follows

that $m = m' = m''$ and this voter is a dictator! This completes the proof for the case of three candidates.

Now assume there are many candidates, say $A, B, C, \dots, Z$, and assume (as before) that the voting system obeys unanimity and independence. If we focus on the first three of these candidates, A, B, C , and ignore the rest, we have a three-candidate voting system (that still obeys unanimity and independence). So there must be a single voter m who is a local dictator for A, B , for B, C , and for A, C . By considering just B, C, D , we have another three-candidate voting system, so there is a single voter m' who is a local dictator for B, C , for C, D , and for B, D . Since m and m' are both the local dictator for B, C , we must have $m = m'$. Continuing in this fashion, we see that m will be a local dictator for every pair of candidates, so m is a dictator. $\square$

Exercises

- (1) Why is every equitable division between two players envy-free?
- (2) In this problem we will use Nash arbitration to divide a cake among two players. Assume that each player i has a utility function u_i that assigns the entire cake a value of 1, an empty piece a value of 0, and every nonempty piece some positive value. Consider all possible ways of dividing the cake into two pieces P_1, P_2 and for each such division plot the point $(u_1(P_1), u_2(P_2))$ in the plane. This corresponds to the division where piece P_1 is given to player 1 and piece P_2 is given to player 2. Let S be the set of all such points in the plane, and assume that S is closed and convex. Apply Nash arbitration to S with status quo point $(0, 0)$ to select a solution point (x^*, y^*) . Show that this solution point corresponds to an equitable division of the cake.
- (3) This question explores Split and Choose for many players.
 - (a) Extend the Split and Choose procedure to work with three players. Hint: As a first step, have players 1 and 2 apply the usual Split and Choose procedure to divide the cake among themselves. Then arrange for player 3 to get a portion from player 1 and a portion from player 2.

- (b) Extend your procedure to n players for every $n \geq 2$.
 - (c) In n -player Split and Choose, prove that each player has a strategy that guarantees a piece he or she values at $\geq \frac{1}{n}$.
 - (d) Explain why in 3-player Split and Choose, players 1 and 2 do not have strategies that guarantee them a portion which they consider to have value at least that of any other player's portion.
- (4) Four travelers have rented two rooms in a hotel, and each room has 2 beds. Accordingly, they will divide into two groups of two roommates. Each traveler has a linear ranking of the other three in order of with whom he would most like to share a bedroom. Ideally, the group will be split into two groups of two so that for every pair XY of travelers who are not sharing a room, either X prefers his roommate to Y or Y prefers his roommate to X . Find a system of preferences so that such a division is impossible.
- (5) For every $n \geq 1$ construct a system with n men and n women each of which have individual preferences (as in Section 12.2) with the property that there is a unique stable total marriage.
- (6) Complete the proof of Theorem 12.12 by proving the second part.
- (7) Consider a system with m men and n women each with individual preferences among the members of the opposite sex, but now with $m \neq n$. We can still run the Gale-Shapley algorithm to produce a set of couples M . Show that whenever m is a man and w is a woman which do not form a couple (i.e. $\{w, m\}$ is not in M), either m is in a couple with a woman he prefers to w or w is in a couple with a man she prefers to m .
- (8) A certain item is going to be auctioned off to one of the n players $1, \dots, n$. Each player makes a bid in secret and the highest bidder gets the item. Assume that player i has a value V_i which he feels the item is worth. If this player gets the item at a cost of C , his payoff is $V_i - C$ and if he does not get the item, he gets a payoff of 0.
- (a) First suppose the highest bidder (who wins the item) pays the price he bid for the item. What would you advise a player to do if he values the item at say \$10?
 - (b) Now suppose that the highest bidder (who wins the item) pays the price bid by the second highest bidder. Show that in this

auction, when every player bids the true price at which he values the item, we have an equilibrium. That is, no single player can improve his payoff by changing his bid while everyone else keeps his bid fixed.

- (9) In this exercise we devise a procedure for dividing a cake among three players which gives envy-free divisions (under optimal play). For simplicity we will describe how the players should act in order to guarantee an envy-free division. First, player 1 divides the cake into three pieces A, B, C which player 1 considers to be of equal value. Suppose that player 2 ranks the pieces $A \geq B \geq C$. The next step is to have player 2 reduce the size of piece A by removing a small amount R so that player 2 considers the remaining portion A' of A to have equal value to B (and thus still at least that of C).
- (a) Show that the pieces A', B, C can be distributed to the three players in an envy-free manner.
 - (b) Observe that player 1 does not get the piece A' , and moreover, player 1 will not be envious of the player who did get A' no matter how the remaining portion of cake R is divided.
 - (c) Use the above observation to divide the remaining piece of cake R in such a way that the total portion of cake each player gets (including the first piece A', B , or C) gives an envy-free division.

Appendix A

On Games and Numbers

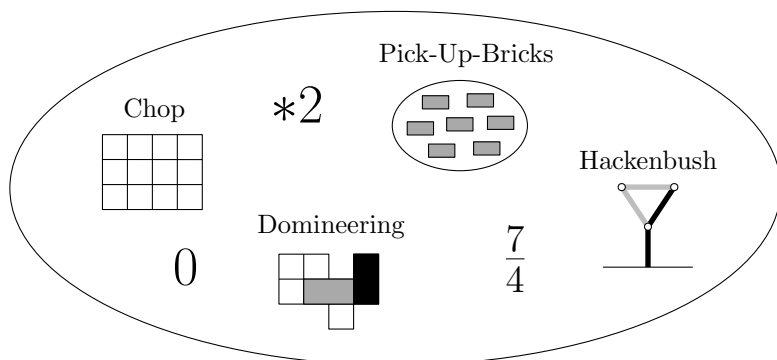


Figure A.1. Conway's universe

Around 1970 John Conway was attempting to find a unified way to describe all possible positions in all possible normal-play games. He developed a beautiful recursive framework to do this, which has become the basic foundation for combinatorial game theory. In addition to its impact in game theory, this perspective has led to exciting developments in set theory and logic, most notably the surreal numbers. This appendix introduces Conway's wonderful idea for describing positions and gives just a hint of the surreal numbers. Our main goal is to provide a bridge between our development of combinatorial game theory and the more

sophisticated treatments found elsewhere. In particular, this appendix aims to prepare our reader for the combinatorial game theory classic *Winning Ways for Your Mathematical Plays* [7] and its modern update *Lessons in Play: An Introduction to Combinatorial Game Theory* [1].¹

Games. Our grand aim is to make a kind of “universe of positions” which contains every position in every normal-play game. The notion of equivalence for positions introduced in Chapter 2 will be a key ingredient here, but we will carry this notion even further now. For starters, let’s consider the positions in Figure A.2 and note that they are all equivalent. (This is easy to check since in each such position there is just one move available for either player, and that move brings the game to a terminal position.)

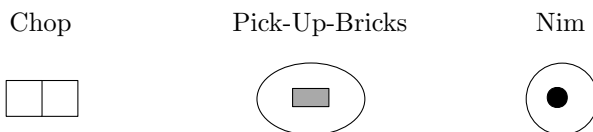


Figure A.2. Equivalent positions

Instead of treating the above positions as different positions which are equivalent, we will now treat all of them as one and the same position. In other words, we will consider any two equivalent positions to be equal. From a notational perspective, there is very little difference here... instead of writing $\equiv$ we will now write $=$. This nonetheless signals a very important change in perspective. Whereas before we had a set of positions equivalent to $\ast 1$, we will now consider all of these to be the same position. So, just as we regard $1 - 1$ and 0 as different expressions for the same number, we will now treat equivalent positions (which are not identical) as different expressions for one and the same position.

This unification aids our attempt to establish the universe of positions. Still, to construct all positions, we are going to need a powerful way of defining them. The key tool we will use is based on the $\{|\}$ notation we introduced in Chapter 2. Recall that we use this notation for

¹As a warning to our reader, let us note that in addition to the substantive differences between our development and that found in these other books, there are also some notational differences. For instance, what we have called the “type” of a position is generally known as its “outcome class”.

a position γ to indicate positions to which Louise can move from γ and positions to which Richard can move from γ , as in the example from Hackenbush shown in Figure A.3.

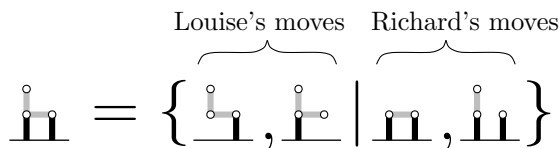


Figure A.3. Position notation

More generally, we write $\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$ to indicate that in the position γ , Louise can move to $\alpha_1, \dots, \alpha_m$ and Richard can move to $\beta_1, \dots, \beta_n$. In our investigation of combinatorial games so far, we have used the notation $\{\mid\}$ to describe positions in a game we have already defined. We will now turn things around. Instead of starting with a known game and using this notation to describe its positions, we will now use $\{\mid\}$ as a way of *constructing* new positions. So, if $\alpha_1, \dots, \alpha_m$ and $\beta_1, \dots, \beta_n$ are existing positions, then we *define*

$$\{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$$

to be a new position from which Louise can move to $\alpha_1, \dots, \alpha_m$ and Richard can move to $\beta_1, \dots, \beta_n$. Although this shift in perspective may seem small at first, we will see that the operator $\{\mid\}$ has some extremely surprising consequences when taken to its logical extreme.

We start by considering the very simplest position we could create using $\{\mid\}$ —a position where neither player has any available moves. We call this position

$$0 = \{\mid\}.$$

In Chapter 3, we defined $*0$ to be a Nim position with no stones, and in Chapter 4, we defined $\bullet 0$ to be a Hackenbush position with no edges. More generally, all of our normal-play games have “zero” positions where neither player has any moves left. All of these positions are equivalent to one another and to any other position of type P. In our new unified manner of thinking, we will now consider all of these to be the same position, called 0.

Using the position 0 we just constructed together with our operator $\{\mid\}$ permits us to create a few more positions. For instance, we can make a new position in which Louise can move to 0 but Richard has no available move. We call this position

$$1 = \{0 \mid\}.$$

As you can readily verify, this position plays just like the Hackenbush position consisting of one black edge, which we denoted $\bullet 1$. It also plays just like a Cut-Cake position consisting of a single 1×2 array. Now we will treat all of those positions equivalent to $\bullet 1$ as a single position called 1. In a similar vein we have the position

$$-1 = \{\mid 0\},$$

which has no move for Louise and a single move for Richard, that brings the position to 0. The position -1 is our unified version of the Hackenbush position $\bullet(-1)$. We can also use the construct $\{\mid\}$ to make a position where each player has just one available move and whomever moves brings the game to 0. This position is named

$$*1 = \{0 \mid 0\}.$$

Indeed, this position plays just the same as a game of Nim with just one pile of 1 stone (i.e the nimber $*1$). Namely, either player has just one move, and if they make this move, they bring the game to the zero position. In our unified language, $*1$ is equal to any of the positions from Figure A.2.

To keep things in order, it will be helpful to introduce a chronology to our construction of our positions. The first position we constructed, 0, has the simplest structure: $0 = \{\mid\}$, and it is not based on any other positions. We will think of this position as born on day 0. The next three positions, 1, -1 , $*1$, all have definitions which depend only on the position 0, so these positions will all be born on day 1. Continuing in this manner, we will construct new positions on each day $n + 1$ based on the existing positions from days 0, 1, $\dots$, n . On day 2 we can create a position using our operator $\{\mid\}$ where each of Louise and Richard are permitted to move to any subset of $\{0, 1, -1, *1\}$. Here are some new positions born

on day 2 which we name with numbers:

$$\begin{aligned} -2 &= \{ \mid -1, 0, 1 \}, & 2 &= \{-1, 0, 1 \mid \}, \\ -\frac{1}{2} &= \{-1 \mid 0, 1 \}, & \frac{1}{2} &= \{-1, 0 \mid 1 \}. \end{aligned}$$

A little checking reveals that each of the above positions q is equal to the dyadic position $\bullet q$ we encountered previously. Also on day 2 we can make the position $\{-1 \mid 1\}$, but it follows from the Simplicity Principle that this is equal to 0, so it isn't actually anything new. Another familiar position born on day 2 is the nimber

$$*2 = \{0, *1 \mid 0, *1\},$$

which we encountered before. However, there are new positions born on day 2 which we have not yet seen in our study of combinatorial game theory. Here are two nice ones:

$$\uparrow = \{0 \mid *1\} \quad \text{and} \quad \downarrow = \{*1 \mid 0\}.$$

We can carry this process on and on and this leads to a rich universe which consists of all positions born on some day n . It is easy to see that this universe will contain every possible position γ from every normal-play game. Indeed, to check this we only need to focus on the recursive structure of γ . If $\gamma = \{\alpha_1, \dots, \alpha_m \mid \beta_1, \dots, \beta_n\}$, then we will be ready to construct the position γ using the $\{\mid\}$ operator as soon as we have constructed all of the α_i and β_j positions to which the players can move. In turn, each of these α_i and β_j positions can be constructed by $\{\mid\}$. This line of recursion leads to simpler and simpler positions before eventually arriving at 0. For instance, the nimber $*n$ will be born on day n since it has the recursive description

$$*n = \{*0, *1, \dots, *(n-1) \mid *0, *1, \dots, *(n-1)\}.$$

This universe of positions is the basic setting for the modern study of combinatorial game theory. The sum operation we introduced for games in Chapter 2 also works in this setting to add any two positions α and β from this universe since the only thing required for this definition is the recursive structure of α and β . Beyond addition, there is also negation (which generalizes negation in Hackenbush) and an inequality $\leq$ (related to the notion of advantage introduced in Chapter 4) which can

be used to compare positions. A driving goal in combinatorial game theory is to better understand this universe of positions together with the behaviour of $+$, $-$, and $\leq$.

Numbers. Some of the positions we have constructed using $\{\mid\}$ are named with dyadic numbers. In fact, we are encountering the dyadic numbers here in the same manner as we did in Chapter 4 (even the concept of birthday is recurring). Let's pause to recall the key ingredients from our earlier development. In Chapter 4 we introduced a process which assigned a birthday to each dyadic number. We imagined the process stepping forward in time, starting at day 0 on which the number 0 is born. Then for every nonnegative integer n , if the numbers $a_1 < a_2 < \dots < a_\ell$ have been born on days $0, 1, \dots, n$, the new numbers born on day $n + 1$ are as follows:

- the largest integer which is less than a_1 ,
- the smallest integer which is greater than a_ℓ ,
- the number $(a_i + a_{i+1})/2$ for every $1 \leq i \leq \ell - 1$.

The dyadic numbers are appearing now as names of positions in precisely the same way. On day 0 we have the first number $0 = \{\mid\}$, and on day $n + 1$ we create new number positions out of the existing number positions created on days $0, 1, \dots, n$. If on days $0, 1, \dots, n$ we have already created the numbers $a_1 < a_2 < \dots < a_\ell$, then on day $n + 1$ we use $\{\mid\}$ to construct the following new number positions:

$$\{\mid a_1, \dots, a_\ell\}, \{a_1 \mid a_2, \dots, a_\ell\}, \dots, \{a_1, \dots, a_{\ell-1} \mid a_\ell\}, \{a_1, \dots, a_\ell \mid\}.$$

So, each of these positions is made using the $\{\mid\}$ notation and positions made on the previous days. We name these new positions using the same scheme as above. That is, we define as follows:

- $\{\mid a_1, \dots, a_\ell\}$ is the largest integer smaller than a_1 .
- $\{a_1, \dots, a_\ell \mid\}$ is the smallest integer greater than a_ℓ .
- $\{a_1, \dots, a_i \mid a_{i+1}, \dots, a_\ell\}$ is the number $\frac{a_i + a_{i+1}}{2}$.

As you can see, the dyadic numbers are appearing now just as they did before, only now we are regarding them as special types of positions.

More precisely, each dyadic number is made up of earlier dyadic numbers, where Louise's every move is to something less than all of Richard's moves.

Why have we given $\{-1, 0 \mid 1\}$ the name $\frac{1}{2}$ instead of another number such as $\frac{1}{3}$? The answer comes from the rule for adding positions. If you add the position $\{-1, 0 \mid 1\}$ to itself, the resulting position is equal to 1. So, if we want our naming convention to work properly, we need to give $\{-1, 0 \mid 1\}$ the name $\frac{1}{2}$. Indeed, these positions are named with numbers precisely so that our addition operation for positions aligns with the usual addition on these numbers. In fact, the key results from Chapter 4 concerning dyadic positions in Hackenbush can be reinterpreted in this setting to prove that addition and negation work exactly as we are claiming for these positions which are numbers.

Remarkably, this process of constructing numbers (and also positions) can be continued to infinity and beyond! An in-depth understanding of this requires some background in set theory, but we can nevertheless provide a glimpse of how it works. Just as on each finite day $n + 1$ we make new numbers based on the numbers we have built on the days $0, 1, \dots, n$, we can imagine a day infinity which comes after every finite day. On day infinity we create, as before, new numbers using our $\{\mid\}$ operator, only now we will be using all of the numbers we created on the finite days to build our new numbers. Since the numbers constructed on finite days are precisely the dyadic rationals, this infinite day will give rise to a new number for each possible location of the $\mid$ in $\{\mid\}$ to split the dyadic rationals. As a warning, our convention for naming numbers no longer applies here, so it may be unclear what to call these new numbers we form. Nevertheless, the $\{\mid\}$ operator is still perfectly valid, so we are certainly capable of constructing them. We encourage the reader to pause to consider what new numbers might be formed on day infinity.

Although day infinity might seem to be a mind-boggling notion already, this process continues on to even larger and larger infinite days! The resulting number system is known as the surreal numbers, and they are indeed deserving of this title. In addition to the dyadic numbers, every real number is formed in this process, so the surreal numbers contain the real numbers. Beyond this, the surreal numbers also contain

infinite numbers which are larger than any real number as well as infinitesimal numbers which are bigger than 0 but smaller than any positive real number. The definitions of addition and negation we already have for positions still work here, and Conway also introduced a rule for multiplication. Wonderfully, if the simplest infinite number, ω , is multiplied by the simplest infinitesimal number, $\frac{1}{\omega}$, the result is the number 1.

Just as we have continued this process to infinity and beyond in constructing the surreal numbers, the same process can be continued more generally with positions. Doing so yields an even more vast universe of positions which contains all of the positions with finite birthdays, all of the surreal numbers, and much, much more! Figure A.4 shows how these new universes relate to the classical sets $\mathbb{N}$, $\mathbb{Q}$, and $\mathbb{R}$. We encourage the reader interested in exploring this universe further to investigate *Winning Ways for Your Mathematical Plays* [7]. In addition, a delightful introduction to the surreal numbers is Knuth's novelette *Surreal Numbers* [31].

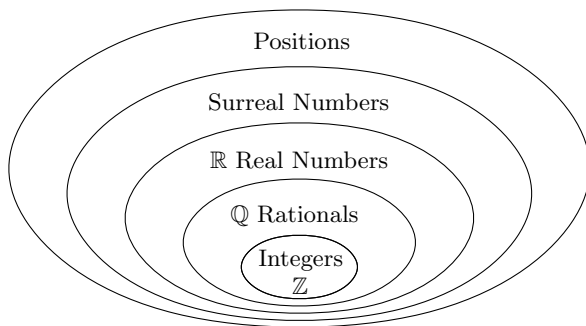


Figure A.4. Numbers and positions

Appendix B

Linear Programming

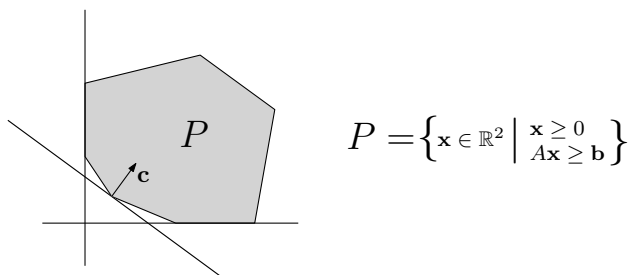


Figure B.1. A two-dimensional LP

Inequalities based on linear functions are ubiquitous both in mathematics and in real-world applications; linear programming is the theory concerned with optimizing a linear function subject to such constraints. This mathematical theory has brought about some very efficient algorithms for solving linear programs, and in turn, these algorithms have had broad impact on our world. Indeed, fast methods for solving linear programs lie at the heart of optimization theory and are routinely called on when a complicated process is being optimized. In particular, linear programs are used extensively in manufacturing, design, telecommunications, scheduling, and transportation. The key mathematical result

at the center of this theory is called linear programming duality. This duality is very closely related with the Minimax Theorem for zero-sum games, as we shall soon see. In particular, these efficient algorithms used to solve linear programming problems can be used to find von Neumann solutions to zero-sum matrix games.

Basic Theory

Imagine a company which is producing feed for cattle. The feed is made up of three basic ingredients: grain, soy, and corn. In the chart below we have listed the nutritional information as well as the cost per unit for these three ingredients:

Source	Protein	Carbohydrate	Fat	Cost
grain	12g	130g	10g	\$.05
soy	60g	50g	36g	\$.09
corn	16g	160g	2g	\$.04

Each serving of feed is required to have at least 200 grams of protein, 1,200 grams of carbohydrate, and 100 grams of fat. The key question is, How much of each of these three ingredients should be used in each serving so as to meet all of the nutritional constraints but minimize cost?

This is an example of a linear programming problem, and one which is indeed quite similar to some genuine real-world applications. It turns out that the optimal answer is to put roughly 3.5 units of grain, 1.6 units of soy, and 3.9 units of corn into each serving. Although these numbers are not particularly meaningful, the fact that we have fast computational tools to solve such problems certainly is. As you can probably imagine, significant optimization problems of this type are ever-present in industry, so solving them is serious business.

Let's take another step with this example to formalize it mathematically. Let x, y, z denote the number of units of grain, soy, and corn to use in each serving of feed. Of course, it is impossible to use a negative quantity of one of our ingredients, so we have the nonnegativity constraints

$$x, y, z \geq 0.$$

We can now express the constraints for the required amount of protein, carbohydrate, and fat (per serving) in terms of these variables as follows:

$$12x + 60y + 16z \geq 200,$$

$$130x + 50y + 160z \geq 1,200,$$

$$10x + 36y + 2z \geq 100.$$

Finally, the goal is to minimize cost, which we can express as follows:

$$\text{Minimize } .05x + .09y + .04z.$$

As you can see, our example problem boils down to minimizing a linear function subject to a collection of linear inequality constraints. This is an example of a linear programming problem and the focus of this appendix. Let us take a step back now and define the problem in general. Let $x_1, \dots, x_n$ be nonnegative variables (i.e. $x_1, \dots, x_n \geq 0$). We will assume that our goal is to optimize the following linear function of these variables (here $c_1, \dots, c_n$ are fixed coefficients):

$$\min c_1x_1 + c_2x_2 + \dots + c_nx_n.$$

However, we will have a collection of linear constraints given as follows:

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n \geq b_1,$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n \geq b_2,$$

$$\vdots$$

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n \geq b_m.$$

We call any problem of this form a *linear program*¹. It will be helpful for us to use vector notation to encode this problem more succinctly. To do

so, define the matrix $A = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{bmatrix}$ and the vectors $\mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$

and $\mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$. Now treating $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ as a variable, we can express our

linear program (LP for short) as follows:

$$\boxed{\text{LP}} \quad \begin{array}{l} \text{Minimize } \mathbf{c}^T \mathbf{x} \\ \text{subject to } \mathbf{Ax} \geq \mathbf{b} \\ \text{and } \mathbf{x} \geq 0. \end{array}$$

¹This is one of many equivalent ways to formulate linear program.

Just as zero-sum matrix games have two competing players with one selecting columns and the other rows, linear programs have dual problems which switch the roles of the rows and columns. If the LP was based on an $m \times n$ matrix A , then we have a variable x_i for each column of A and a constraint for each row of A . In the dual problem, there will be a variable y_j for each row of A and a constraint for each column. The dual of our LP given above is as follows:

$$\boxed{\text{Dual}} \quad \begin{array}{l} \text{Maximize } \mathbf{b}^\top \mathbf{y} \\ \text{subject to } \mathbf{y}^\top A \leq \mathbf{c}^\top \\ \text{and } \mathbf{y} \geq 0. \end{array}$$

You will notice that the vector $\mathbf{b} \in \mathbb{R}^m$ was associated with the constraints in the LP whereas the vector $\mathbf{c} \in \mathbb{R}^n$ was associated with the objective. In the dual, the roles of $\mathbf{b}$ and $\mathbf{c}$ are switched, so $\mathbf{b}$ is associated with the objective and $\mathbf{c}$ is associated with the constraints.

We say that a vector $\mathbf{x}$ which satisfies the constraints for the LP, namely $\mathbf{x} \geq 0$ and $A\mathbf{x} \geq \mathbf{b}$, is *feasible*. Similarly, we call $\mathbf{y}$ *feasible* if it satisfies the constraints for the dual of $\mathbf{y} \geq 0$ and $\mathbf{y}^\top A \leq \mathbf{c}^\top$. It is possible that either the LP or the dual has no feasible vectors, and in this case, we will say that the corresponding program is *infeasible*. It is also possible for either the LP or the dual to be feasible but have arbitrarily good feasible points (i.e. for the LP there might exist feasible vectors $\mathbf{x}$ for which $\mathbf{c}^\top \mathbf{x}$ is arbitrarily small). In this case we will say that the corresponding program is *unbounded*.

Our first property relating the LP and the dual is analogous to one of our lemmas concerning zero-sum matrix games. Namely, Lemma 5.10 tells us that for a zero-sum matrix game, with mixed strategies $\mathbf{p}$ for Rose and $\mathbf{q}$ for Colin, the guarantee which $\mathbf{p}$ gives Rose must be less than or equal to the guarantee which $\mathbf{q}$ gives Colin.

Proposition B.1 (Weak Duality). *If $\mathbf{x}$ is a feasible vector for the LP and $\mathbf{y}$ is feasible in the dual, then*

$$\mathbf{c}^\top \mathbf{x} \geq \mathbf{y}^\top \mathbf{b}.$$

Proof. Since our vectors are feasible, we have $\mathbf{x}, \mathbf{y} \geq 0$ and $A\mathbf{x} \geq \mathbf{b}$ and $\mathbf{c}^\top \geq \mathbf{y}^\top A$. Combining these gives us

$$\mathbf{c}^\top \mathbf{x} \geq (\mathbf{y}^\top A)\mathbf{x} = \mathbf{y}^\top (A\mathbf{x}) \geq \mathbf{y}^\top \mathbf{b}. \quad \square$$

Although the above proposition is rather quick to prove, it has some significant consequences. First off, anytime we have found a feasible vector $\mathbf{y}$ for the dual problem, it gives us a lower bound on the minimum that we can hope to achieve in the LP. Similarly, any feasible vector $\mathbf{x}$ for the LP gives us an upper bound on the maximum that we can hope to achieve in the dual. In particular, this tells us that whenever one of our programs is feasible, the other cannot be unbounded.

Von Neumann's Minimax Theorem is the central result in the theory of zero-sum matrix games. This theorem asserts that for every such game, Rose and Colin have mixed strategies with matching guarantees. In the world of linear programming, the most important theoretical result is the following strong duality theorem which exhibits a similar phenomena.

Theorem B.2 (Strong Duality). *If both an LP and its dual are feasible, then the optimum solutions $\mathbf{x}$ and $\mathbf{y}$ satisfy $\mathbf{c}^T \mathbf{x} = \mathbf{y}^T \mathbf{b}$.*

We will postpone the proof of this result until the last part of the appendix. Next we turn our attention to the relationship between linear programming and zero-sum matrix games.

A Connection to Game Theory

In this section we will reveal that the problem of finding optimal strategies in zero-sum matrix games is a special case of linear programming. In fact, finding optimal strategies for Rose and Colin corresponds to optimizing dual linear programs, and we can deduce the Minimax Theorem for zero-sum games from the strong LP duality theorem of linear programming.

Fix a zero-sum matrix game represented by an $m \times n$ matrix A and imagine that Colin is trying to find a mixed strategy with the minimum guarantee. At first glance this may not appear to be a linear program, but there are some useful techniques which will help us transform it into one. Let's begin by fixing a real number t and considering whether Colin has

a mixed strategy $\mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$ which gives him a guarantee of at most t . Can

we encode this problem using only linear inequalities? In order for $\mathbf{q}$ to

be a mixed strategy, the variables $q_1, \dots, q_n$ must be nonnegative, so we have the constraints

$$\begin{aligned} q_1 &\geq 0, \\ &\vdots \\ q_n &\geq 0. \end{aligned}$$

The description of our LP already has nonnegativity constraints for the variables, so in fact these constraints will be automatic. Also, to be a mixed strategy, we must have $q_1 + \dots + q_n = 1$. This can be encoded with the following two inequalities:

$$\begin{aligned} q_1 + q_2 + \dots + q_n &\geq 1, \\ -q_1 - q_2 + \dots - q_n &\geq -1. \end{aligned}$$

Finally, we want to add the constraints that $\mathbf{q}$ gives Colin a guarantee which is at most t . We can express this as follows:

$$-A\mathbf{q} + \begin{bmatrix} t \\ \vdots \\ t \end{bmatrix} \geq \mathbf{0}.$$

Above we have collected a system of linear inequalities which encode the problem of finding a mixed strategy for Colin with a guarantee of at most t . Of course, Colin is interested in finding the minimum guarantee. To do this, we will just think of t as a variable. Indeed, finding an optimal strategy for Colin is precisely the problem of minimizing t subject to the above constraints (where we view $q_1, \dots, q_n$ and t as variables and the other parameters as fixed). This is almost the form of our usual LP with one slight hitch. We are now treating t as a variable which should be permitted to be either positive or negative. However, our LP formulation insists that the variables be nonnegative. To get around this problem, we will use two nonnegative variables t^+ and t^- to represent t . More precisely, we let $t = t^+ - t^-$ and then replace every occurrence of t by $t^+ - t^-$. This allows us to mimic a variable which can be any real number by two nonnegative ones. Now we can cast Colin's optimization problem

as the following LP using the variables $q_1, \dots, q_n, t^+, t^-$:

$$\begin{array}{ll}
 \text{Colin} & \text{subject to} \\
 & \begin{array}{c} \text{Minimize } t^+ - t^- \\
 \begin{array}{|c|c|c|} \hline -A & 1 & -1 \\ \hline 1 & 0 & 0 \\ -1 & 0 & 0 \\ \hline \end{array}
 \begin{array}{|c|} \hline q_1 \\ \vdots \\ q_n \\ t^+ \\ t^- \\ \hline \end{array}
 \geq
 \begin{array}{|c|} \hline 0 \\ \vdots \\ 0 \\ 1 \\ -1 \\ \hline \end{array}
 \end{array}
 \end{array}$$

and $q_1, \dots, q_n, t^+, t^- \geq 0$.

To give this the look of our usual LP, we only need to name our terms.

Set A^+ to be the matrix in the above constraint, and define $\mathbf{b} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -1 \end{bmatrix}$

and $\mathbf{c} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -1 \end{bmatrix}$ of the appropriate dimensions. Now we may rewrite this

LP which finds a mixed strategy for Colin with the minimum guarantee as follows:

$$\begin{array}{ll}
 \text{Colin} & \begin{array}{l} \text{Minimize } \mathbf{c}^\top \mathbf{x} \\ \text{subject to } A^+ \mathbf{x} \geq \mathbf{b} \\ \text{and } \mathbf{x} \geq 0. \end{array}
 \end{array}$$

Once we have found the solution $\mathbf{x} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \\ t^+ \\ t^- \end{bmatrix}$, the vector $\mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$ gives a

mixed strategy for Colin which has the minimum guarantee of $t^+ - t^-$.

As we will show, the dual of this LP (below) is precisely the problem of finding a mixed strategy for Rose with maximum guarantee:

$$\begin{array}{ll}
 \text{Rose} & \begin{array}{l} \text{Maximize } \mathbf{b}^\top \mathbf{y} \\ \text{subject to } \mathbf{y}^\top A^+ \leq \mathbf{c}^\top \\ \text{and } \mathbf{y} \geq 0. \end{array}
 \end{array}$$

To illuminate the situation, let us set $\mathbf{y}^\top = [p_1 \ \cdots \ p_m \ s^+ \ s^-]$ and uncompress the above program. For notational simplicity, we replace the equation $\mathbf{y}^\top A^+ \leq \mathbf{c}^\top$ by the equivalent transpose equation $(A^+)^\top \mathbf{y} \leq \mathbf{c}$. This gives us the following:

$$\begin{array}{c}
 \text{Rose} \quad \text{subject to} \quad \begin{array}{c} \text{Maximize } s^+ - s^- \\ \begin{array}{|c|c|c|} \hline -A^\top & 1 & -1 \\ \hline 1 & 0 & 0 \\ -1 & 0 & 0 \\ \hline \end{array} \end{array} \leq \begin{array}{|c|} \hline p_1 \\ \vdots \\ p_m \\ s^+ \\ s^- \\ \hline \end{array} \leq \begin{array}{|c|} \hline 0 \\ \vdots \\ 0 \\ 1 \\ -1 \\ \hline \end{array}
 \end{array}$$

and $p_1, \dots, p_m, s^+, s^- \geq 0$.

Set $s = s^+ - s^-$ and $\mathbf{p} = \begin{bmatrix} p_1 \\ \vdots \\ p_m \end{bmatrix}$. Now the nonnegativity constraints on

$p_1, \dots, p_m$ combined with the last two rows in the above matrix equation are equivalent to the statement that the vector $\mathbf{p}$ is a mixed strategy for Rose. The earlier rows in this matrix equation are equivalent to the

equation $-A^\top \mathbf{p} + \begin{bmatrix} s \\ \vdots \\ s \end{bmatrix} \leq \mathbf{0}$, which is precisely the statement that $\mathbf{p}$ has a guarantee of at least s . The objective of this dual program is to maximize $s = s^+ - s^-$, so, as claimed, this program is precisely that of finding a mixed strategy for Rose with the maximum guarantee.

From a theoretical standpoint, we can now see that the Minimax Theorem follows by applying strong linear programming duality to the above systems. In fact these two problems are very close in spirit, and the strong linear programming duality theorem was recognized by von Neumann as an extension of his work on games.

From a computational standpoint, if we want to find solutions to zero-sum matrix games, we can now use algorithms which solve general linear programs. This is quite valuable since there has been a massive effort to develop efficient algorithms for this purpose (motivated by the dazzling array of important linear programming problems). The first significant algorithm to solve linear programs is called the simplex al-

gorithm and was devised by Dantzig. More recent developments have lead to algorithms with good theoretical and experimental performance guarantees.

LP Duality

In the remainder of this appendix we will give a proof of strong linear programming duality. This result will call upon some key geometric principles which we introduce next. We will use lowercase boldface letters to denote column vectors and we will frequently write matrices for which the dimensions must be determined from the context.

Let A be an $m \times n$ matrix. We define the *polyhedral cone associated with A* to be the set

$$C = \{Ax \mid x \geq 0\}.$$

More generally, we call any set of this form a *polyhedral cone*. Our next lemma exhibits two nice properties of cones: They are closed under multiplication by positive scalars, and they are closed under addition. These properties imply that such a set C will have a “cone-like” shape starting at $\mathbf{0}$ and continuing outward toward infinity.

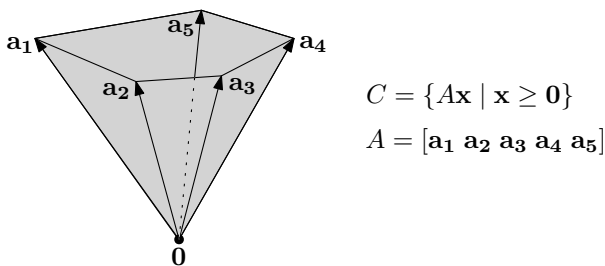


Figure B.2. A polyhedral cone in $\mathbb{R}^3$ based on a 3×5 matrix A

Proposition B.3. *Let C be a polyhedral cone, let $\mathbf{y}, \mathbf{y}' \in C$, and let $t \geq 0$. Then*

- (1) $t\mathbf{y} \in C$,
- (2) $\mathbf{y} + \mathbf{y}' \in C$.

Proof. Since $\mathbf{y}, \mathbf{y}' \in C$, we may choose $\mathbf{x}, \mathbf{x}' \geq \mathbf{0}$ so that $A\mathbf{x} = \mathbf{y}$ and $A\mathbf{x}' = \mathbf{y}'$. Now, the first part follows from the equation $t\mathbf{y} = t(A\mathbf{x}) = A(t\mathbf{x})$ and the observation that $t\mathbf{x} \geq \mathbf{0}$. The second part follows from $\mathbf{y} + \mathbf{y}' = A\mathbf{x} + A\mathbf{x}' = A(\mathbf{x} + \mathbf{x}')$ and the observation that $\mathbf{x} + \mathbf{x}' \geq \mathbf{0}$. $\square$

The above proposition immediately yields another useful fact about polyhedral cones.

Corollary B.4. *Every polyhedral cone is convex.*

Proof. Let $\mathbf{y}, \mathbf{y}'$ be points in the polyhedral cone C and let $0 \leq t \leq 1$. The previous proposition implies that $t\mathbf{y} + (1 - t)\mathbf{y}' \in C$ and it follows that C is convex. $\square$

The next lemma gives us a convenient separation property for polyhedral cones. In words, this lemma states that for a point $\mathbf{b}$ and a polyhedral cone C , either $\mathbf{b}$ is in C or there is a hyperplane passing through the origin with $\mathbf{b}$ strictly on one side and C on the other. Our proof relies on the property that two vectors in $\mathbb{R}^n$ make an acute angle whenever their dot product is positive.

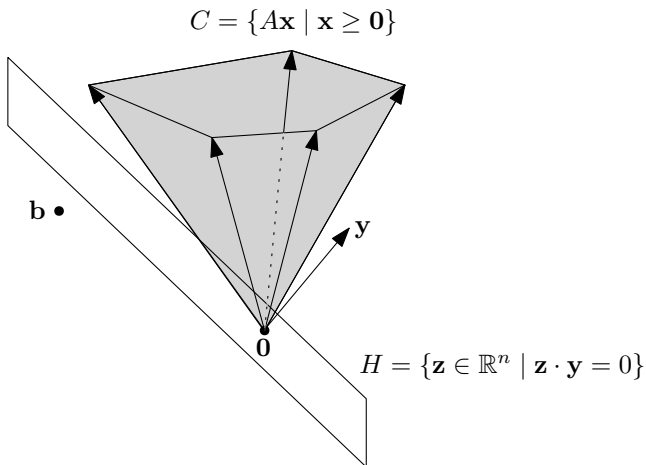
Lemma B.5 (Farkas Lemma). *Let A be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$. Then one of the following holds:*

- (1) *There exists $\mathbf{x} \geq \mathbf{0}$ so that $A\mathbf{x} = \mathbf{b}$.*
- (2) *There exists $\mathbf{y}$ so that $\mathbf{y}^\top A \geq \mathbf{0}$ and $\mathbf{y}^\top \mathbf{b} < 0$.*

Proof. Define the polyhedral cone $C = \{A\mathbf{x} \mid \mathbf{x} \geq \mathbf{0}\}$. If $\mathbf{b}$ lies in the cone C , then we have our first outcome. Otherwise, let $\mathbf{z}$ be a closest vector in C to $\mathbf{b}$.² Define the vector $\mathbf{y} = \mathbf{z} - \mathbf{b}$ and the scalar $d = \mathbf{z} \cdot \mathbf{y}$. Next, we proceed with two claims.

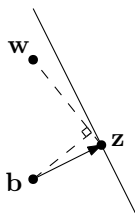
²The fact that such a closest vector exists follows from the fact that polyhedral cones are closed.

Claim 1. *The vector $\mathbf{b}$ satisfies $\mathbf{b} \cdot \mathbf{y} < d$.*



The above equation is equivalent to $\mathbf{b} \cdot (\mathbf{z} - \mathbf{b}) < \mathbf{z} \cdot (\mathbf{z} - \mathbf{b})$ which we may rearrange to get $0 < (\mathbf{z} - \mathbf{b}) \cdot (\mathbf{z} - \mathbf{b})$. This holds because $\mathbf{z} - \mathbf{b}$ is nonzero.

Claim 2. *For every $\mathbf{w} \in C$ we have $\mathbf{w} \cdot \mathbf{y} \geq d$.*



Suppose (for a contradiction) that $\mathbf{w} \in C$ satisfies $\mathbf{w} \cdot \mathbf{y} < d$. This assumption is equivalent to $\mathbf{w} \cdot (\mathbf{z} - \mathbf{b}) < \mathbf{z} \cdot (\mathbf{z} - \mathbf{b})$. Rearranging, we find $(\mathbf{w} - \mathbf{z}) \cdot (\mathbf{z} - \mathbf{b}) < 0$, or, equivalently, $(\mathbf{w} - \mathbf{z}) \cdot (\mathbf{b} - \mathbf{z}) > 0$. This last equation tells us that the vectors $\mathbf{w} - \mathbf{z}$ and $\mathbf{b} - \mathbf{z}$ make an acute angle. By the convexity of C , the line segment from $\mathbf{w}$ to $\mathbf{z}$ (as shown in the figure) must also be contained in C . However, this contradicts the assumption that $\mathbf{z}$ is a closest point in C to $\mathbf{b}$. This contradiction establishes the claim.

If there exists $\mathbf{w} \in C$ with $\mathbf{w} \cdot \mathbf{y} < 0$, then by taking large multiples of $\mathbf{w}$ we can find points in C which have arbitrarily small dot products with $\mathbf{y}$. However, this contradicts the second claim. We conclude that every point in C has nonnegative dot product with $\mathbf{y}$. Since each column of the matrix A is contained in C (if we choose $\mathbf{x}$ to be a vector with i^{th} entry 1 and all other entries 0, then $A\mathbf{x}$ is the i^{th} column of A), we conclude that $\mathbf{y}^T A \geq 0$. To complete the proof, we only need to show that $\mathbf{y} \cdot \mathbf{b} < 0$. To see this, we apply the second claim to the vector $\mathbf{0}$ (which is in every polyhedral cone) and apply the first claim as follows:

$$0 = \mathbf{0} \cdot \mathbf{y} \geq d > \mathbf{b} \cdot \mathbf{y}. \quad \square$$

With the Farkas Lemma in hand, we are now ready to prove the strong duality theorem for linear programming.

Proof of Theorem B.2. Assume that A is an $m \times n$ matrix, $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^n$ and consider the following matrix equation (here we regard $\mathbf{x}, \mathbf{q} \in \mathbb{R}^n$ and $\mathbf{y}, \mathbf{p} \in \mathbb{R}^m$ and $t \in \mathbb{R}$ as variables):

$$\begin{bmatrix} A & -I & 0 & 0 & \mathbf{0} \\ 0 & 0 & A^T & I & \mathbf{0} \\ \mathbf{c}^T & \mathbf{0}^T & -\mathbf{b}^T & \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{p} \\ \mathbf{y} \\ \mathbf{q} \\ t \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{c} \\ 0 \end{bmatrix}.$$

First suppose there exists a solution to this equation with all of our variables nonnegative: $\mathbf{x}, \mathbf{p}, \mathbf{y}, \mathbf{q}, t \geq 0$. The first row of the matrix in our equation gives us $A\mathbf{x} - \mathbf{p} = \mathbf{b}$ so $A\mathbf{x} \geq \mathbf{b}$ and $\mathbf{x}$ is a feasible solution to the LP. Similarly, the second row gives us $A^T \mathbf{y} + \mathbf{q} = \mathbf{c}$ so $A^T \mathbf{y} \leq \mathbf{c}$ and $\mathbf{y}$ is a feasible solution to the dual. Now the third row gives us $\mathbf{c}^T \mathbf{x} - \mathbf{b}^T \mathbf{y} + t = 0$ so our feasible solutions satisfy $\mathbf{c}^T \mathbf{x} \leq \mathbf{y}^T \mathbf{b}$. This combines with weak duality to yield the desired result.

If there does not exist a nonnegative solution to the above matrix equation, then by the Farkas Lemma, there is a vector $[\mathbf{u}^T \quad \mathbf{v}^T \quad \lambda]$ so that

$$[\mathbf{u}^T \quad \mathbf{v}^T \quad \lambda] \begin{bmatrix} A & -I & 0 & 0 & \mathbf{0} \\ 0 & 0 & A^T & I & \mathbf{0} \\ \mathbf{c}^T & \mathbf{0}^T & -\mathbf{b}^T & \mathbf{0} & 1 \end{bmatrix} \geq 0$$

and

$$\begin{bmatrix} \mathbf{u}^\top & \mathbf{v}^\top & \lambda \end{bmatrix} \begin{bmatrix} \mathbf{b} \\ \mathbf{c} \\ 0 \end{bmatrix} < 0.$$

We will show that this leads us to a contradiction (thus completing the proof). By separating the inequality coming from the above matrix product into five inequalities (one for each column) and doing some elementary manipulations (e.g. multiplying by -1 and transposing), we obtain the following equivalent system of inequalities:

$$(-\mathbf{u})^\top A \leq \lambda \mathbf{c}^\top,$$

$$(-\mathbf{u}) \geq \mathbf{0},$$

$$A\mathbf{v} \geq \lambda \mathbf{b},$$

$$\mathbf{v} \geq \mathbf{0},$$

$$\lambda \geq 0,$$

$$(-\mathbf{u})^\top \mathbf{b} > \mathbf{v}^\top \mathbf{c}.$$

If $\lambda > 0$, then our first two equations imply that the vector $\mathbf{y} = \frac{1}{\lambda}(-\mathbf{u})$ is a feasible solution for the dual, and the next two imply that $\mathbf{x} = \frac{1}{\lambda}\mathbf{v}$ is a feasible solution for the LP. However, the last equation gives us $\mathbf{y}^\top \mathbf{b} = \frac{1}{\lambda}(-\mathbf{u})^\top \mathbf{b} > \frac{1}{\lambda}\mathbf{v}^\top \mathbf{c} = \mathbf{x}^\top \mathbf{c}$, which contradicts weak duality.

In the remaining case, $\lambda = 0$ and we will show that either the primal or dual problem is unbounded (which contradicts the assumption that they are both feasible). Since $(-\mathbf{u})^\top \mathbf{b} > \mathbf{v}^\top \mathbf{c}$, we must have either $(-\mathbf{u})^\top \mathbf{b} > 0$ or $\mathbf{v}^\top \mathbf{c} < 0$. In the latter case, choose a feasible solution $\mathbf{x}$ to the LP, choose a (large) positive integer N , and consider the vector $\mathbf{x}_N = \mathbf{x} + N\mathbf{v}$. This vector is feasible for the dual since it is non-negative and $A\mathbf{x}_N = A(\mathbf{x} + N\mathbf{v}) = A\mathbf{x} + N(A\mathbf{v}) \geq \mathbf{c} + \mathbf{0}$. Furthermore, since $\mathbf{v}^\top \mathbf{c} < 0$, by choosing N sufficiently large, we can make $\mathbf{x}_N^\top \mathbf{c} = \mathbf{x}^\top \mathbf{c} + N(\mathbf{v}^\top \mathbf{c})$ as small as we desire. It follows that our LP is unbounded, giving us a contradiction. On the other hand, if $(-\mathbf{u})^\top \mathbf{b} > 0$, then a similar argument shows that the dual is unbounded (choose a feasible vector $\mathbf{y}$; then for every $N > 0$ the vector $\mathbf{y}_N = \mathbf{y} + N(-\mathbf{u})$ is feasible, and by choosing N sufficiently large, $\mathbf{b}^\top \mathbf{y}_N$ can be made as large as we desire). This completes the proof. $\square$

Appendix C

Nash Equilibrium in High Dimensions

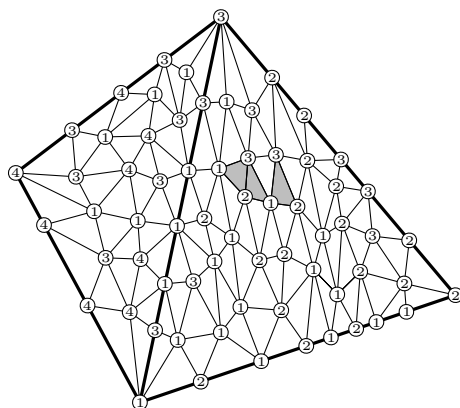


Figure C.1. Sperner in 3D

In Chapter 9 we introduced a number of sophisticated (and beautiful!) ideas in our proof of Nash's Equilibrium Theorem. However, we focused our discussion on the special case of 2×2 matrices, so in many cases we omitted details about how these ideas work in higher dimensions. In this appendix we fill in these details so as to give a complete proof of Nash's amazing theorem.

Our proof of Nash's Equilibrium Theorem has four key ingredients as shown again in Figure C.2. Each of these corresponds to a lemma or theorem which was introduced in Chapter 9 but only proved for small dimensions. Our goal for this appendix is to prove these four results (Lemma 9.5, Theorem 9.11, Lemma 9.17, and Lemma 9.22), thus completing the full proof.

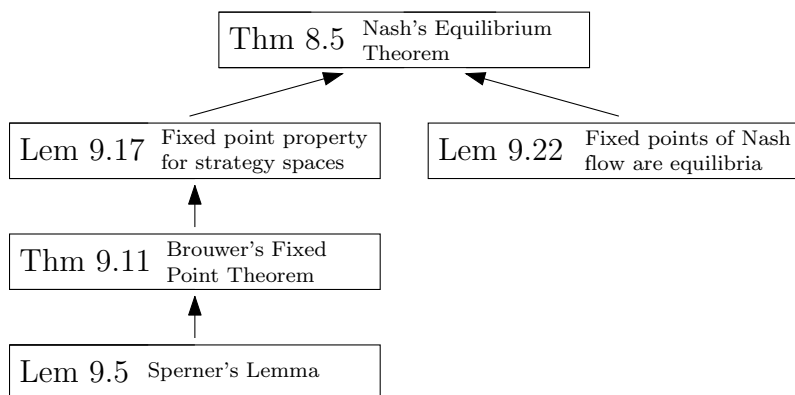


Figure C.2. Steps in proving Nash's Theorem

The first ingredient in our proof of Nash's Equilibrium Theorem is Sperner's Lemma. We proved the 1D and 2D versions of this (Lemmas 9.2 and 9.3) in Chapter 9, but we omitted the proof of the general version. Next we restate this general version and give an inductive proof of it. This argument is a generalization of that for the 2D version.

Lemma 9.5 (Sperner). *Consider an n -simplex given as the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_{n+1}$ that has been subdivided into small simplexes. Suppose that each vertex is given a label from $1, 2, \dots, n + 1$ satisfying the following rule:*

- *If a vertex has label i , then it does not lie in the convex hull of the points $\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}_{i+1}, \dots, \mathbf{x}_{n+1}$.*

Then the number of small simplexes with all $n + 1$ labels is odd.

Proof. We proceed by induction on n . As a base case, we have already proved this for $n = 1$ (and also $n = 2$). So, for the inductive step, we may assume $n \geq 3$ and that the result holds for $n - 1$.

As before, we will imagine each small simplex as a room in a giant house. The walls of our house are the $(n - 1)$ -simplexes which are contained in a room. So each room (as an n -simplex) will have $n + 1$ vertices and $n + 1$ walls. Each wall contains all but one of these vertices, and we could express it as the convex hull of the other n vertices.

Also as in the 2D version, we will be adding doors to some of these walls. Add a door in each wall which has one vertex of each label $1, 2, \dots, n$. Let's think about what a room with a door will look like. This door lies on a wall with one vertex of each label $1, 2, \dots, n$. If the remaining vertex for this room has label $n + 1$, then this room will have just one door. On the other hand, if the remaining vertex has label k for some $1 \leq k \leq n$, then this room will have a second door which contains this other vertex of label k but leaves out the first one of label k . It's easy to see that these will be the only two doors.

So, just as in the 2D case, we can imagine the pathways which go through our house. Some rooms are not in a pathway and we will ignore them, some pathways form cycles, and these too will be ignored. The pathways we will be interested in are those which have a beginning and an end. Here, the first and last door must either be doors to the outside (along the outer n -simplex which is the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_n$) or doors into rooms which have just one door. Since each such pathway has two ends, it follows that the total number of doors to the outside world plus the number of rooms with one door is even. By applying our lemma inductively to the subdivided $(n - 1)$ -simplex with vertices $\mathbf{x}_1, \dots, \mathbf{x}_n$, we find that the number of doors to the outside will be odd. So, the number of rooms with exactly one door must be odd—but this is precisely the number of small simplices with vertices of all labels! $\square$

The next step in our proof of Nash's Equilibrium Theorem is to establish Brouwer's Fixed Point Theorem. This result tells us that the simplex

$$\Delta_n = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1, \dots, x_n \geq 0 \text{ and } x_1 + \dots + x_n \leq 1\}$$

has the fixed point property. In Chapter 9 we used the 2D Sperner Lemma to prove that the triangle Δ_2 has the fixed point property. More generally, we can use the n -dimensional version of Sperner's Lemma to prove that the simplex Δ_n has the fixed point property.

Theorem 9.11 (Brouwer). *The simplex Δ_n has the fixed point property for every $n \geq 1$.*

Proof. Consider an arbitrary continuous function $f : \Delta_n \rightarrow \Delta_n$. Similar to the $n = 2$ case, we shall assign every point in Δ_n a label from the set $1, 2, \dots, n + 1$. Let $\mathbf{x} = (x_1, \dots, x_n) \in \Delta_n$ and assume $f(\mathbf{x}) = \mathbf{x}' = (x'_1, \dots, x'_n)$. We label $\mathbf{x}$ according to the following rule:

$$\mathbf{x} \text{ gets label } \begin{cases} k & \text{if } k \text{ is the smallest index for which } x'_k < x_k, \\ n + 1 & \text{if } x'_i \geq x_i \text{ for every } 1 \leq i \leq n. \end{cases}$$

As before, the only way for a point $\mathbf{x}$ to have points of all $n + 1$ labels arbitrarily close is for $\mathbf{x}$ to be a fixed point. To see this, observe that if the k^{th} coordinate of $f(\mathbf{x})$ is less than that of $\mathbf{x}$, then (by continuity) the same will hold for all points sufficiently close to $\mathbf{x}$, so no point sufficiently close to $\mathbf{x}$ has label $n + 1$. On the other hand, if the k^{th} coordinate of $f(\mathbf{x})$ is greater than that of $\mathbf{x}$, then (by continuity) the same will hold for all points sufficiently close to $\mathbf{x}$, so no point sufficiently close to $\mathbf{x}$ has label k .

Claim. *For every $\ell \geq 0$, there is either a fixed point or a point in Δ_n with distance $\leq \frac{1}{2^\ell}$ to points of every label.*

For $1 \leq i \leq n$ let $\mathbf{x}_i$ be the vertex of Δ_n with a 1 in the i^{th} coordinate and 0 elsewhere, and define $\mathbf{x}_{n+1}$ to be the origin (i.e. the other vertex of Δ_n). Subdivide Δ_n into small simplices so that the center of each of these small simplexes is at a distance $\leq \frac{1}{2^\ell}$ to each of its vertices. Now considering just the labels on the vertices of this subdivision, assuming none of them is a fixed point (in which case we are finished), we see that it satisfies the assumptions of Sperner's Lemma. Indeed, any point which lies in the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_n$ has coordinates summing to 1 so it cannot get the label $n + 1$; similarly, any point which lies in the convex hull of $\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}_{i+1}, \dots, \mathbf{x}_{n+1}$ with $1 \leq i \leq n$ will have i^{th} coordinate equal to zero so it cannot have the label i . Now by Sperner's Lemma we are guaranteed the existence of a small simplex with one vertex of each label, and the center point satisfies the claim.

The remainder of the argument is exactly as in the proof of Theorem 9.10. Namely, we choose a sequence of points $\mathbf{x}_1, \mathbf{x}_2, \dots$ so that $\mathbf{x}_k$ has points of all three labels within distance $\leq \frac{1}{2^k}$. Since Δ_n is compact, we

may choose a convergent subsequence of this, and its limit point $\mathbf{x}^*$ will be a fixed point. $\square$

The next step in our progression toward the proof of the Equilibrium Theorem is to prove that the strategy space $S_{m,n}$ has the fixed point property. Recall that the strategy space $S_{m,n}$ consists of all points $(p_1, \dots, p_{m-1}, q_1, \dots, q_{n-1})$ in $\mathbb{R}^{m+n-2}$ which have all entries nonnegative and satisfy $\sum_{i=1}^{m-1} p_i \leq 1$ and $\sum_{j=1}^{n-1} q_j \leq 1$. The purpose here is that setting $p_m = 1 - \sum_{i=1}^{m-1} p_i$ and $q_n = 1 - \sum_{j=1}^{n-1} q_j$ gives us a natural correspondence between points in $S_{m,n}$ and pairs of mixed strategies $[p_1 \ \cdots \ p_m]$

for Rose and $\begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$ for Colin. In Chapter 9, we proved that $S_{2,2}$ has the fixed point property (Corollary 9.16) by showing it is topologically equivalent to Δ_2 . Next we will generalize this to prove that $S_{m,n}$ has the fixed point property by showing it is topologically equivalent to Δ_{m+n-2} .

Lemma 9.17. *For every pair of positive integers m, n , the strategy space $S_{m,n}$ has the fixed point property.*

Proof. Define S to be the unit square. The proof of Corollary 9.16 introduces a pair of continuous inverse functions $g : \Delta_2 \rightarrow S$ and $g^{-1} : S \rightarrow \Delta_2$. We will use these to define a continuous function $h : \Delta_{m+n-2} \rightarrow S_{m,n}$ which has continuous inverse $h^{-1} : S_{m,n} \rightarrow \Delta_{m+n-2}$. Once we have established the existence of these functions h and h^{-1} , we will have demonstrated that $S_{m,n}$ is topologically equivalent to Δ_{m+n-2} , and then the desired result follows from Proposition 9.15.

To define h , let $(x_1, \dots, x_{m-1}, y_1, \dots, y_{n-1}) \in \Delta_{m+n-2}$ (so these coordinates are all nonnegative and sum to at most 1) and let $x = \sum_{i=1}^{m-1} x_i$ and $y = \sum_{j=1}^{n-1} y_j$. Note that (x, y) will be a point in Δ_2 . Let $g(x, y) = (x', y')$, and note that since $(x', y') \in S$, we have $0 \leq x' \leq 1$ and $0 \leq y' \leq 1$. Now we define

$$h(x_1, \dots, x_{m-1}, y_1, \dots, y_{n-1}) = \left(\frac{x'}{x} x_1, \dots, \frac{x'}{x} x_{m-1}, \frac{y'}{y} y_1, \dots, \frac{y'}{y} y_{n-1} \right).$$

In the case that $x = 0$, we must have $x_i = 0$ and $x' = 0$ (by the definition of g) and we treat the expression $\frac{x'}{x} x_i$ as 0. Similarly, if $y = 0$, we treat

$\frac{y'}{y} y_i$ as 0. Observe that the sum of the first $m - 1$ coordinates of this new point will be $x' \leq 1$ and the sum of the last $n - 1$ coordinates will be $y' \leq 1$. It follows from our definitions that h is a continuous function from Δ_{m+n-2} to $S_{m,n}$, as desired.

Next we construct a function h^{-1} which we will prove to be the inverse of h . For this purpose we let $(x_1, \dots, x_{m-1}, y_1, \dots, y_{n-1}) \in S_{m,n}$ (so these coordinates are nonnegative and both the first $m - 1$ and last $n - 1$ sum to at most 1) and let $x = \sum_{i=1}^{m-1} x_i$ and $y = \sum_{j=1}^{n-1} y_j$. Now let $g^{-1}(x, y) = (x', y')$ and note that since $(x', y') \in \Delta_2$, we have $x', y' \geq 0$ and $x' + y' \leq 1$. Define

$$h^{-1}(x_1, \dots, x_{m-1}, y_1, \dots, y_{n-1}) = \left(\frac{x'}{x} x_1, \dots, \frac{x'}{x} x_{m-1}, \frac{y'}{y} y_1, \dots, \frac{y'}{y} y_{n-1} \right).$$

As before, if, say, $x = 0$, we will have $x' = 0$ and $x_i = 0$ and we treat $\frac{x'}{x} x_i$ as 0. Observe that now the first $m - 1$ coordinates sum to x' and the last $n - 1$ sum to y' so in total the coordinate sum will be $x' + y' \leq 1$. It follows that h^{-1} is a continuous mapping from $S_{m,n}$ to Δ_{m+n-2} , as desired.

The proof that these functions h and h^{-1} are inverses follows from our definitions and the fact that g and g^{-1} are inverse. $\square$

The last ingredient in our proof of Nash's Equilibrium Theorem is Nash flow. In Chapter 9, we defined for every 2×2 matrix game A the Nash flow function $f_A : S_{2,2} \rightarrow S_{2,2}$. The key properties of this function are that it is continuous and every fixed point corresponds to a Nash equilibrium of A . Our goal here will be to define for an arbitrary $m \times n$ matrix game A a Nash flow function $f_A : S_{m,n} \rightarrow S_{m,n}$ which has the same properties.

To define Nash flow, fix a matrix game A and let R and C denote Rose's and Colin's payoff matrices. Consider the point $(p_1, \dots, p_{m-1}, q_1, \dots, q_{n-1}) \in S_{m,n}$ which corresponds to the mixed strategies $\mathbf{p} =$

$$\begin{bmatrix} p_1 & \cdots & p_m \end{bmatrix} \text{ for Rose and } \mathbf{q} = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix} \text{ for Colin. Define the numbers } a_1, \dots, a_m \text{ by the equation } R\mathbf{q} = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \text{ and the numbers } b_1, \dots, b_n \text{ by}$$

$\mathbf{p}C = [b_1 \ \cdots \ b_n]$. So, when Colin plays $\mathbf{q}$, Rose gets a payoff of a_i for choosing row i , and similarly, when Rose plays $\mathbf{p}$, Colin gets a payoff of b_j for choosing column j .

Define $a = \sum_{i=1}^m a_i p_i = \mathbf{p}R\mathbf{q}$ and note that a is precisely Rose's present payoff when playing her mixed strategy $\mathbf{p}$ against Colin's strategy of $\mathbf{q}$. Now we will have Rose modify her mixed strategy to do better against $\mathbf{q}$ (if possible). To do so, we define a new mixed strategy $\mathbf{p}' = [p'_1 \ \cdots \ p'_m]$ according to the rule

$$p'_i = \frac{p_i + (a_i - a)^+}{1 + K},$$

where $K = (a_1 - a)^+ + \cdots + (a_m - a)^+$. It follows immediately from the definitions that these new values p'_i are nonnegative and sum to 1 so $\mathbf{p}' = [p'_1 \ \cdots \ p'_m]$ is indeed a new mixed strategy for Rose.

Similarly, we define a new mixed strategy $\mathbf{q}' = \begin{bmatrix} q'_1 \\ \vdots \\ q'_n \end{bmatrix}$ for Colin by

setting $b = \sum_{j=1}^n b_j q_j = \mathbf{p}C\mathbf{q}$ and then defining

$$q'_j = \frac{q_j + (b_j - b)^+}{1 + L},$$

where $L = (b_1 - b)^+ + \cdots + (b_n - b)^+$. Putting this together, we define the Nash flow by

$$f_A(p_1, \dots, p_{m-1}, q_1, \dots, q_{n-1}) = (p'_1, \dots, p'_{m-1}, q'_1, \dots, q'_{n-1}).$$

We leave to the reader the task of verifying the following essential properties of f_A :

- (1) If $\mathbf{p}$ is a best response to $\mathbf{q}$, then $\mathbf{p}' = \mathbf{p}$.
- (2) If $\mathbf{p}$ is not a best response to $\mathbf{q}$, then $\mathbf{p}' \neq \mathbf{p}$.

The following lemma is an immediate consequence of these properties.

Lemma 9.22. *For every $m \times n$ matrix A , the fixed points of Nash flow $f_A : S_{m,n} \rightarrow S_{m,n}$ are precisely those points that correspond to a Nash equilibrium of A .*

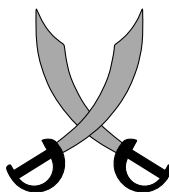
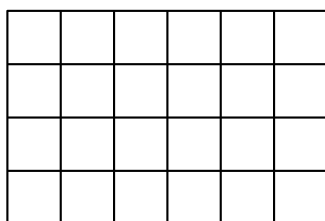
The proof of Nash's Equilibrium Theorem follows immediately from Lemmas 9.17 and 9.22 (as stated in Chapter 9). In short, to prove that

an arbitrary $m \times n$ matrix game A has a Nash equilibrium, we consider the Nash flow of A given by $f_A : S_{m,n} \rightarrow S_{m,n}$. Since Nash flow is continuous, Lemma 9.17 implies that f_A must have a fixed point. Then by Lemma 9.22 we deduce that A has a Nash equilibrium.

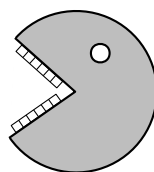
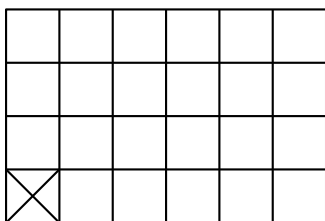
In fact, Nash's Equilibrium Theorem holds even more generally for n -player matrix games. That is, for every n -player matrix game A , there exists a mixed strategy for each player which together form an equilibrium in the sense that each player's mixed strategy is a best response to the choices of all the other players. Although this might sound like a vastly more complicated result, the proof is much the same. Once again the strategy space is topologically equivalent to Δ_ℓ for some ℓ , and once again we can define a (continuous) Nash flow function f_A from the strategy space to itself for which every fixed point is a Nash equilibrium.

Game Boards

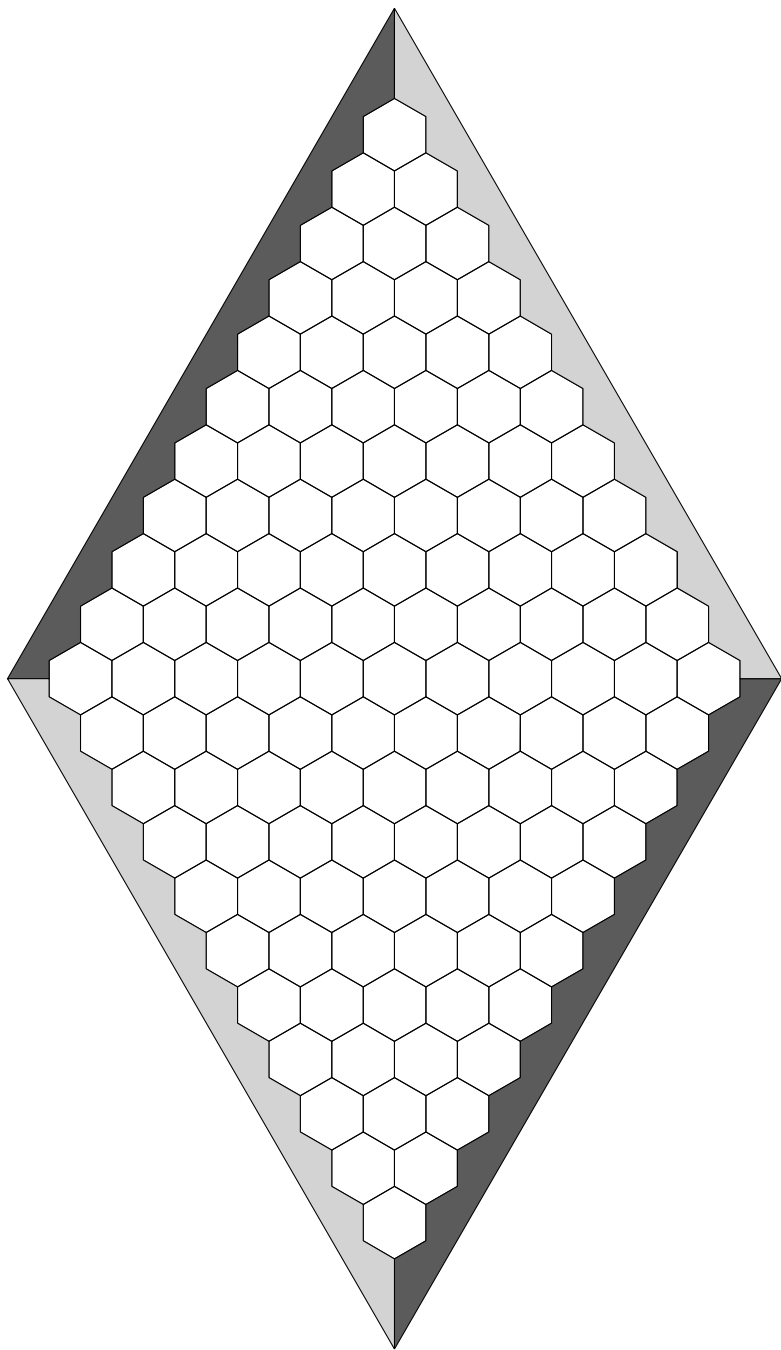
Chop

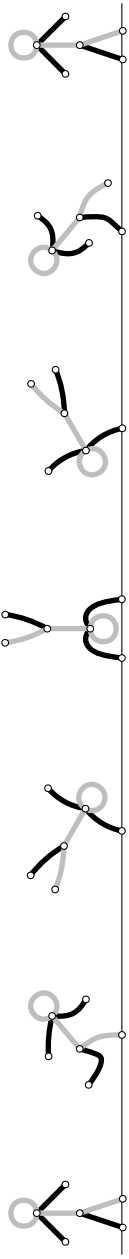


Chomp

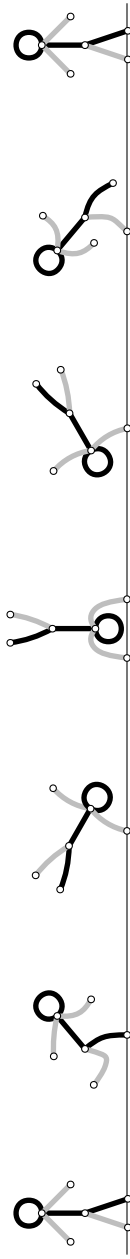


These game boards are available online at www.ams.org/bookpages/stml-80.

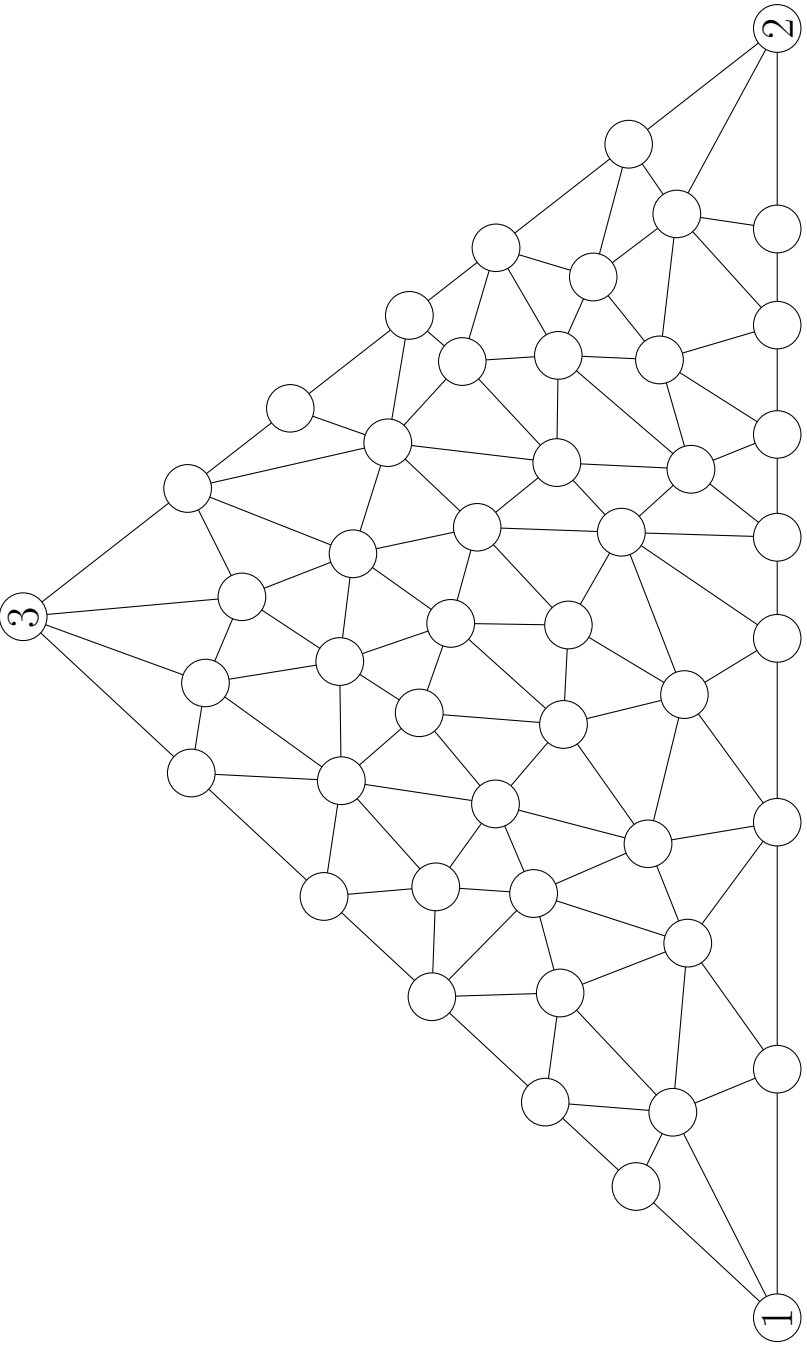




Louise (bLack) to play first



Richard (gRay) to play first



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This book offers a gentle introduction to the mathematics of both sides of game theory: combinatorial and classical. The combination allows for a dynamic and rich treatment of the subject united by a common theme of strategic reasoning.

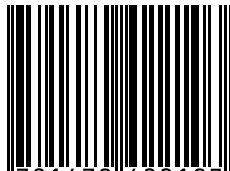
The first four chapters develop combinatorial game theory, beginning with an introduction to game trees and mathematical induction, then investigating the games of Nim and Hackenbush. The analysis of these games concludes with the cornerstones of the Sprague-Grundy Theorem and the Simplicity Principle.

The last eight chapters of the book offer a scenic journey through the mathematical highlights of classical game theory. This contains a thorough treatment of zero-sum games and the von Neumann Minimax Theorem, as well as a student-friendly development and proof of the Nash Equilibrium Theorem. The Folk Theorem, Arrow's voting paradox, evolutionary biology, cake cutting, and other engaging auxiliary topics also appear.

The book is designed as a textbook for an undergraduate mathematics class. With ample material and limited dependencies between the chapters, the book is adaptable to a variety of situations and a range of audiences. Instructors, students, and independent readers alike will appreciate the flexibility in content choices as well as the generous sets of exercises at various levels.



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